How to initialise a second class particle?

Joint with
Attila László Nagy

Márton Balázs

University of Bristol

PAD away day
12 June, 2017.

The models

- Simple exclusion
- Zero range
- Bricklayers

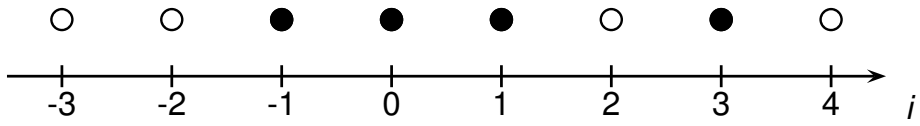
Hydrodynamics

The second class particle

Ferrari-Kipnis for TASEP

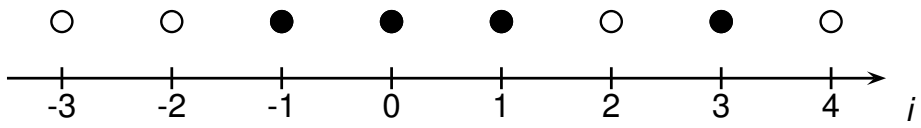
Let's generalise

Totally asymmetric simple exclusion



Bernoulli(ϱ) distribution; $\omega_i = 0$ or 1.

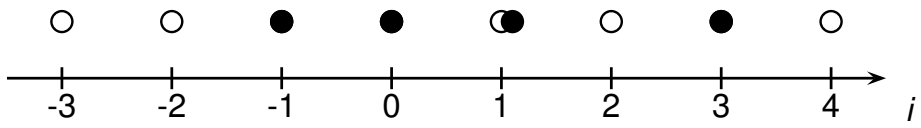
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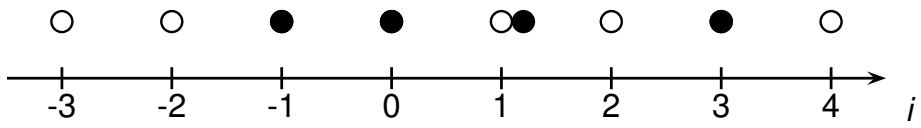
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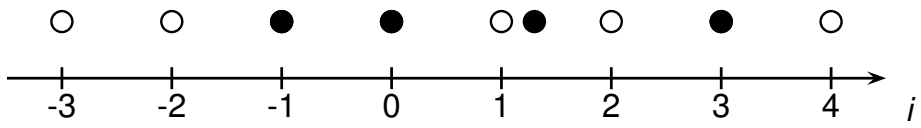
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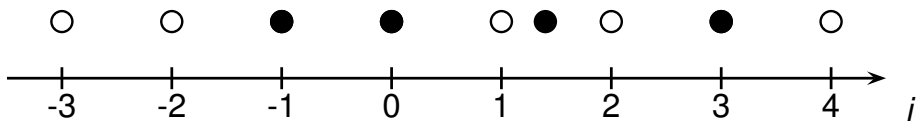
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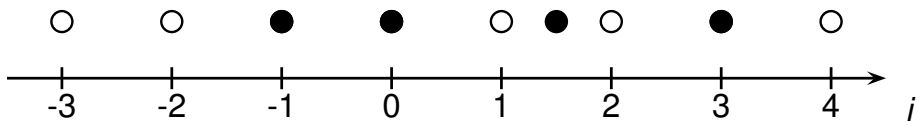
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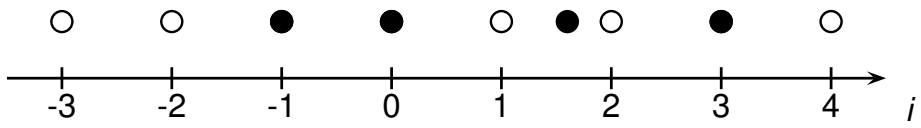
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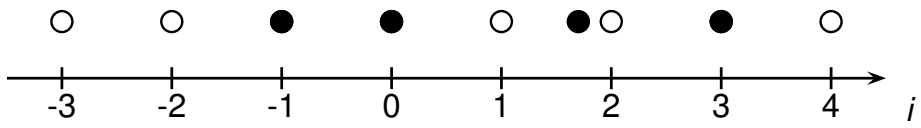
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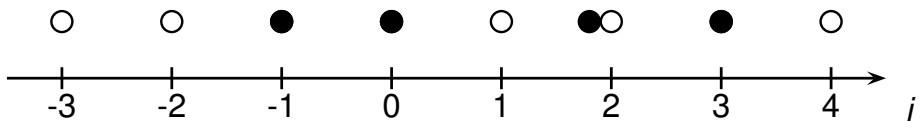
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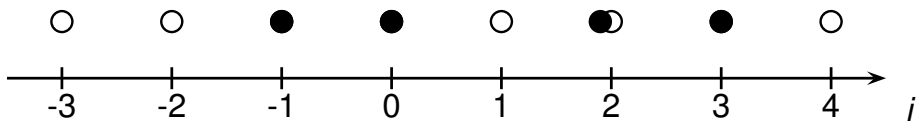
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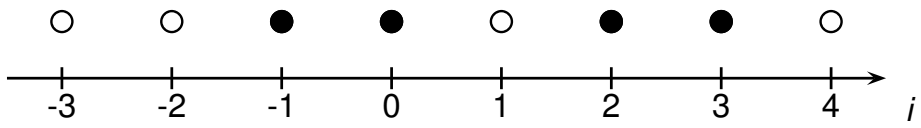
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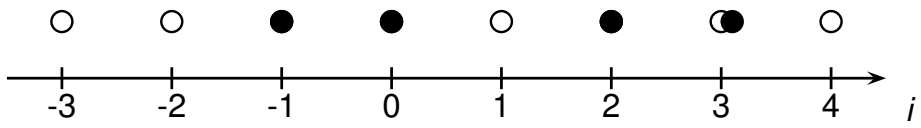
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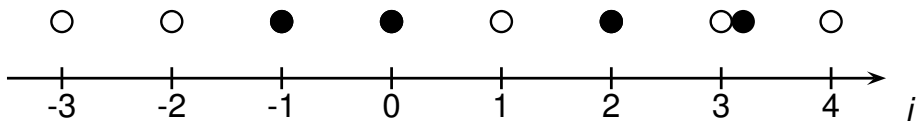
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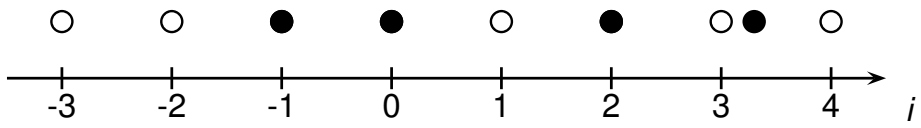
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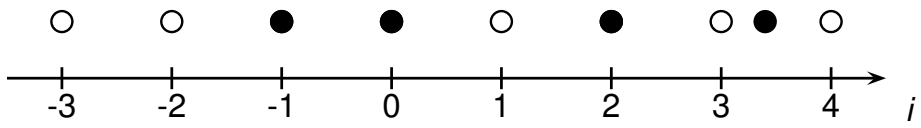
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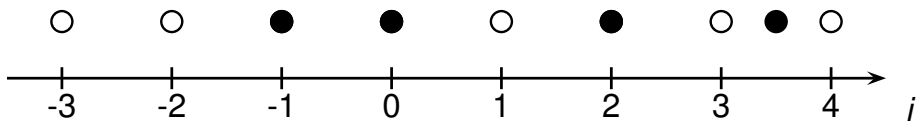
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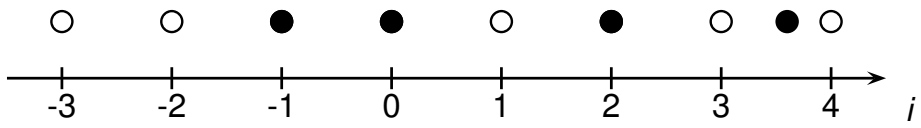
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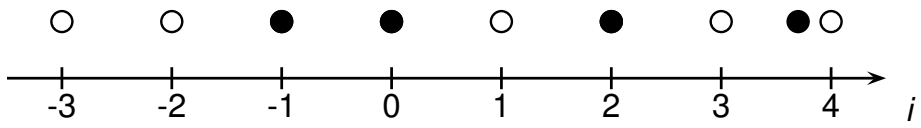
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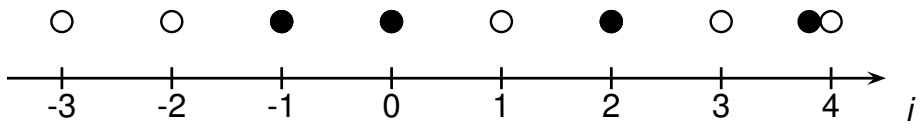
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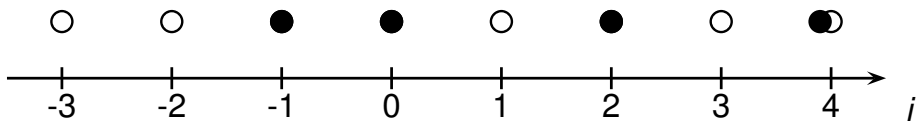
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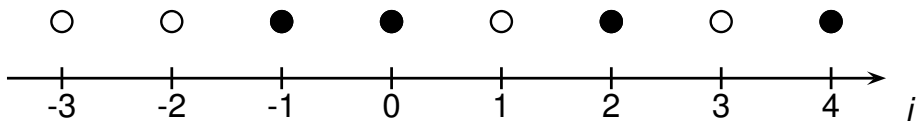
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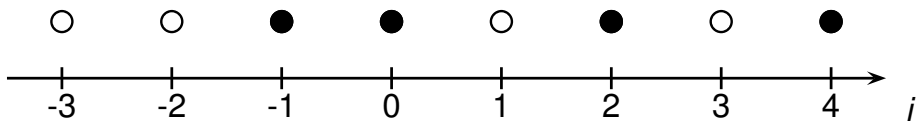
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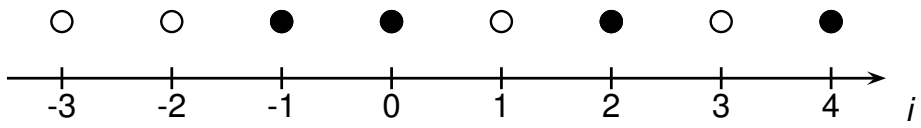
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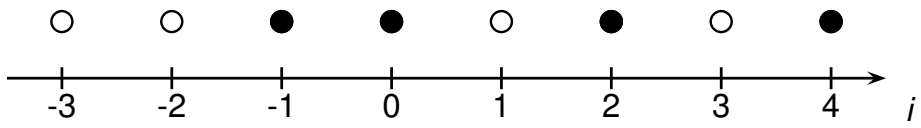
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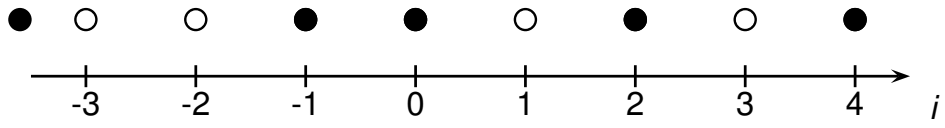
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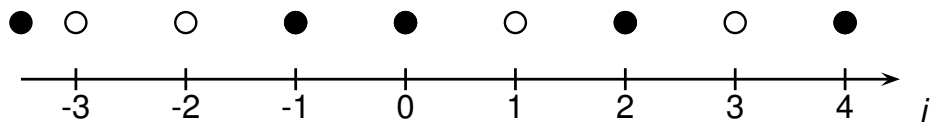
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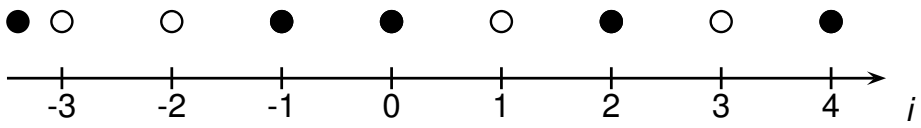
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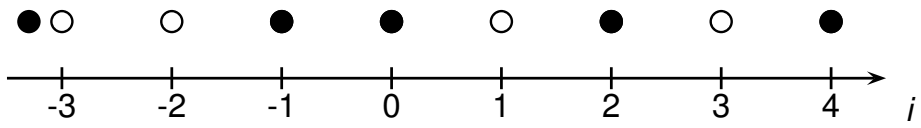
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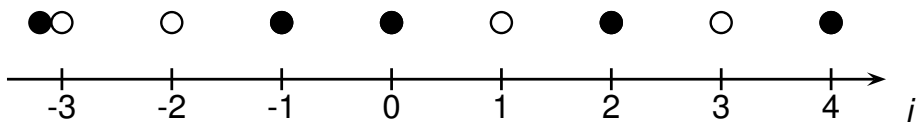
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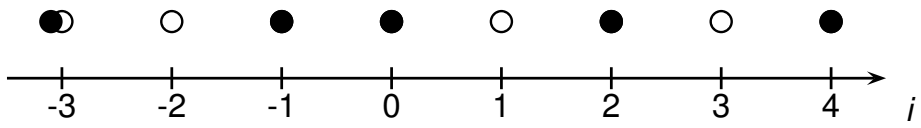
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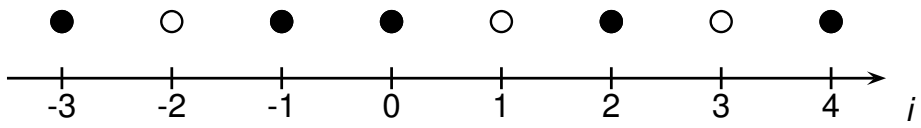
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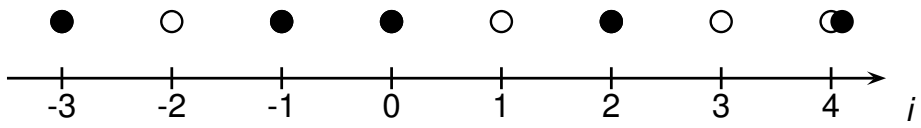
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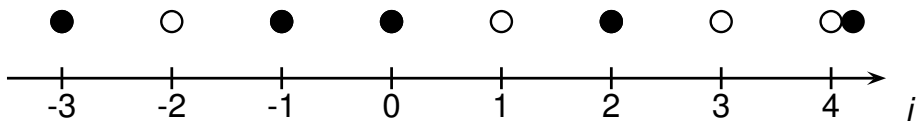
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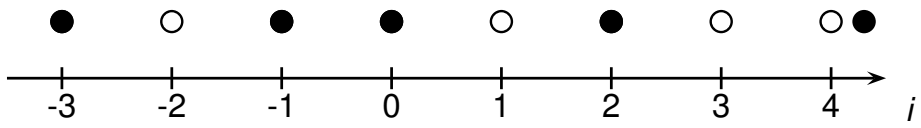
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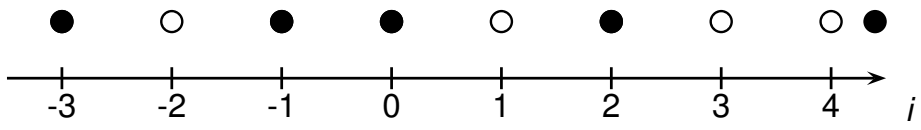
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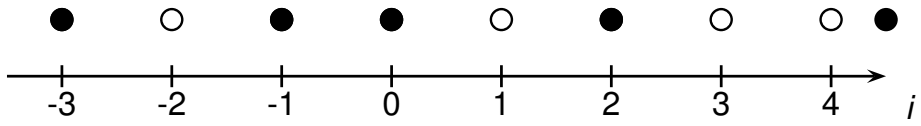
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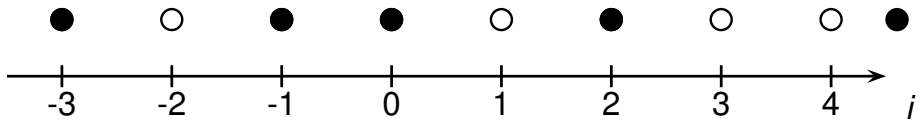
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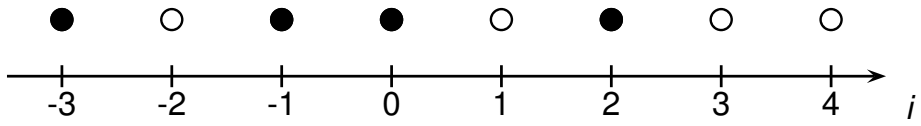
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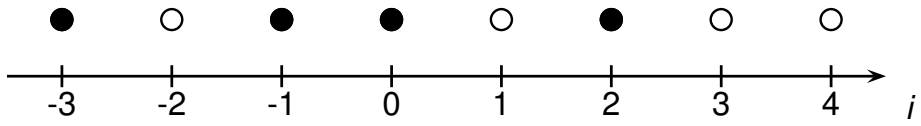
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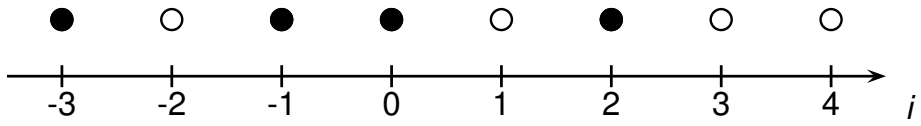
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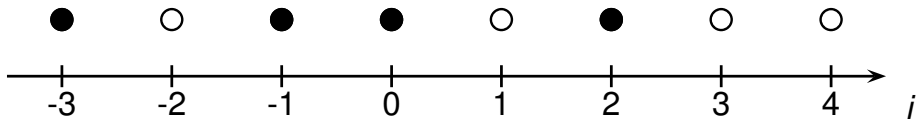
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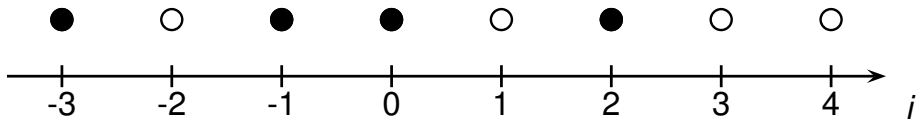
Totally asymmetric simple exclusion



Bernoulli(ϱ) distribution; $\omega_i = 0$ or 1.

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Totally asymmetric simple exclusion



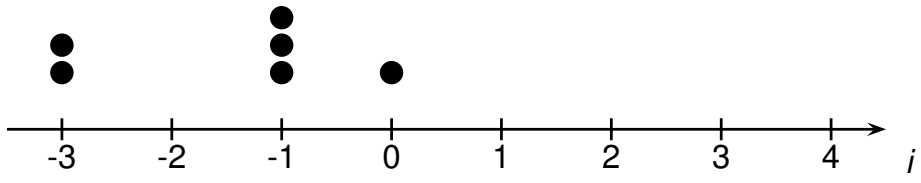
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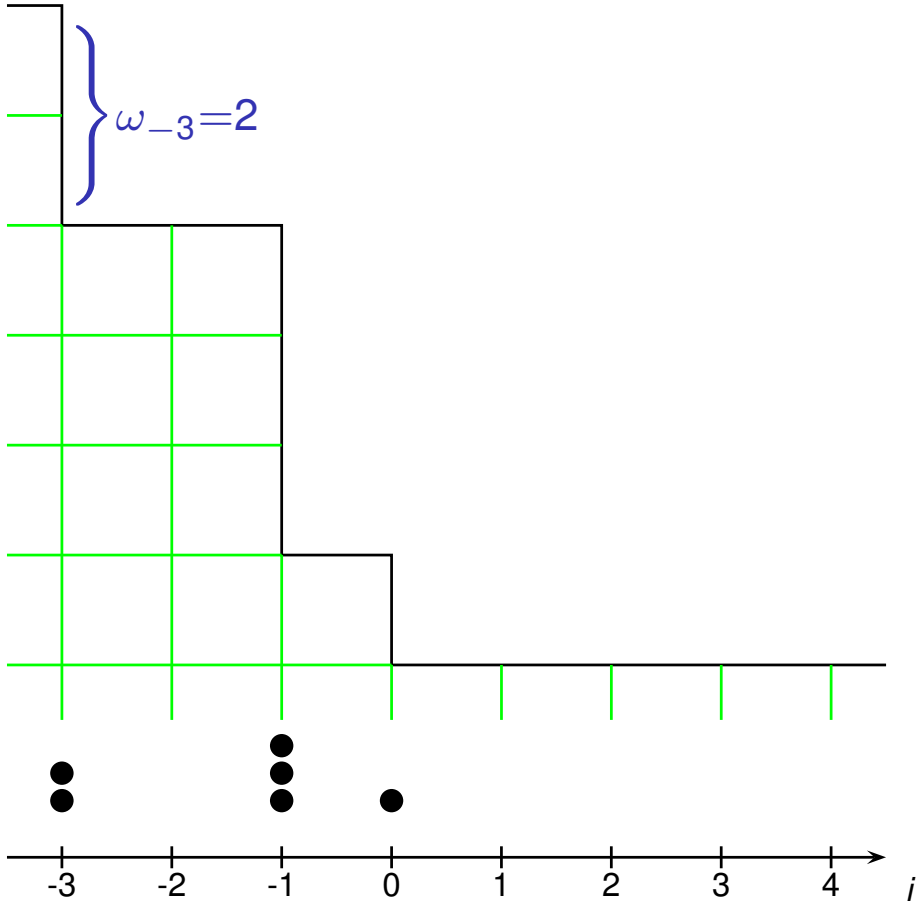
The Bernoulli(ϱ) distribution is time-stationary for any ($0 \leq \varrho \leq 1$). Any translation-invariant stationary distribution is a mixture of Bernoullis.

Totally asymmetric zero range process

$\omega_i \in \mathbb{Z}^+$

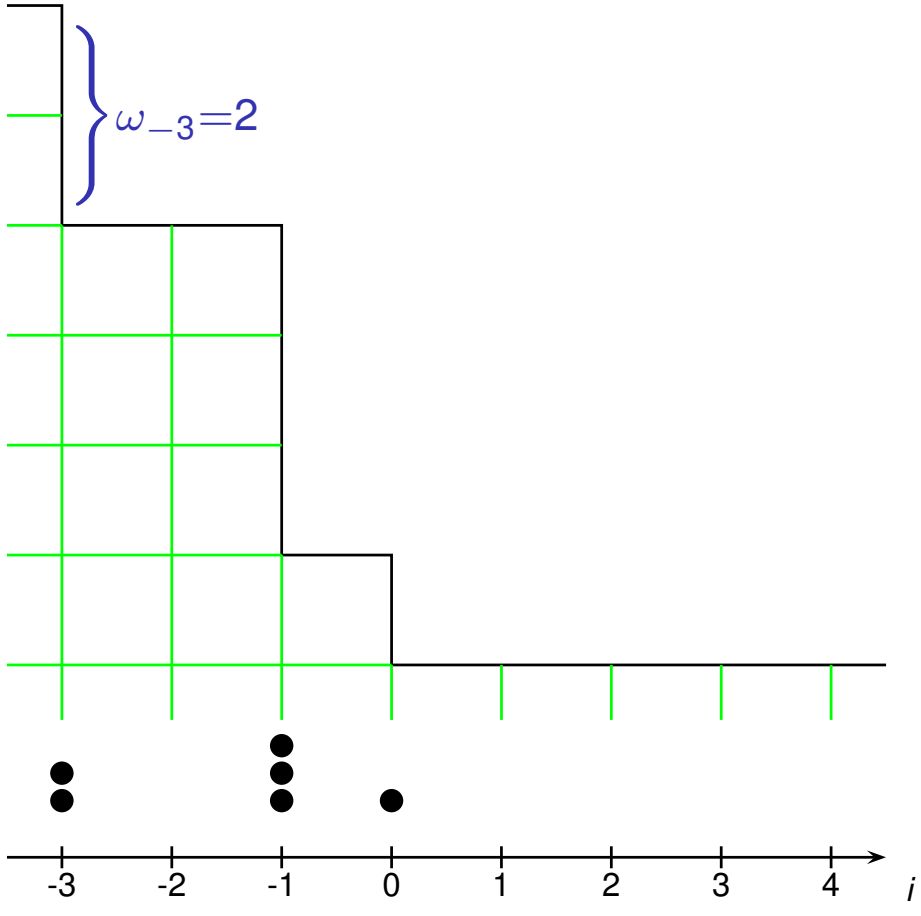


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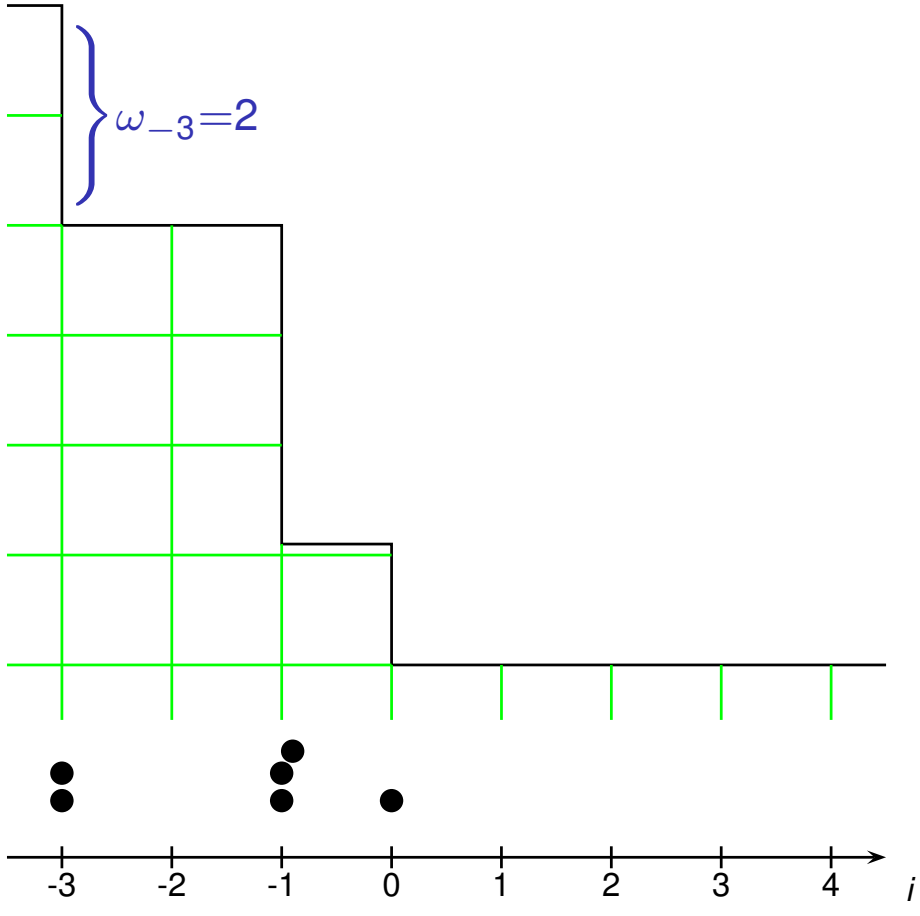
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Particles jump to the right with rate $r(\omega_i)$ (r non-decreasing).

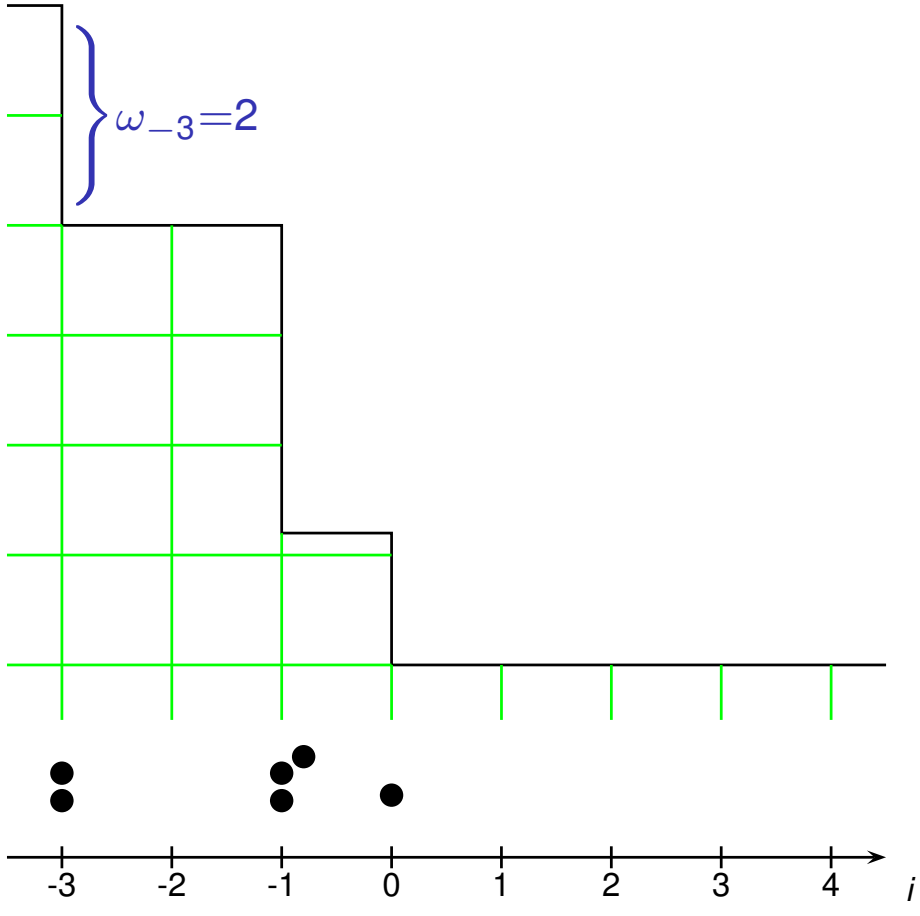
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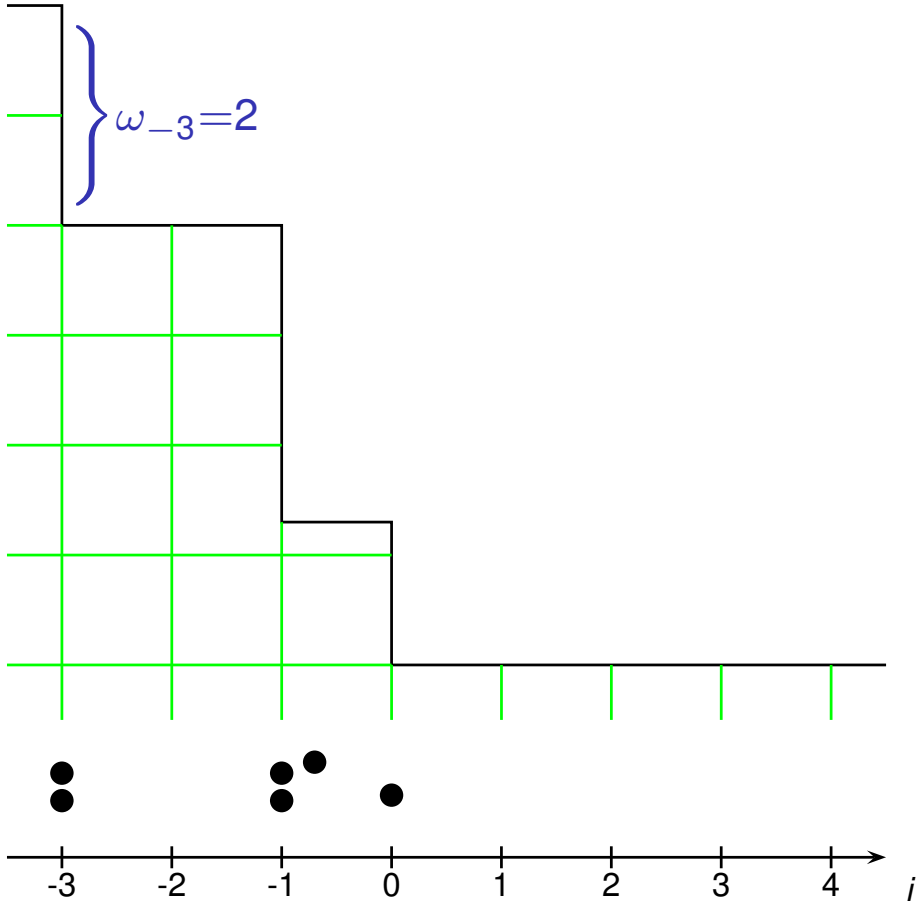
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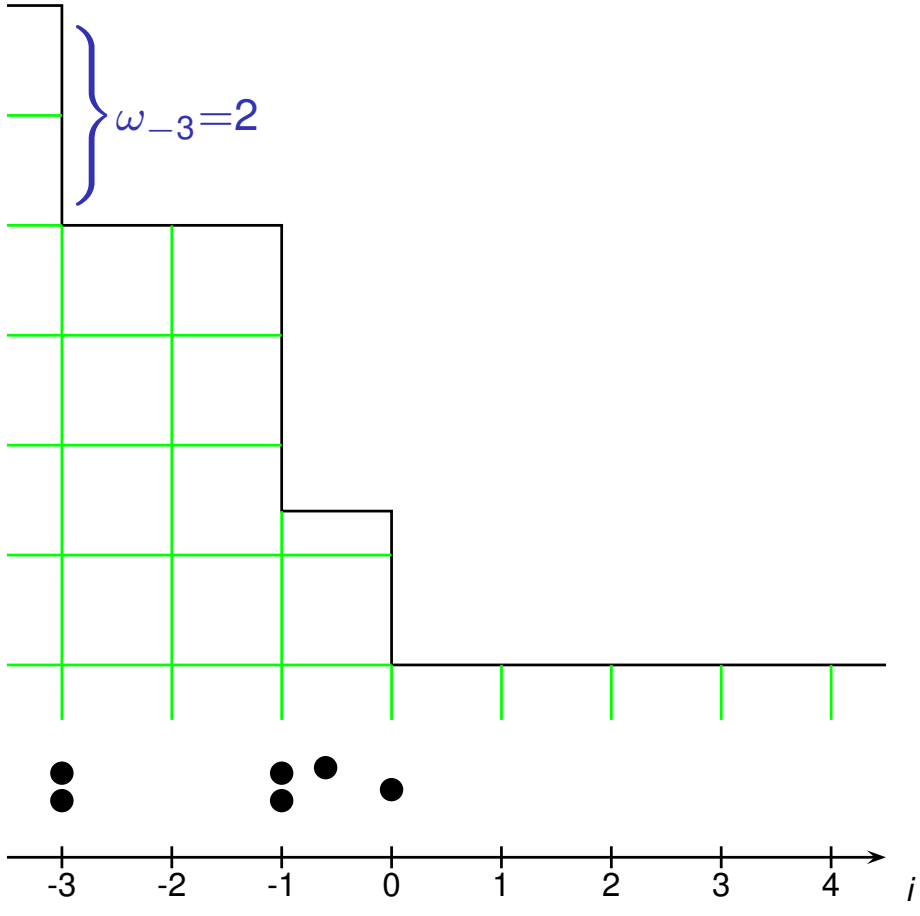
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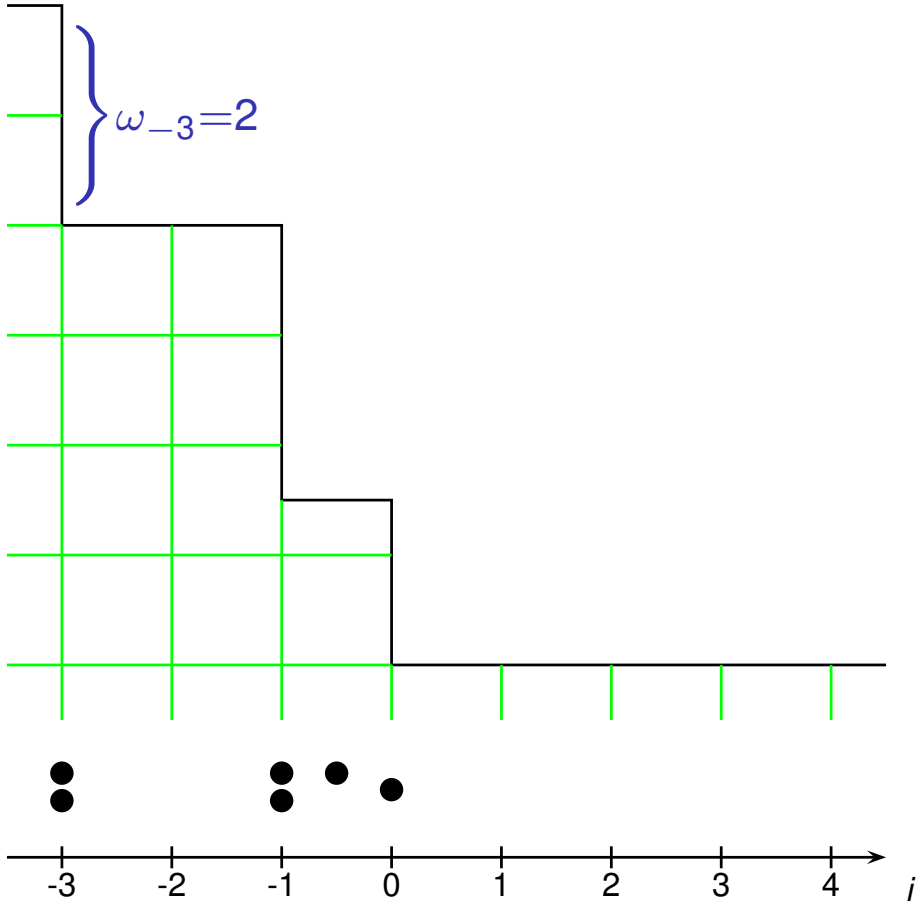
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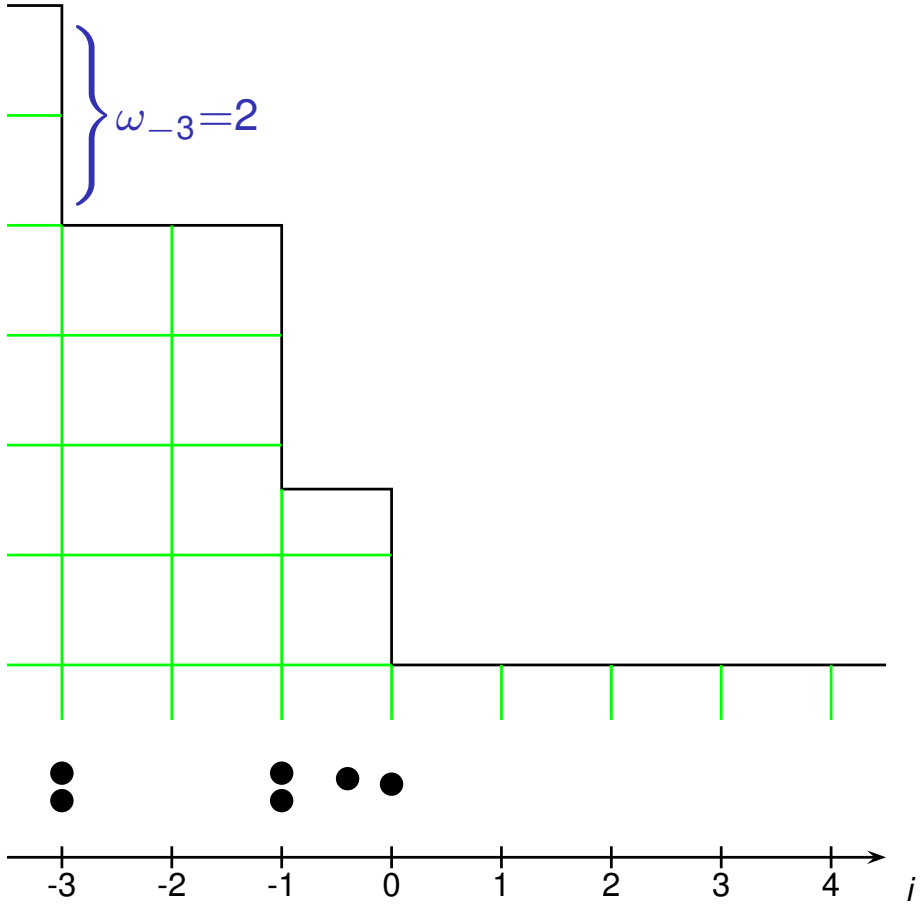
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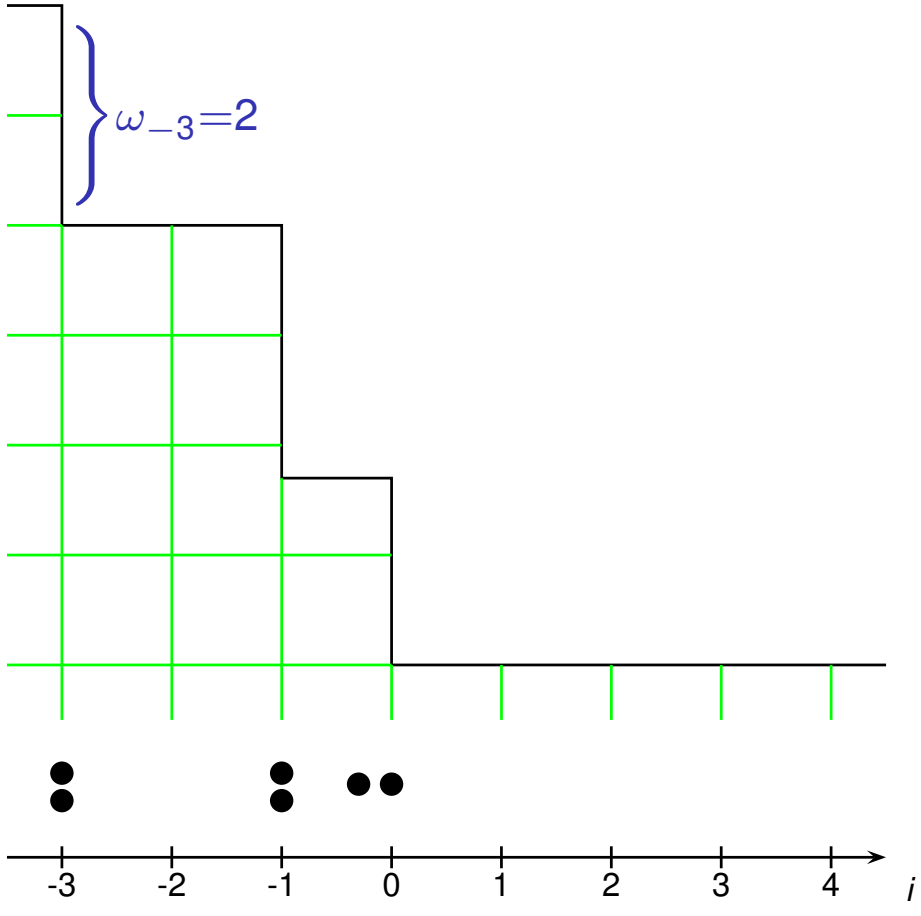
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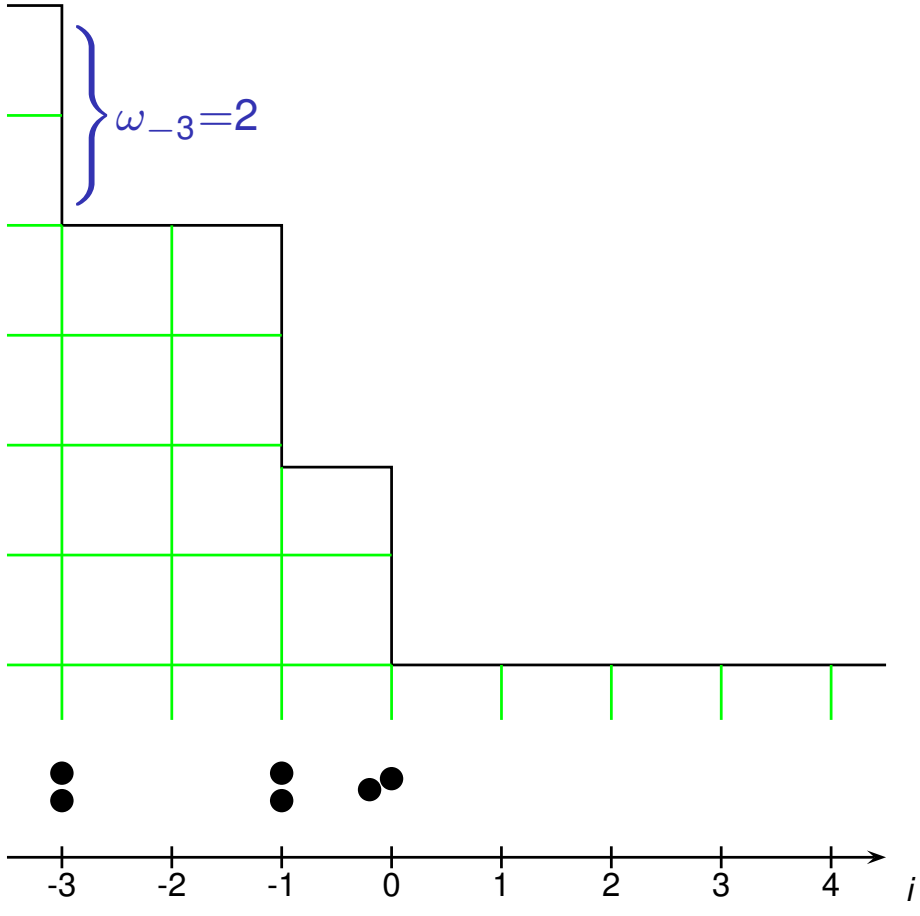
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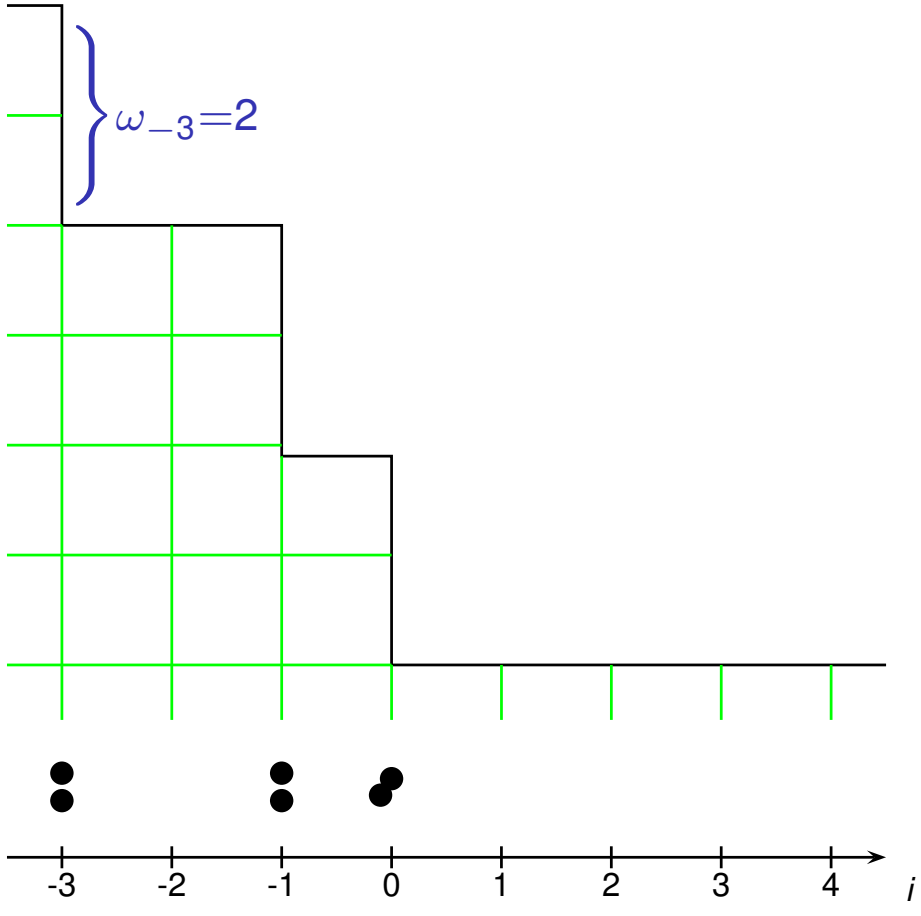
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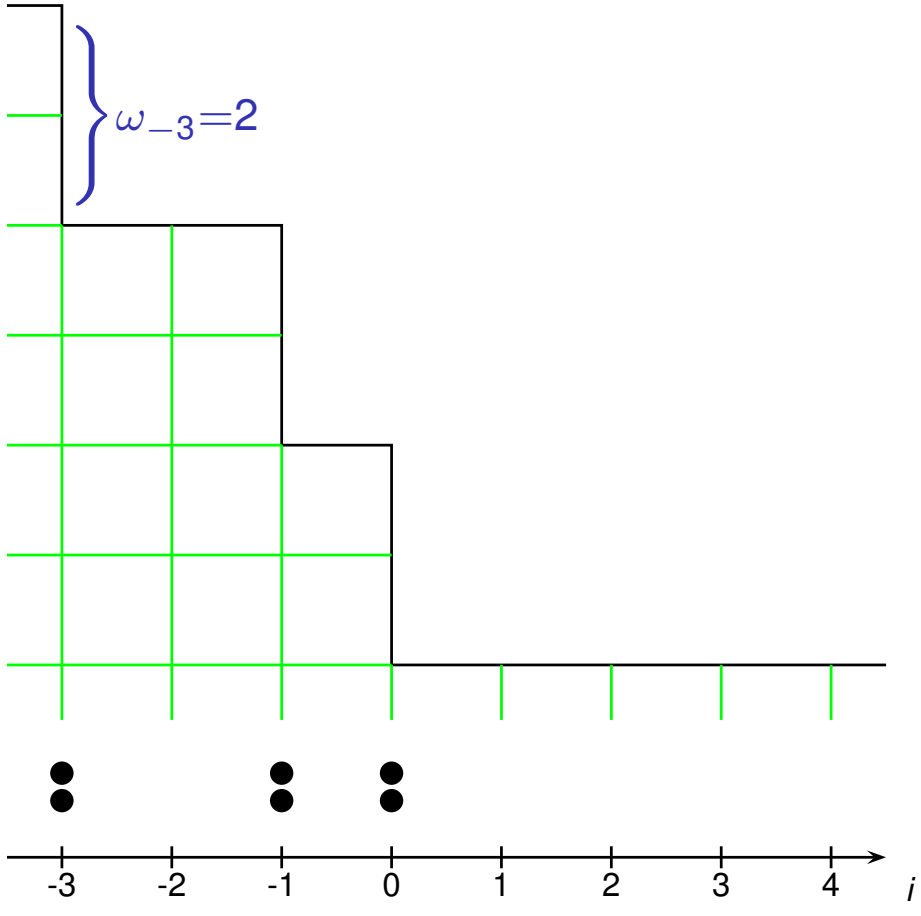
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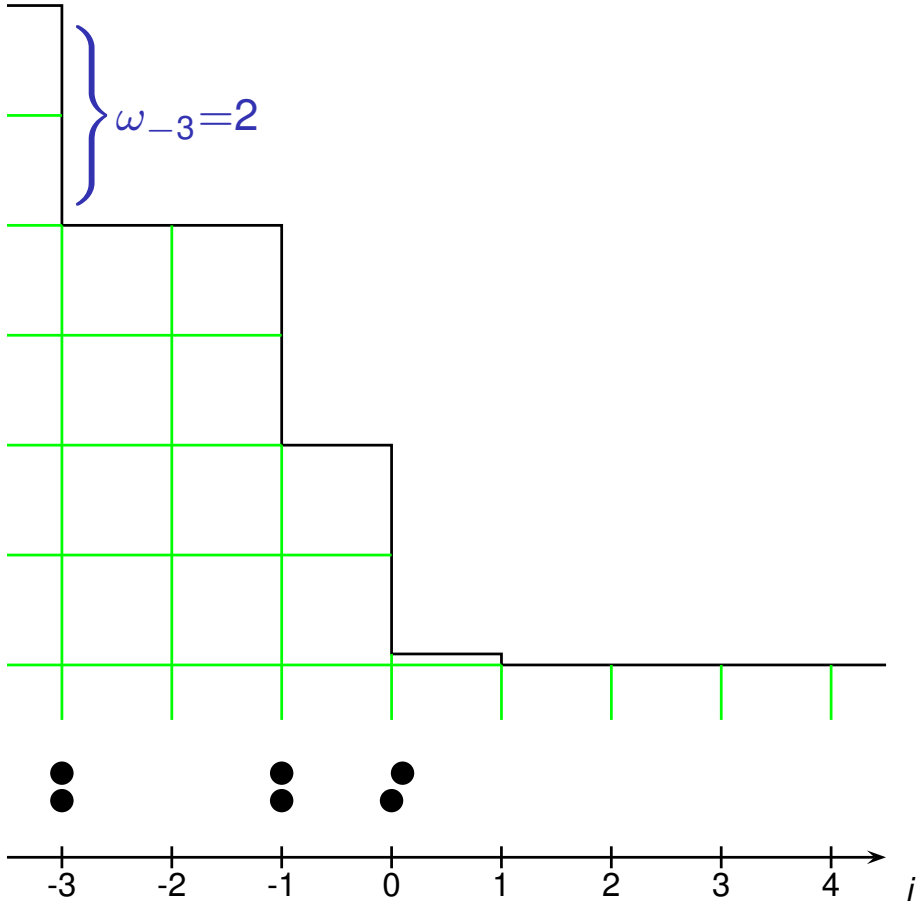
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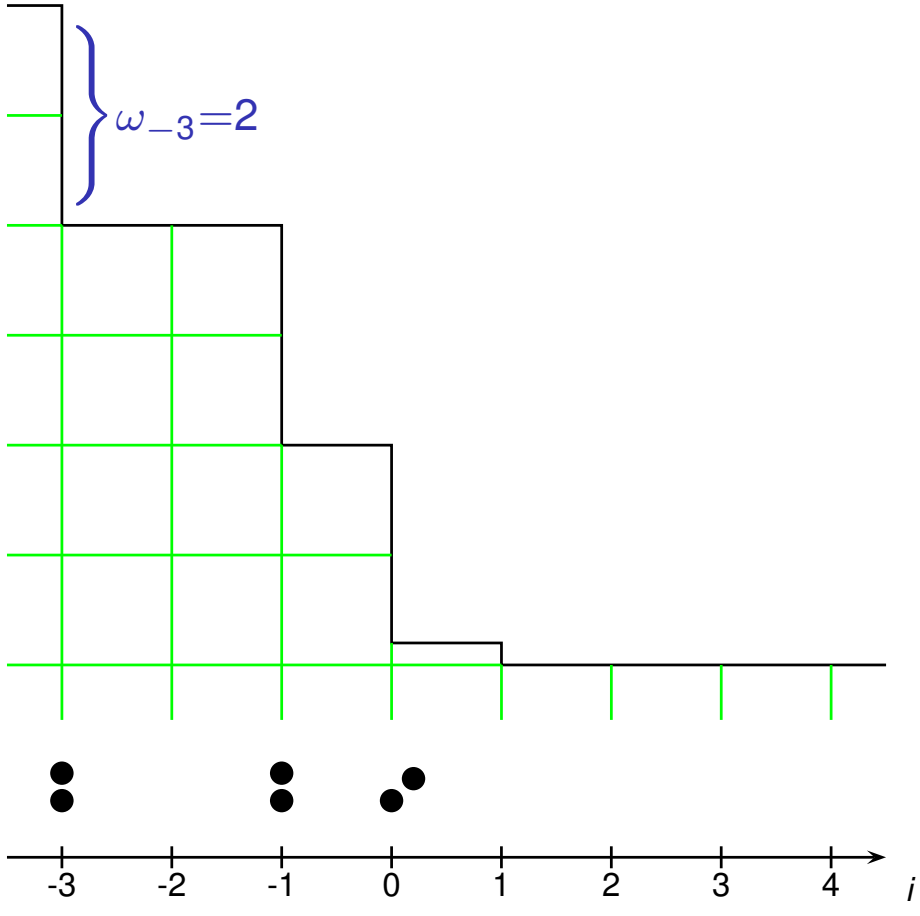
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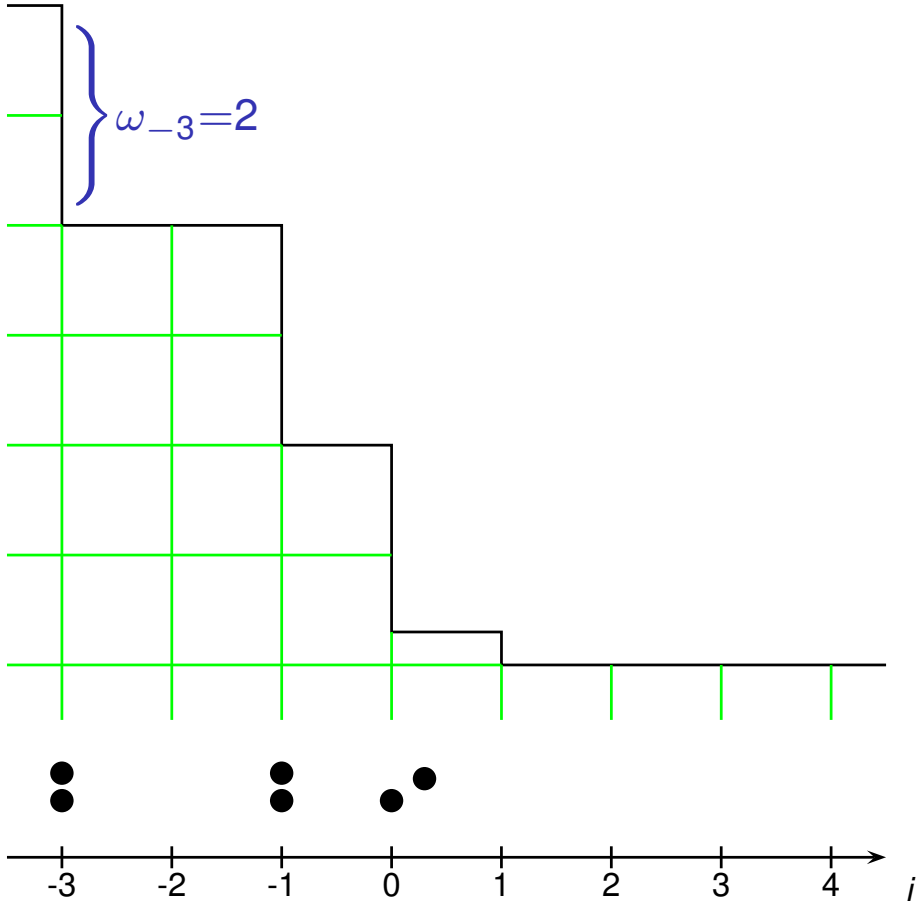
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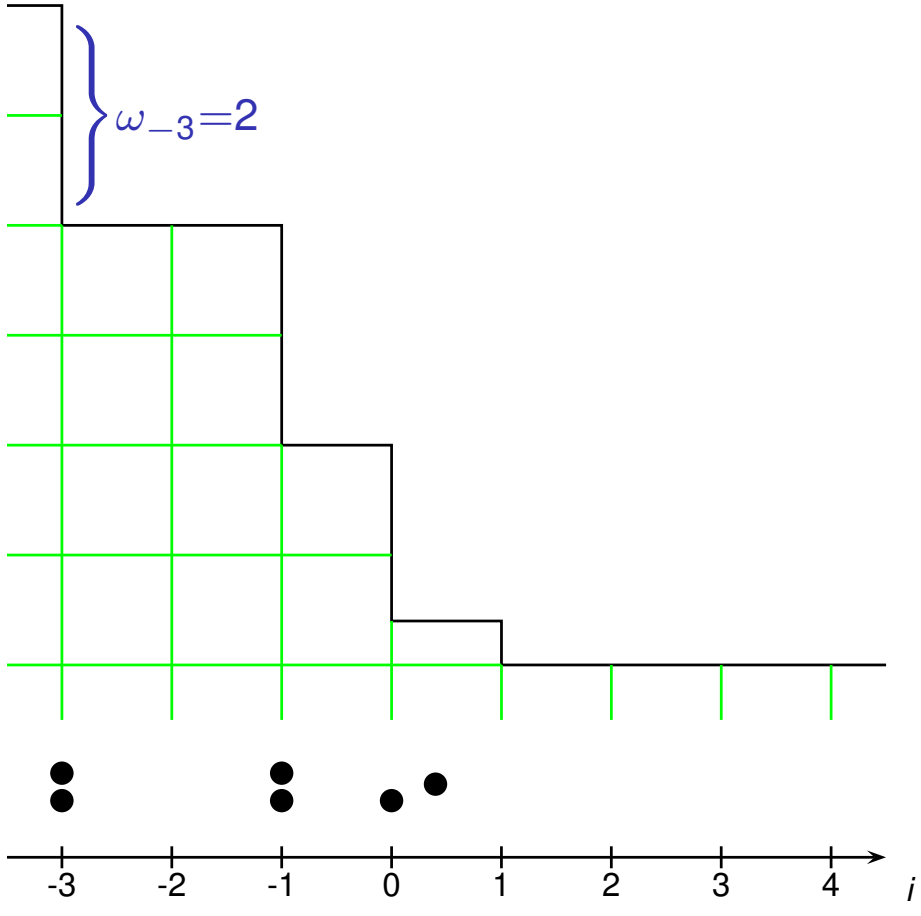
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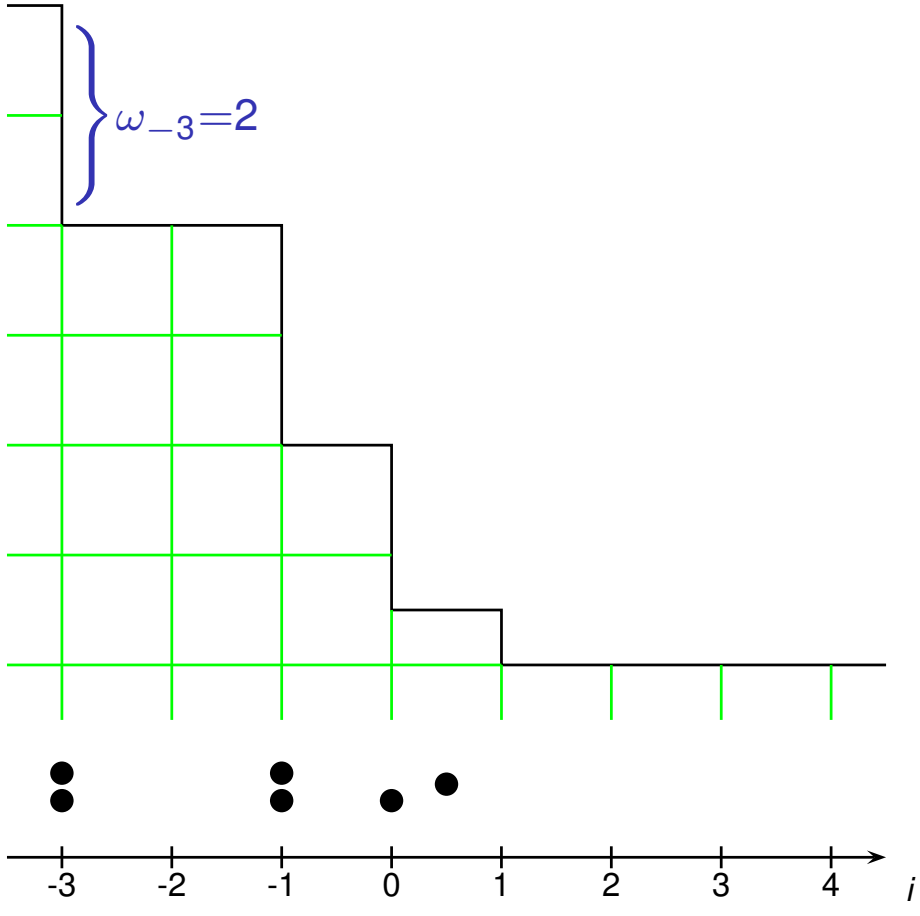
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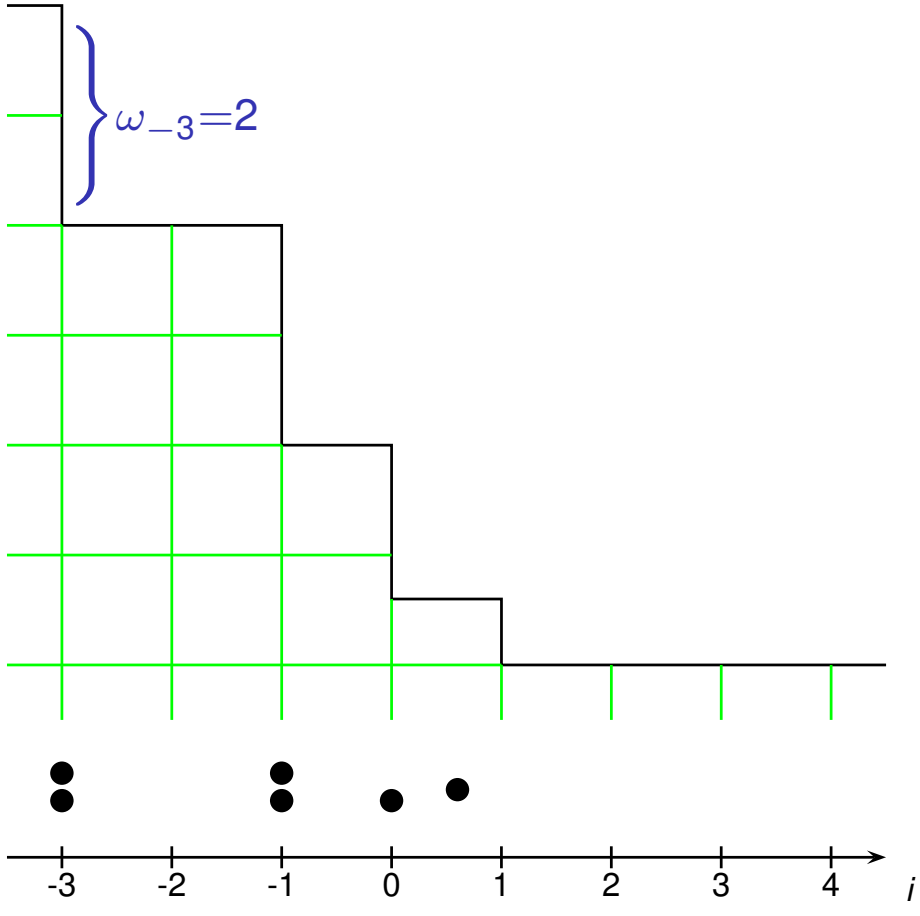
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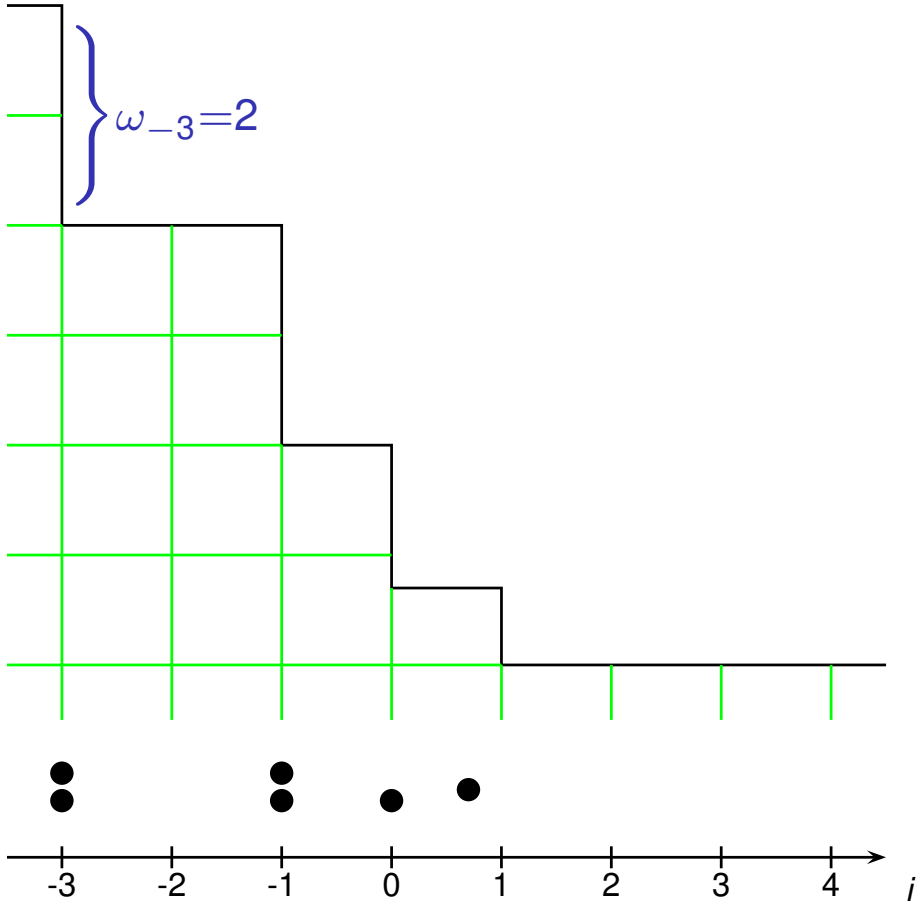
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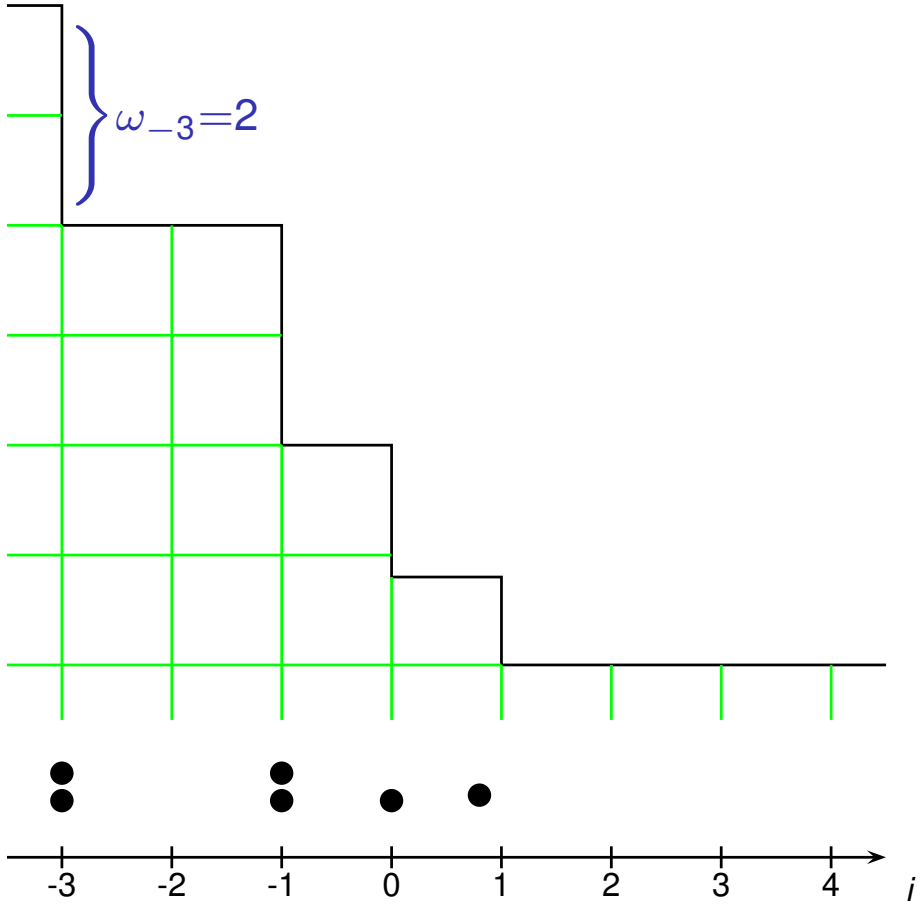
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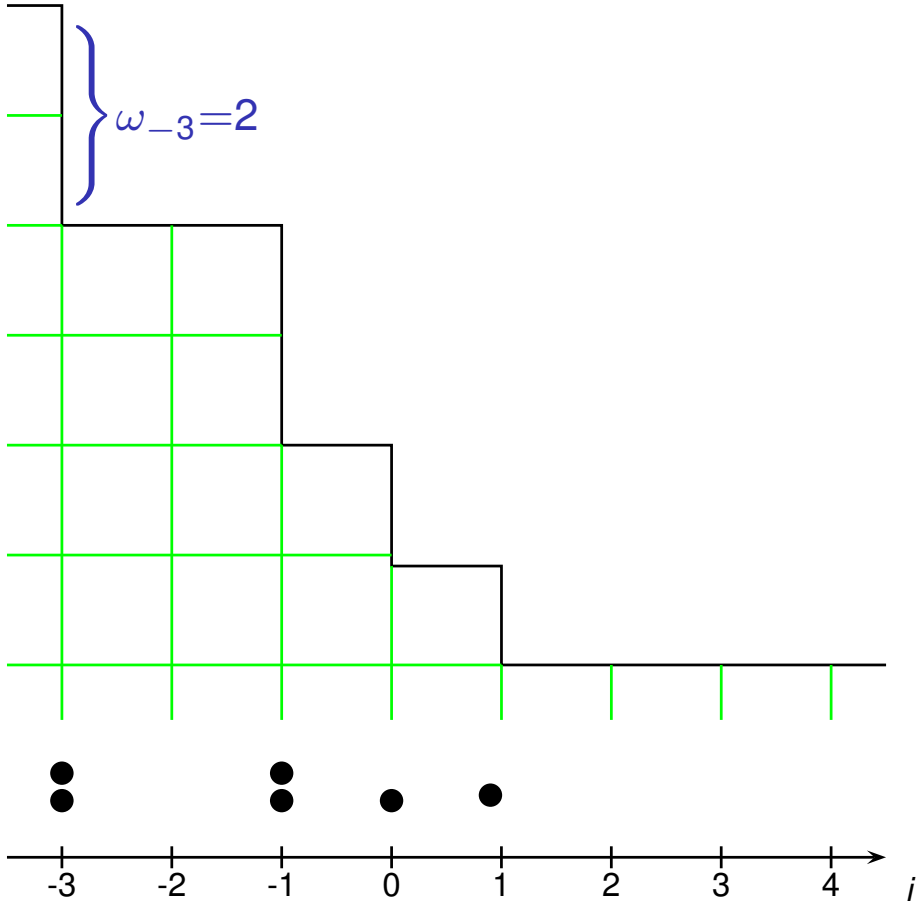
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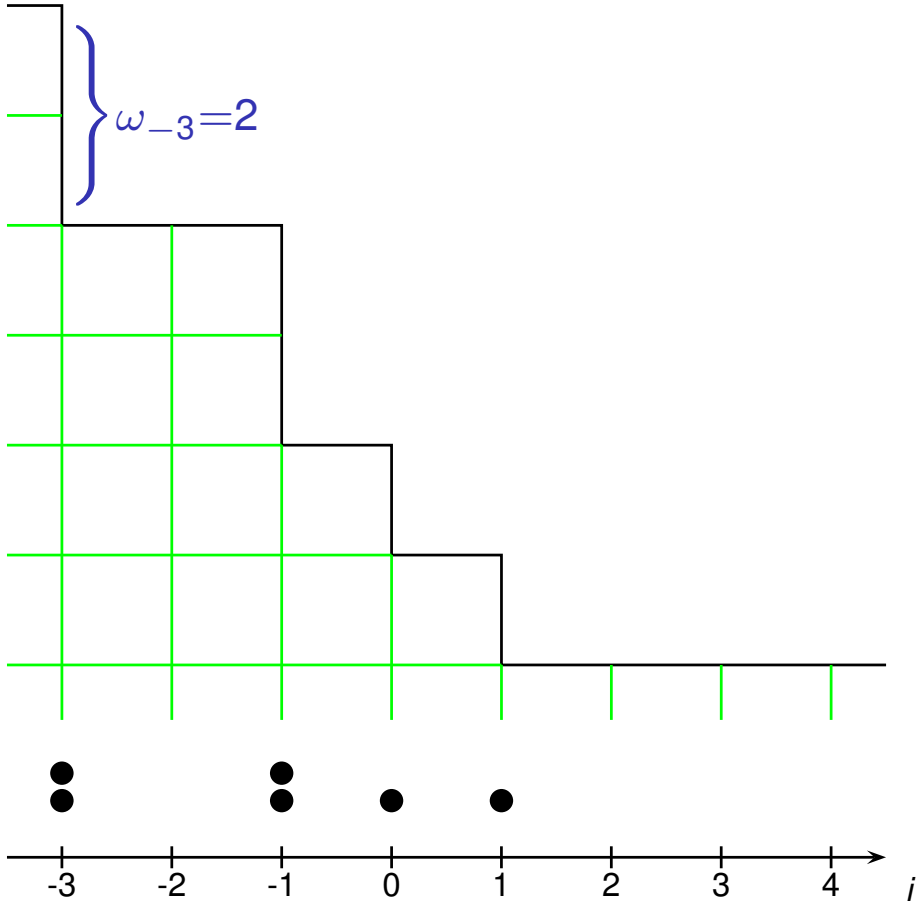
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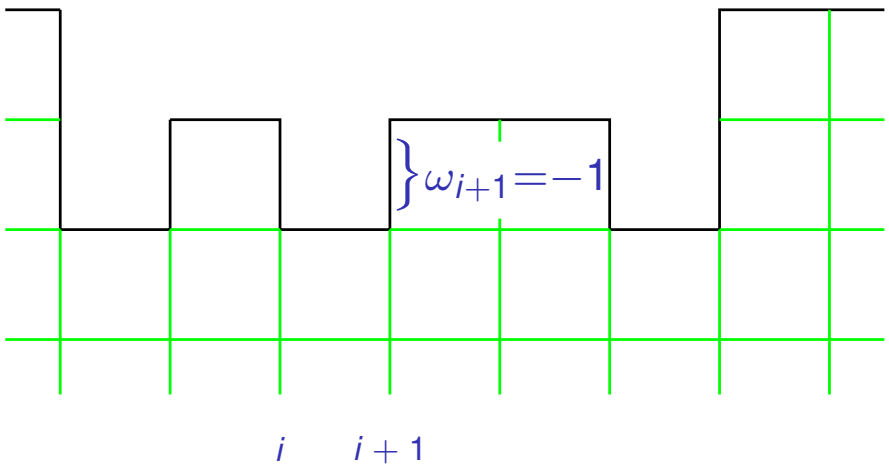
Totally asymmetric zero range process

Extremal translation-invariant stationary distributions are still product, and rather explicit in terms of $r(\cdot)$.

Two special cases:

- $r(\omega_i) = \mathbf{1}\{\omega_i > 0\}$: classical zero range; $\omega_i \sim \text{Geom}(\theta)$.
- $r(\omega_i) = \omega_i$: independent walkers; $\omega_i \sim \text{Poi}(\theta)$.

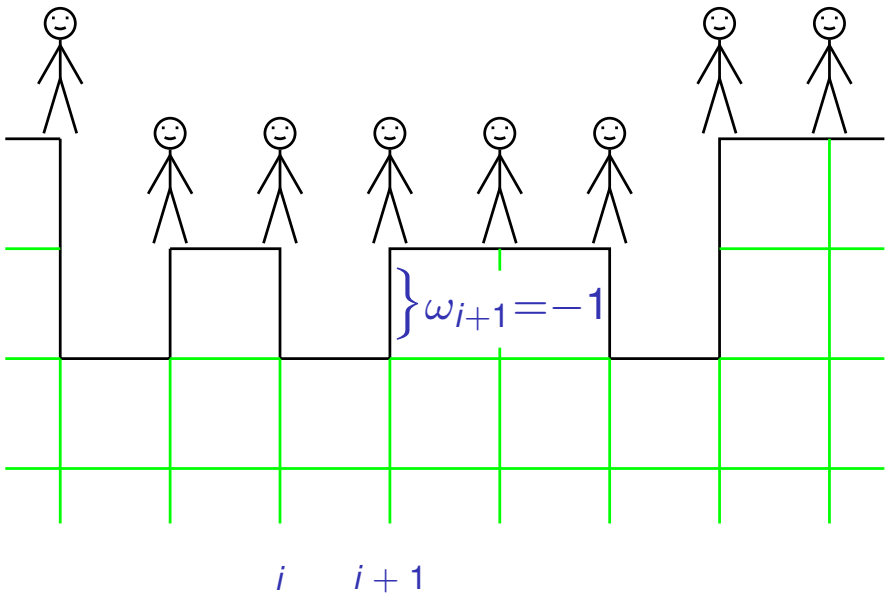
Totally asymmetric bricklayers process



$$\omega_i \in \mathbb{Z}$$

$$\begin{pmatrix} \omega_i \\ \omega_{i+1} \end{pmatrix} \\ || \\ \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Totally asymmetric bricklayers process



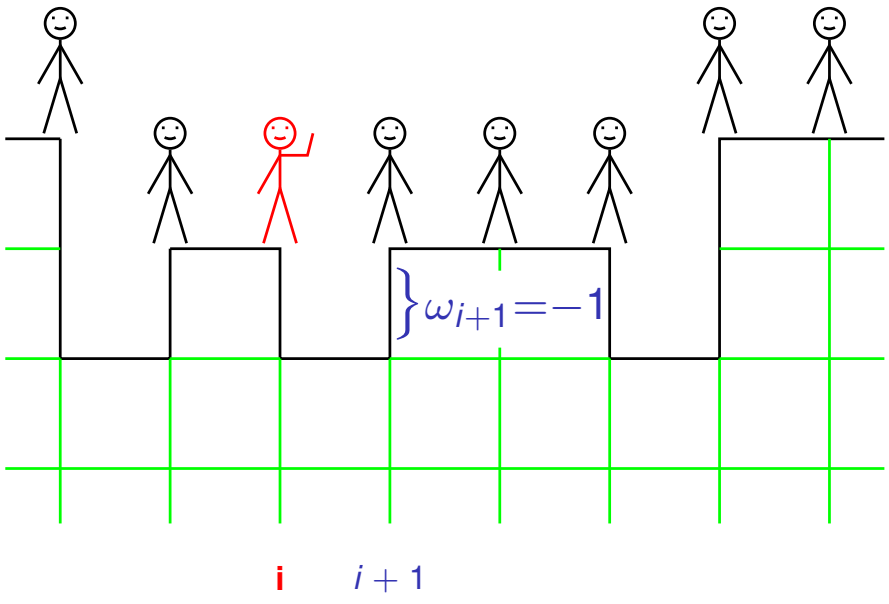
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a brick is added with rate $[r(\omega_i) + r(-\omega_{i+1})]$

$(r(\omega) \cdot r(1-\omega) = 1; \quad r$ non-decreasing).

Totally asymmetric bricklayers process



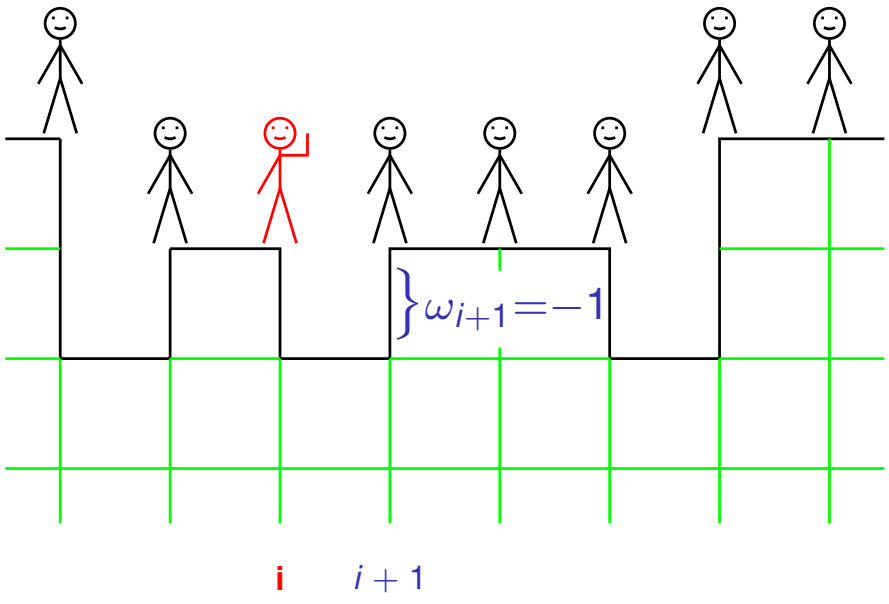
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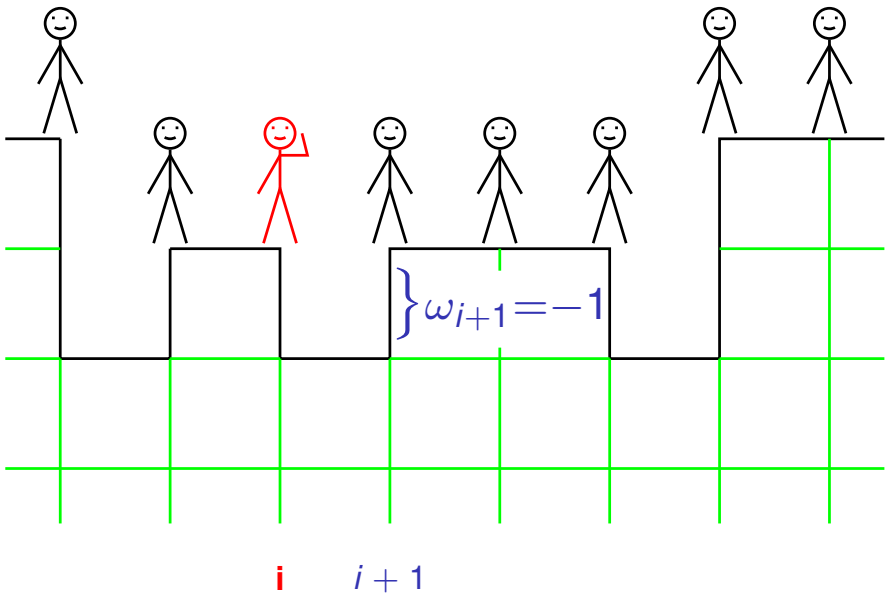
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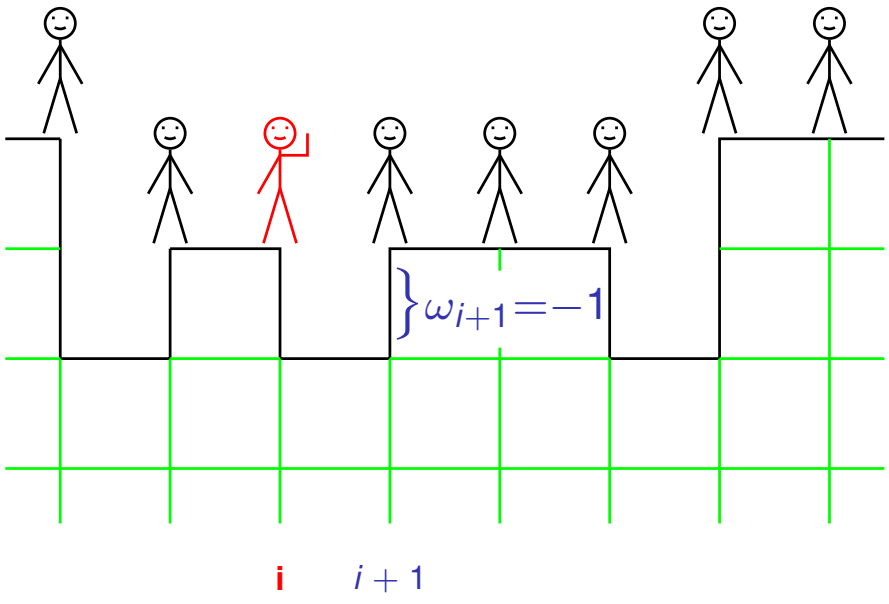
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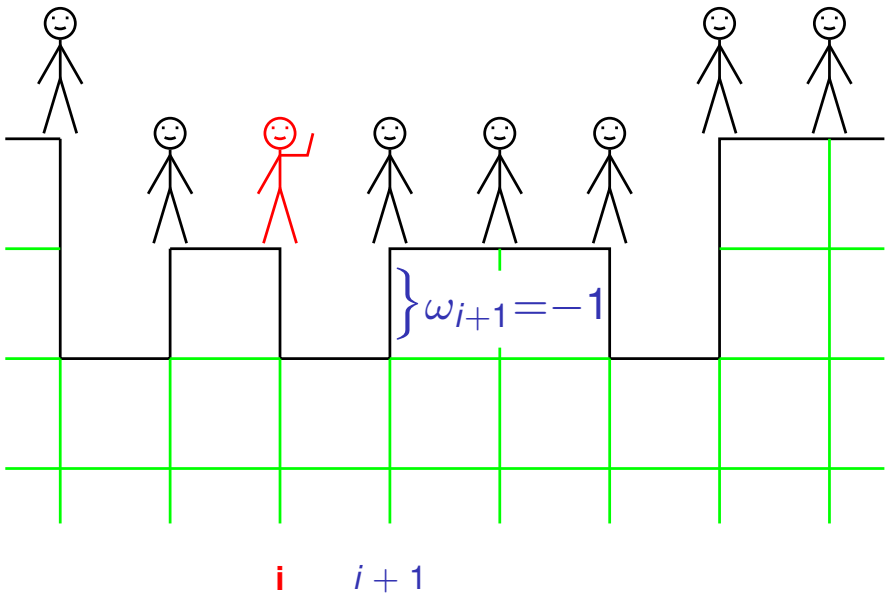
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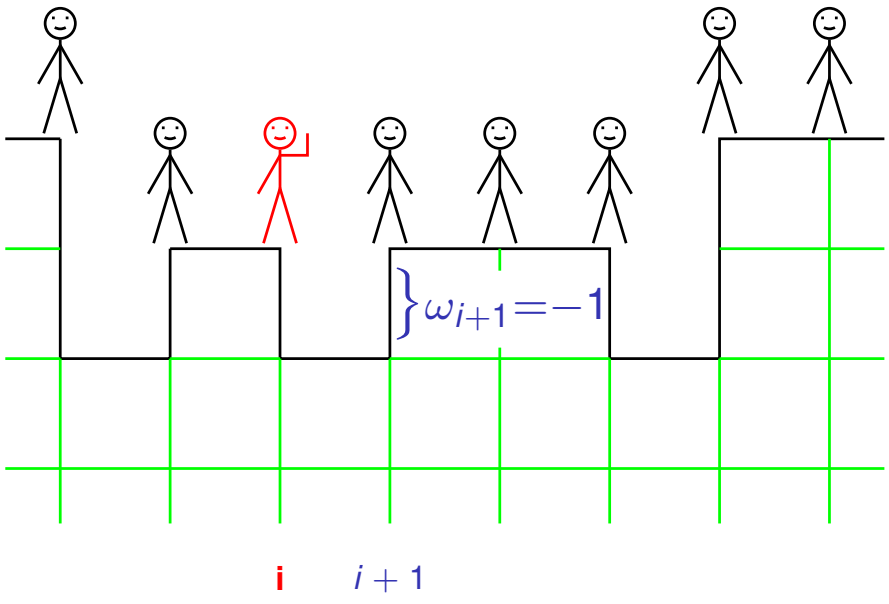
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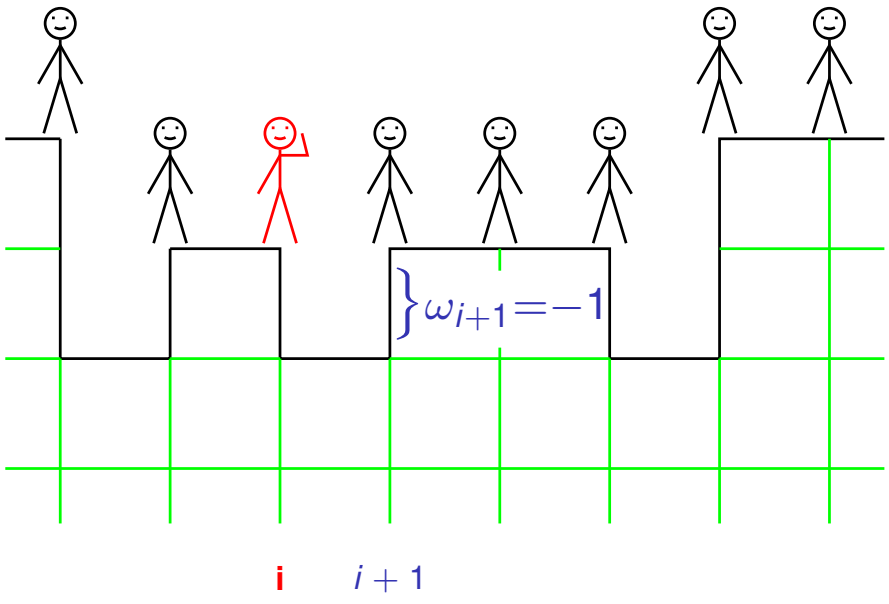
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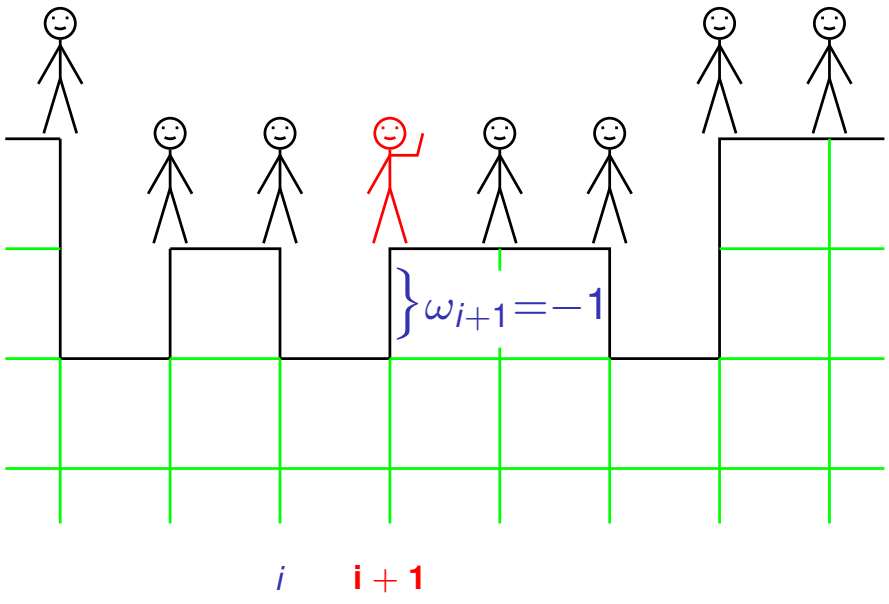
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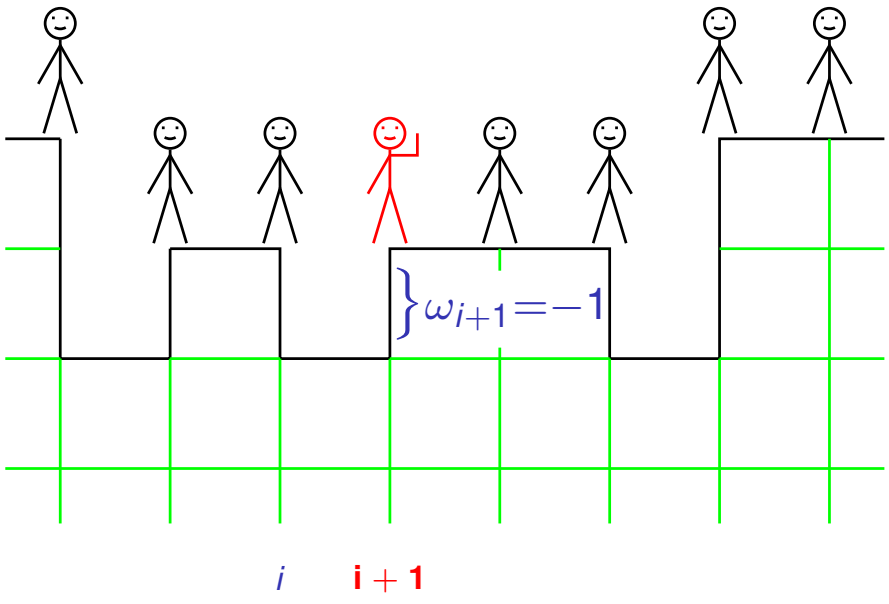
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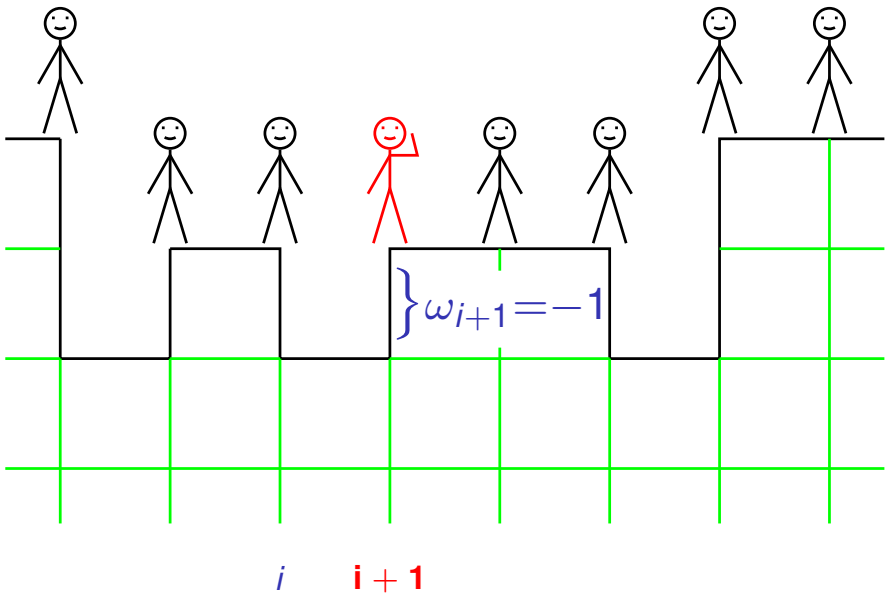
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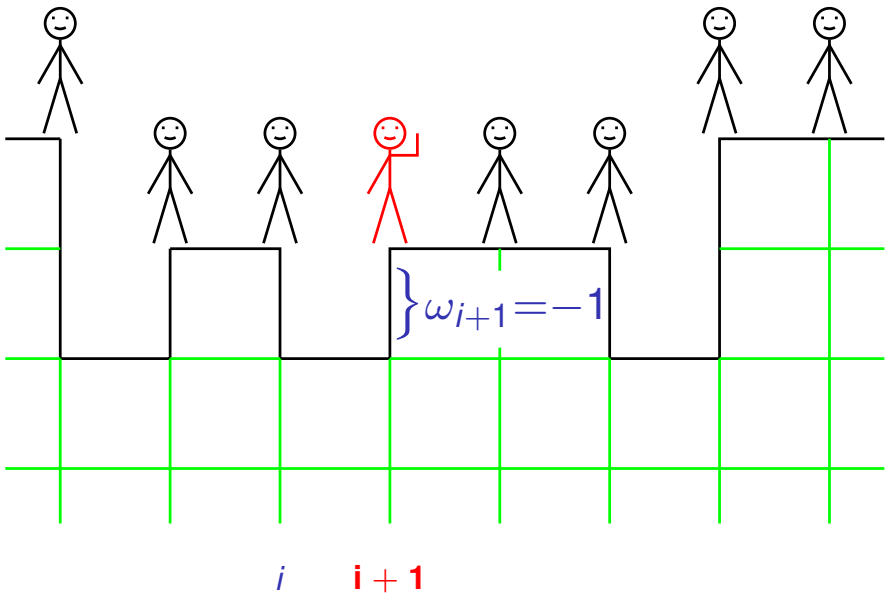
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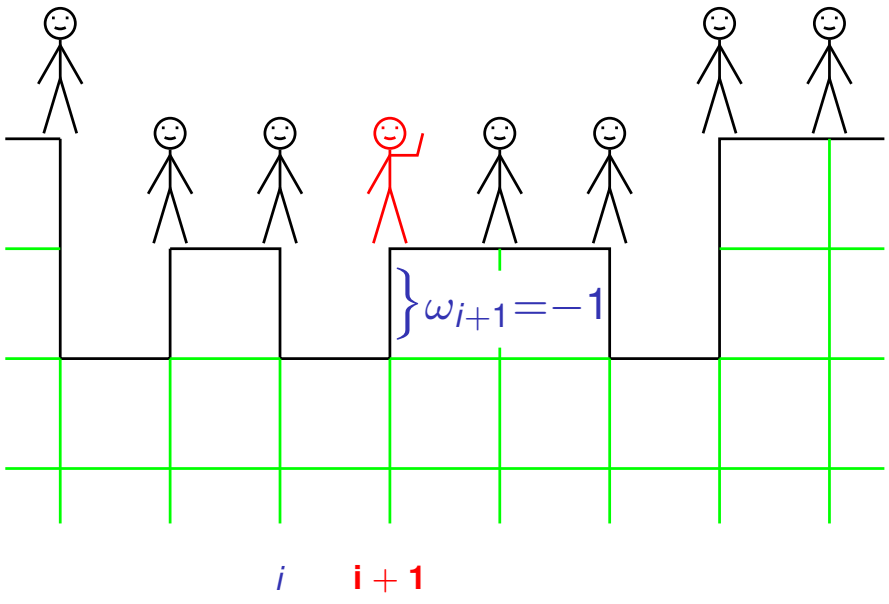
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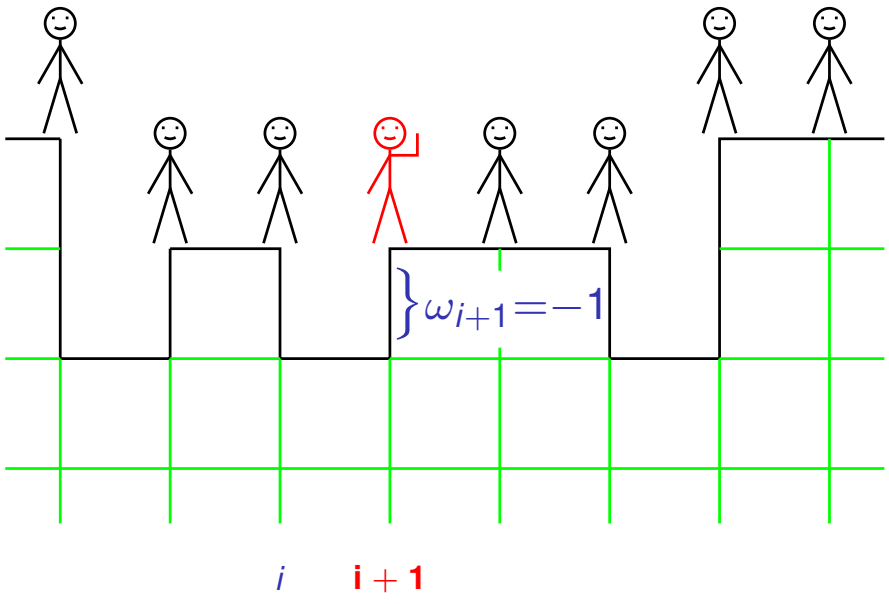
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Totally asymmetric bricklayers process



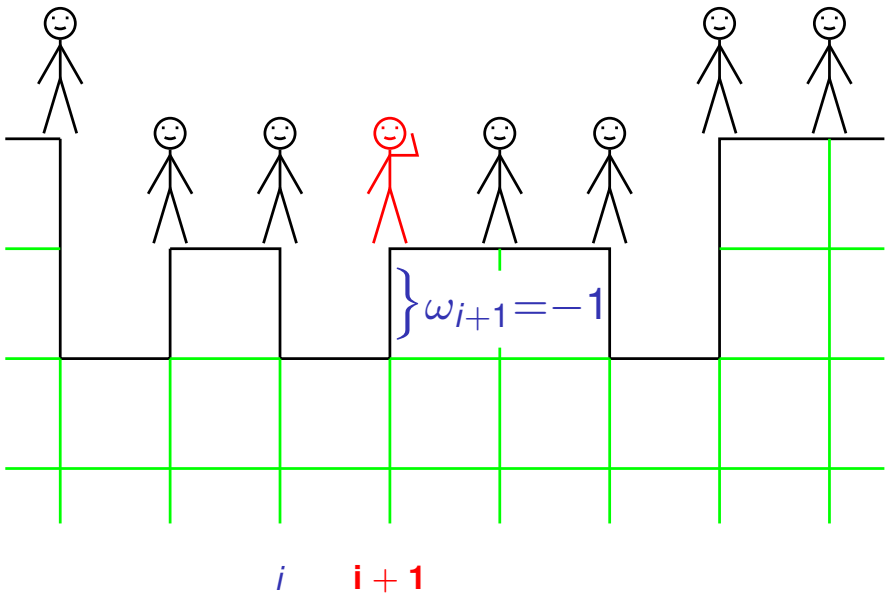
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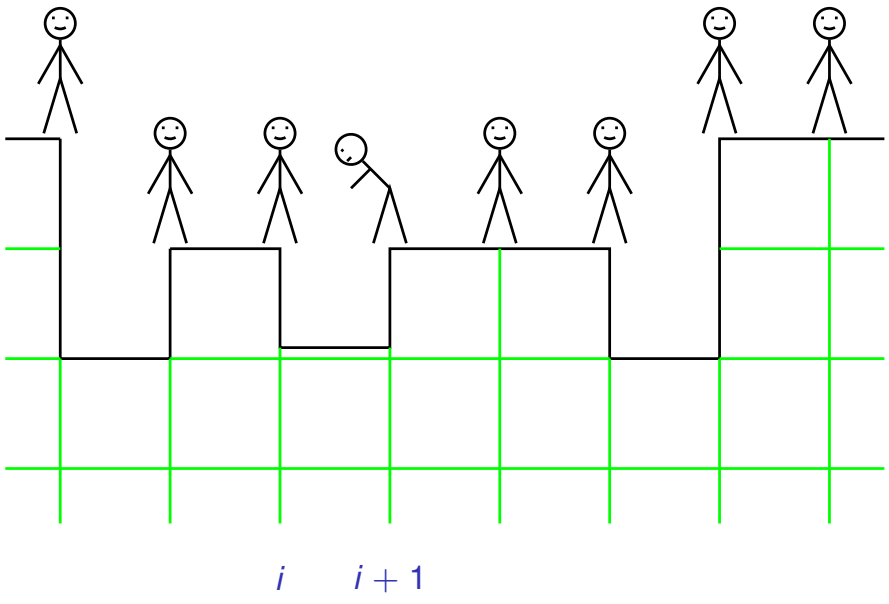
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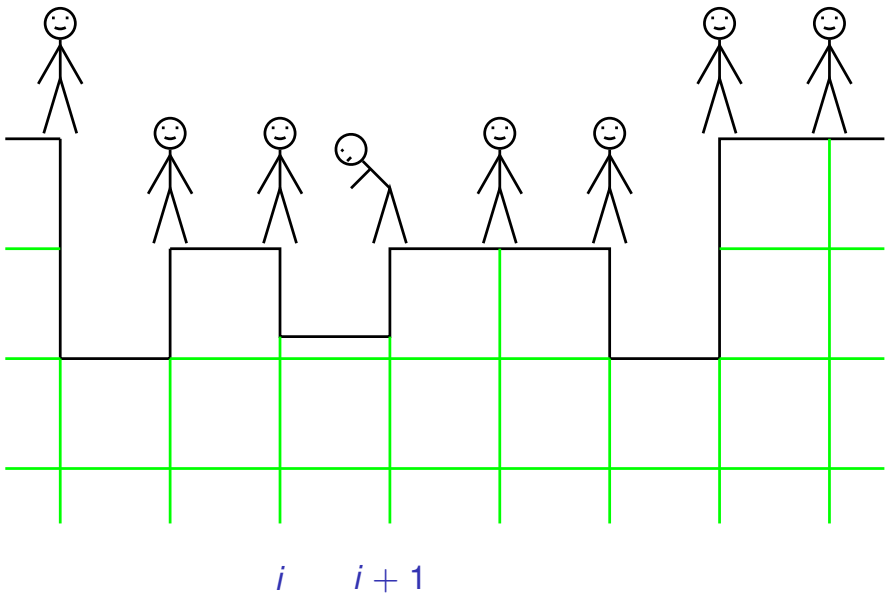
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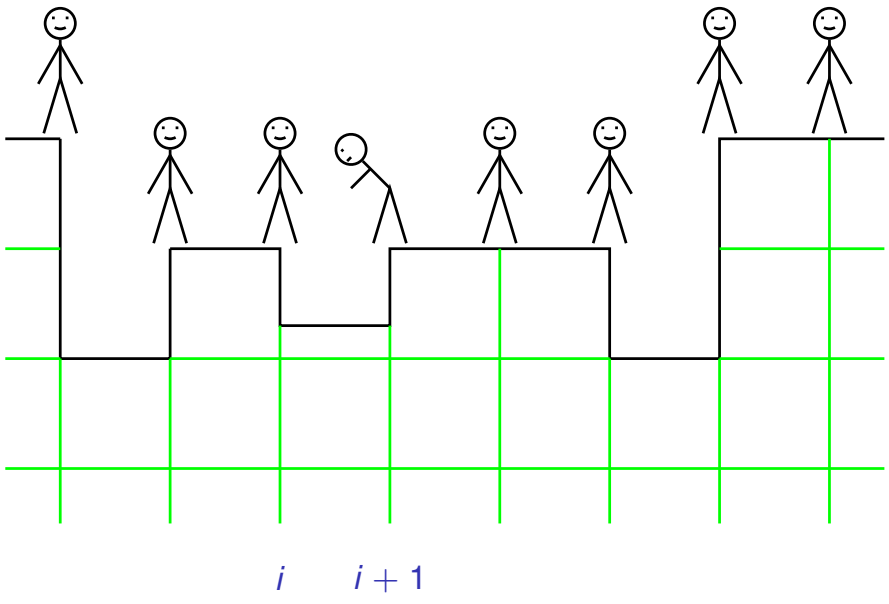
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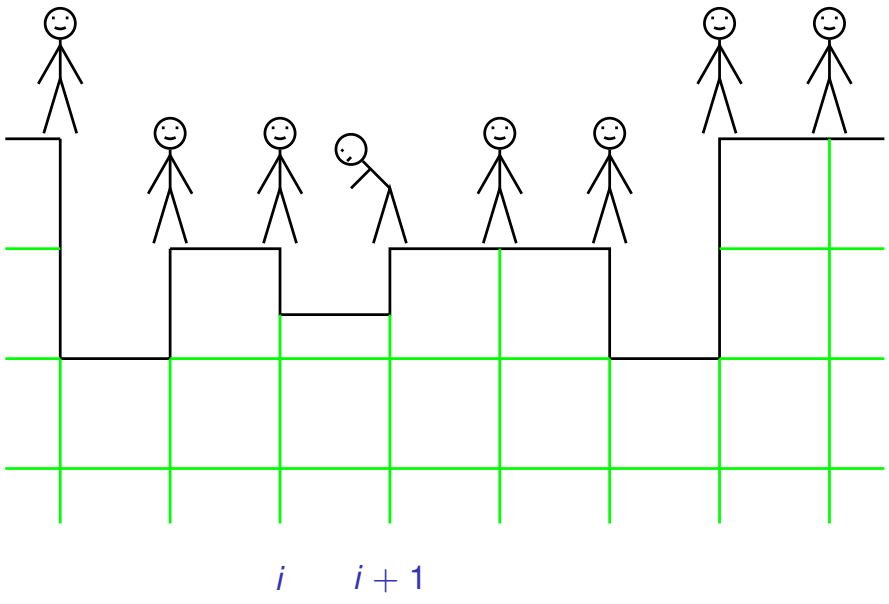
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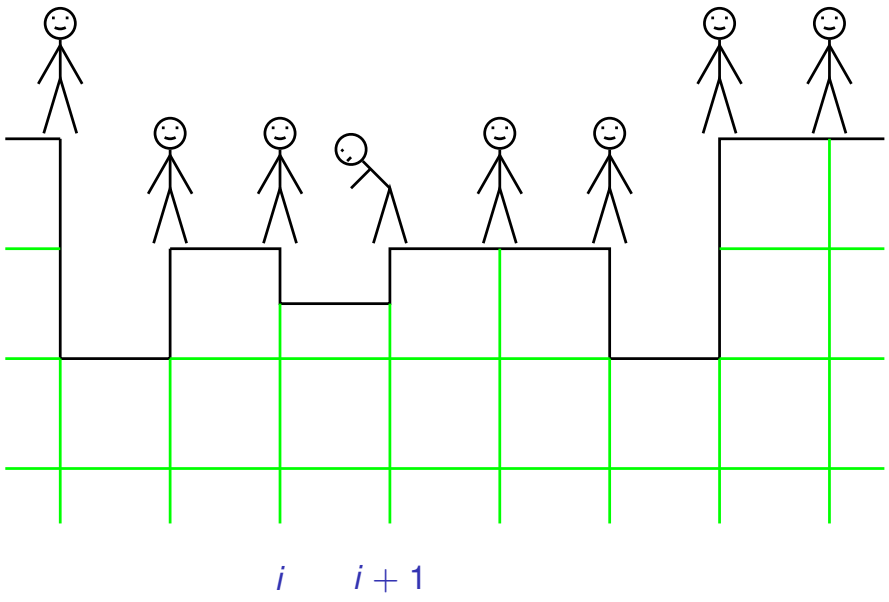
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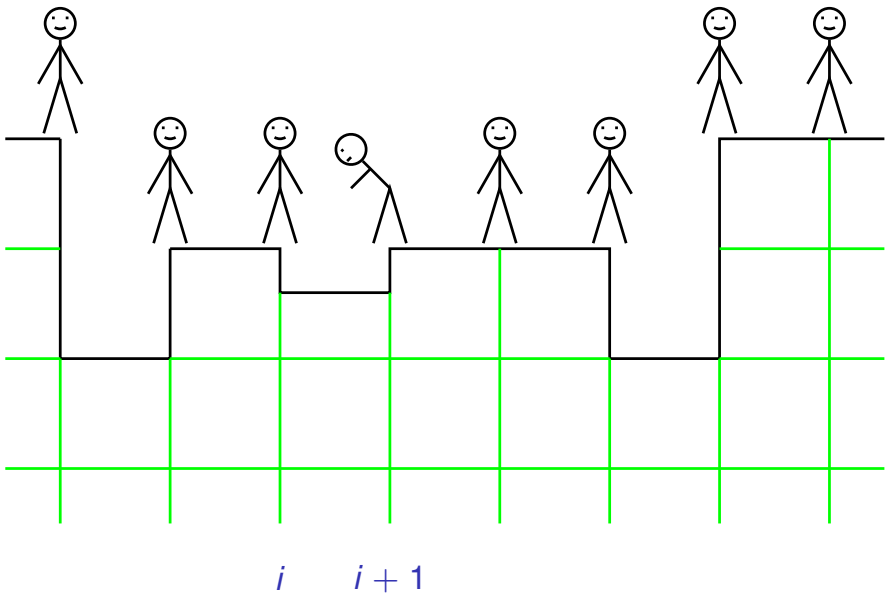
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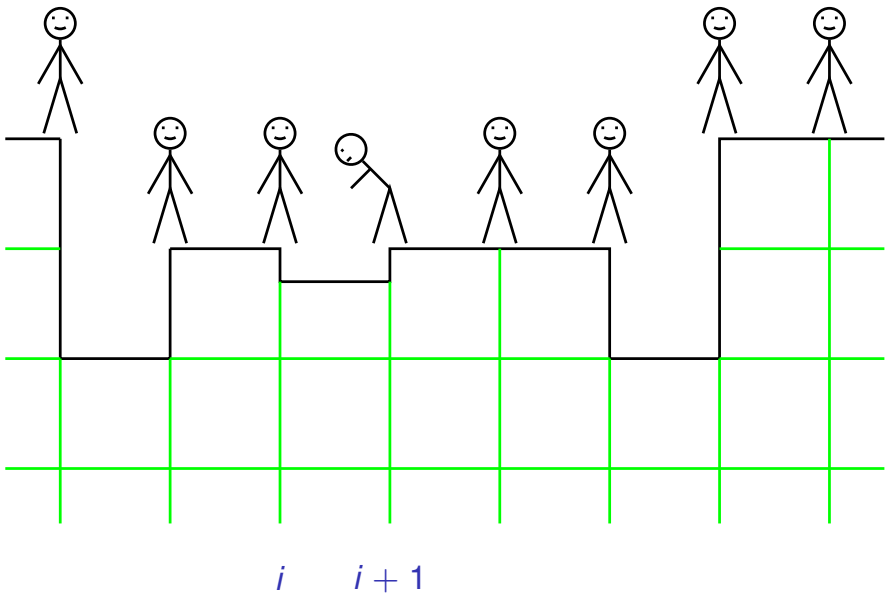
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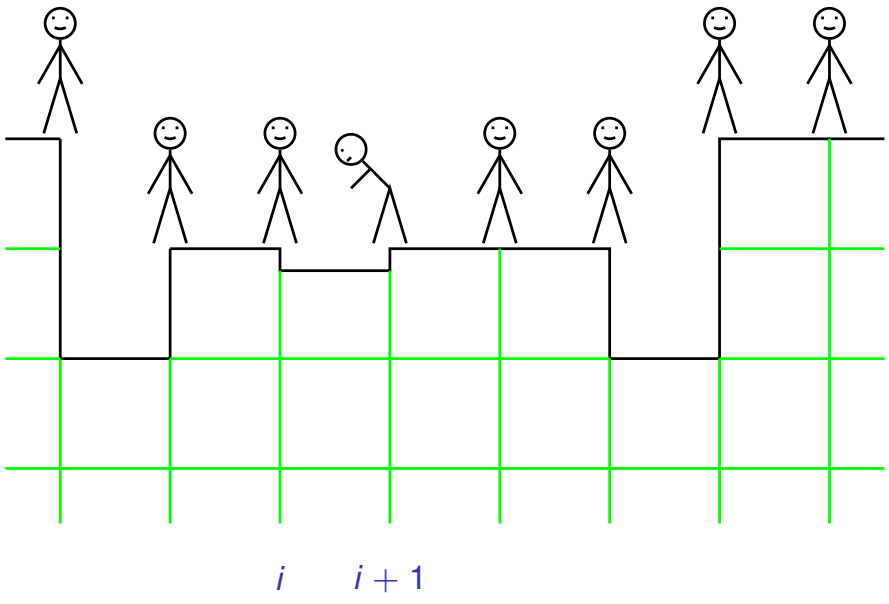
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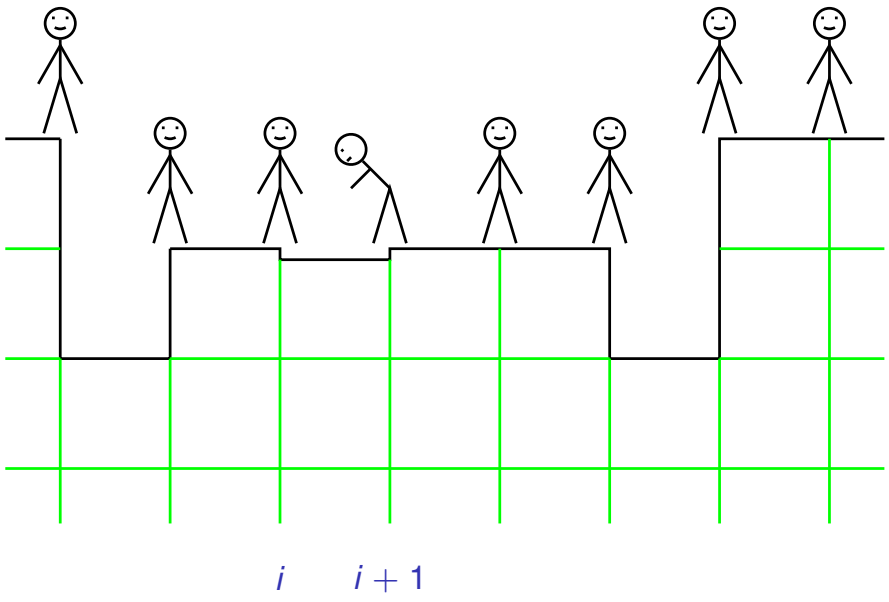
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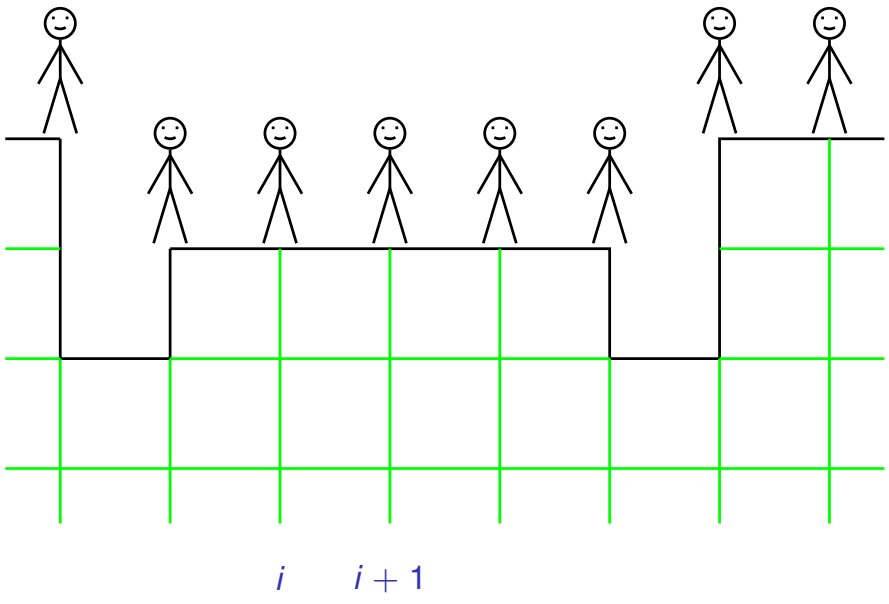
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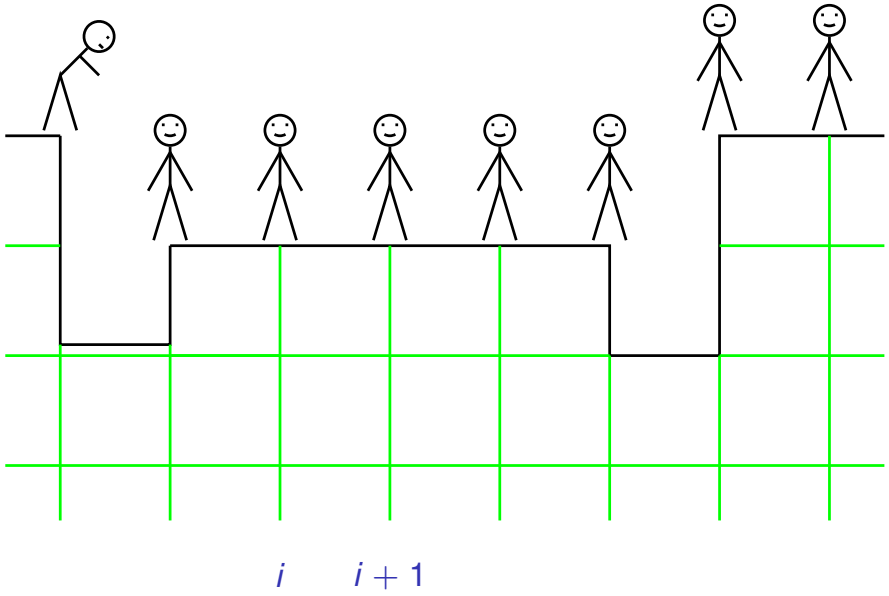
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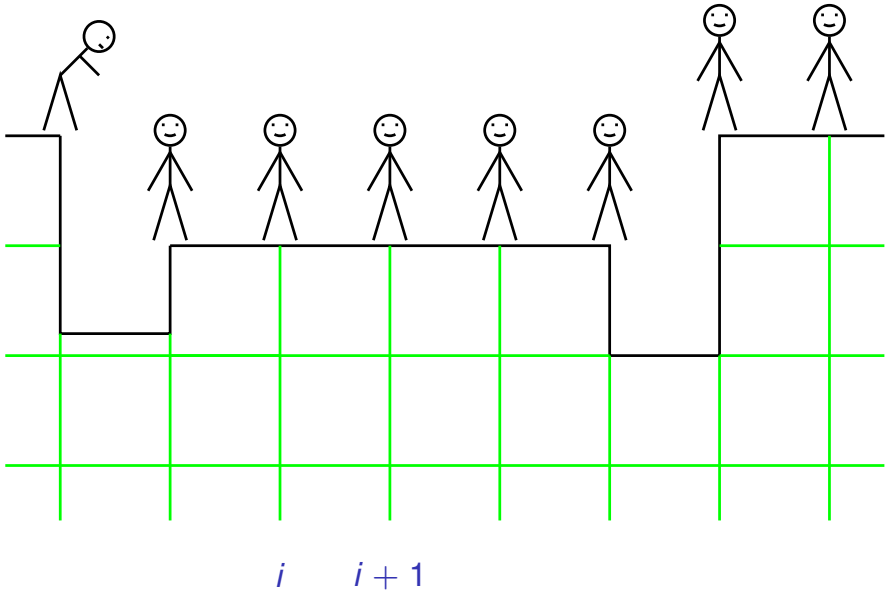
Totally asymmetric bricklayers process



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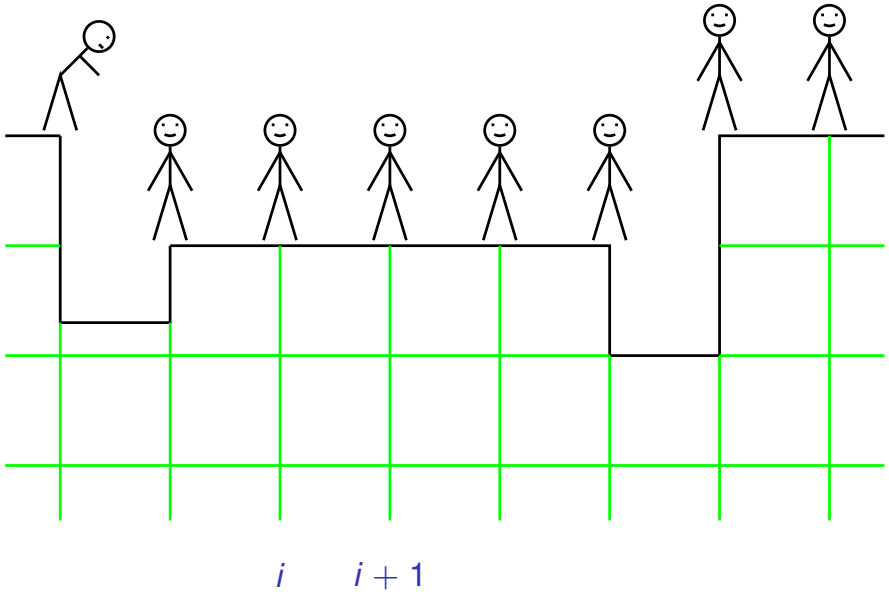
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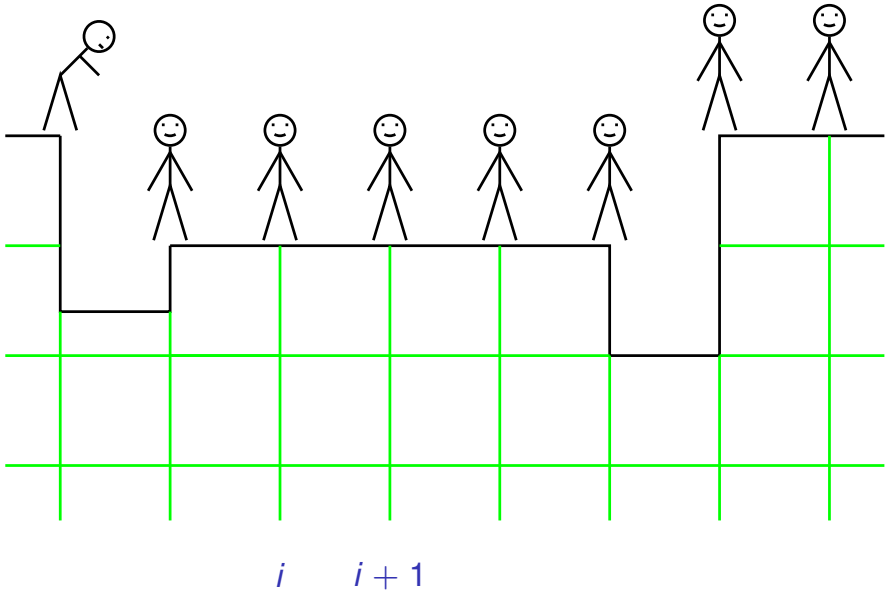
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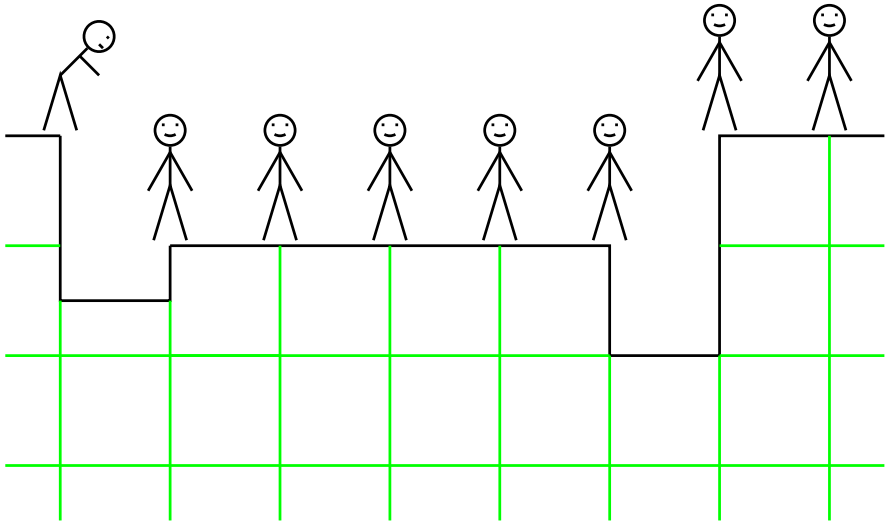
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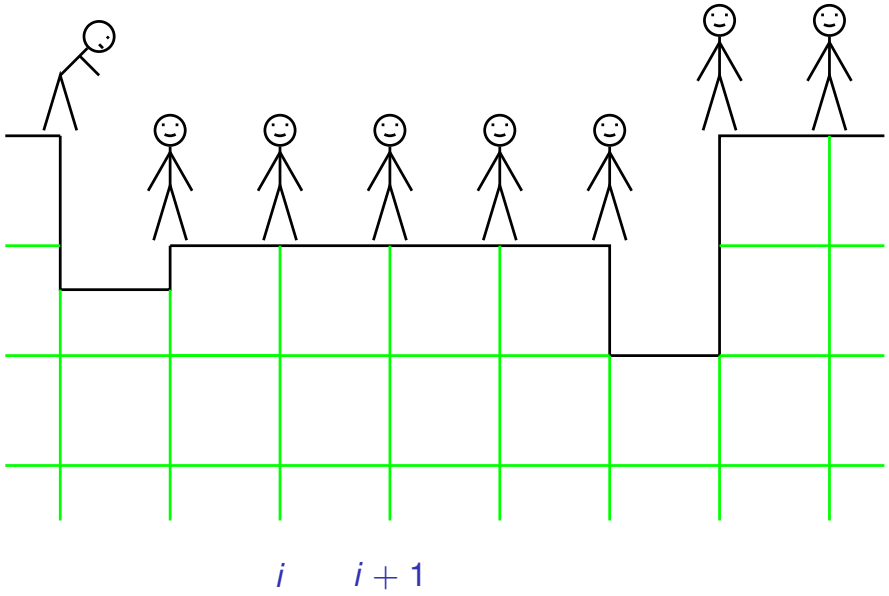


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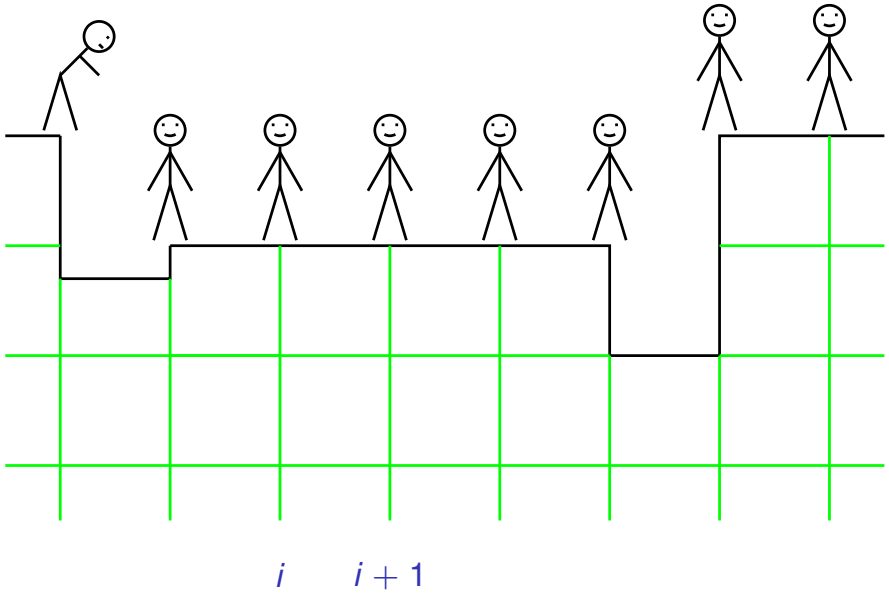
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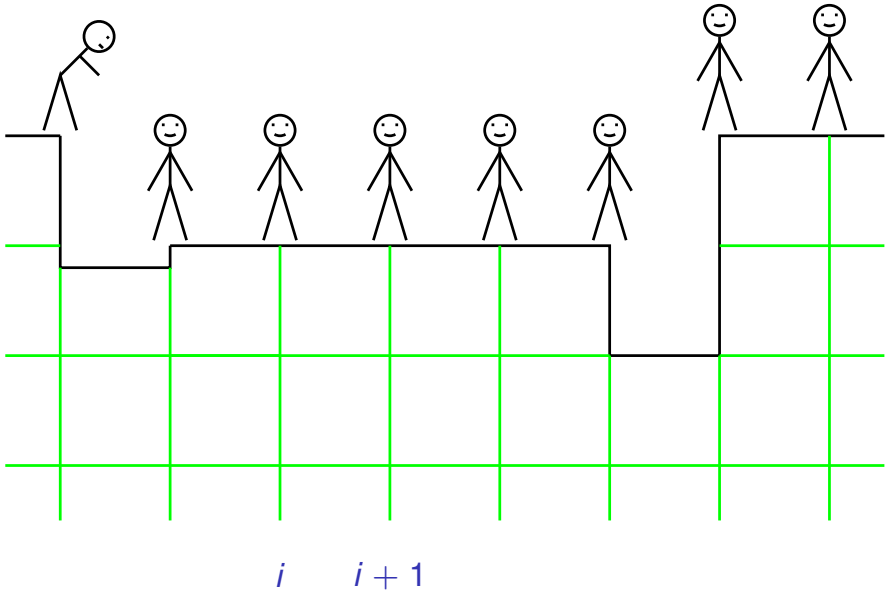
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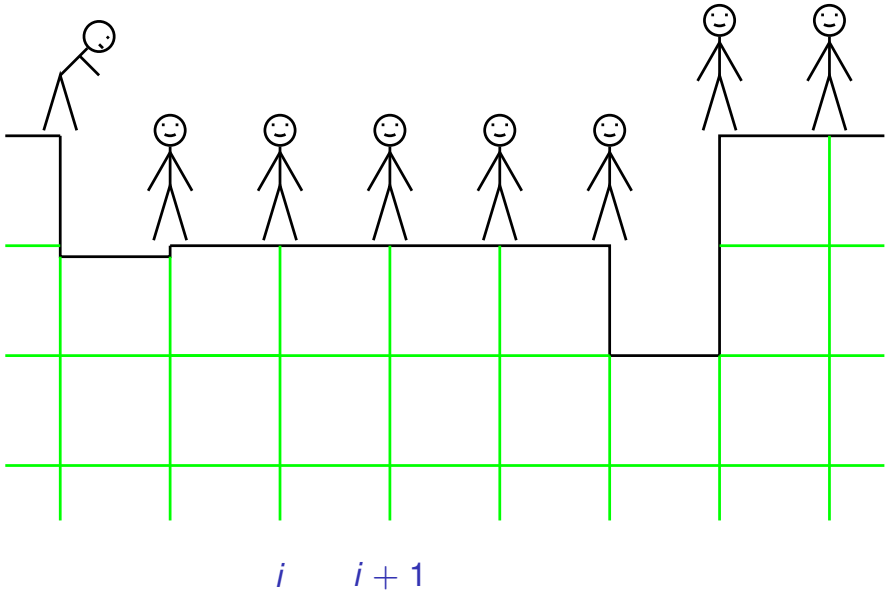
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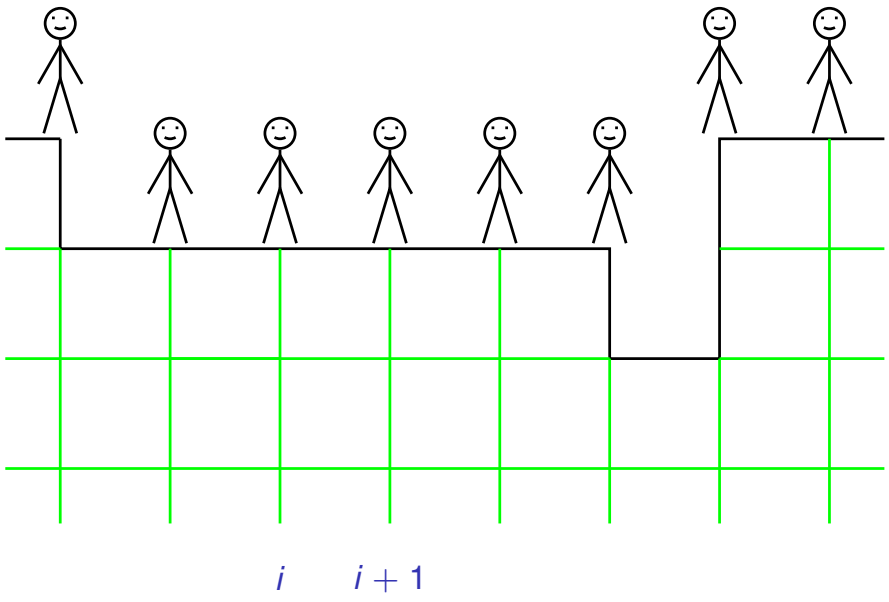
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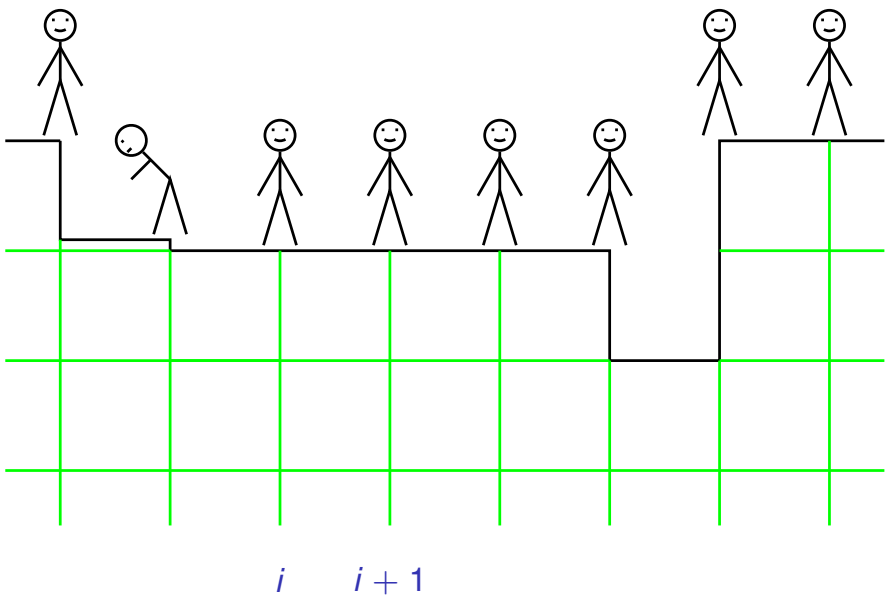
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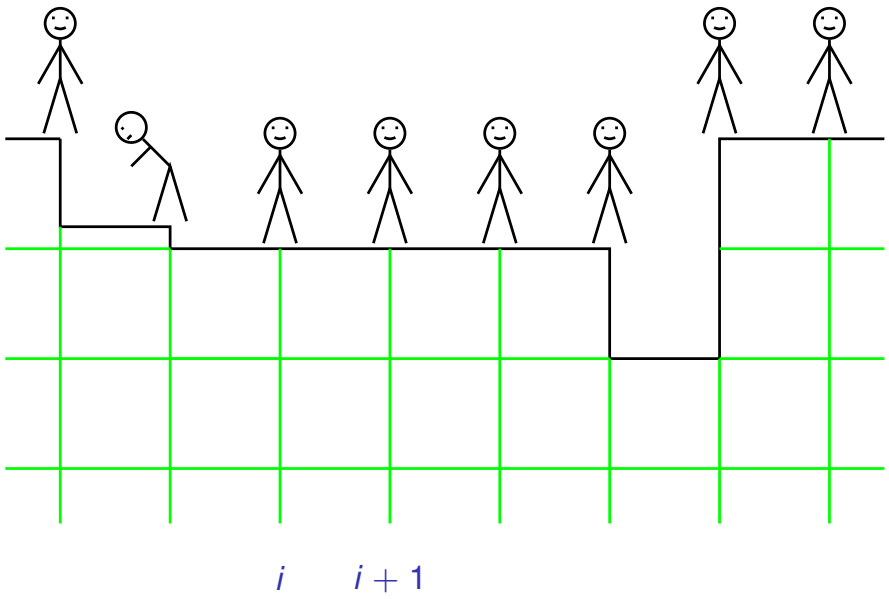
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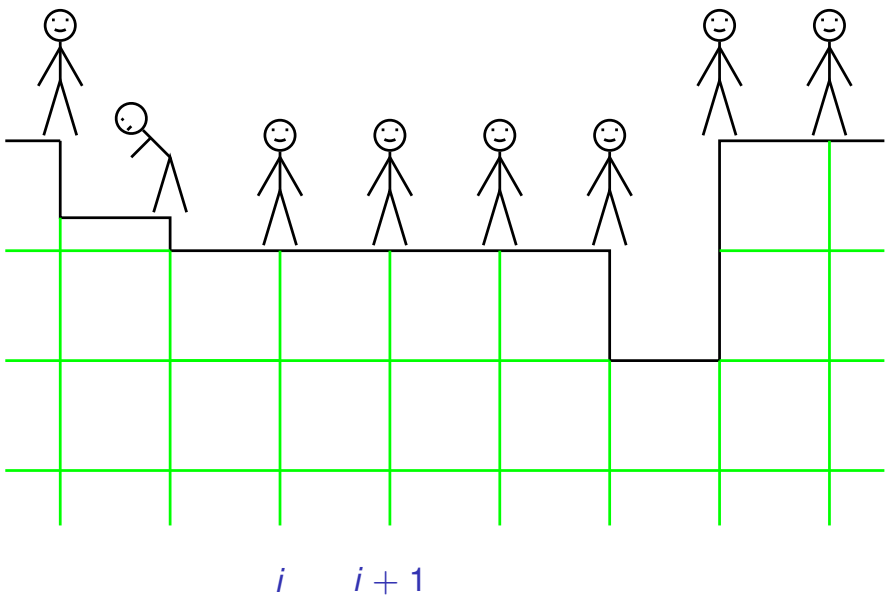
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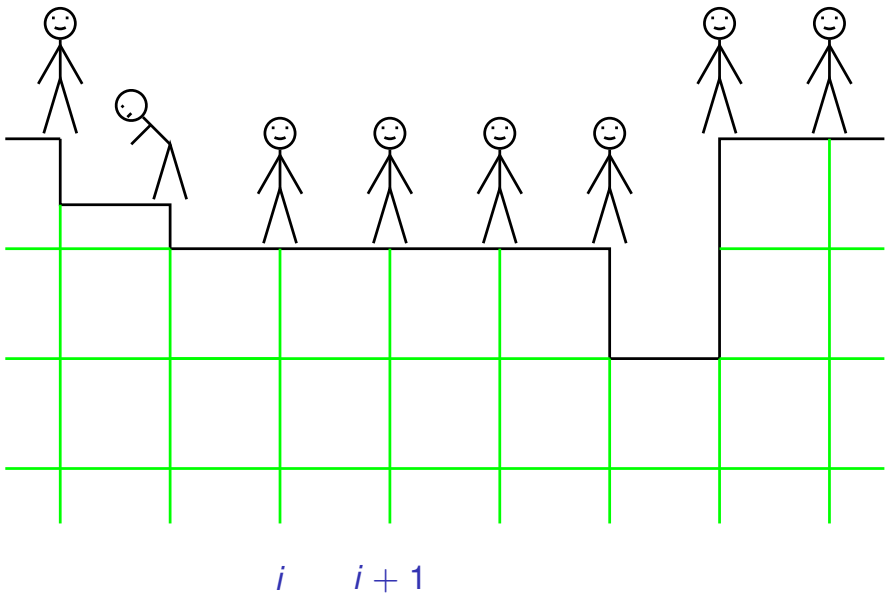


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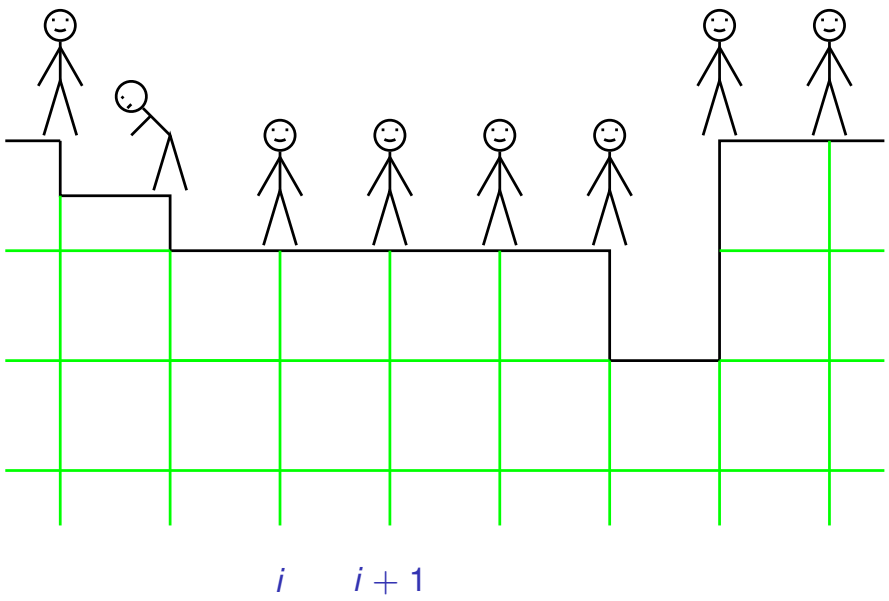
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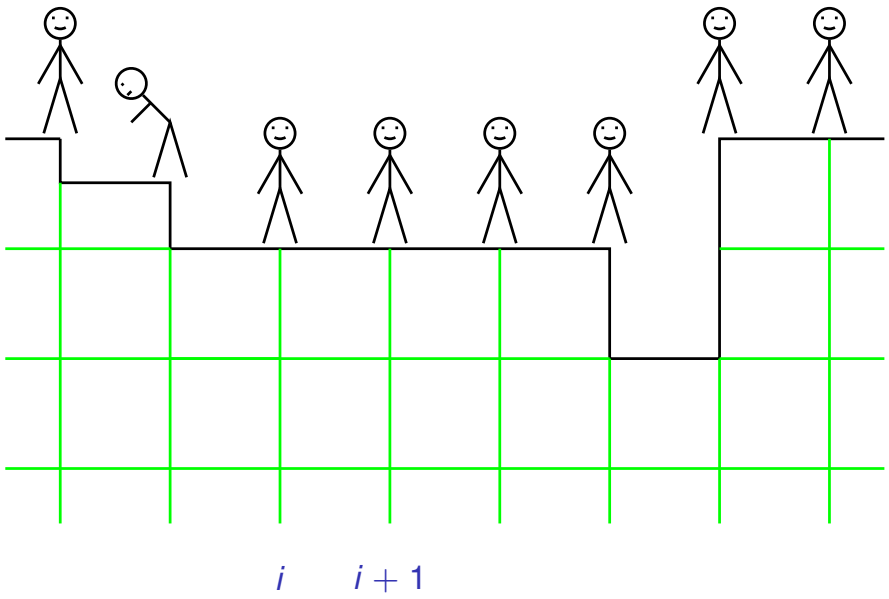


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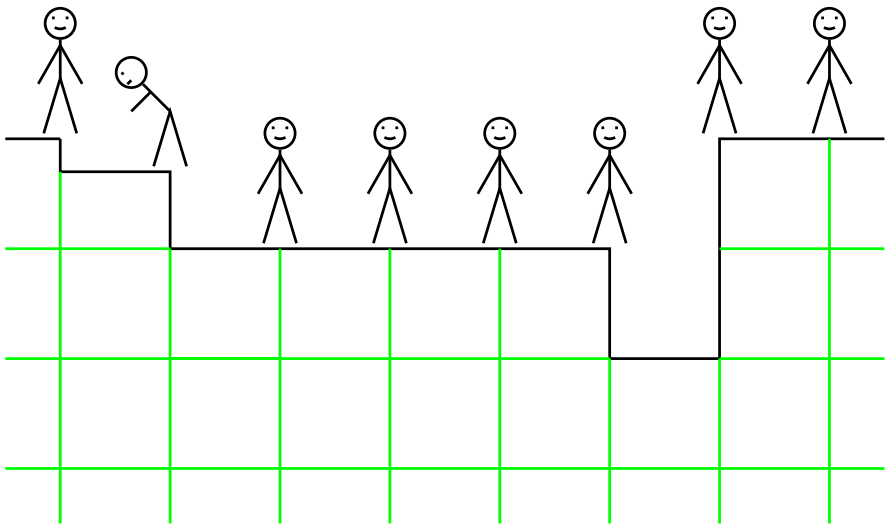
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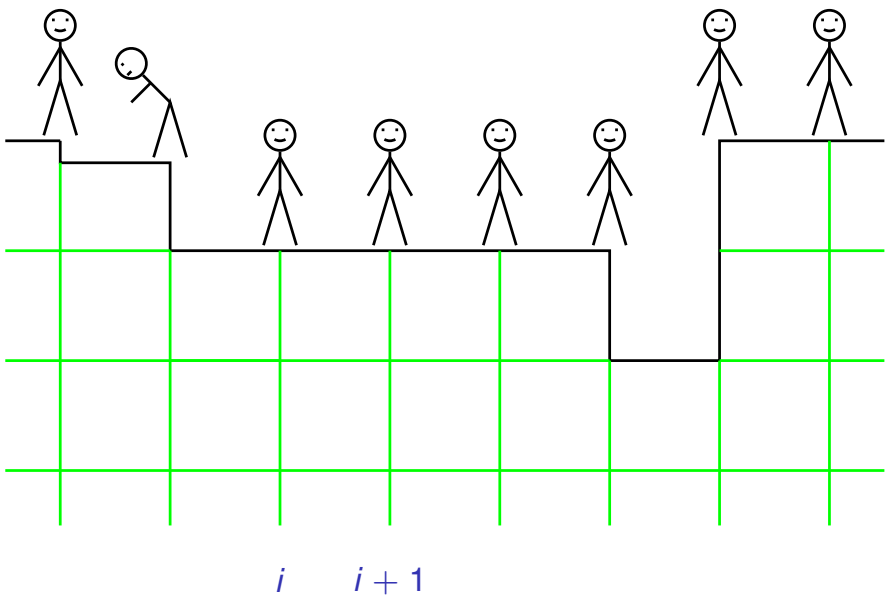
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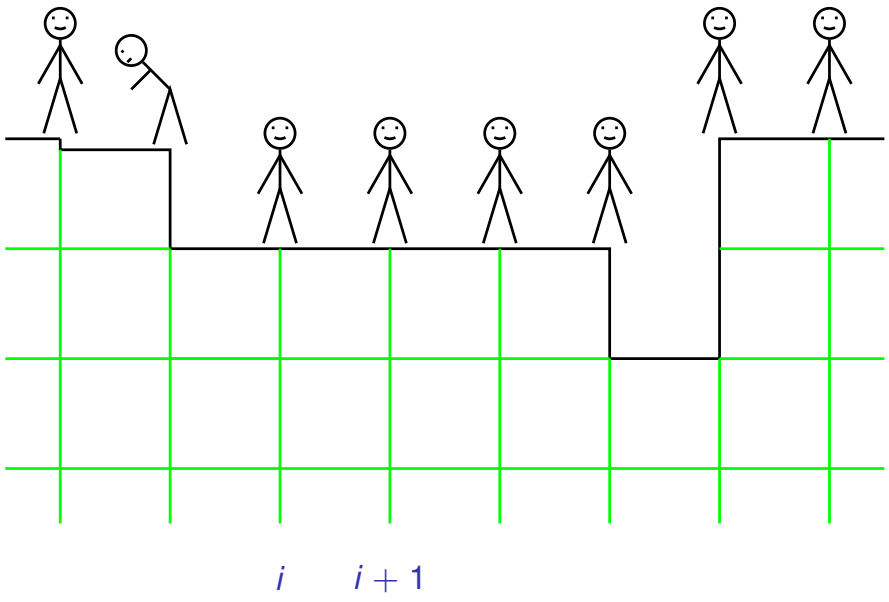
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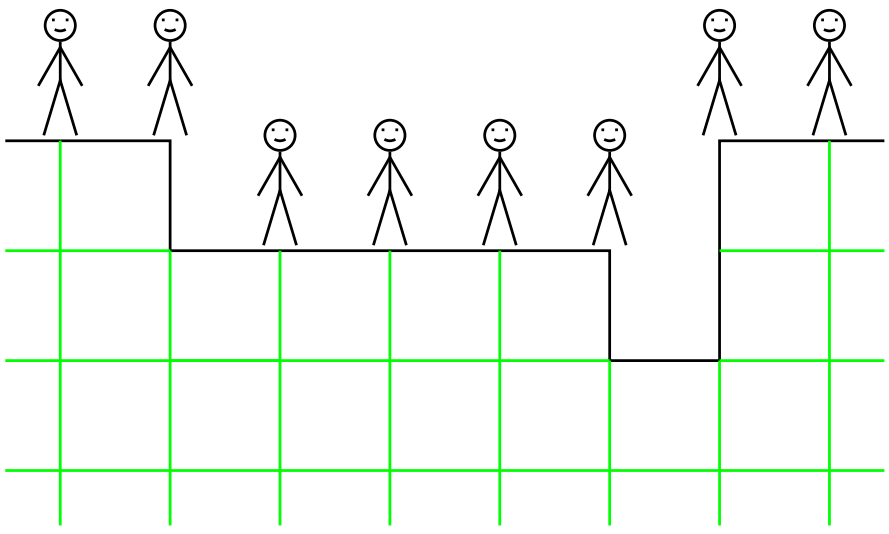
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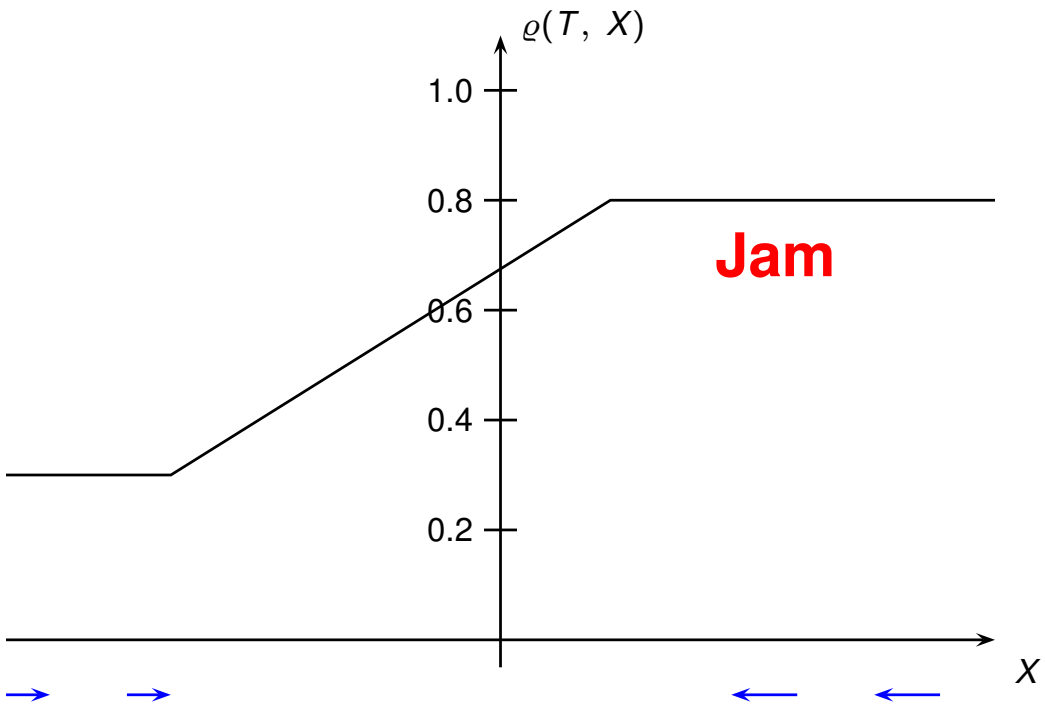
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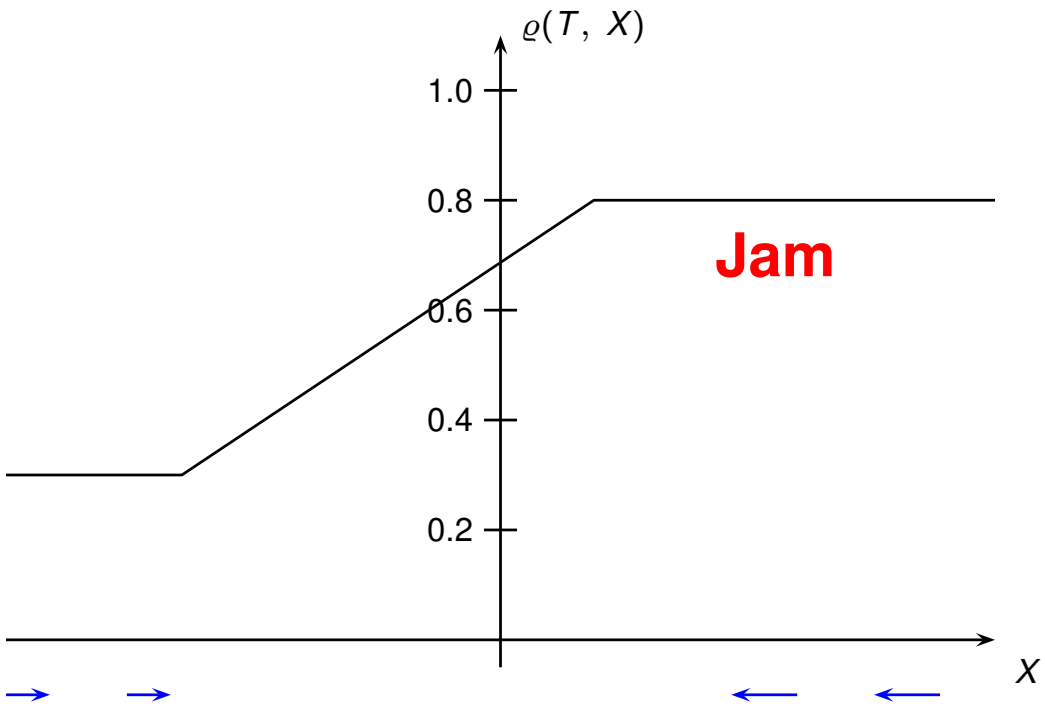
Extremal translation-invariant stationary distributions are still product, and rather explicit in terms of $r(\cdot)$.

A special case: $r(\omega_i) = e^{\beta\omega_i}$: $\omega_i \sim$ discrete Gaussian$(\frac{\theta}{\beta}, \frac{1}{\sqrt{\beta}})$.

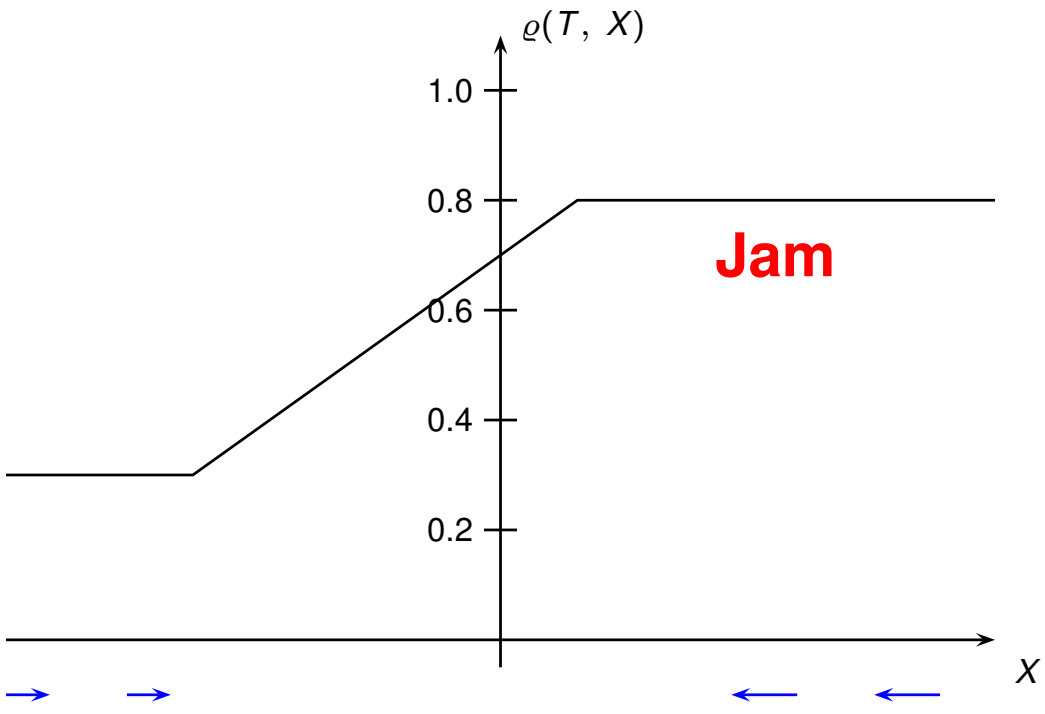
Rescaled version: shock



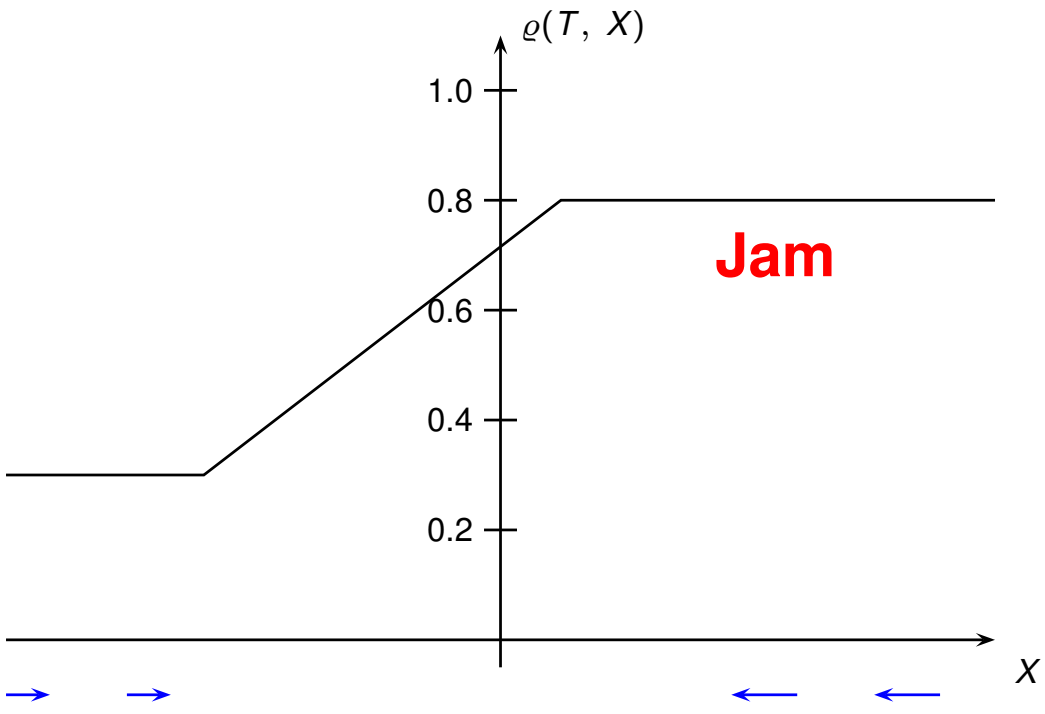
Rescaled version: shock



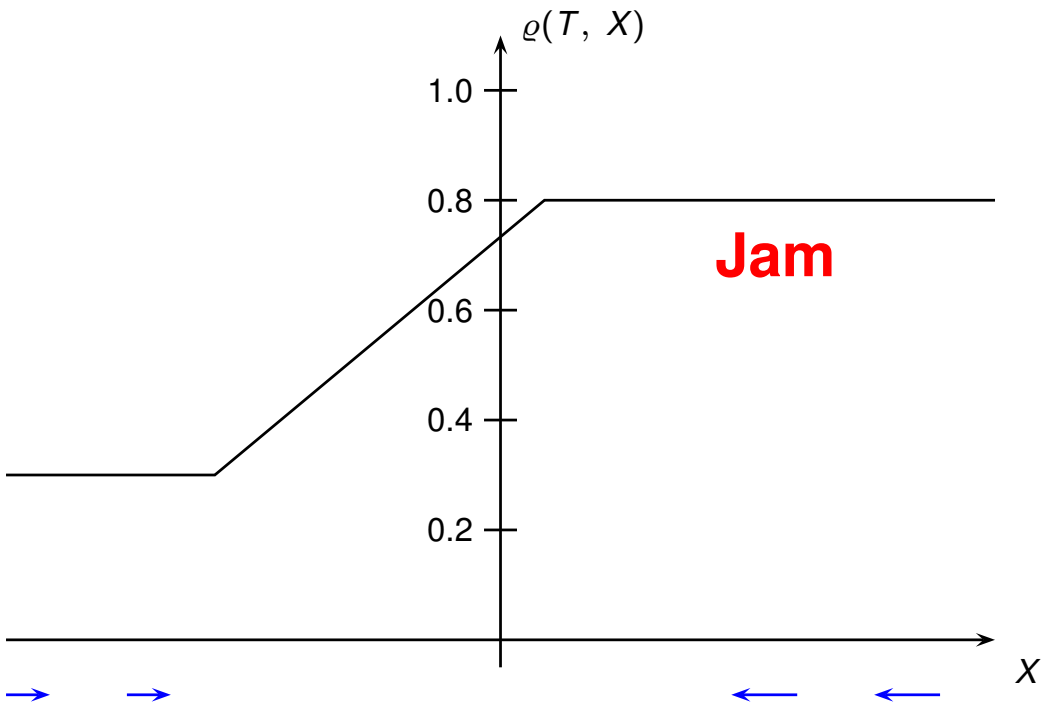
Rescaled version: shock



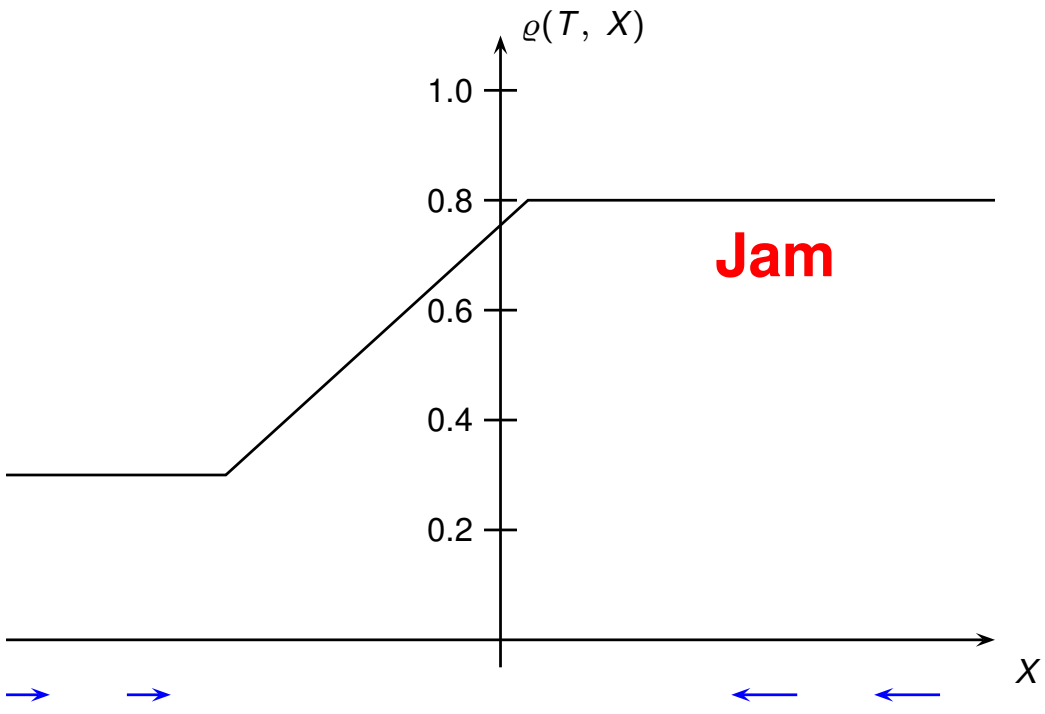
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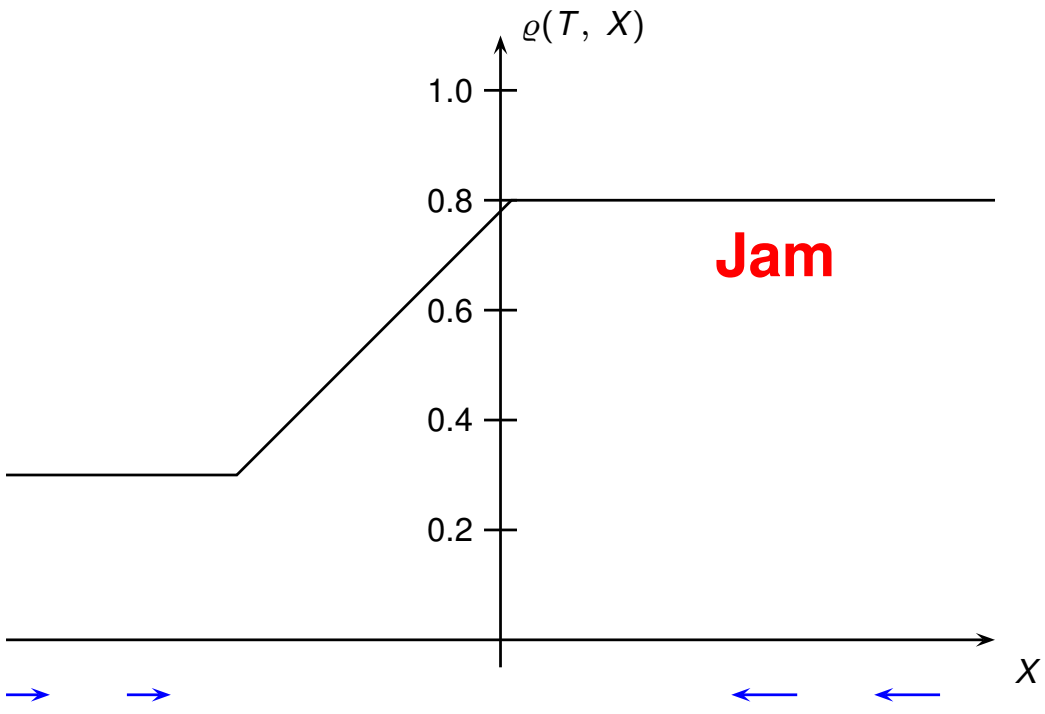
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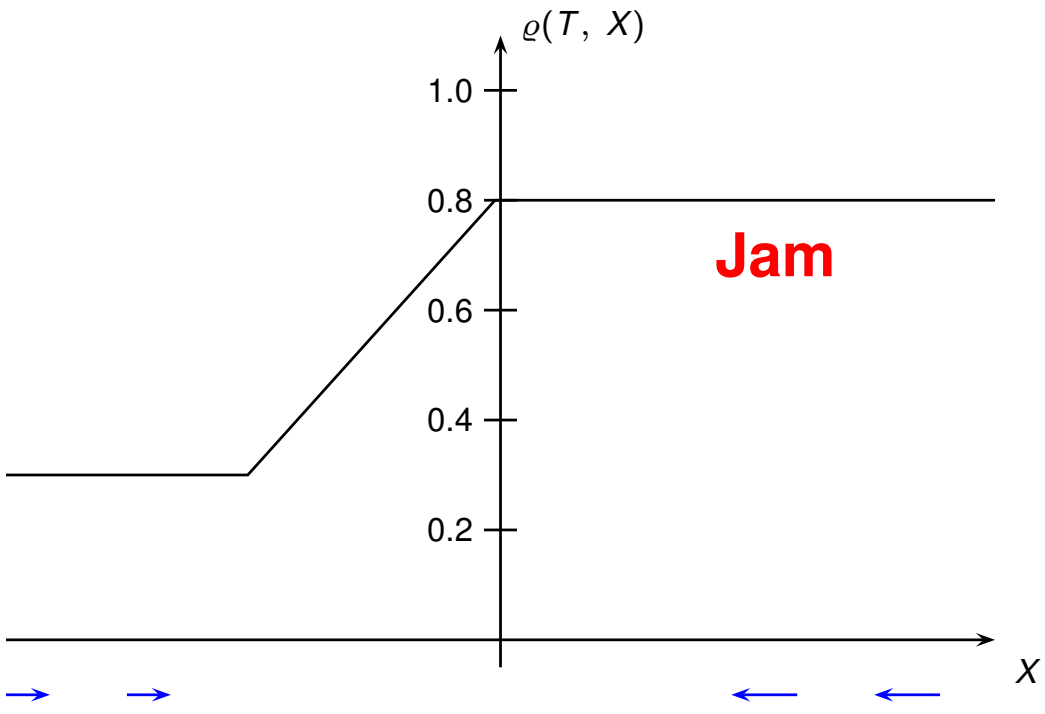
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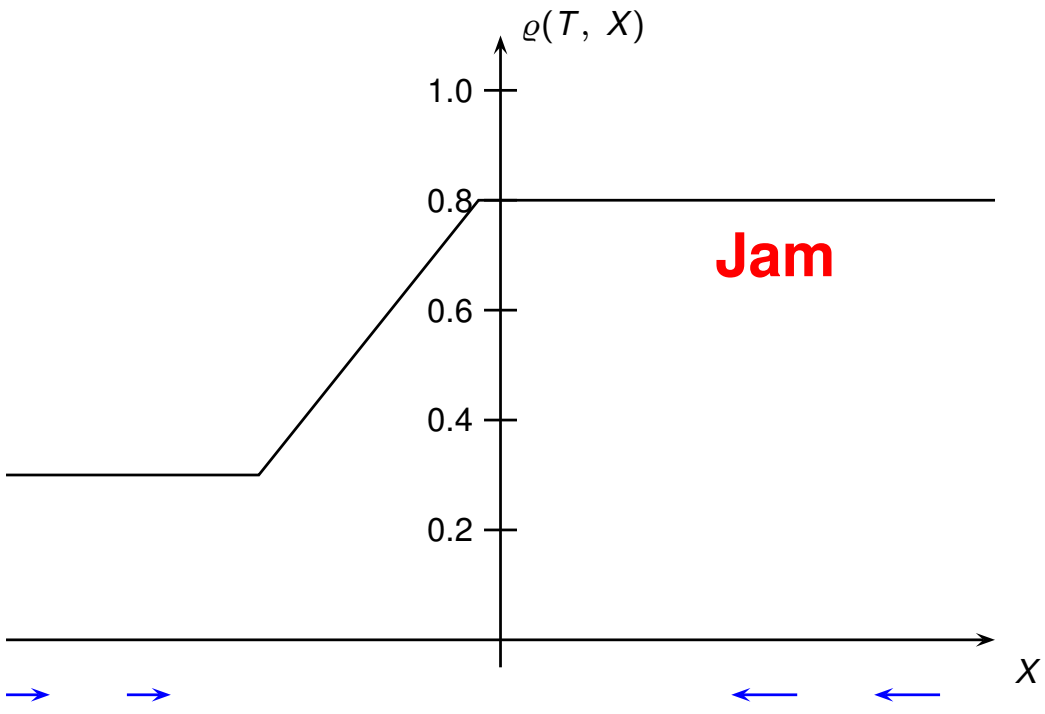
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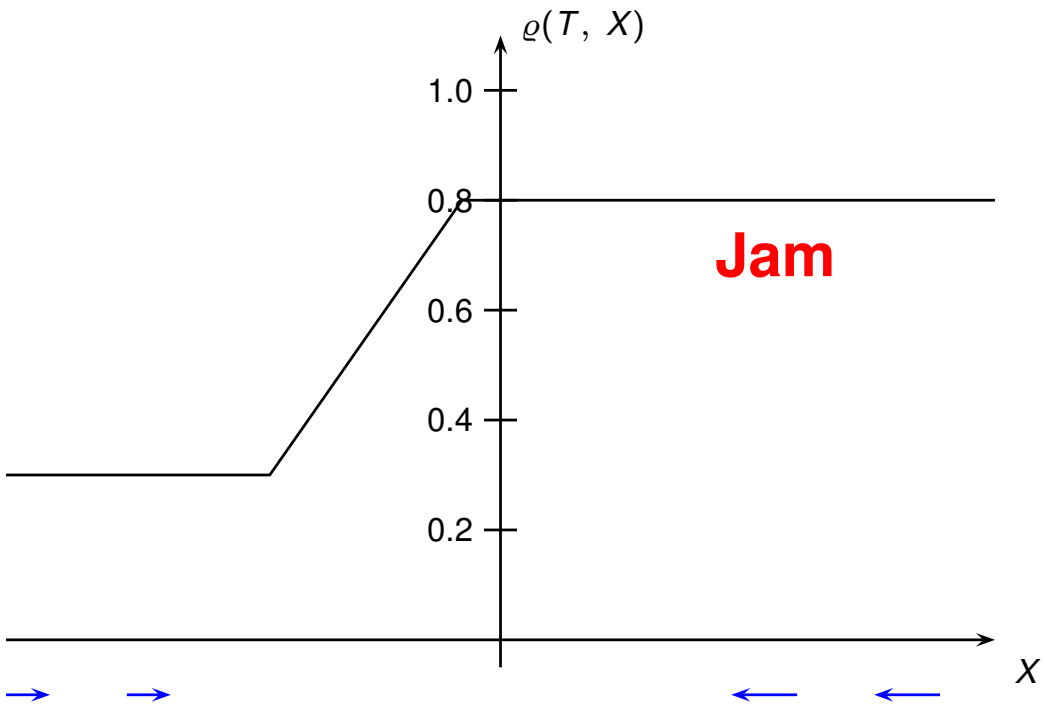
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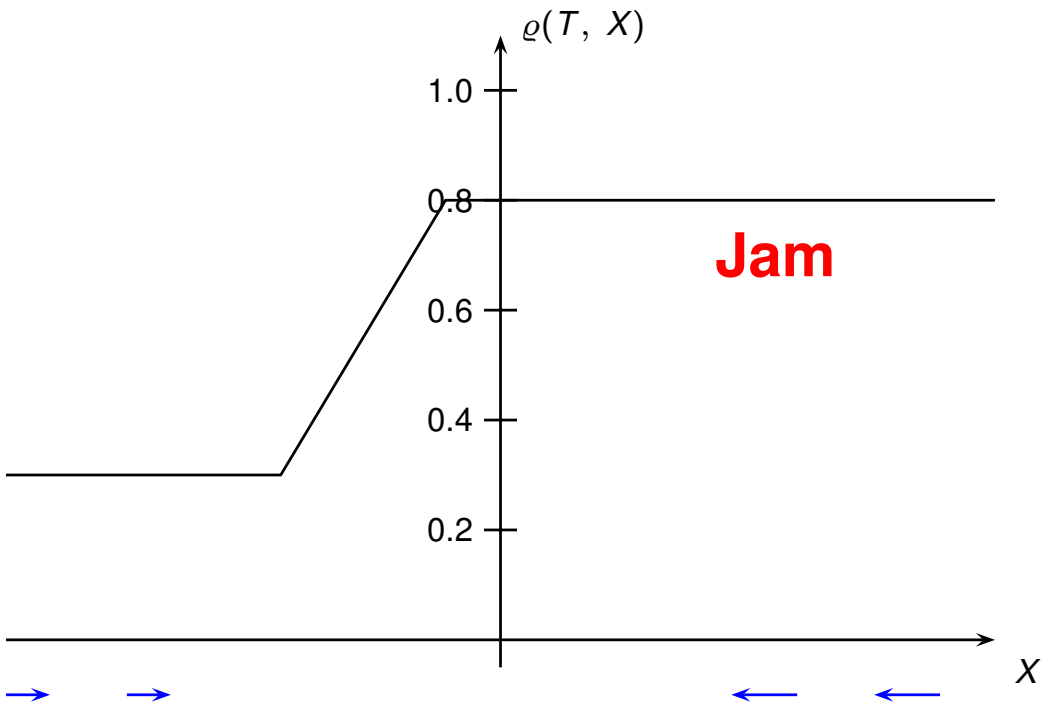
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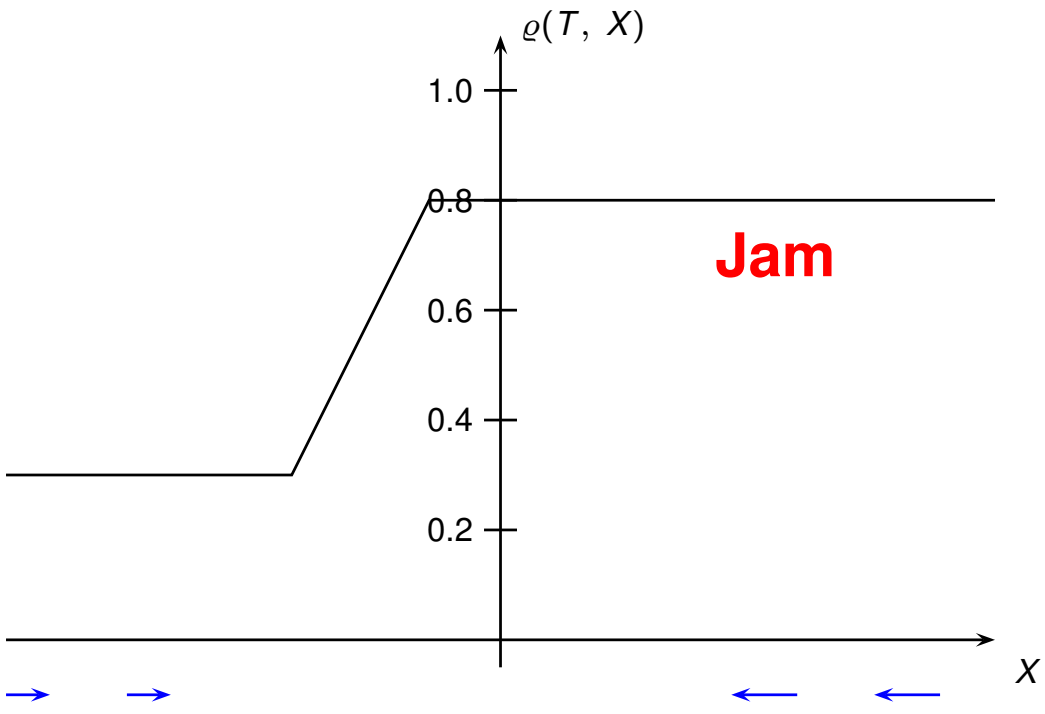
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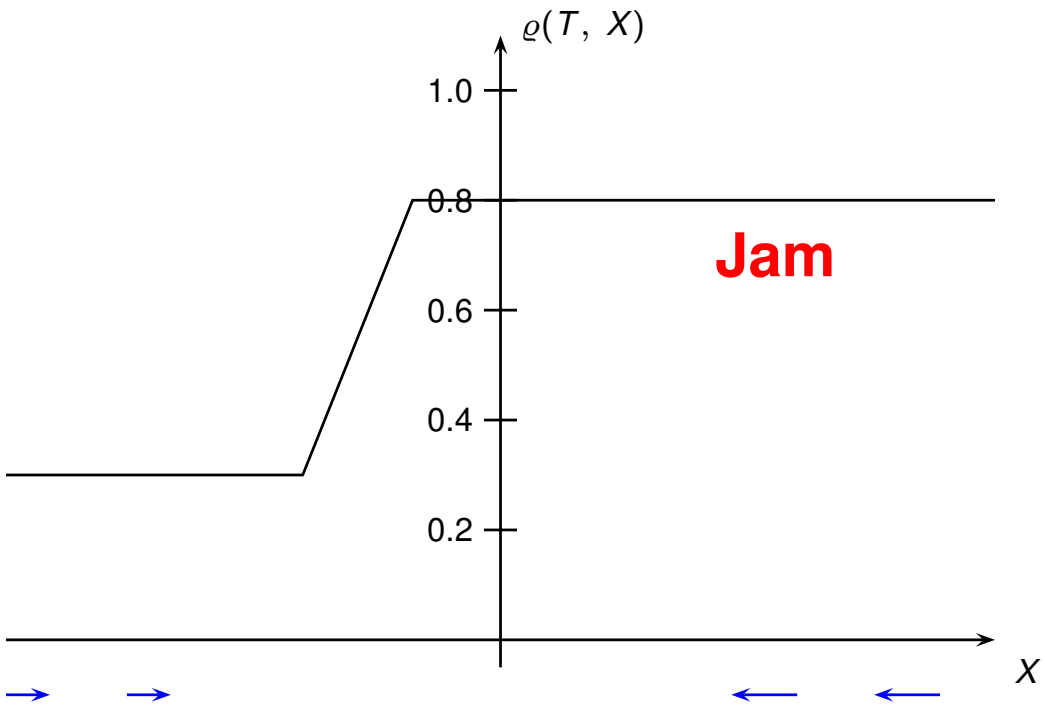
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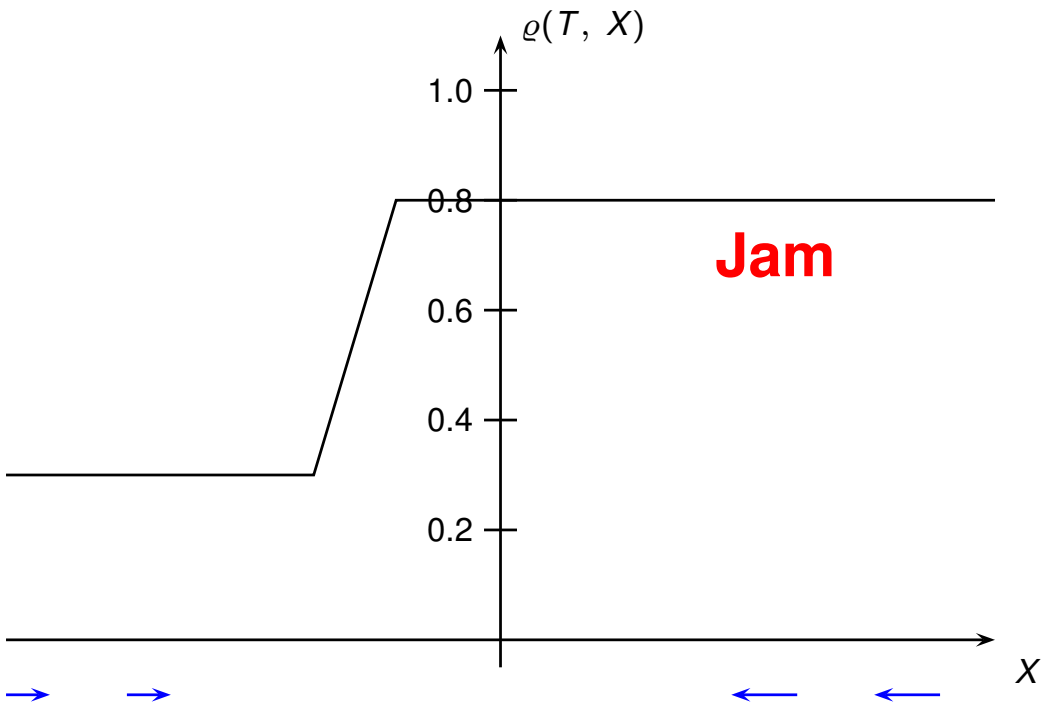
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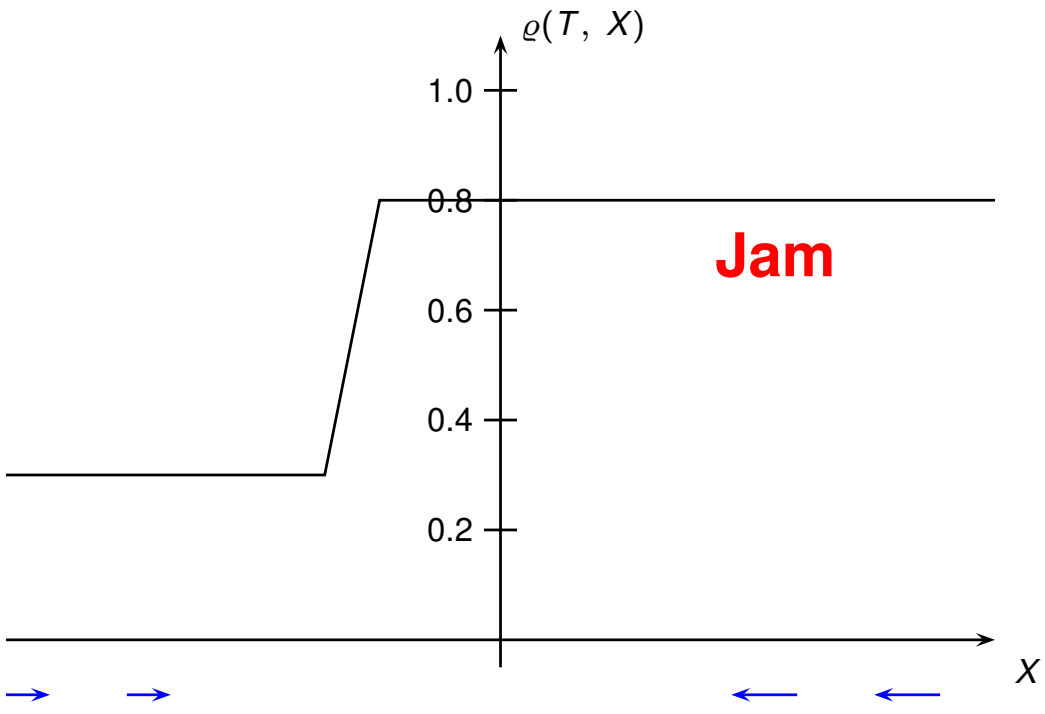
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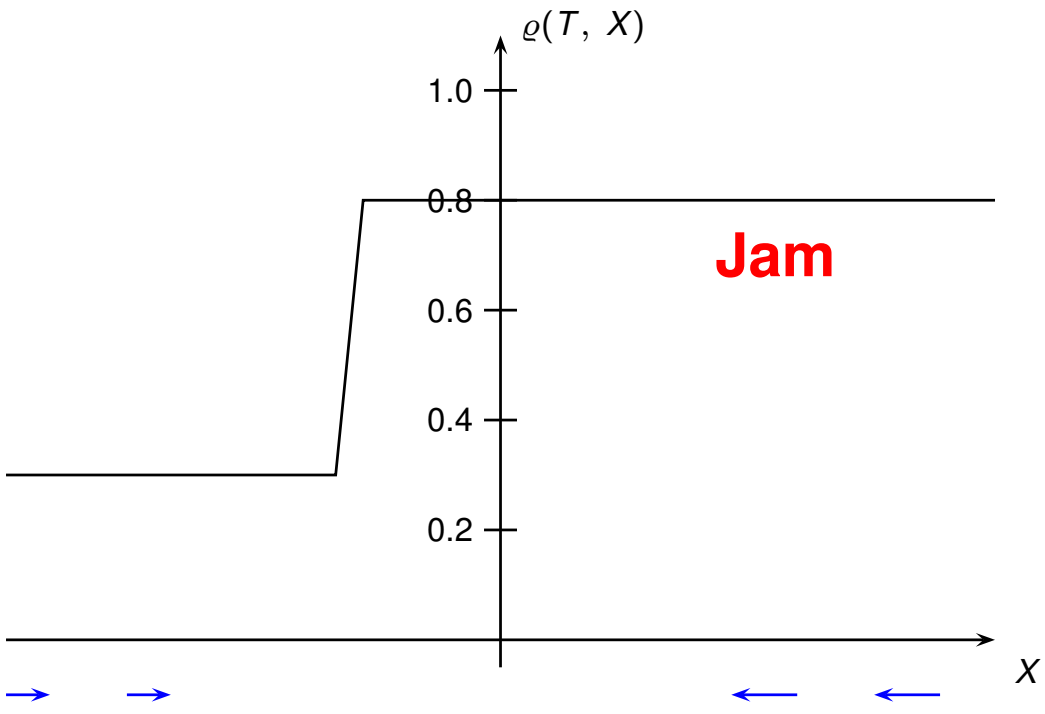
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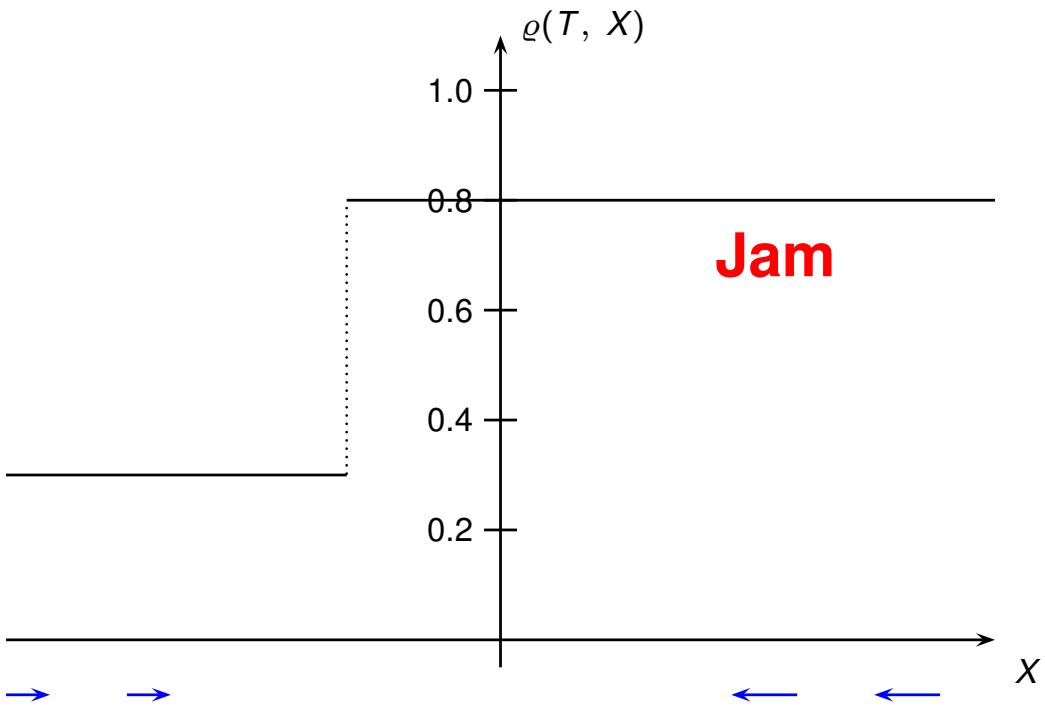
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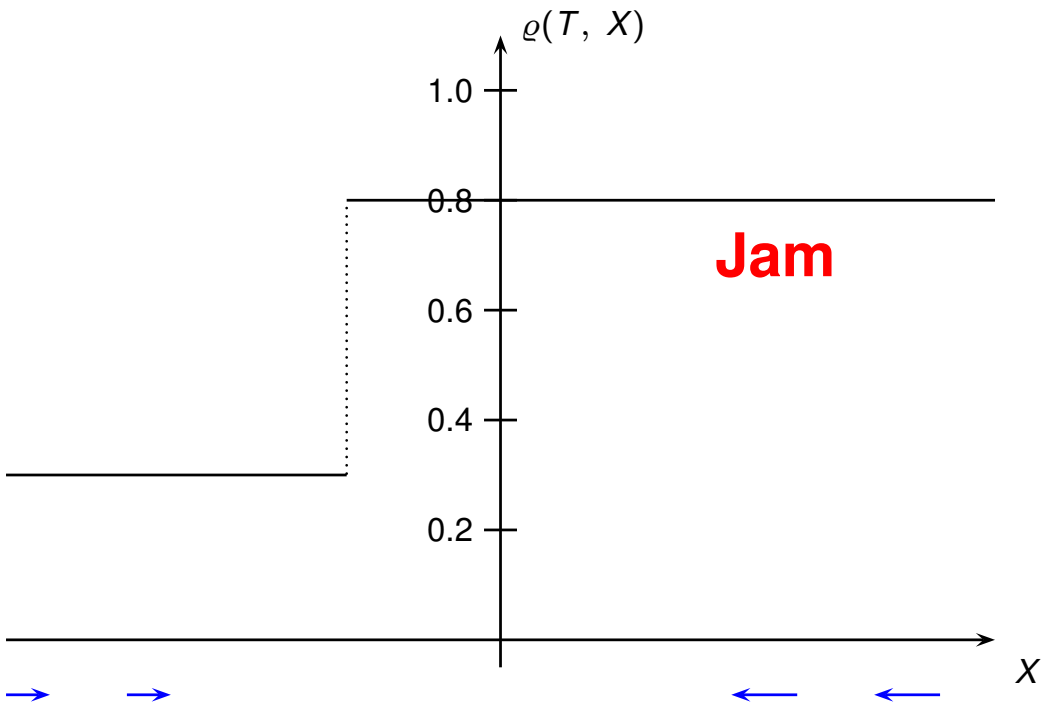
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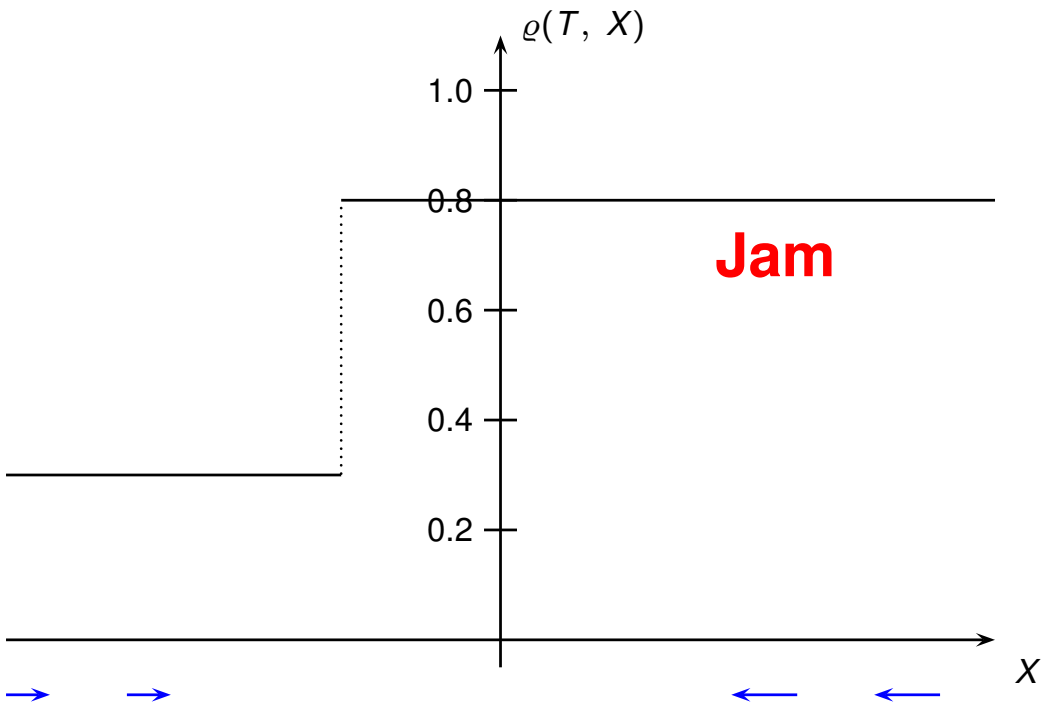
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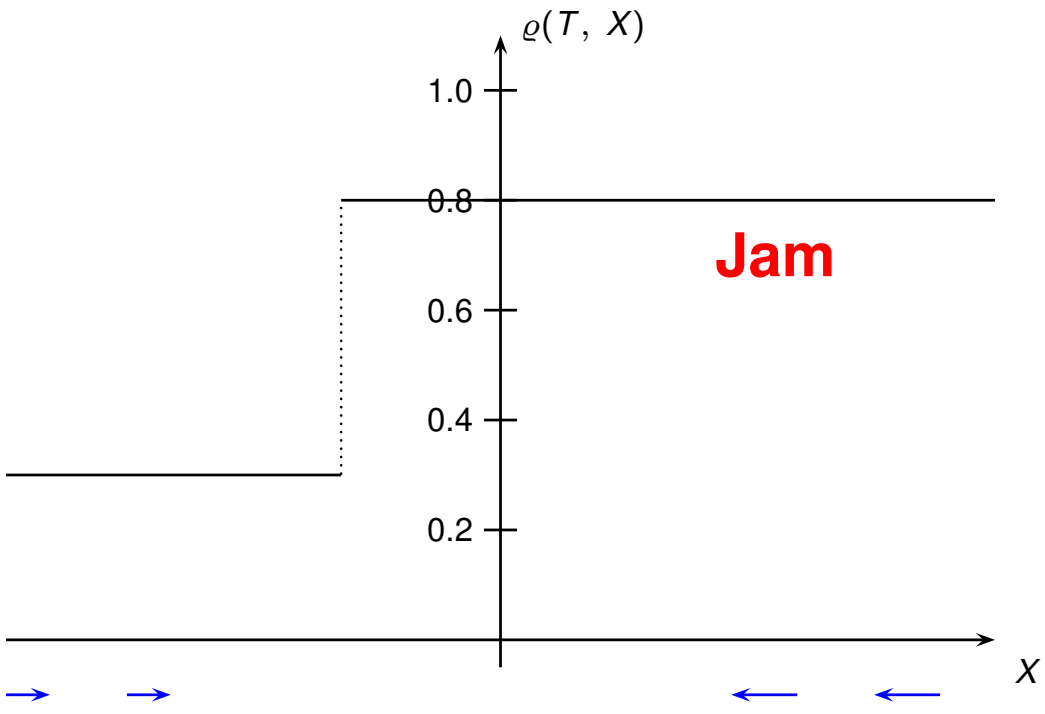
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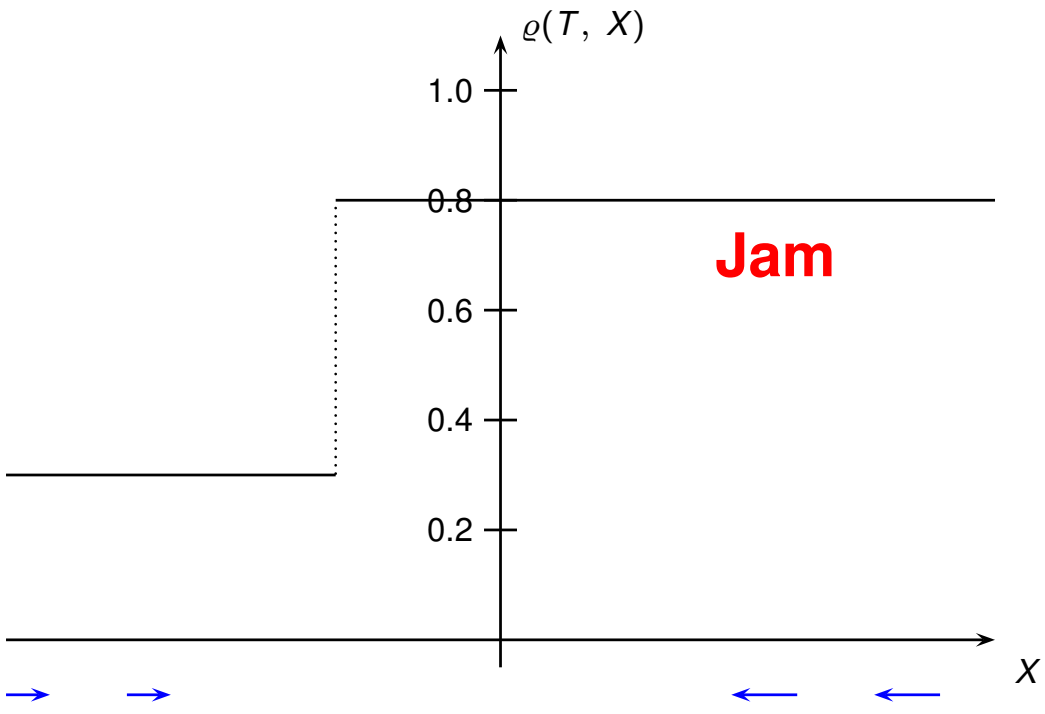
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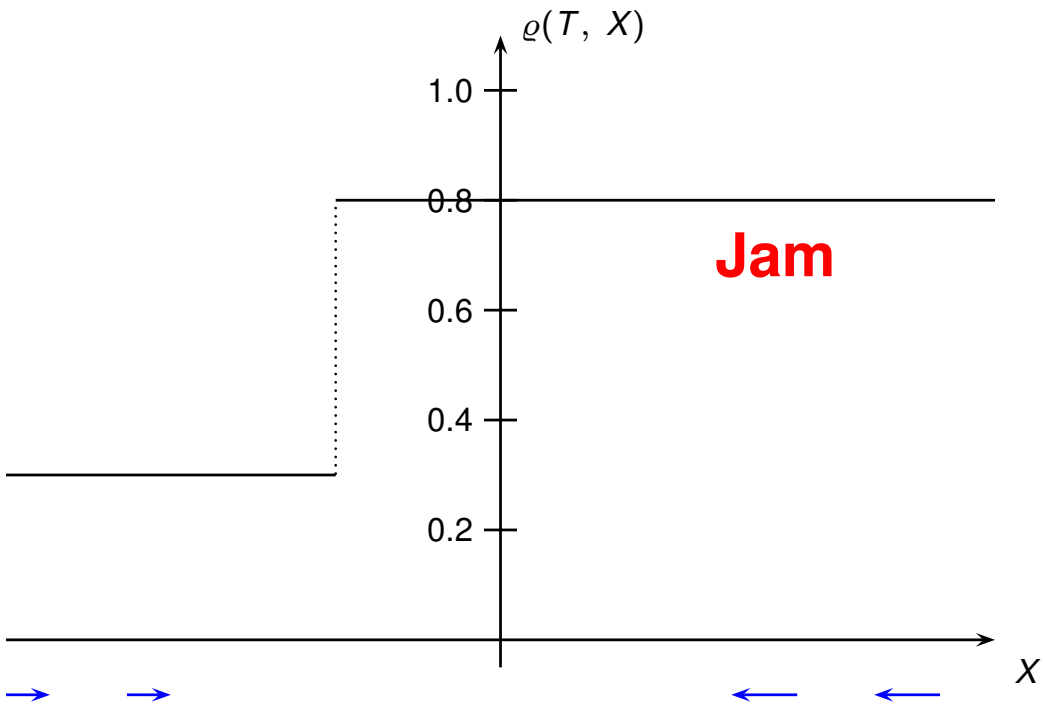
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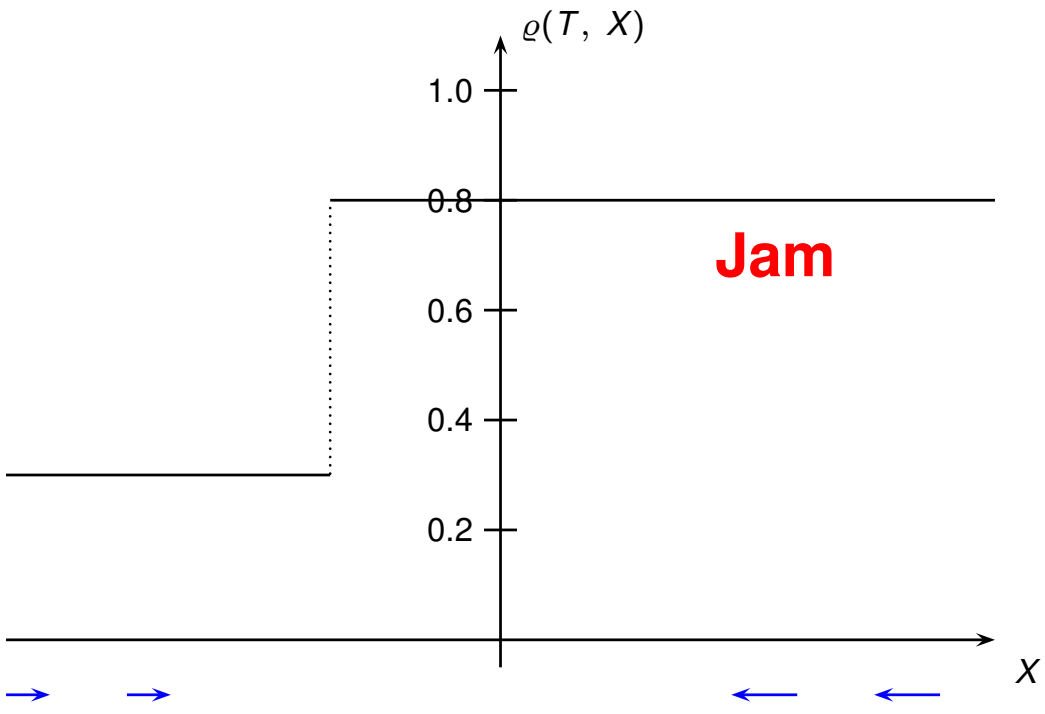
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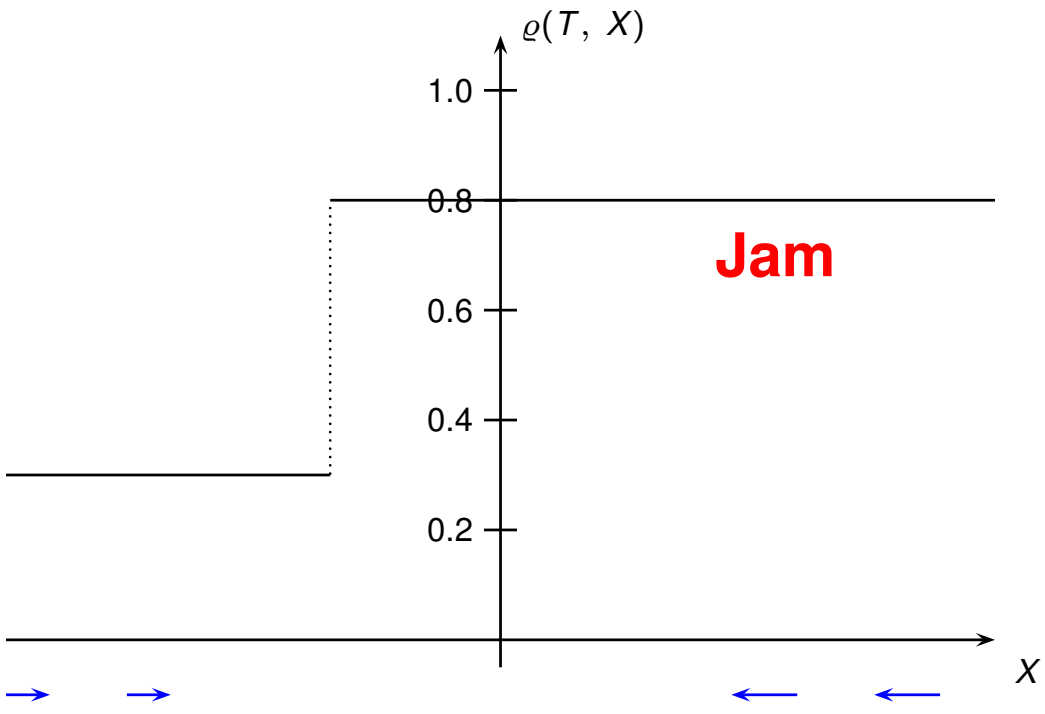
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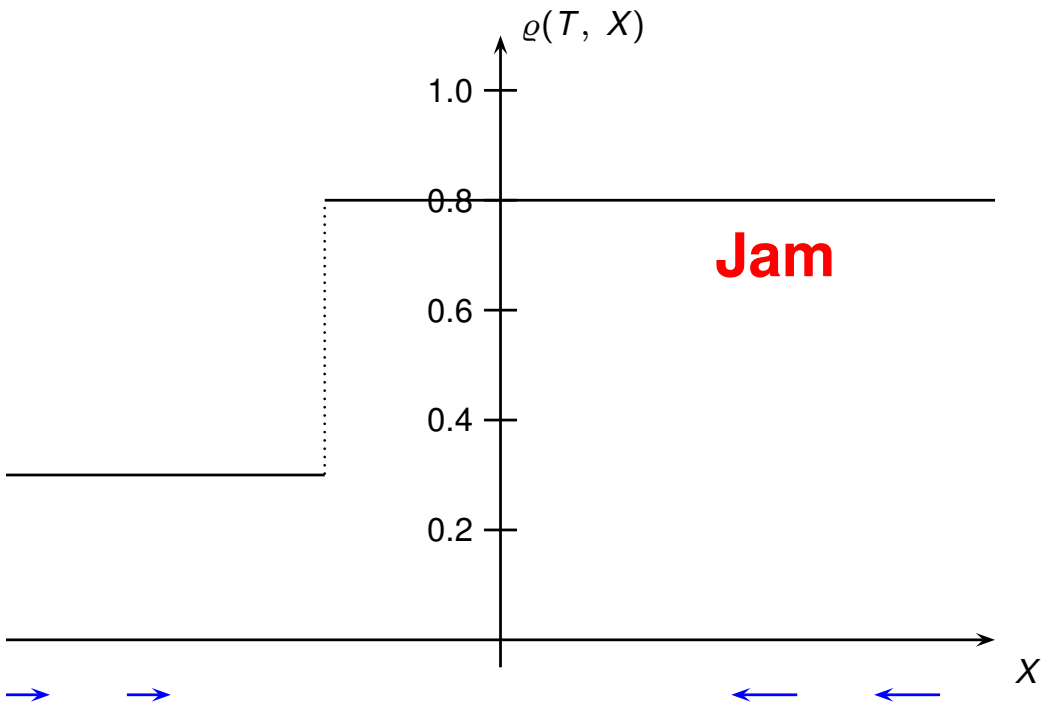
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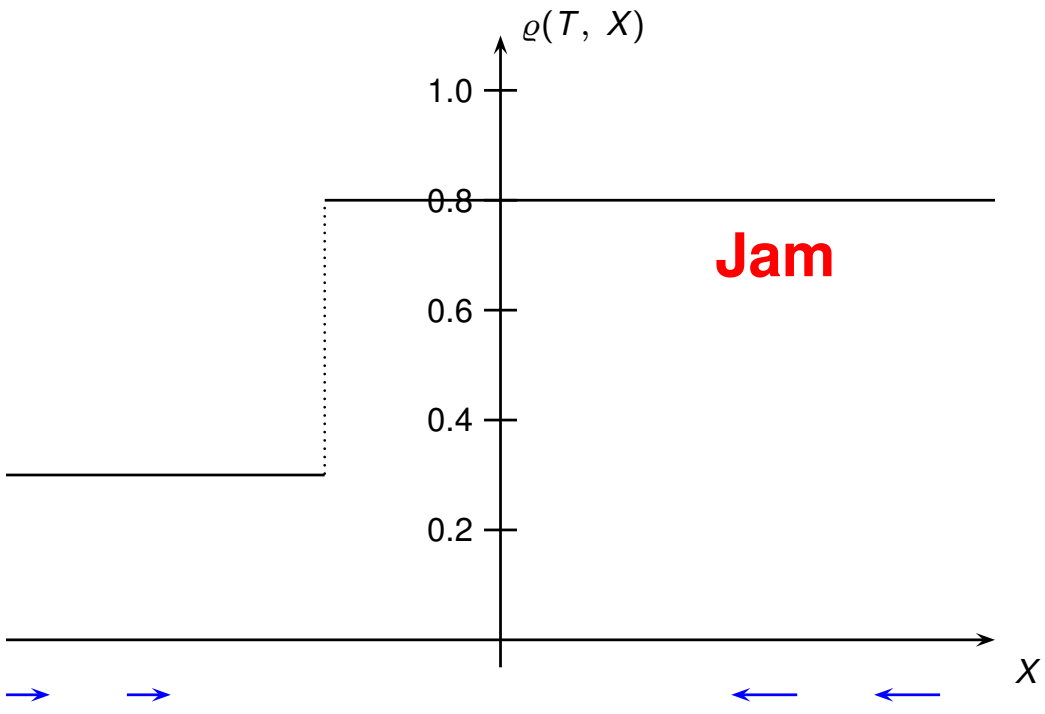
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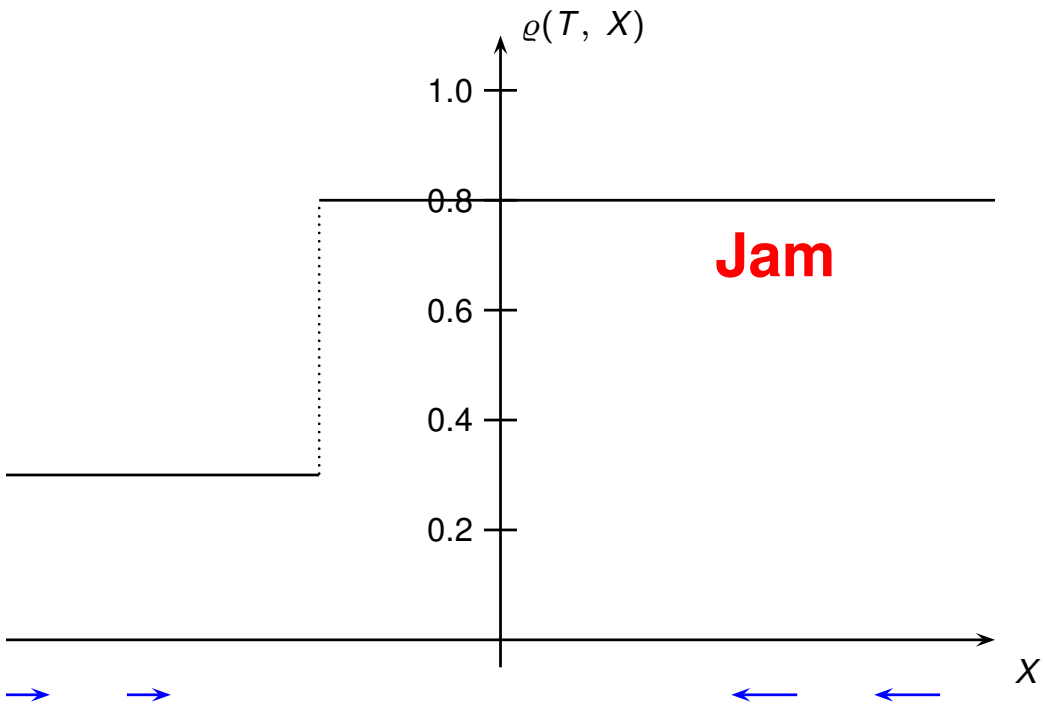
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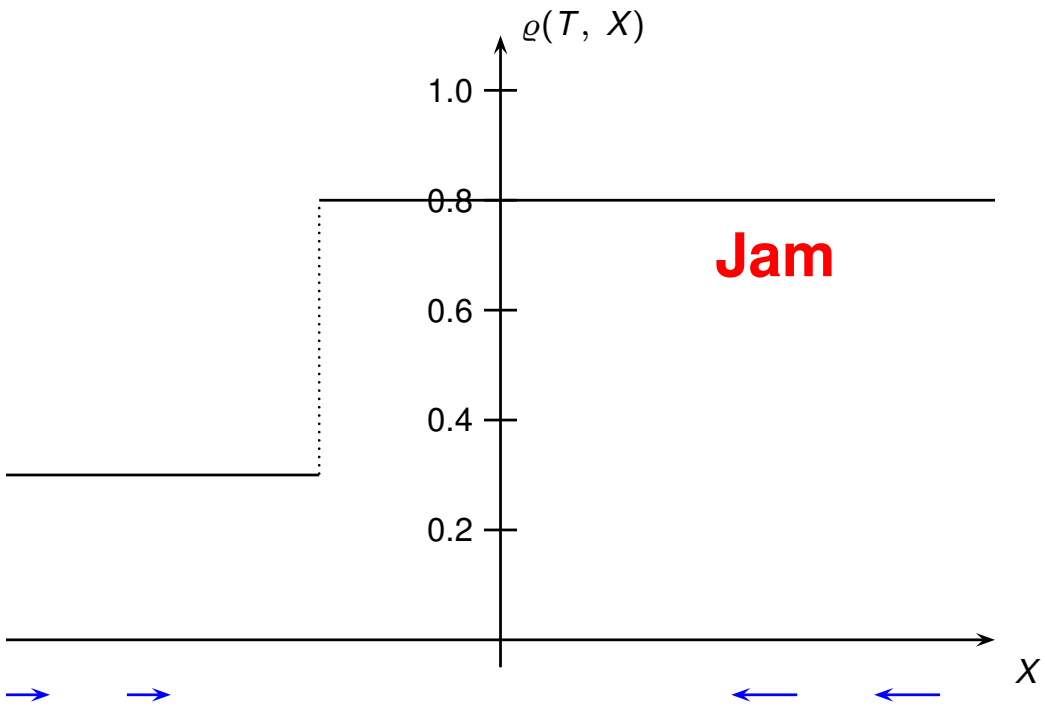
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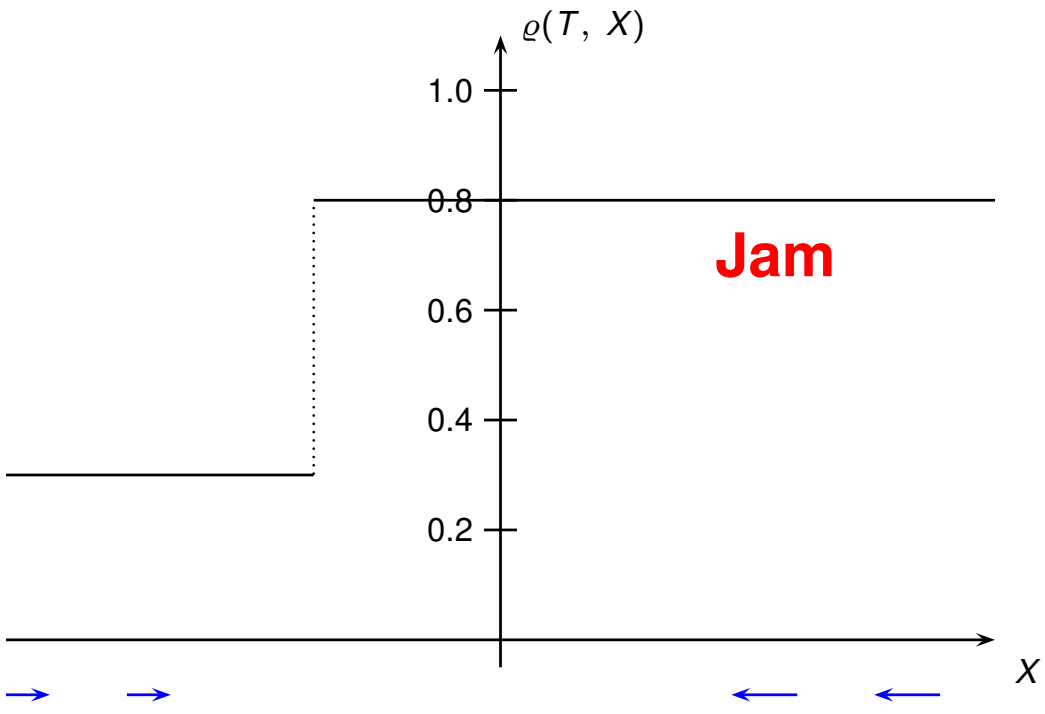
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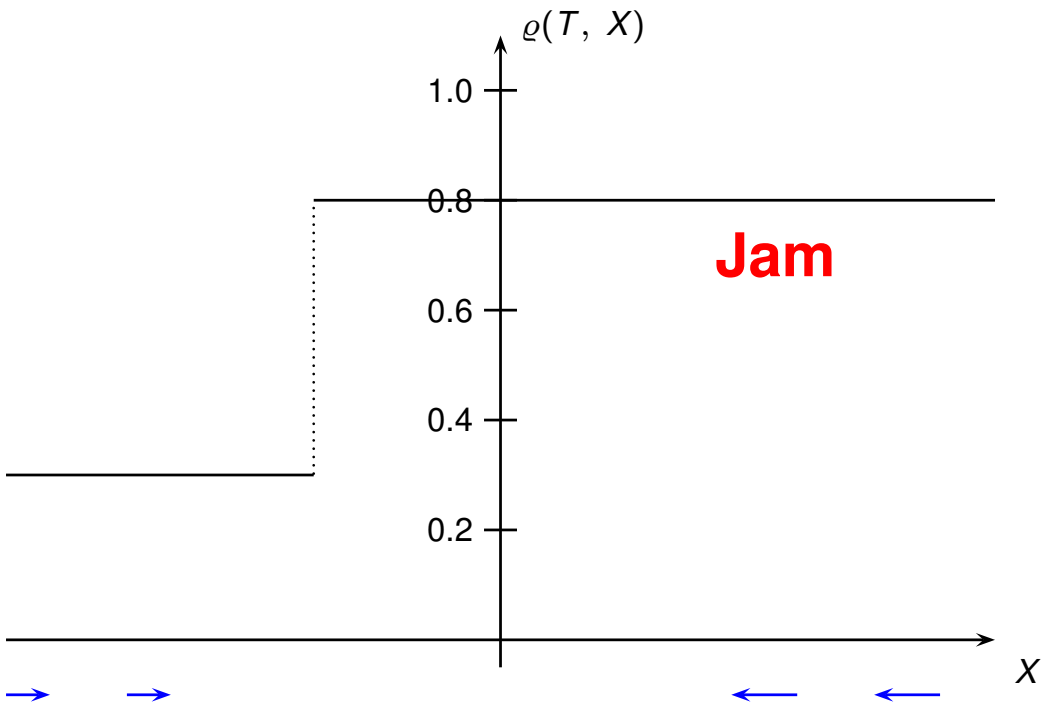
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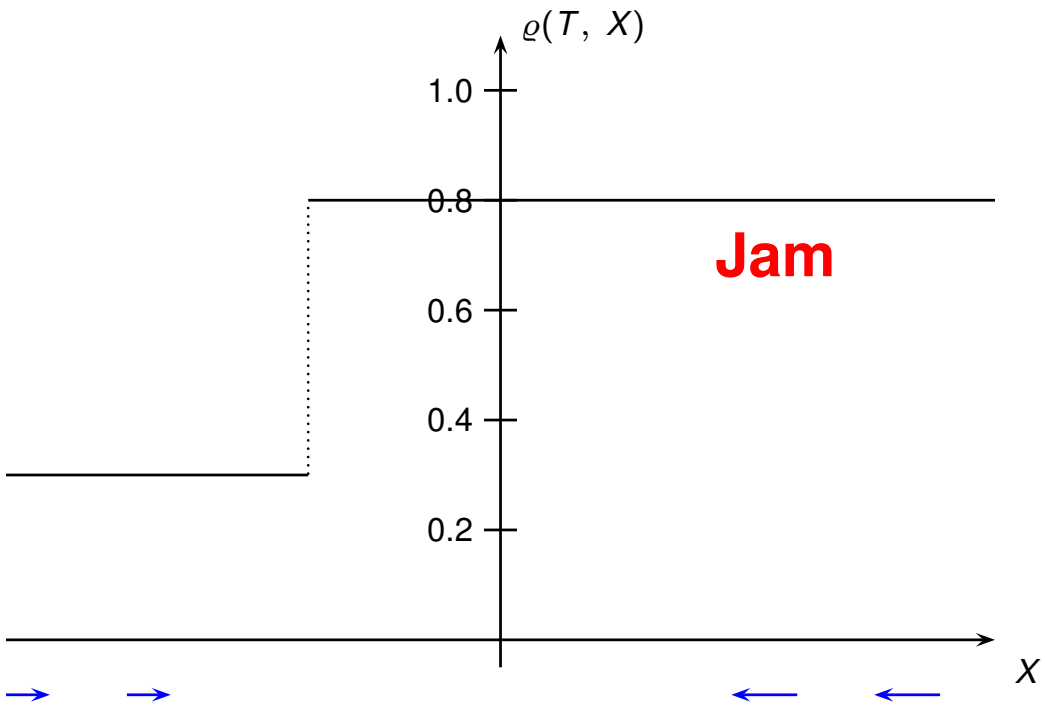
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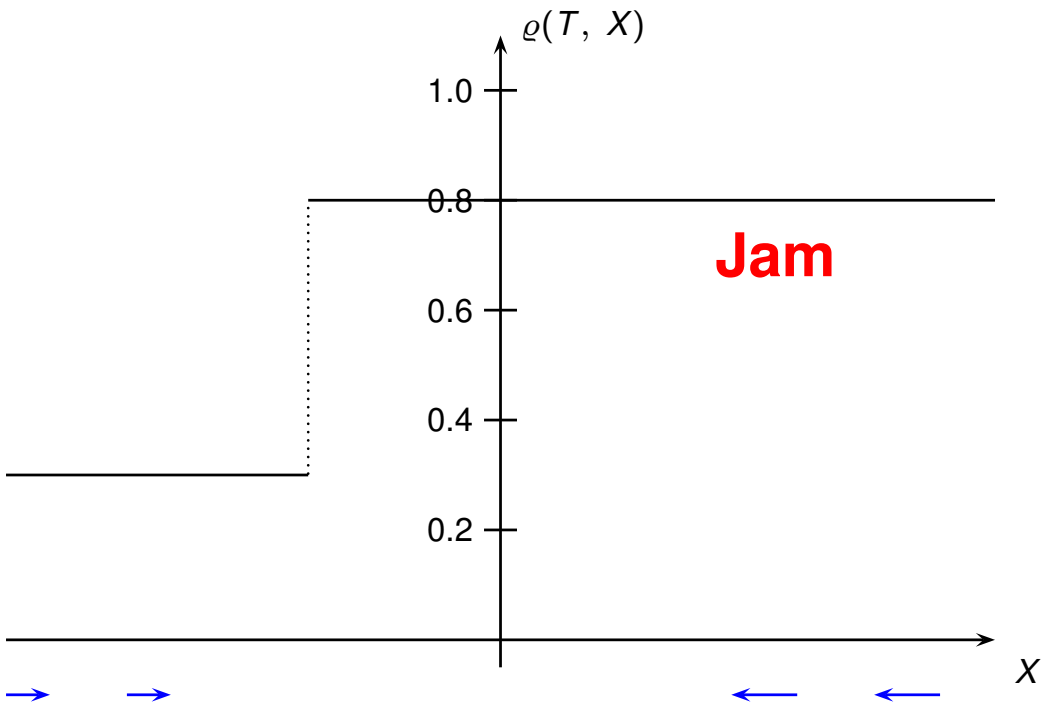
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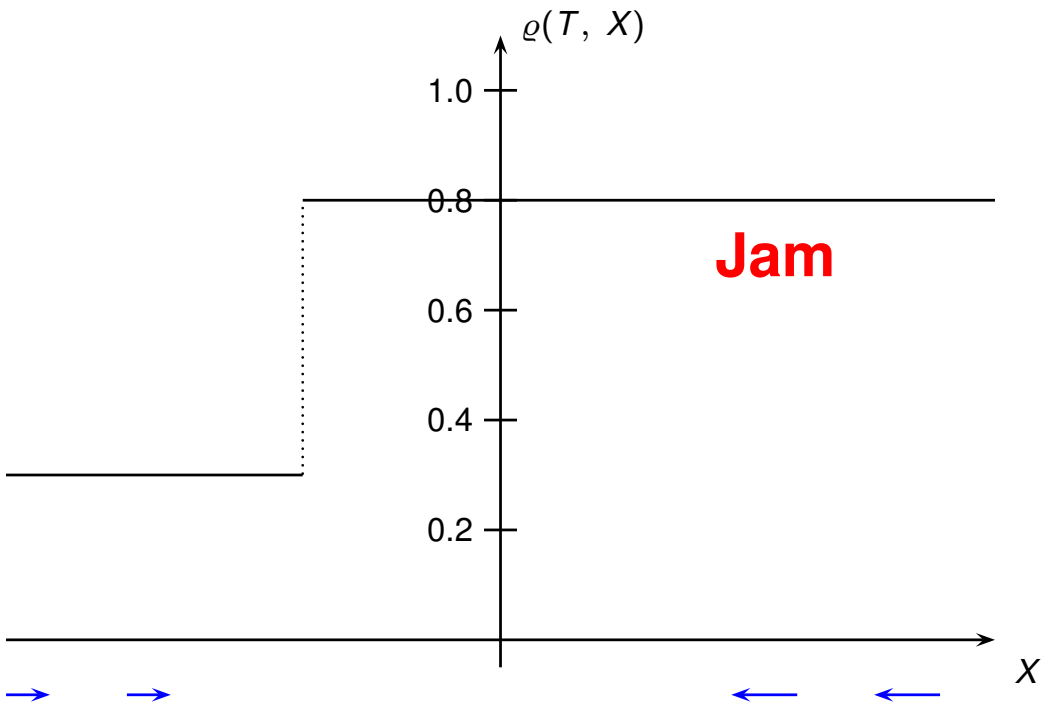
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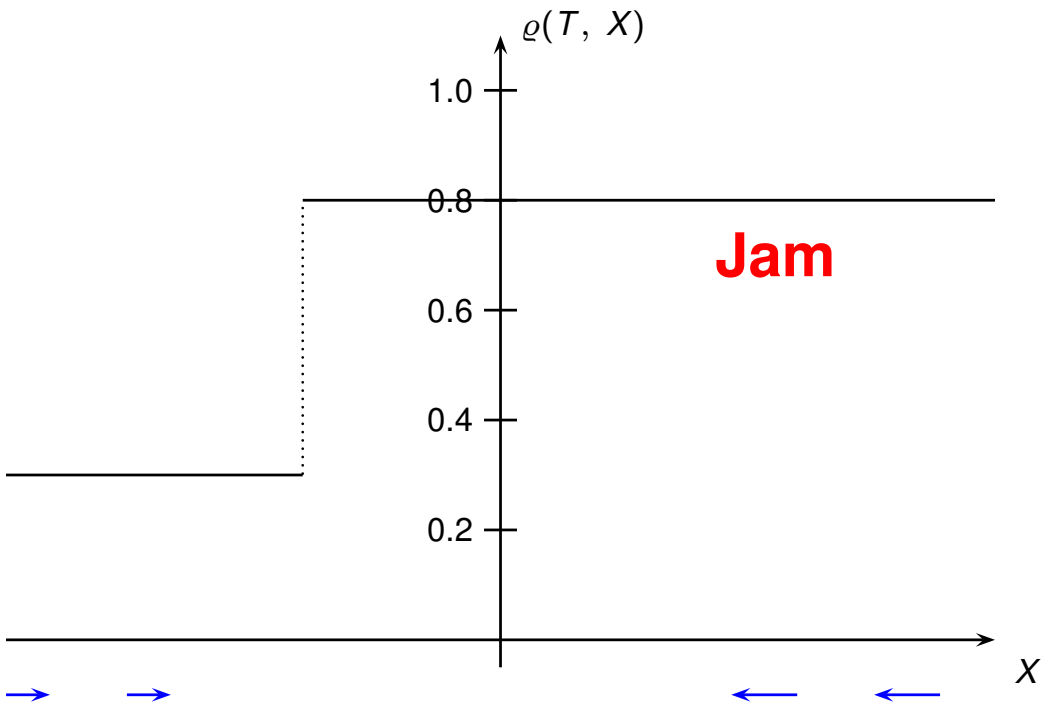
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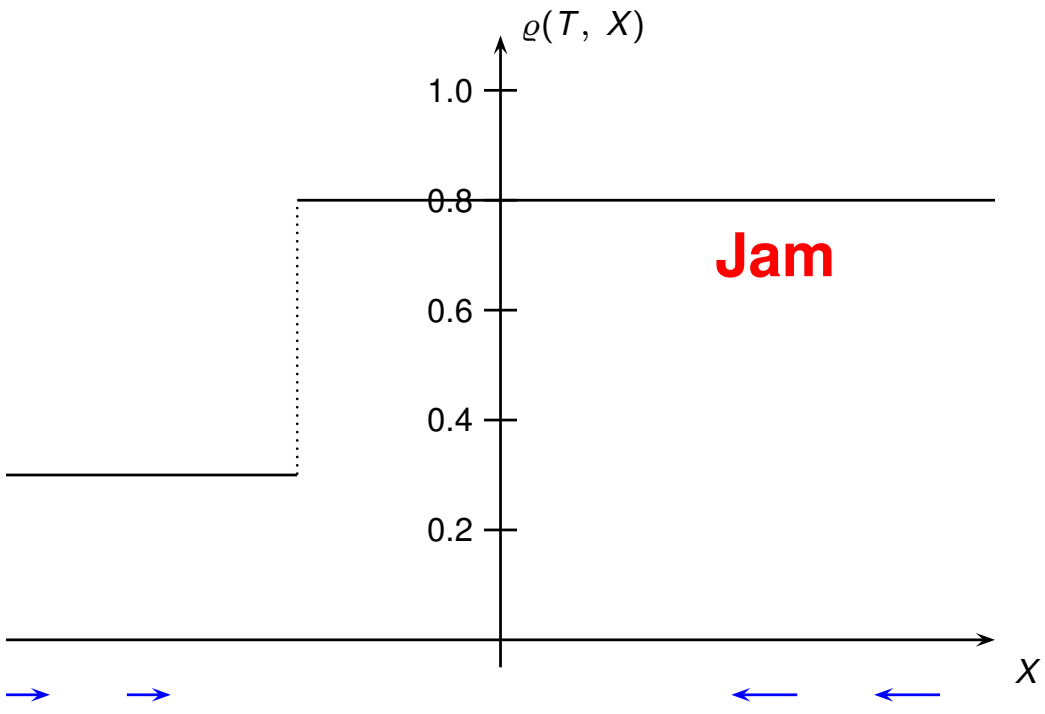
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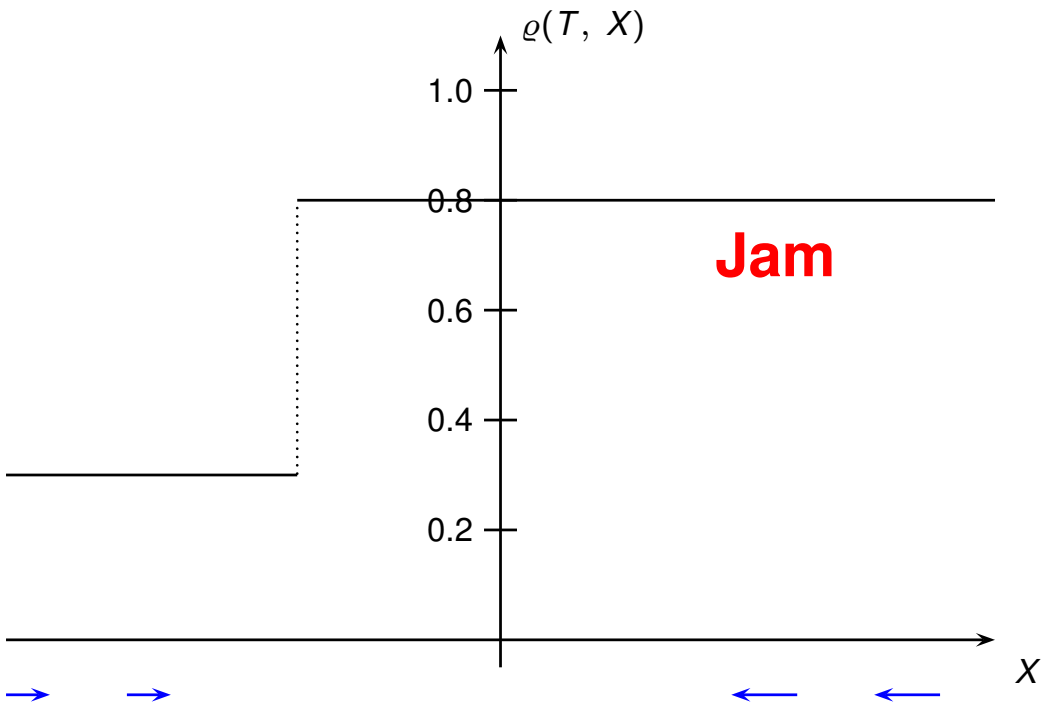
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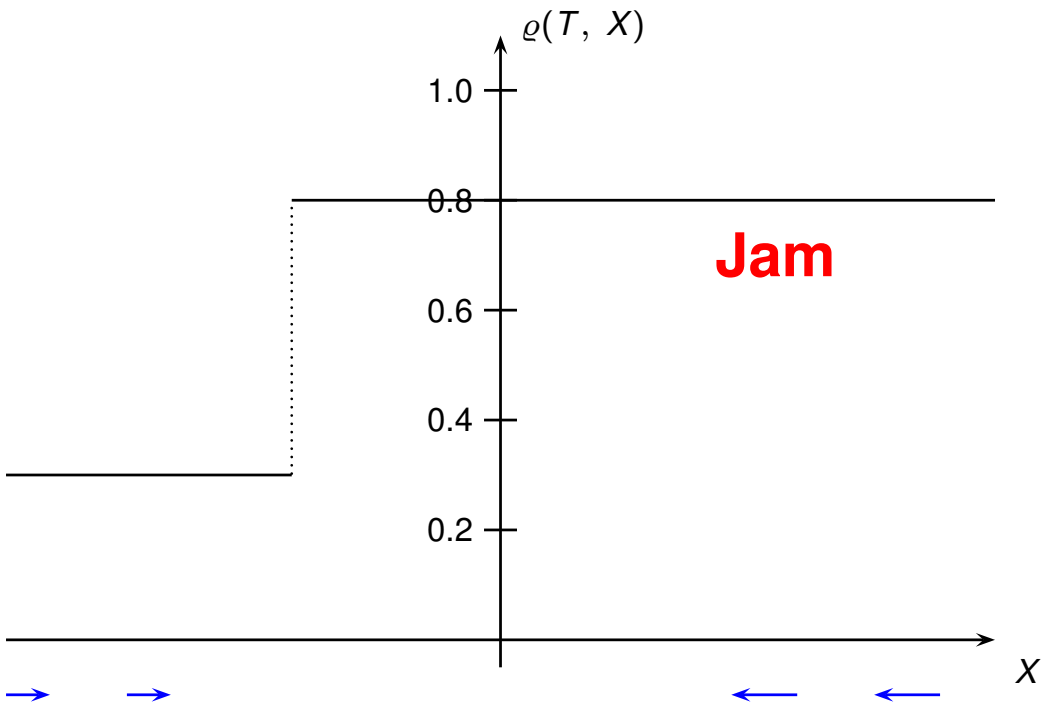
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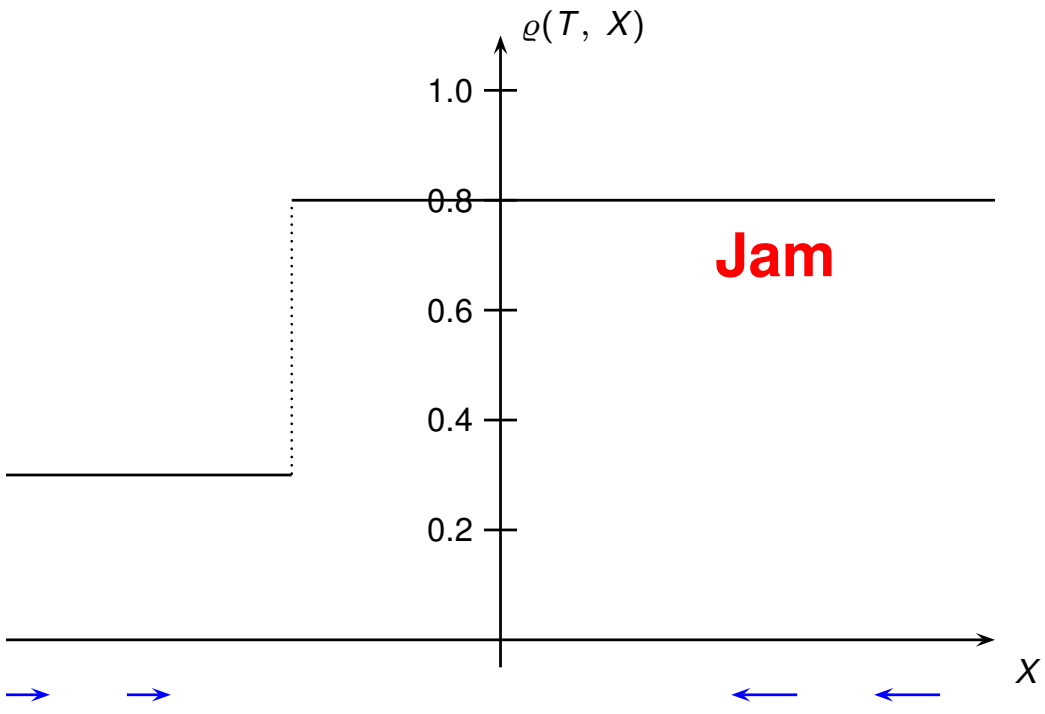
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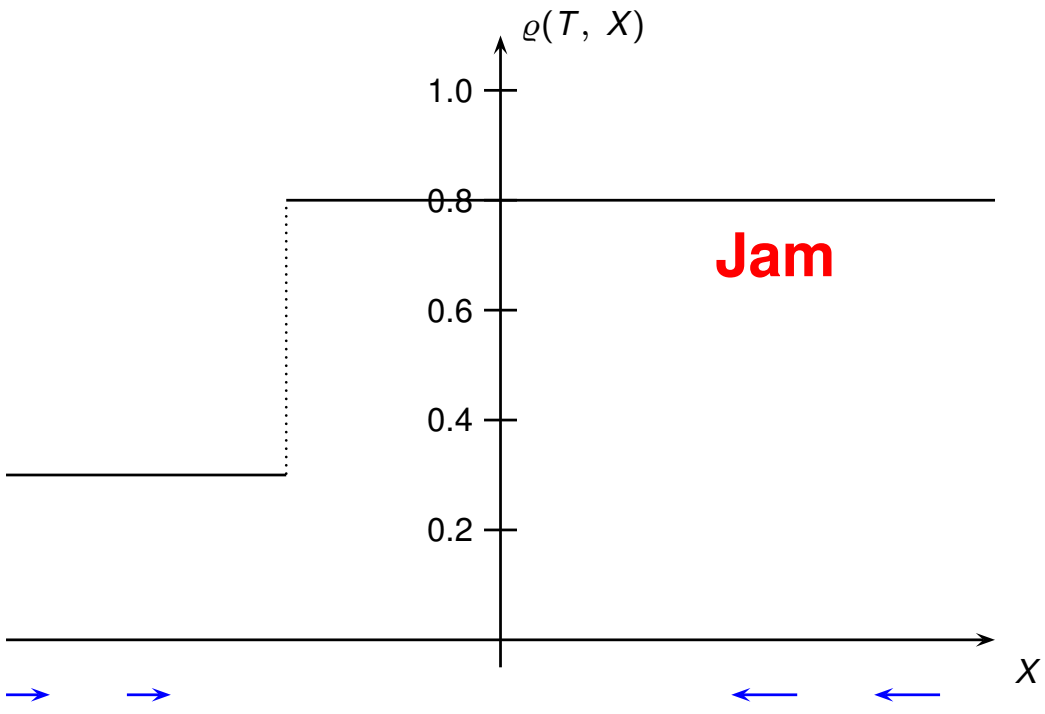
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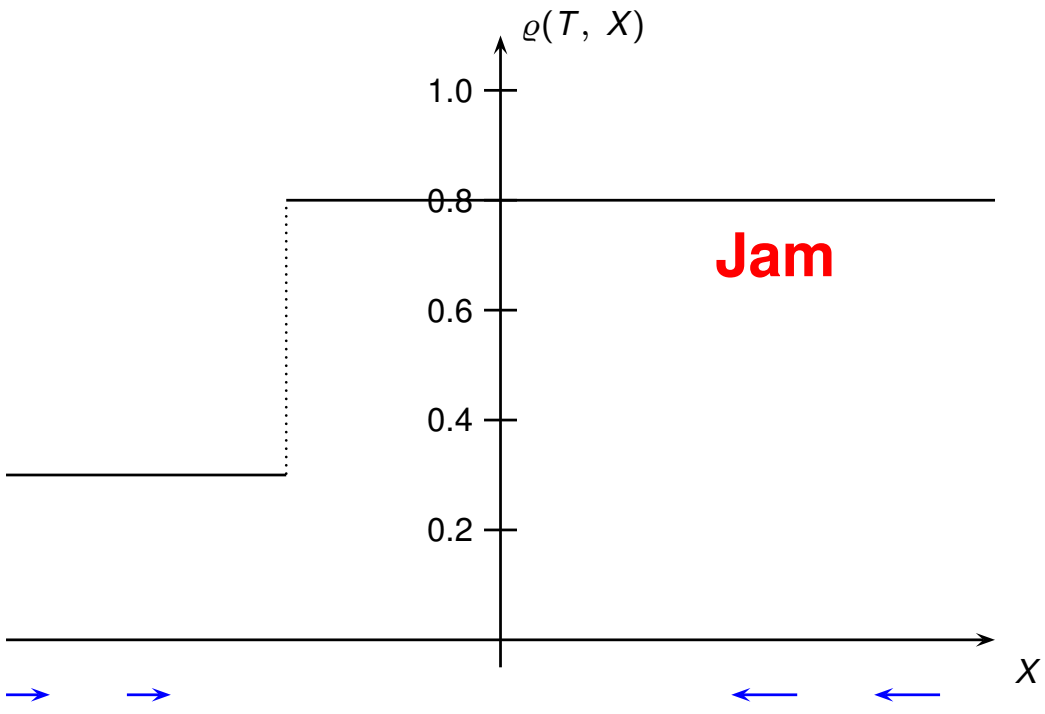
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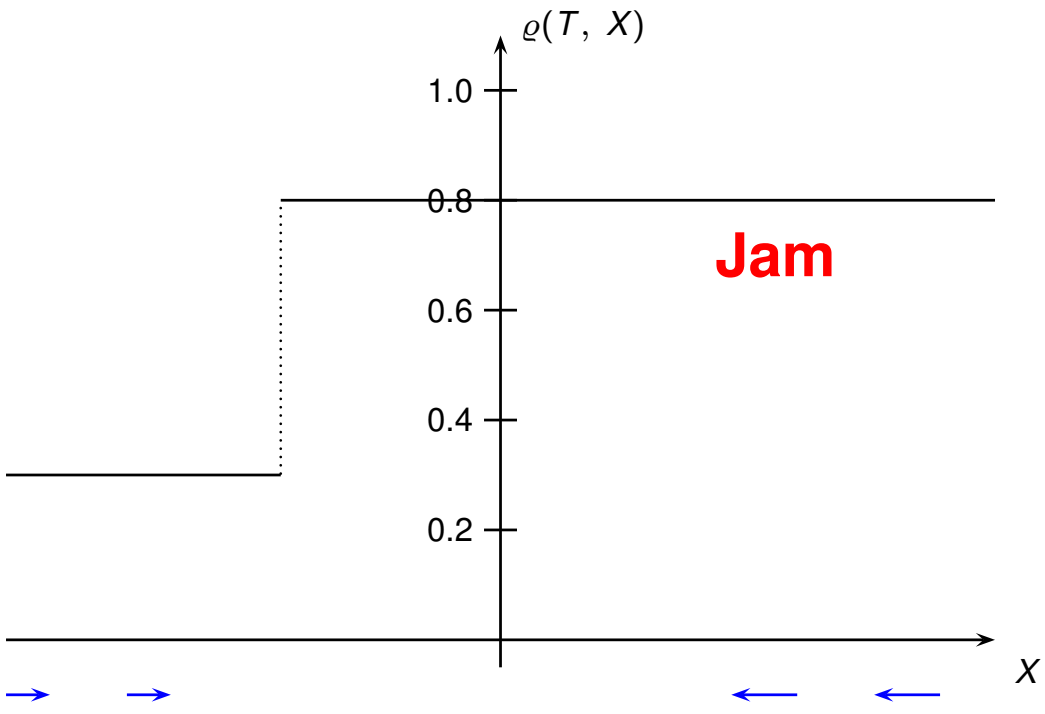
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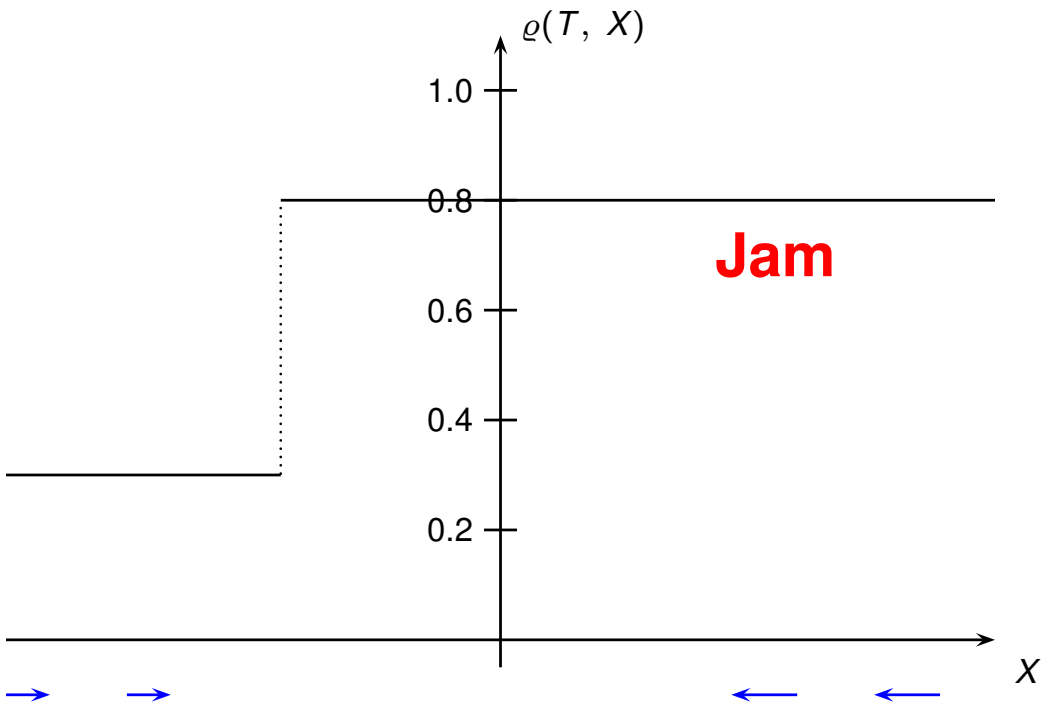
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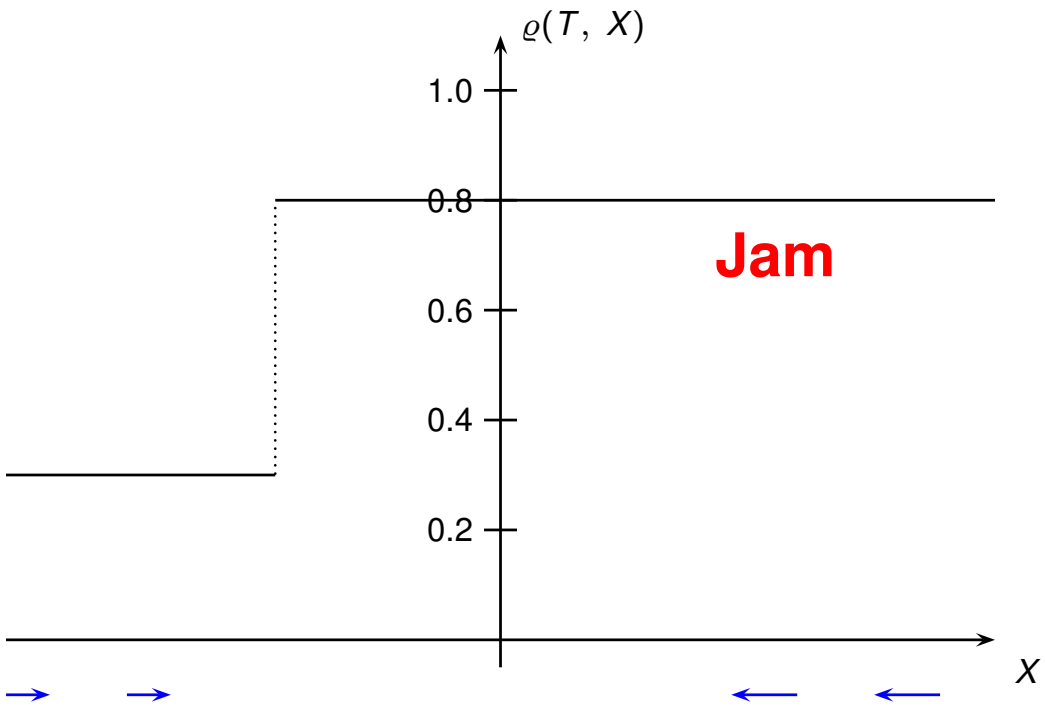
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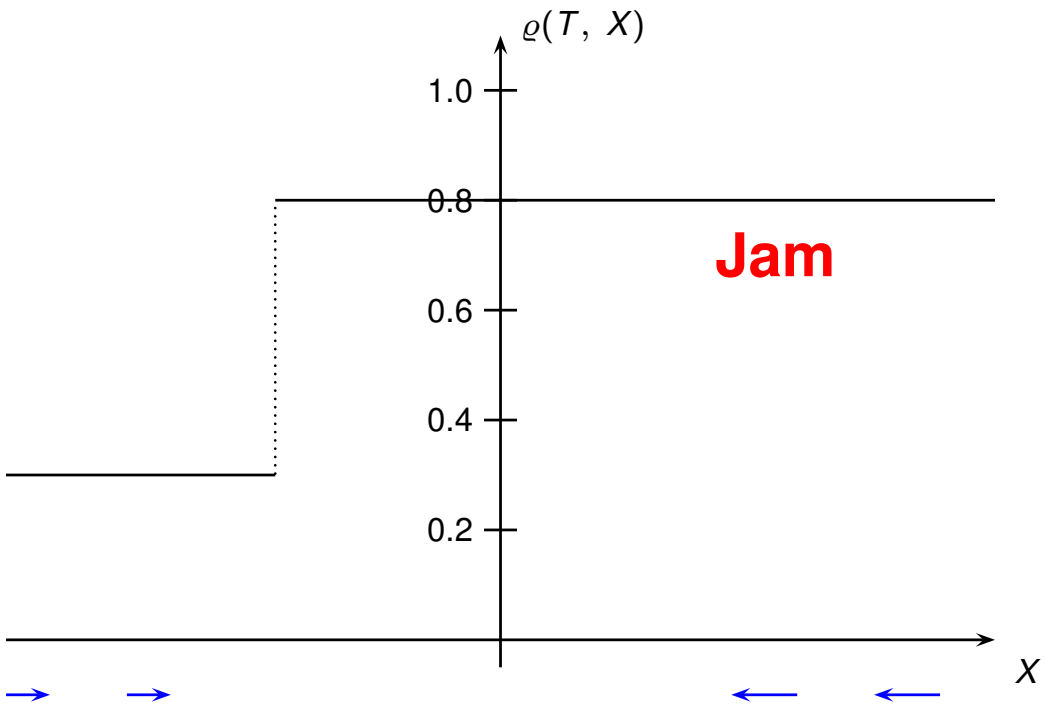
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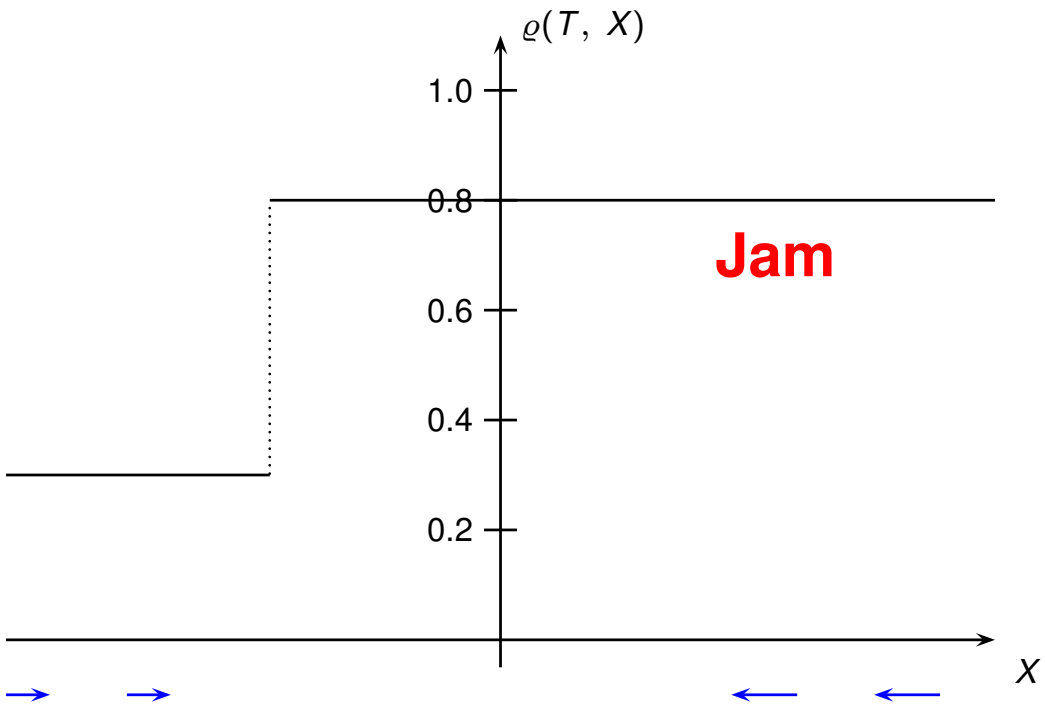
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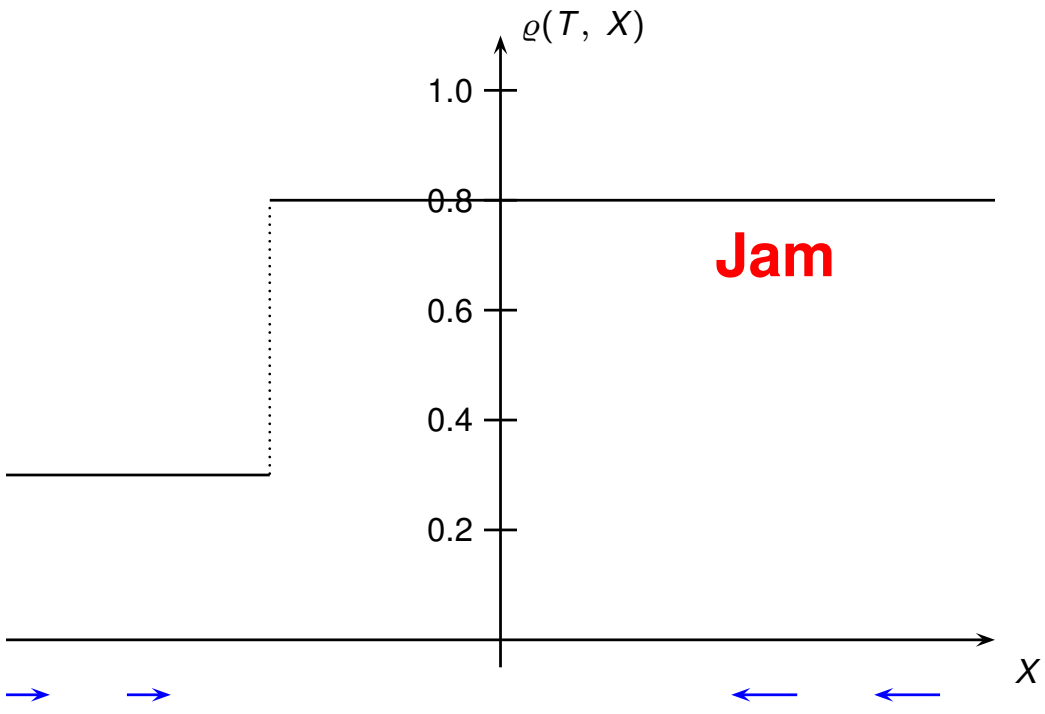
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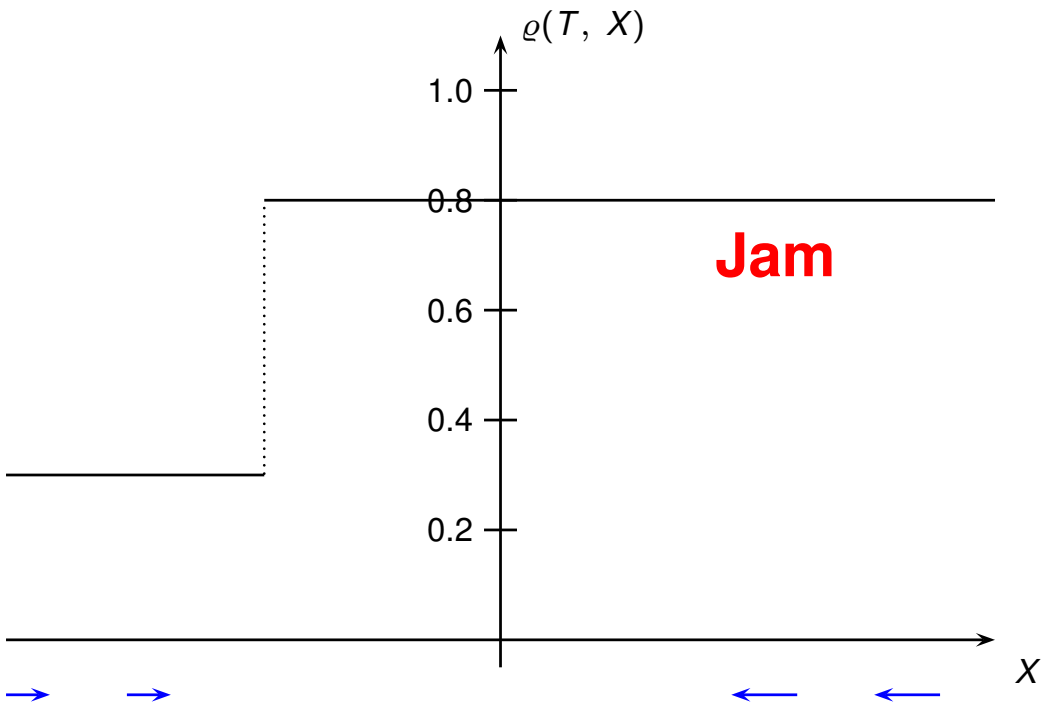
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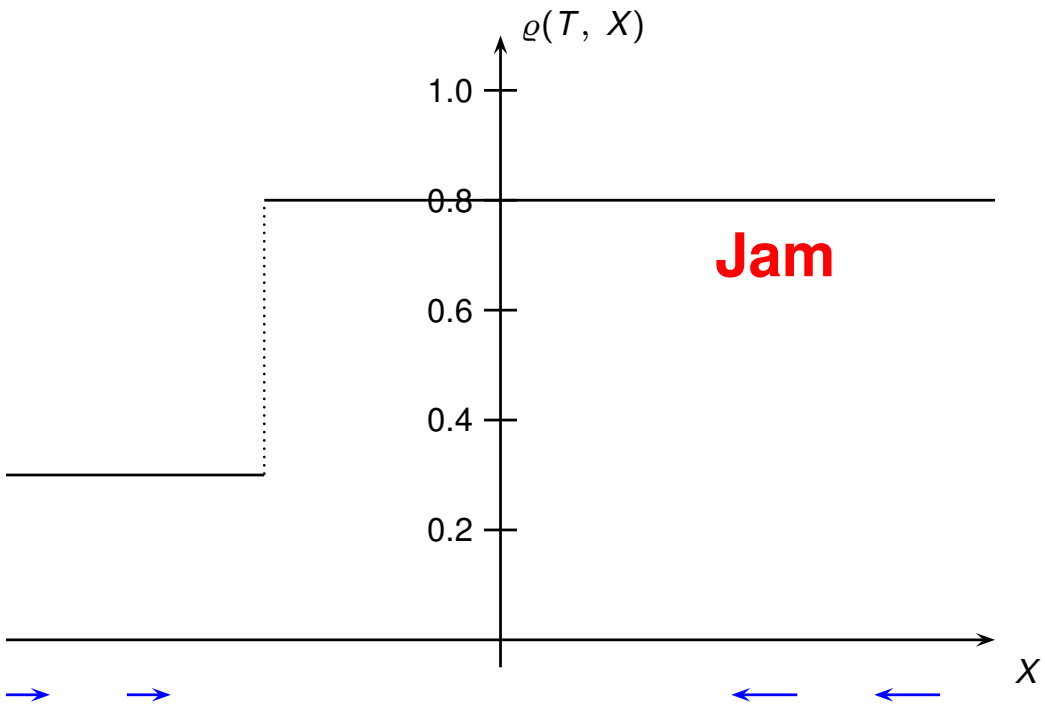
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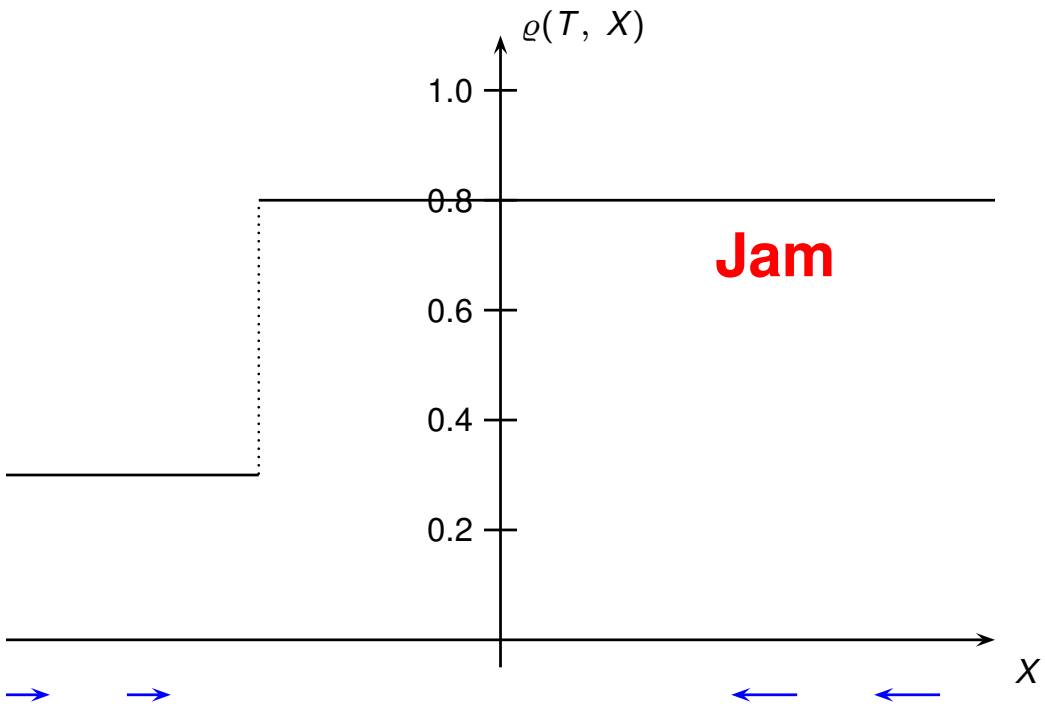
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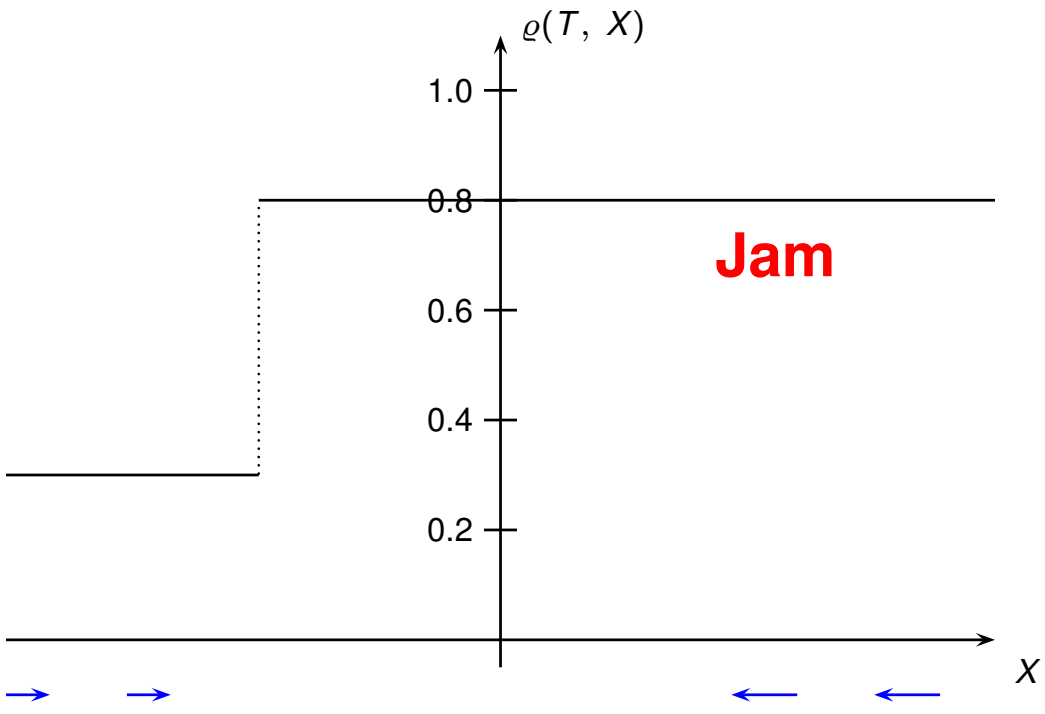
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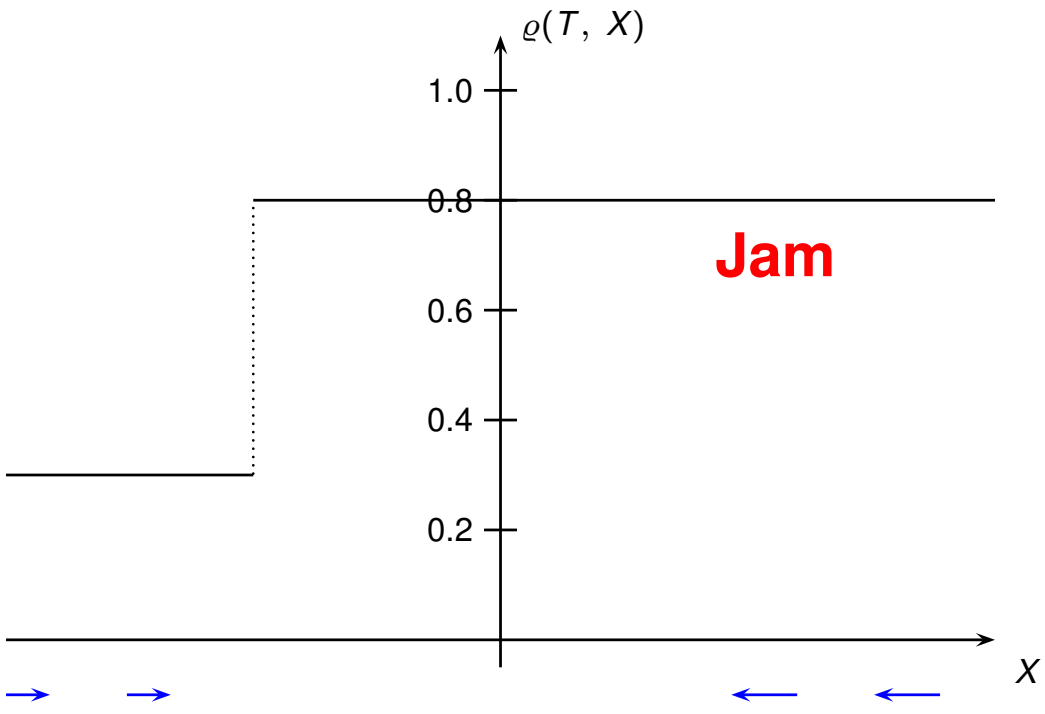
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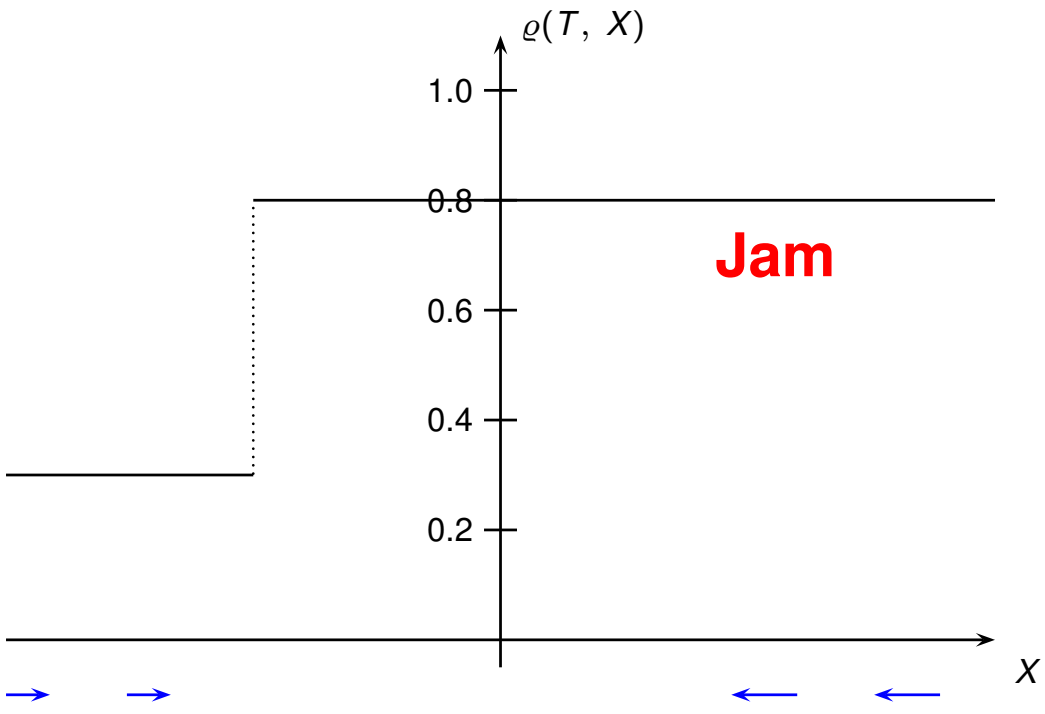
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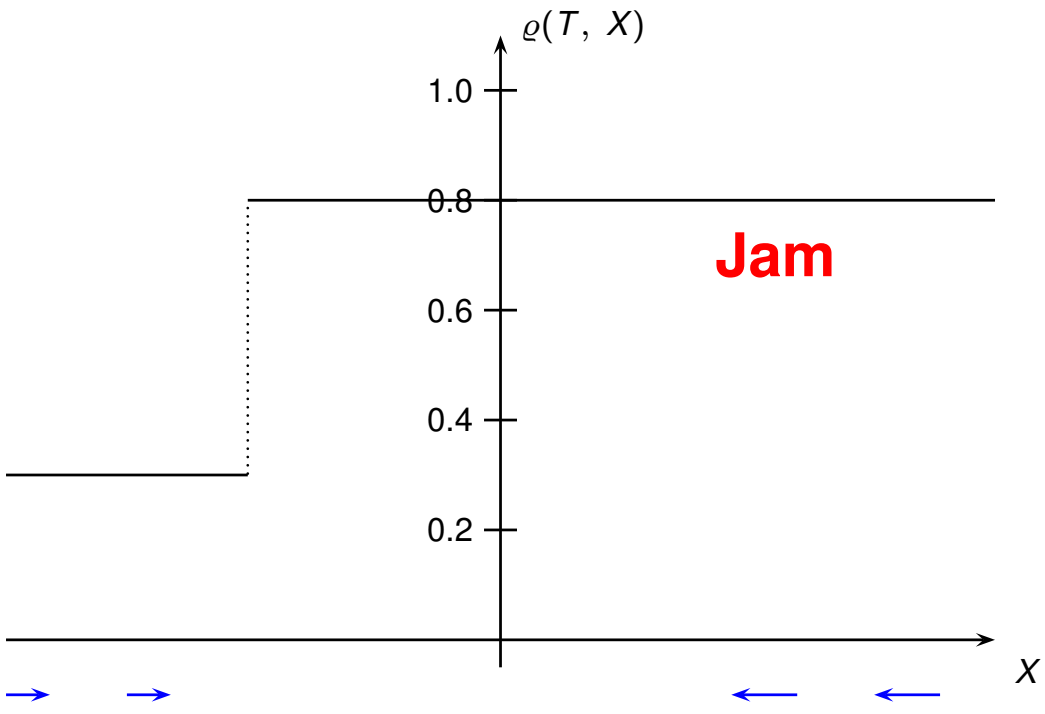
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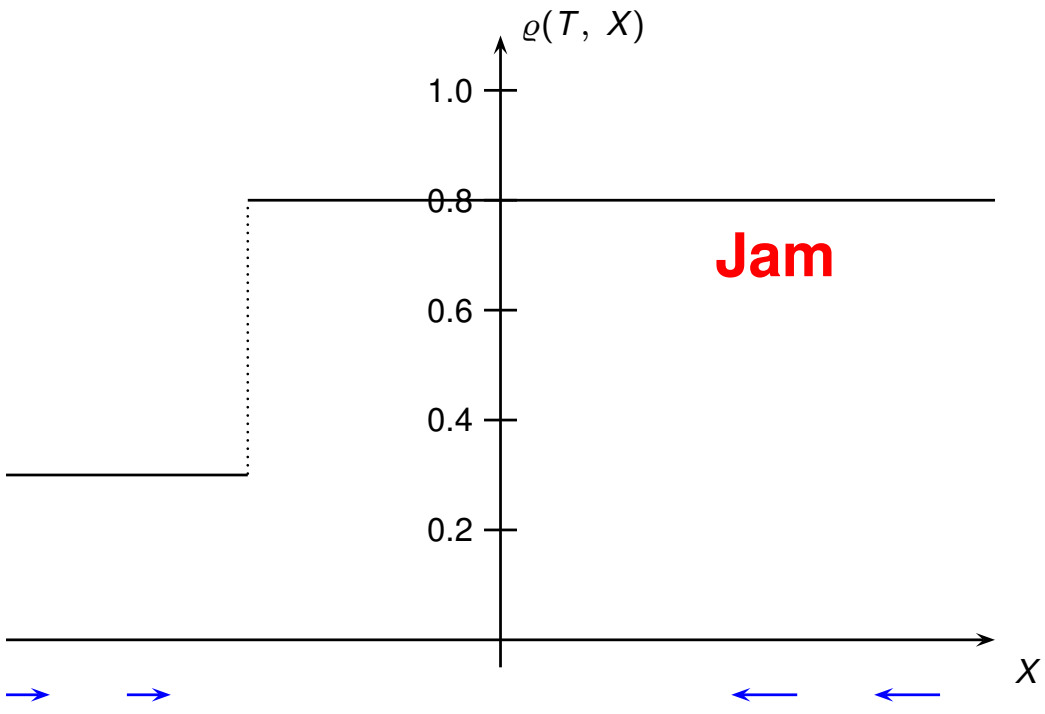
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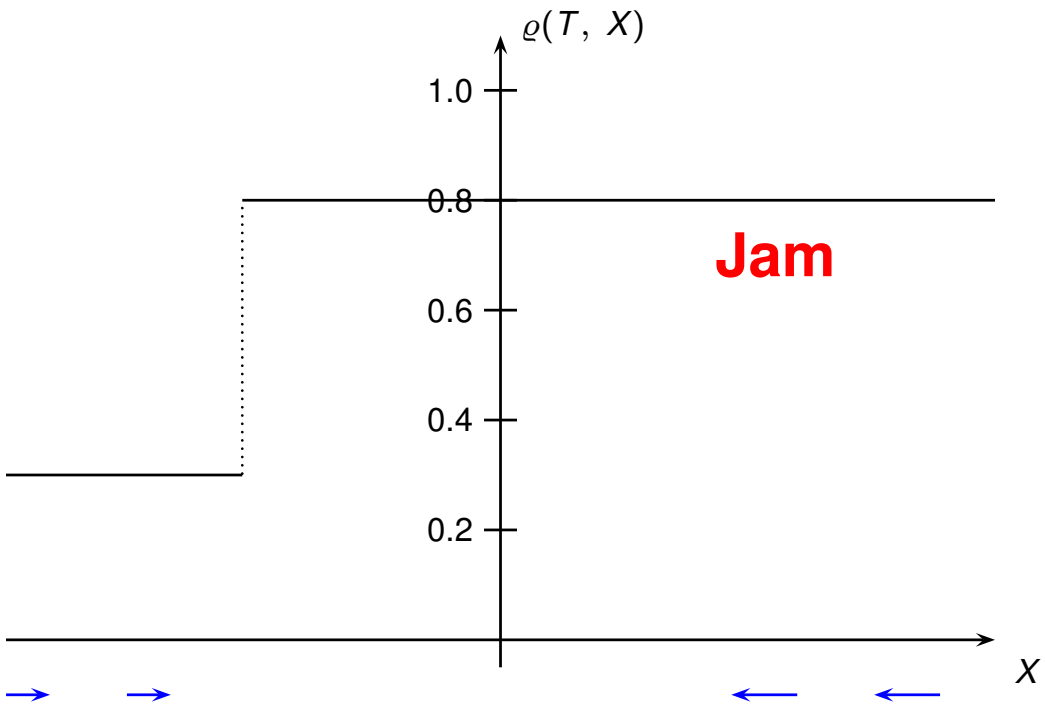
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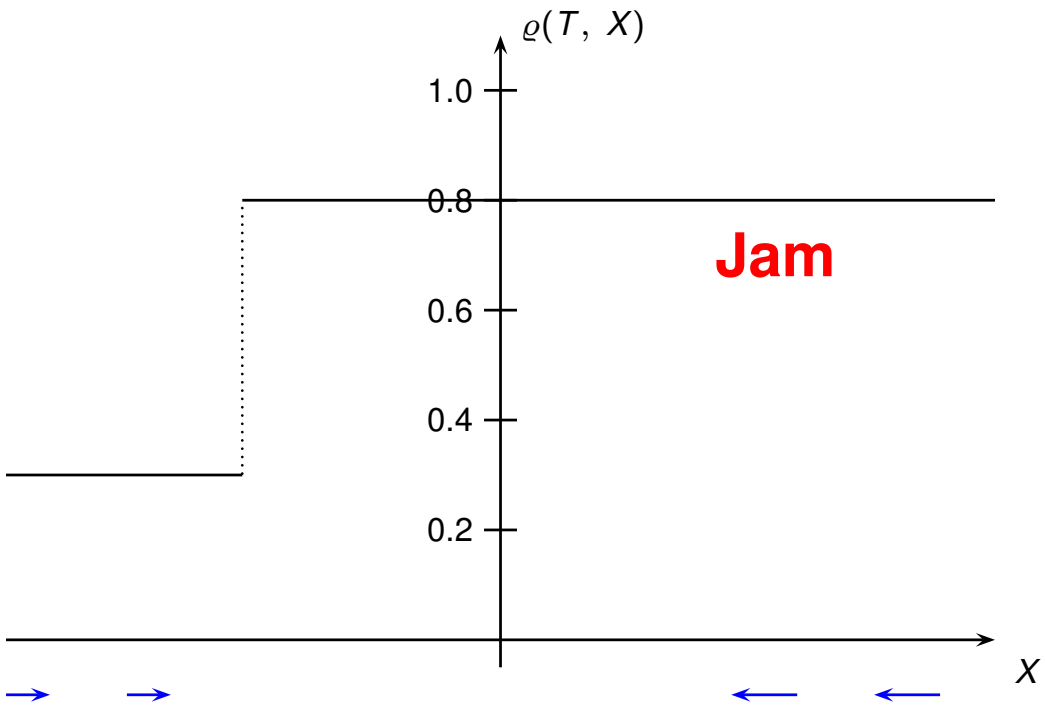
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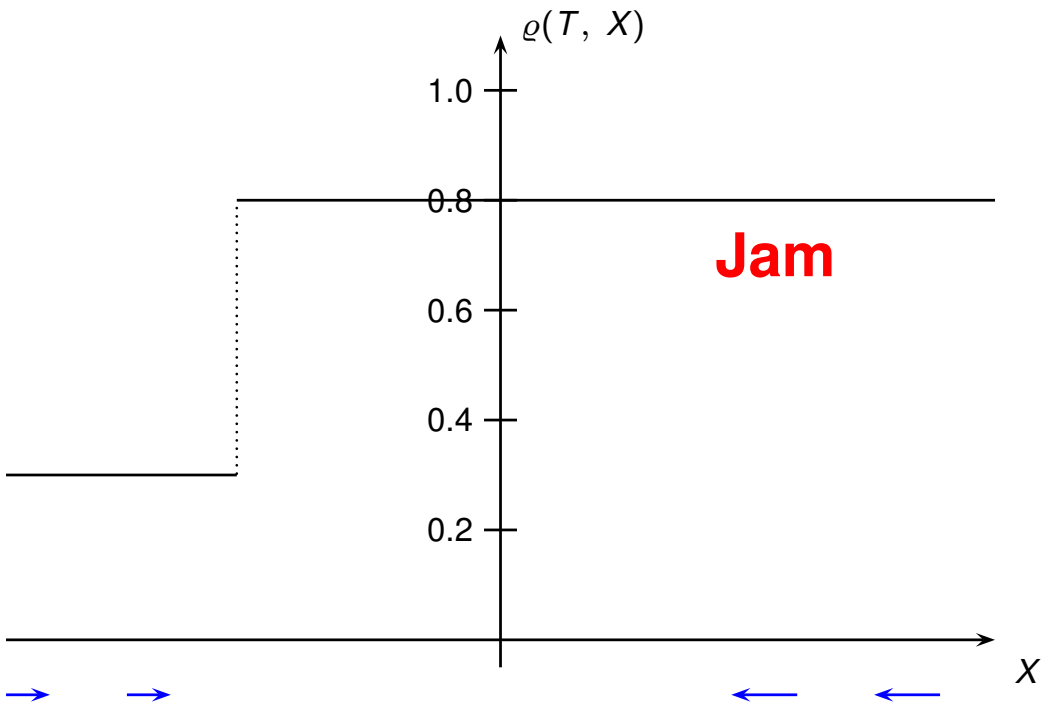
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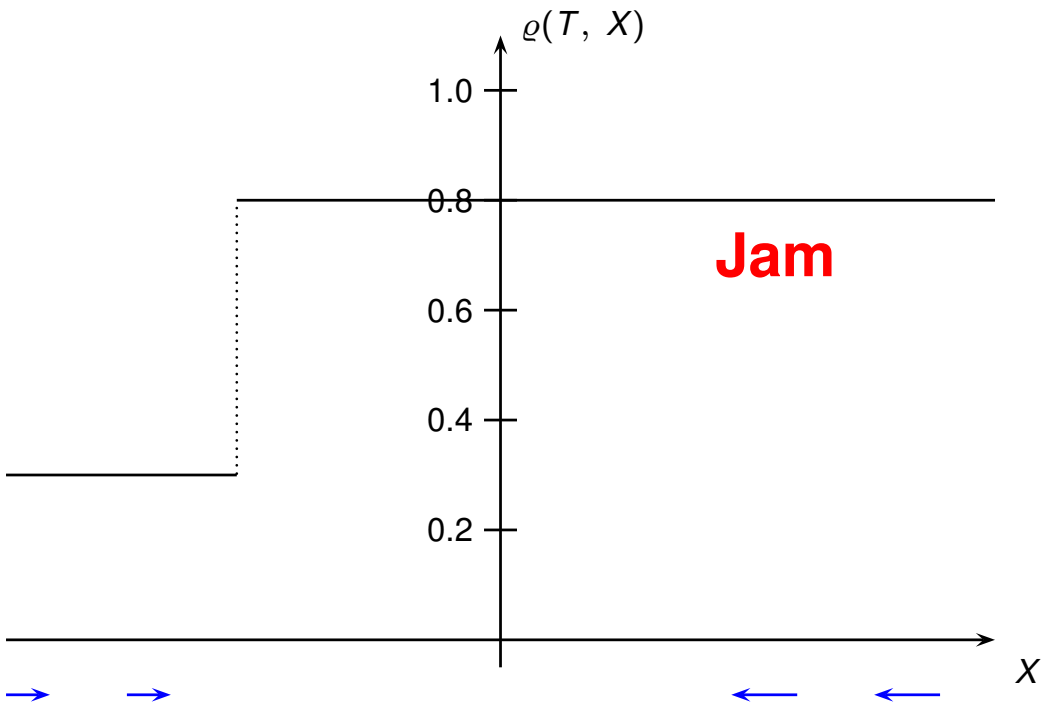
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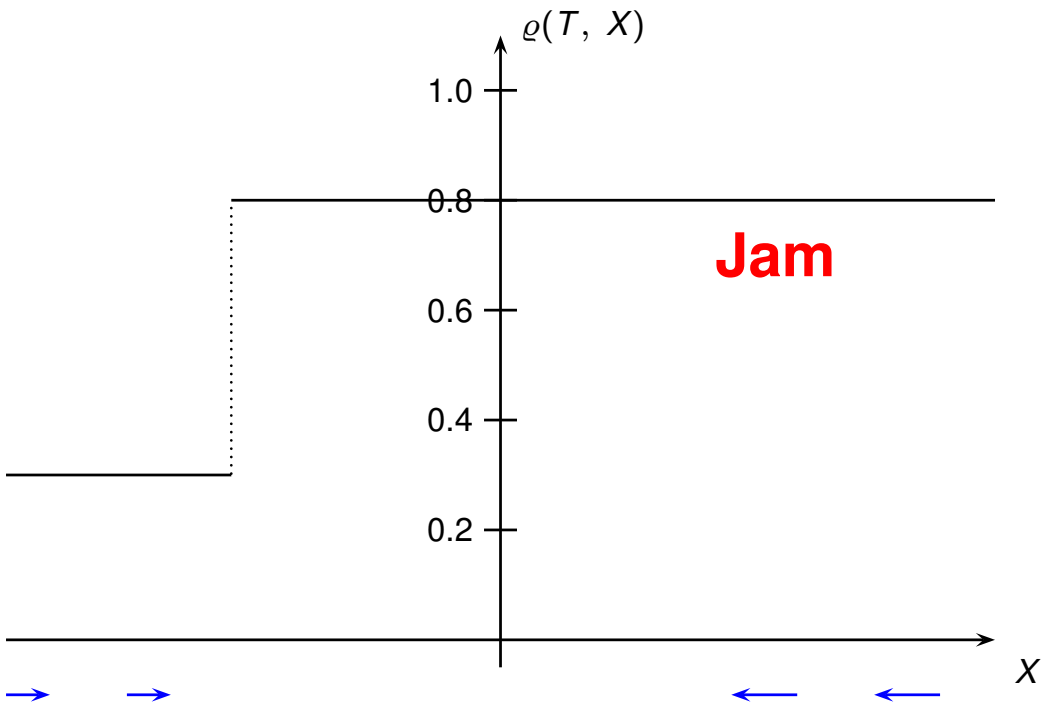
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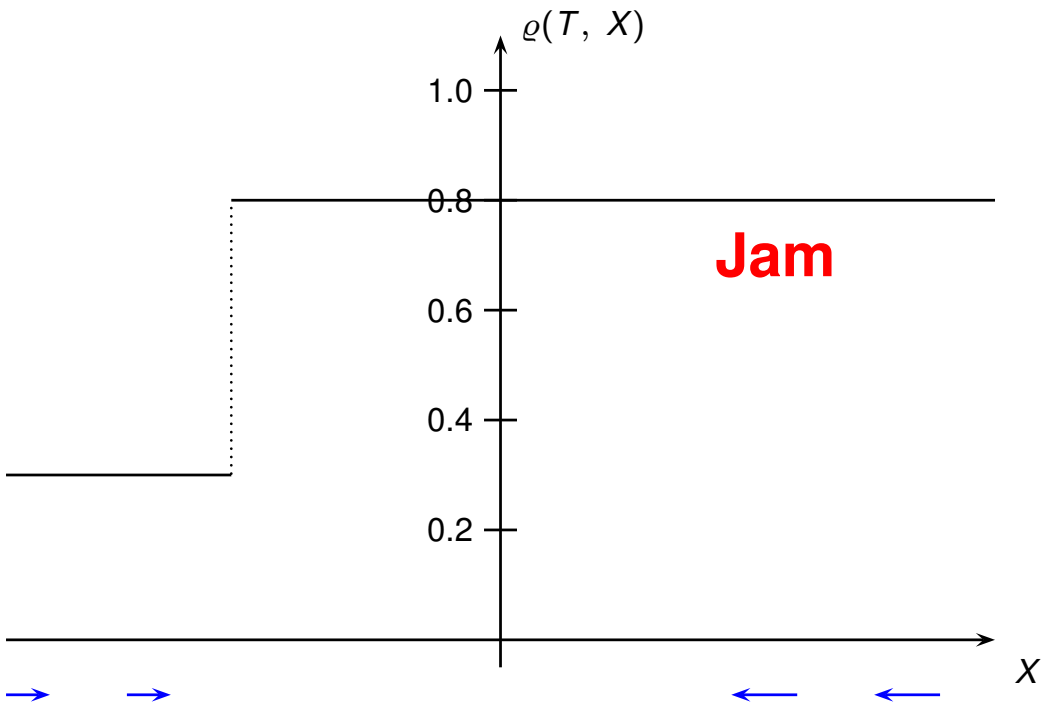
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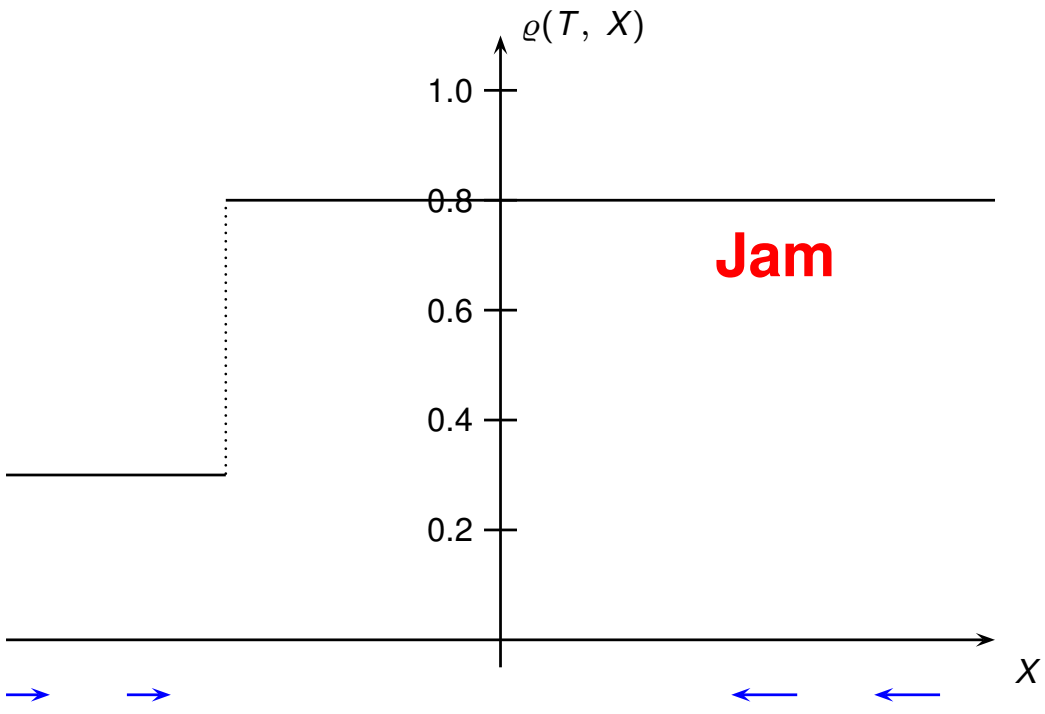
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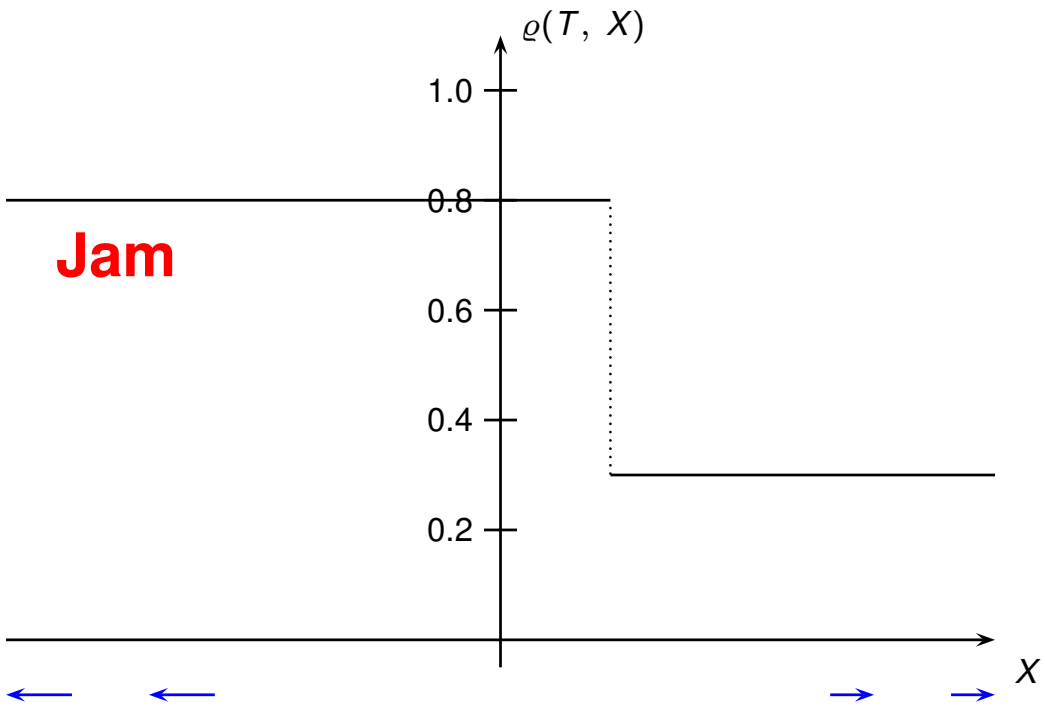
Rescaled version: shock



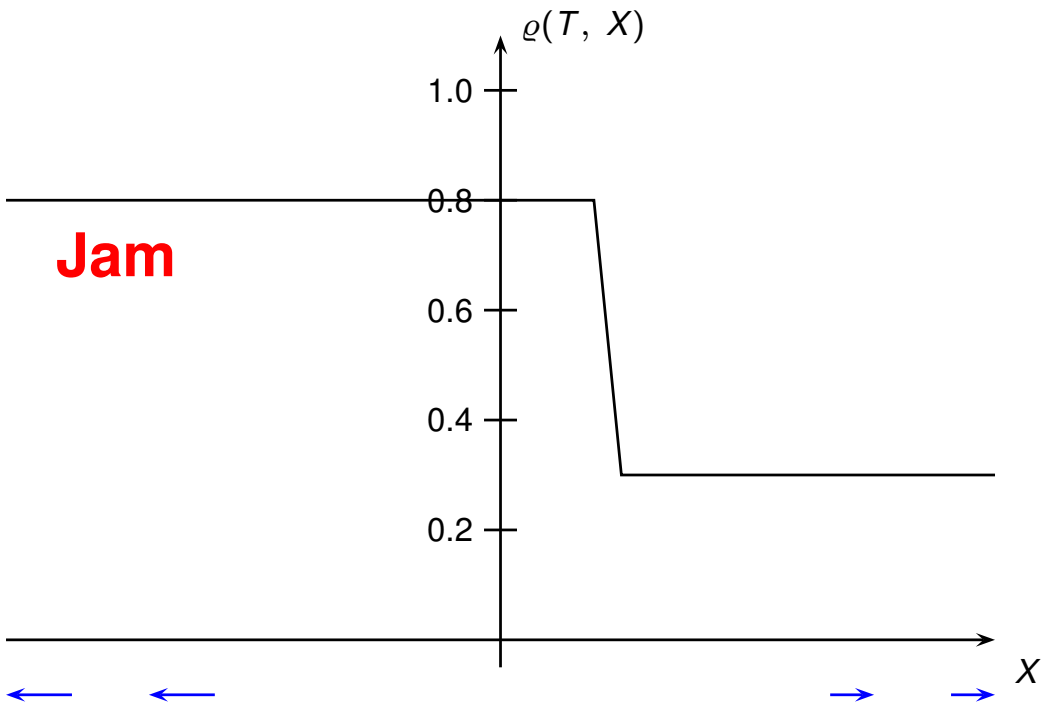
Rescaled version: shock



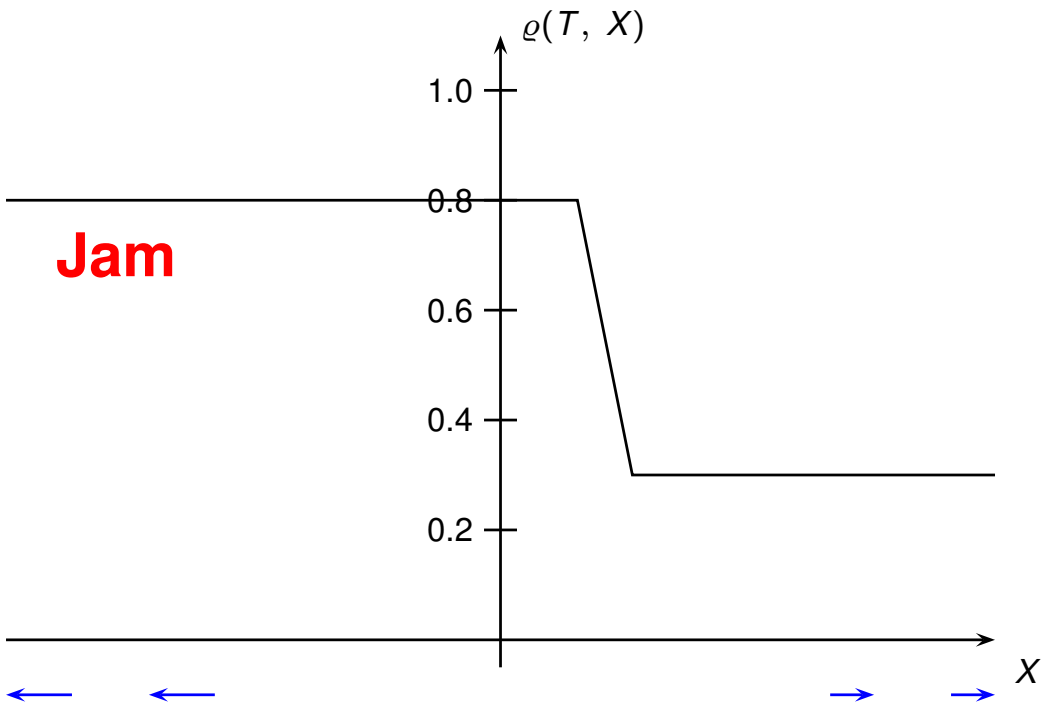
Rescaled version: rarefaction fan



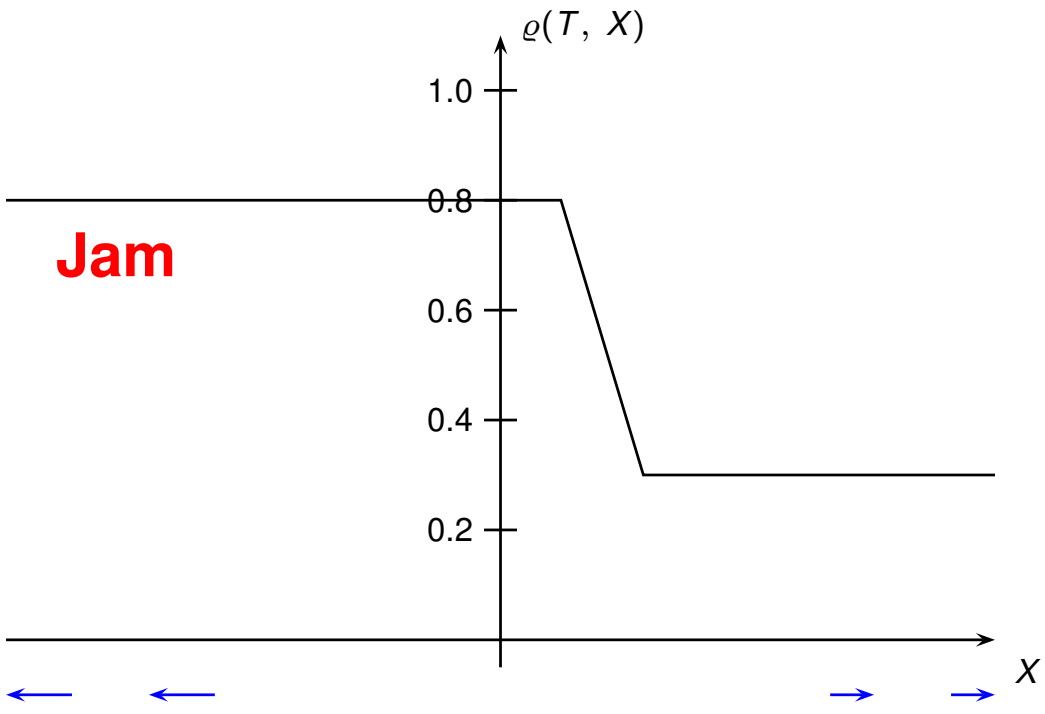
Rescaled version: rarefaction fan



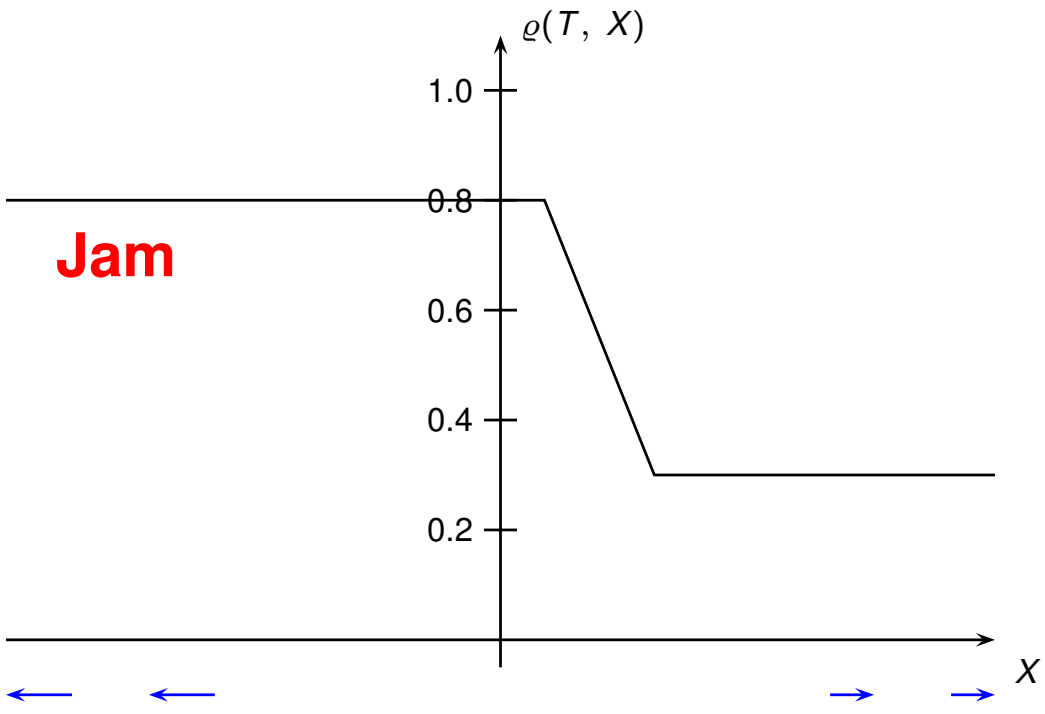
Rescaled version: rarefaction fan



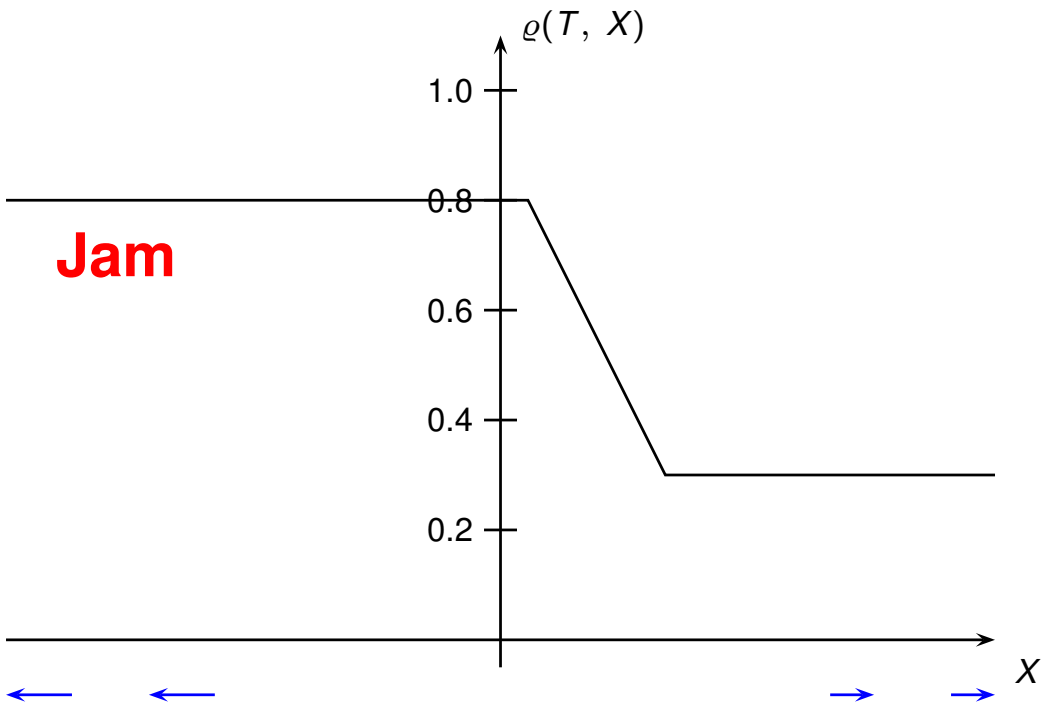
Rescaled version: rarefaction fan



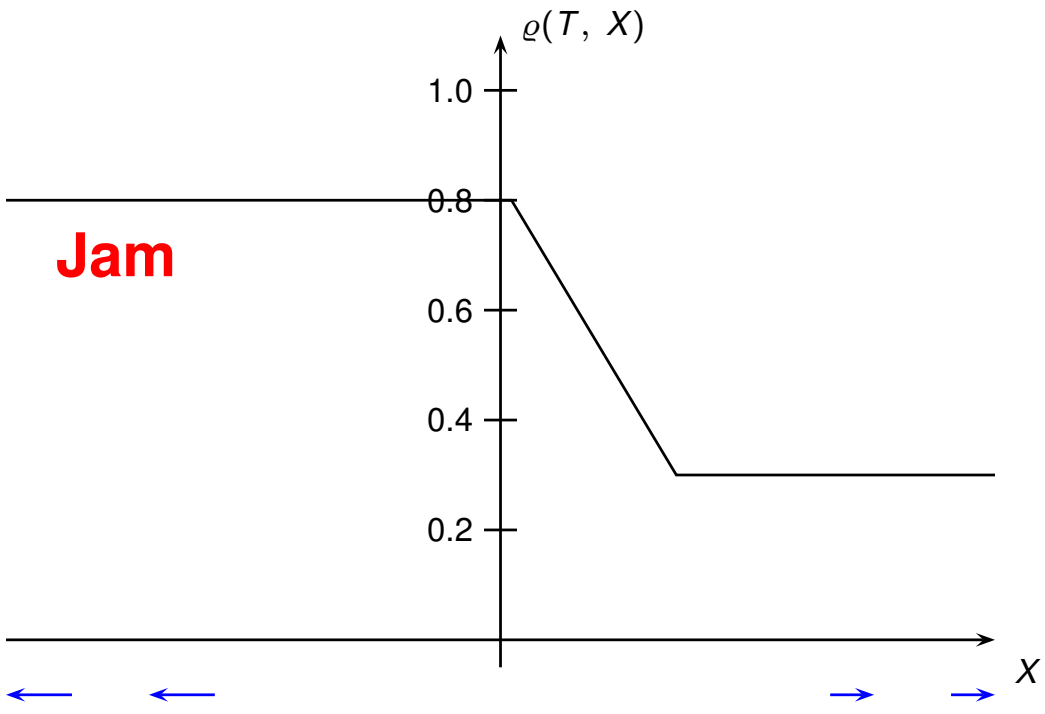
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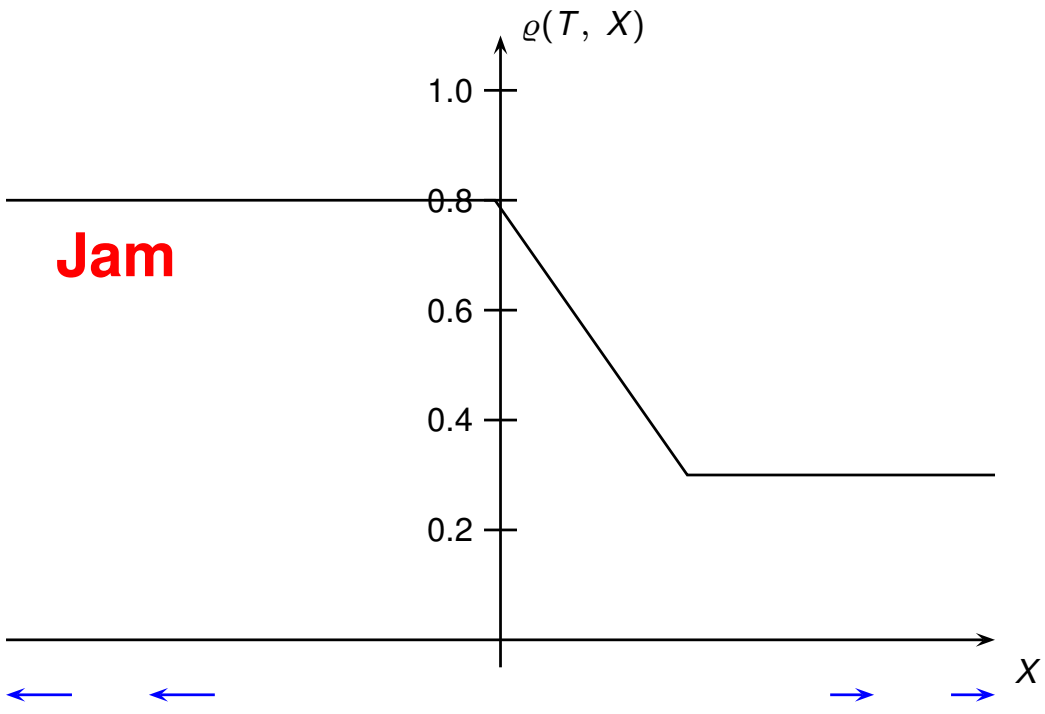
Rescaled version: rarefaction fan



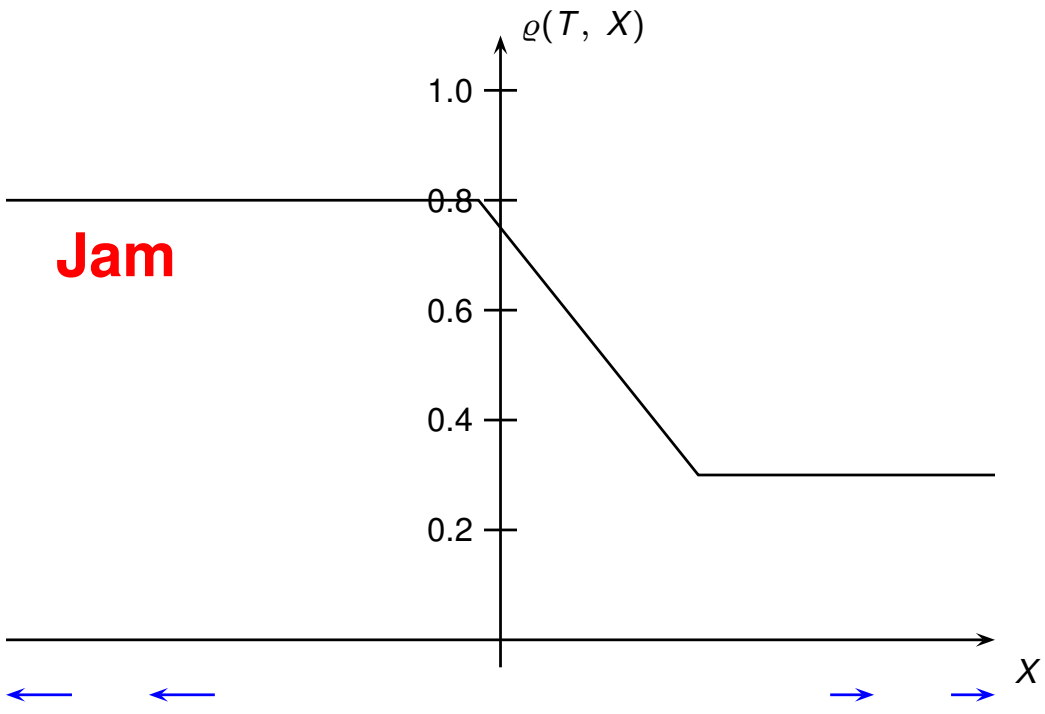
Rescaled version: rarefaction fan



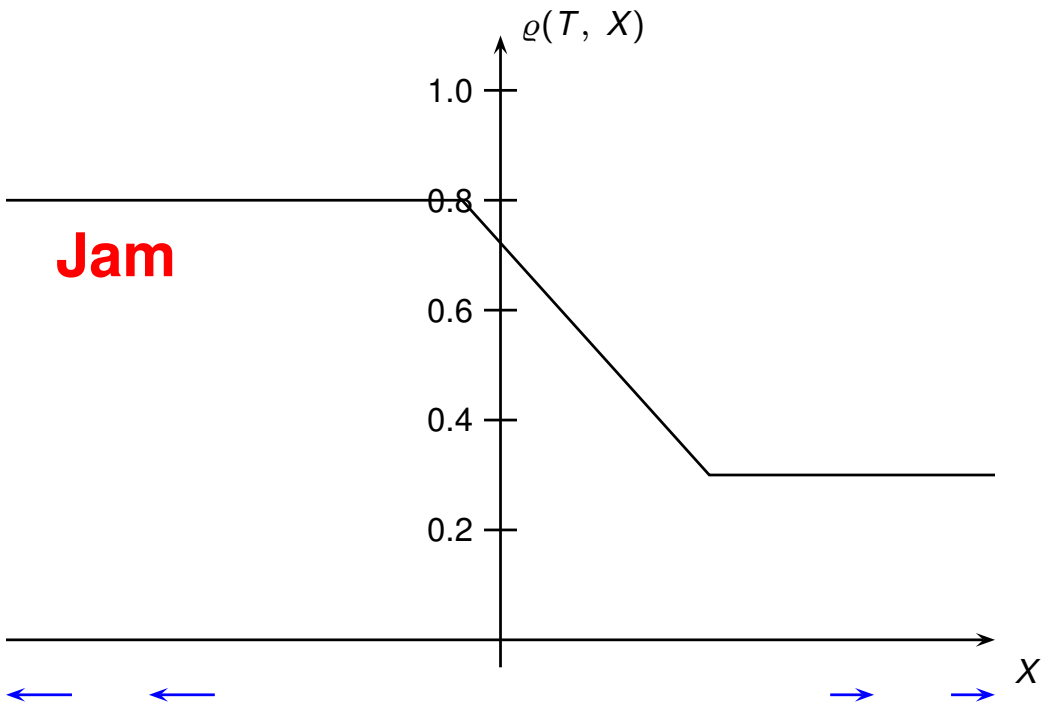
Rescaled version: rarefaction fan



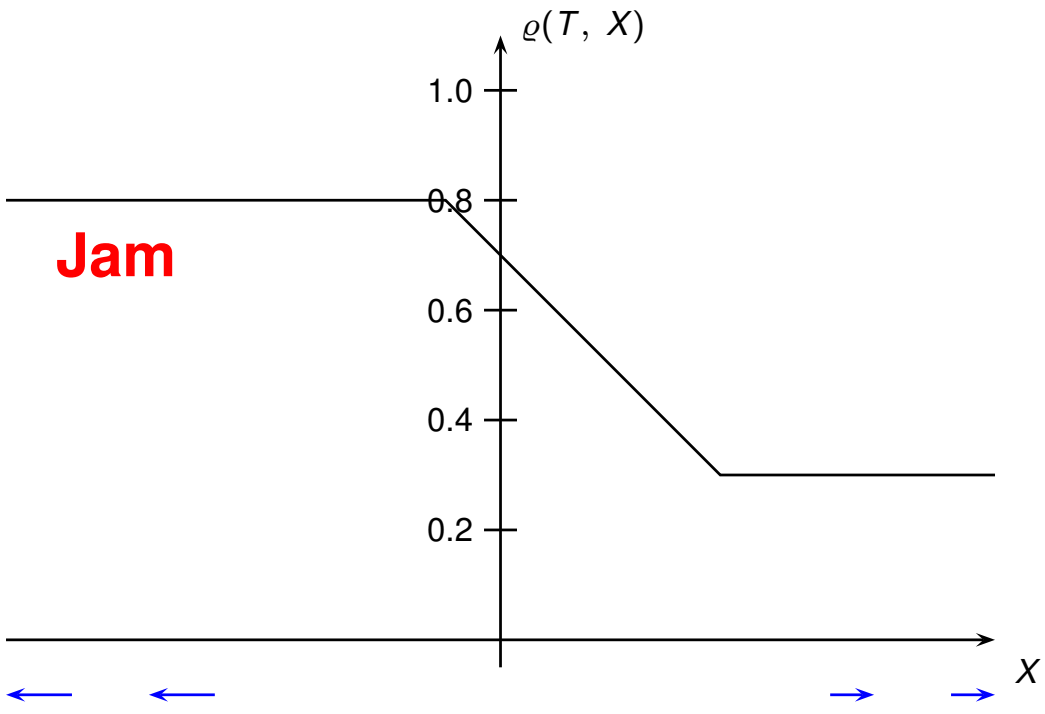
Rescaled version: rarefaction fan



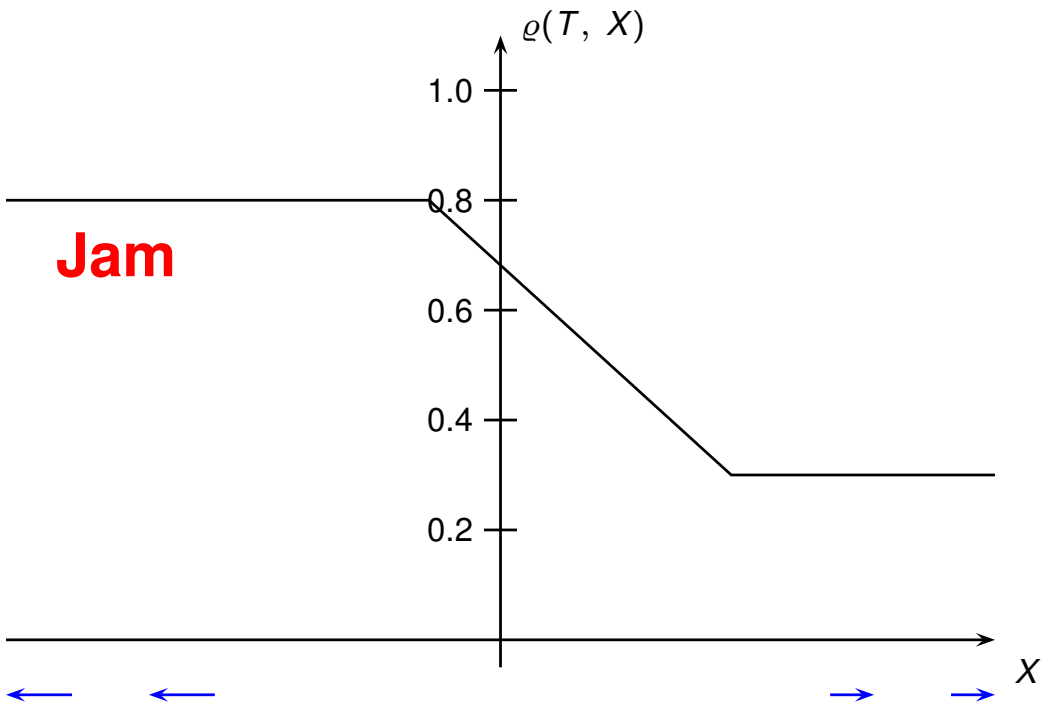
Rescaled version: rarefaction fan



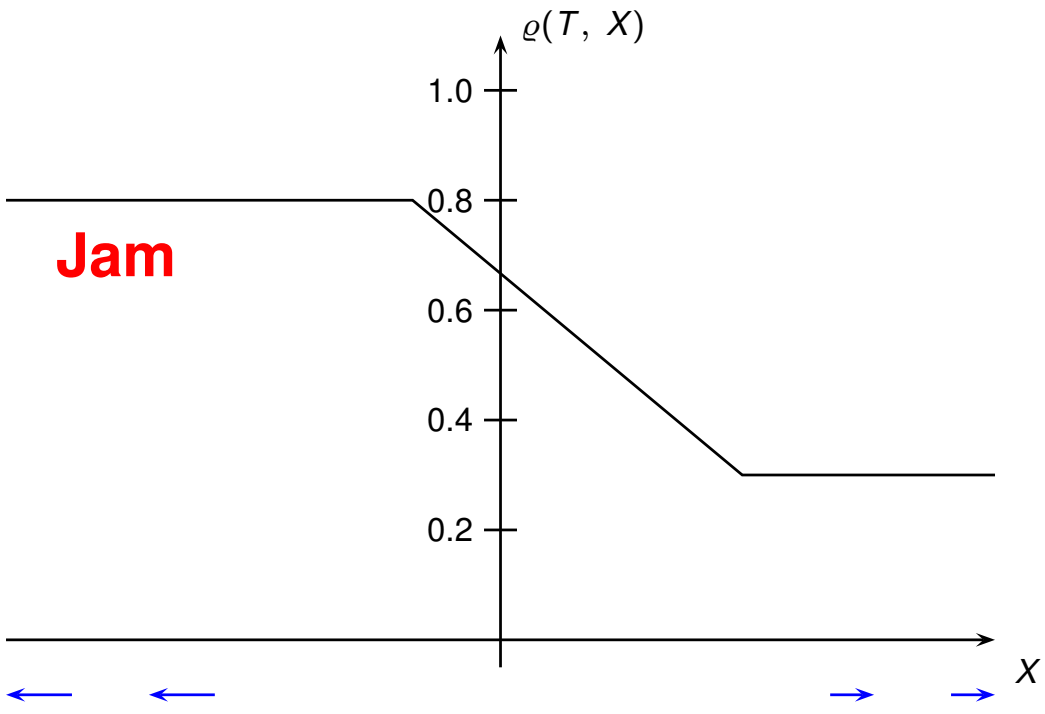
Rescaled version: rarefaction fan



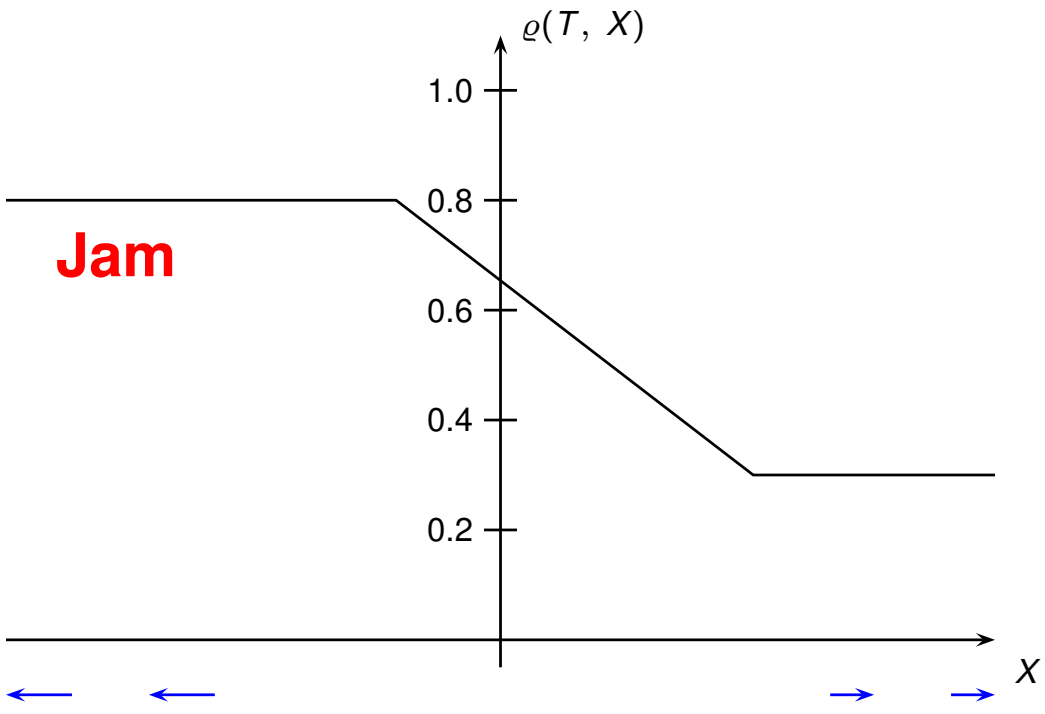
Rescaled version: rarefaction fan



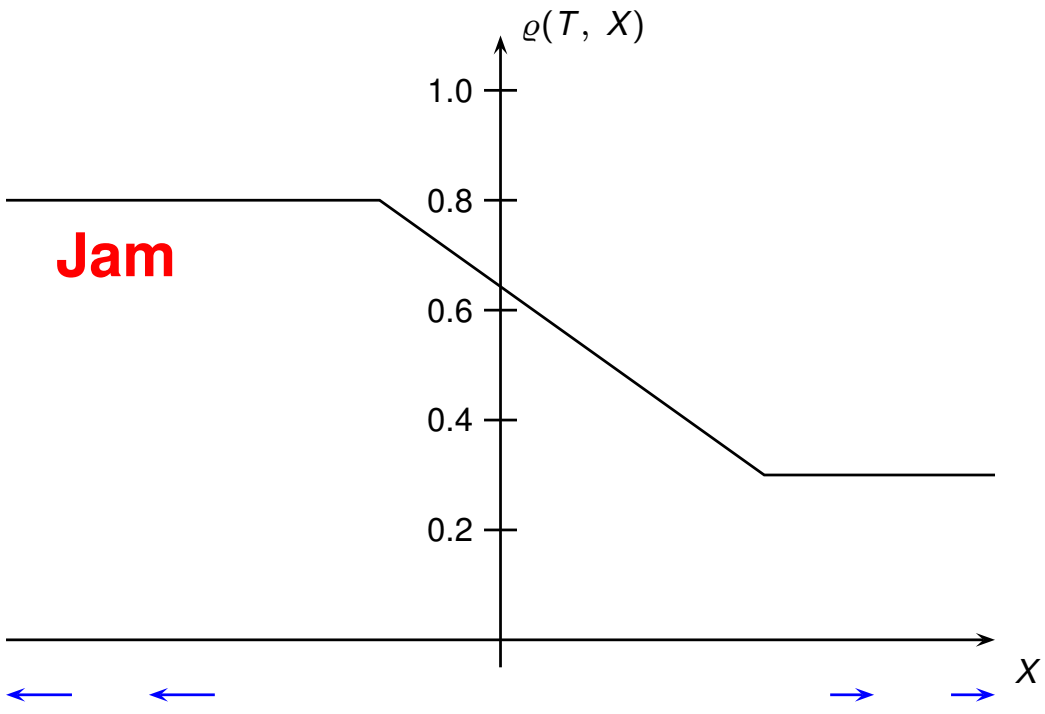
Rescaled version: rarefaction fan



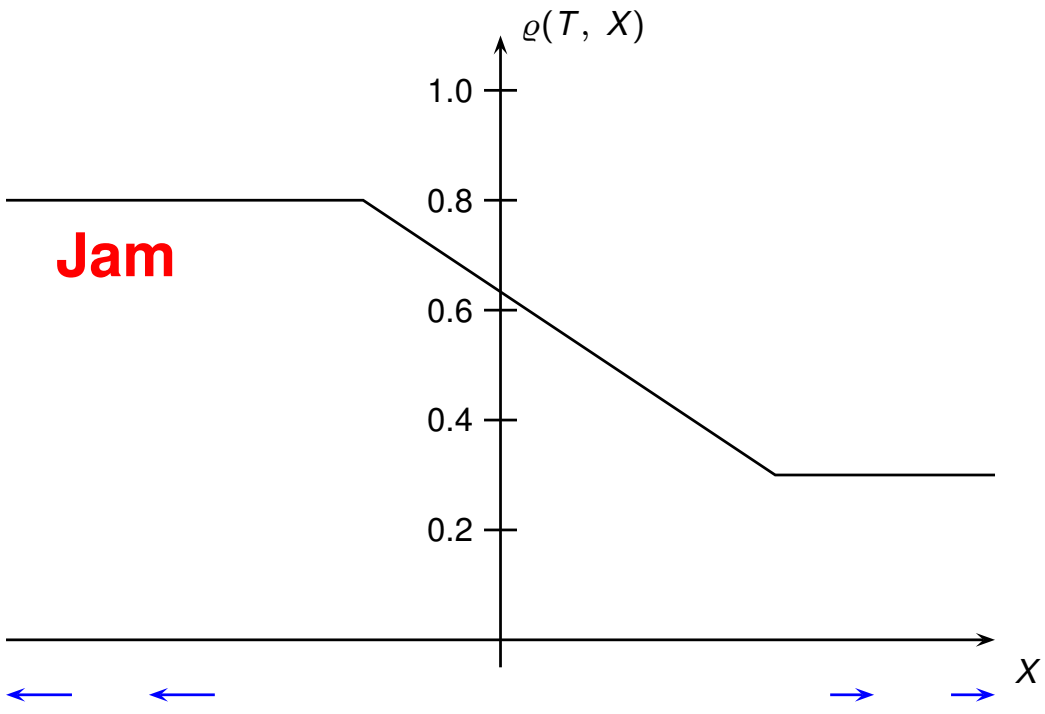
Rescaled version: rarefaction fan



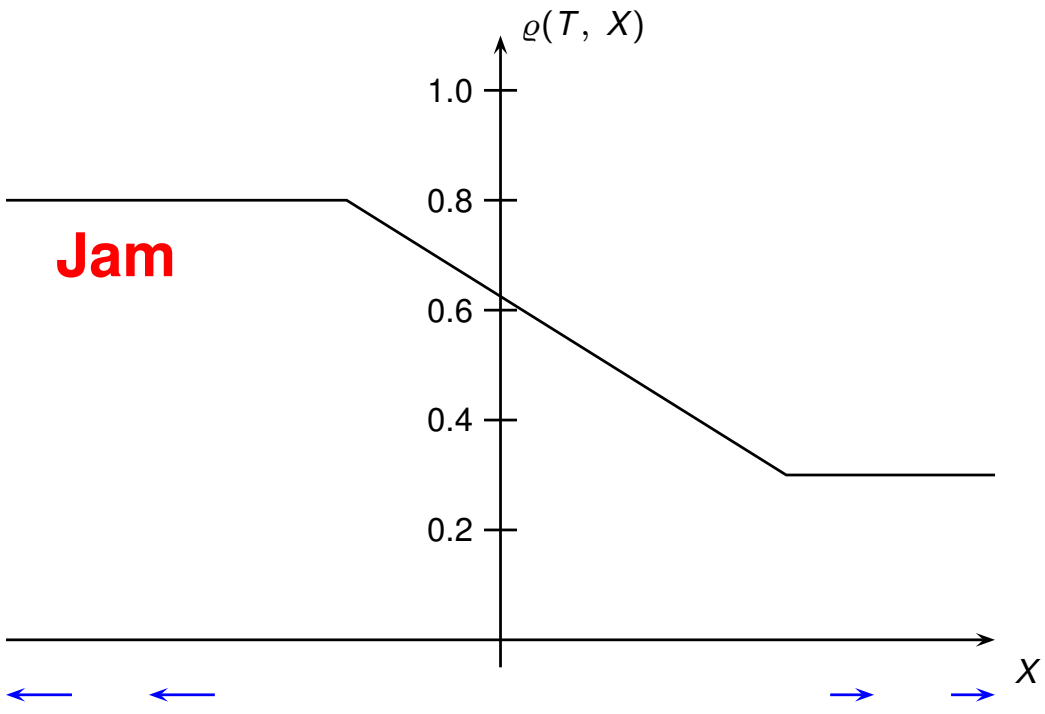
Rescaled version: rarefaction fan



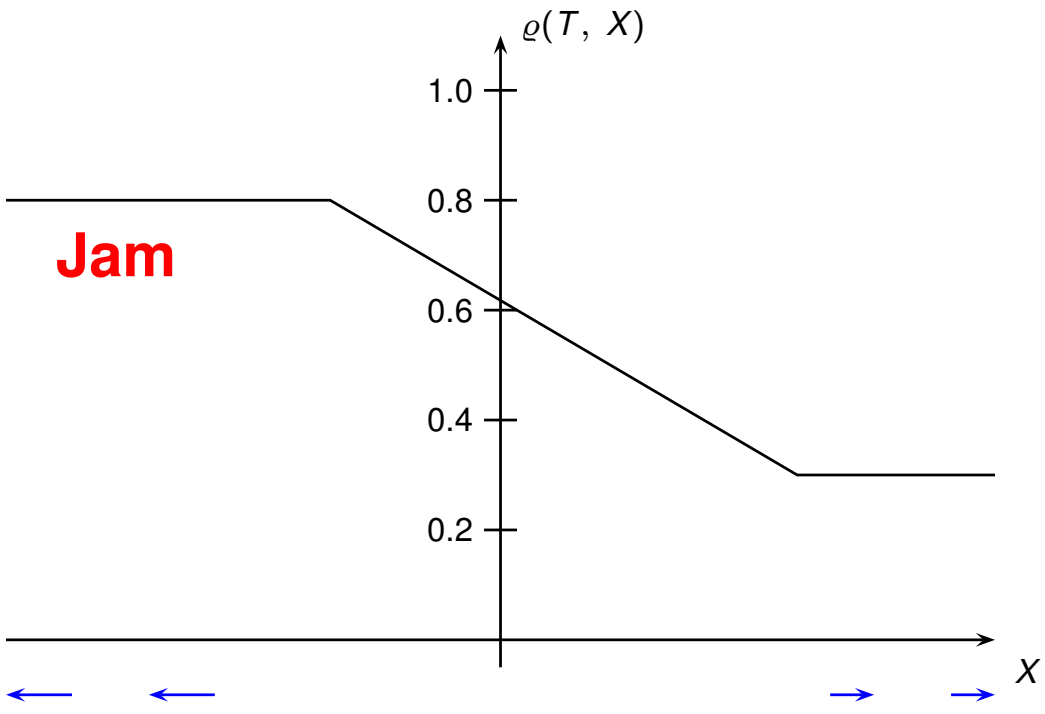
Rescaled version: rarefaction fan



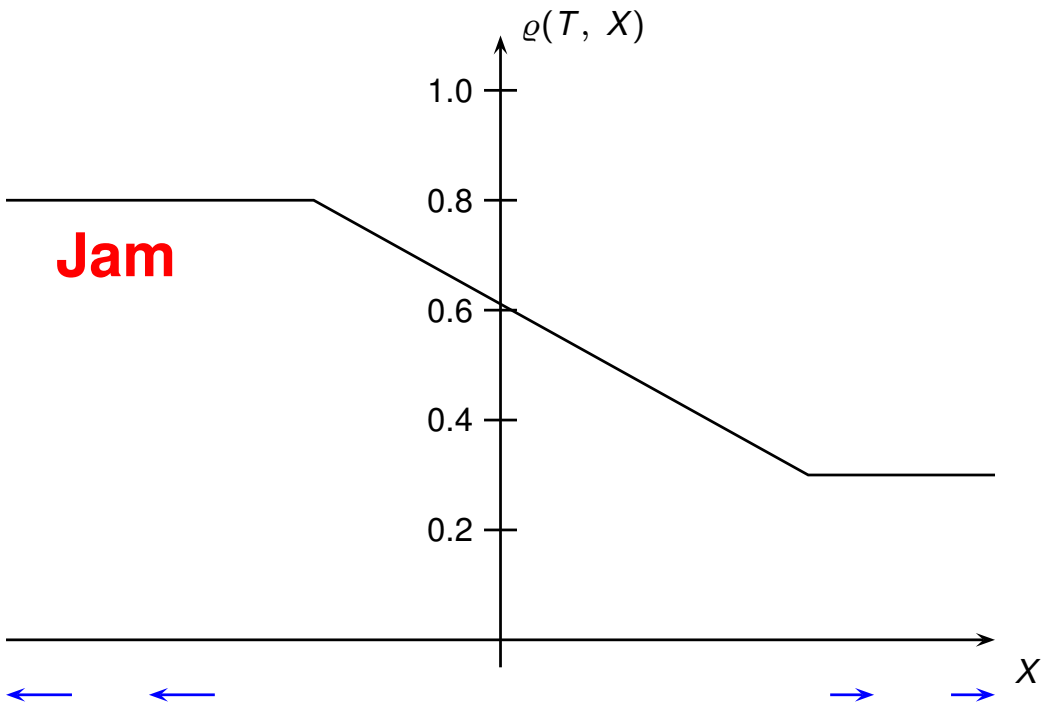
Rescaled version: rarefaction fan



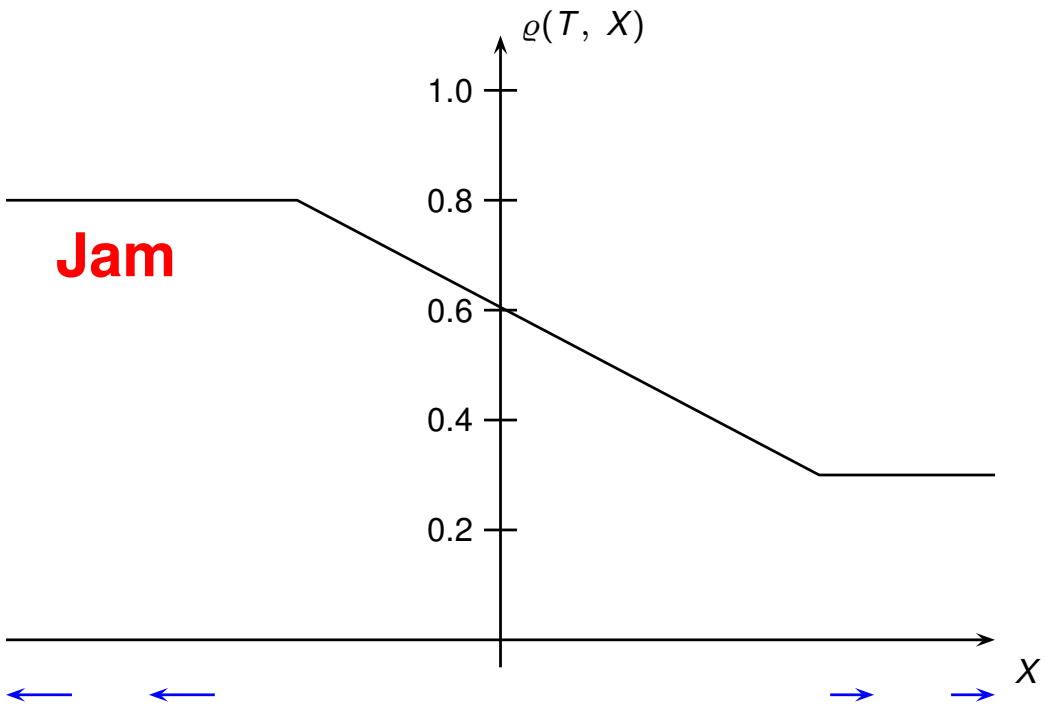
Rescaled version: rarefaction fan



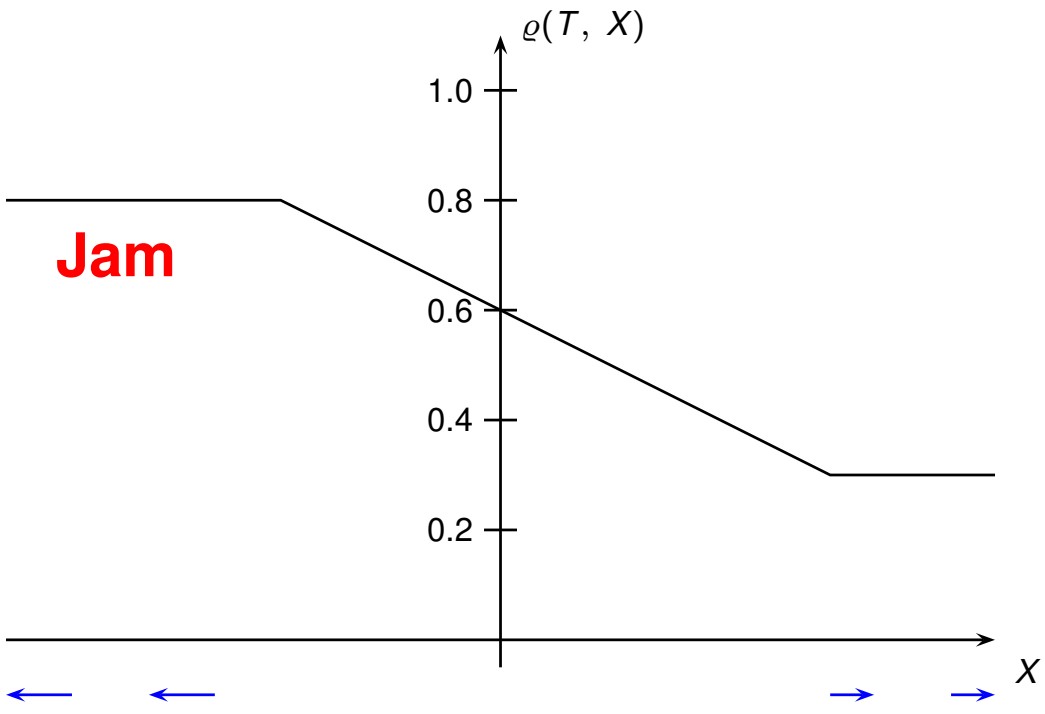
Rescaled version: rarefaction fan



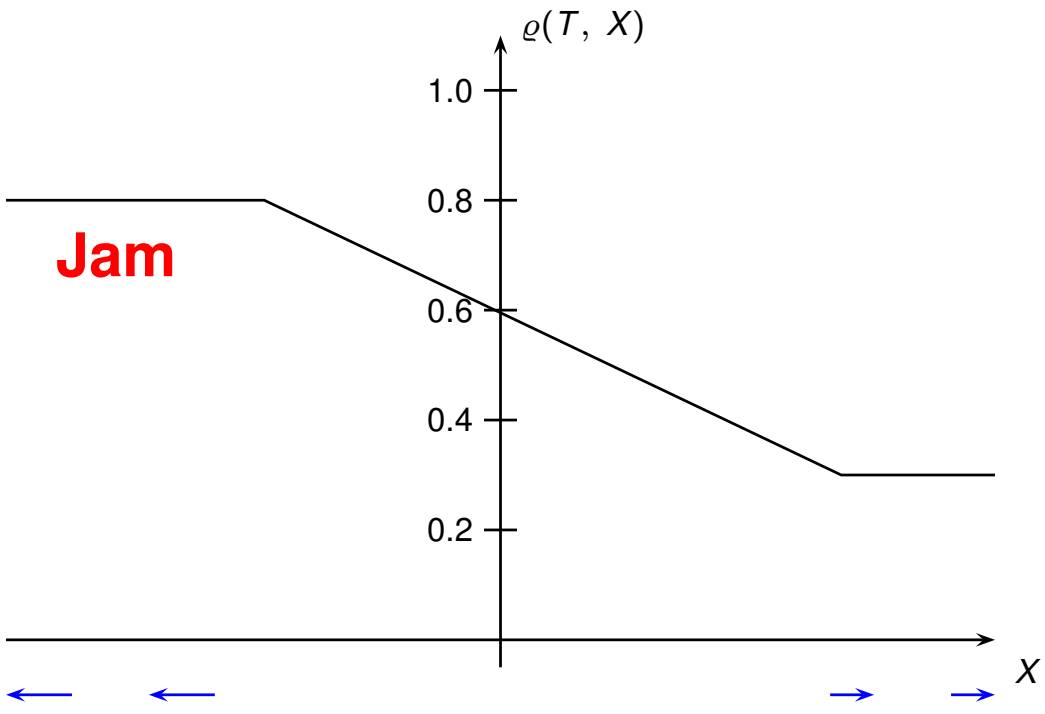
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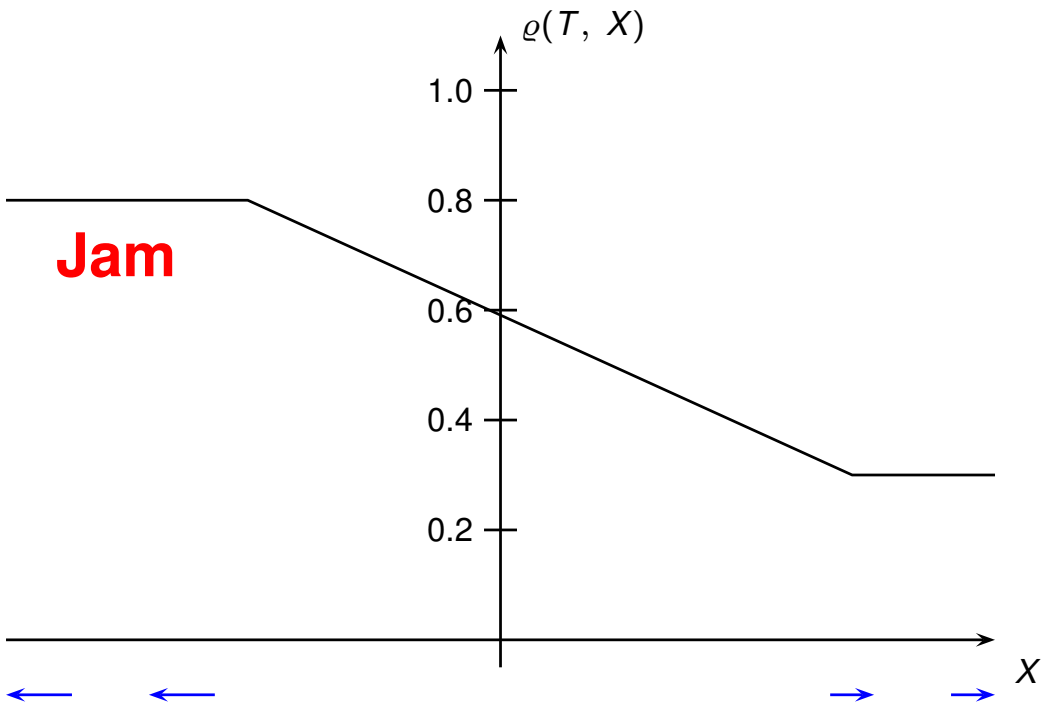
Rescaled version: rarefaction fan



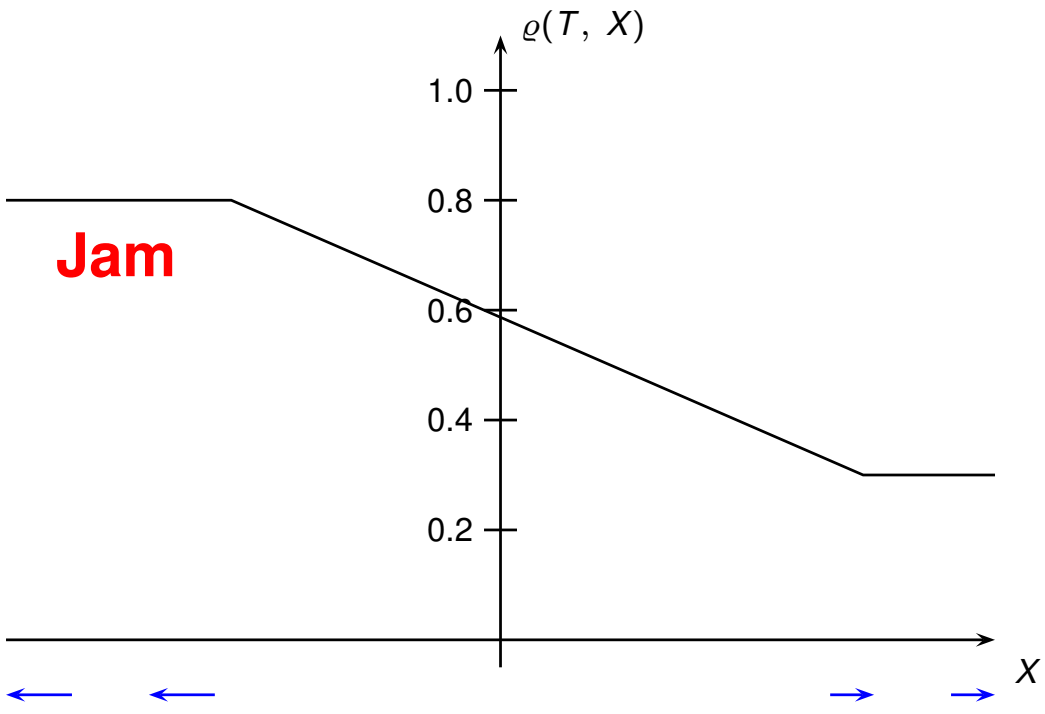
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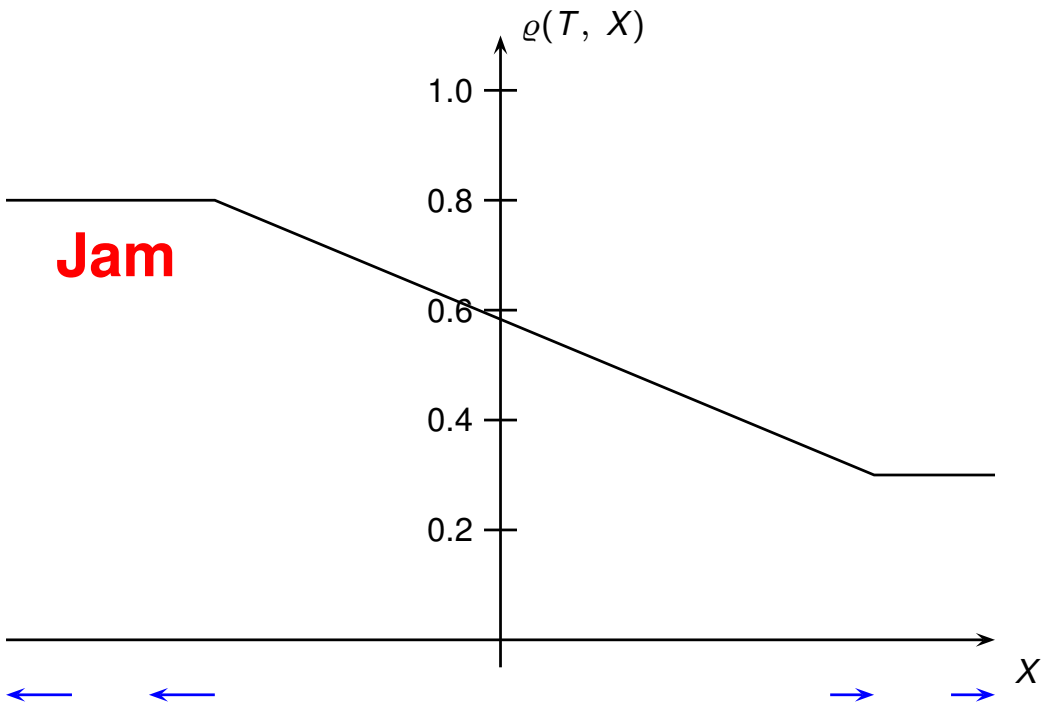
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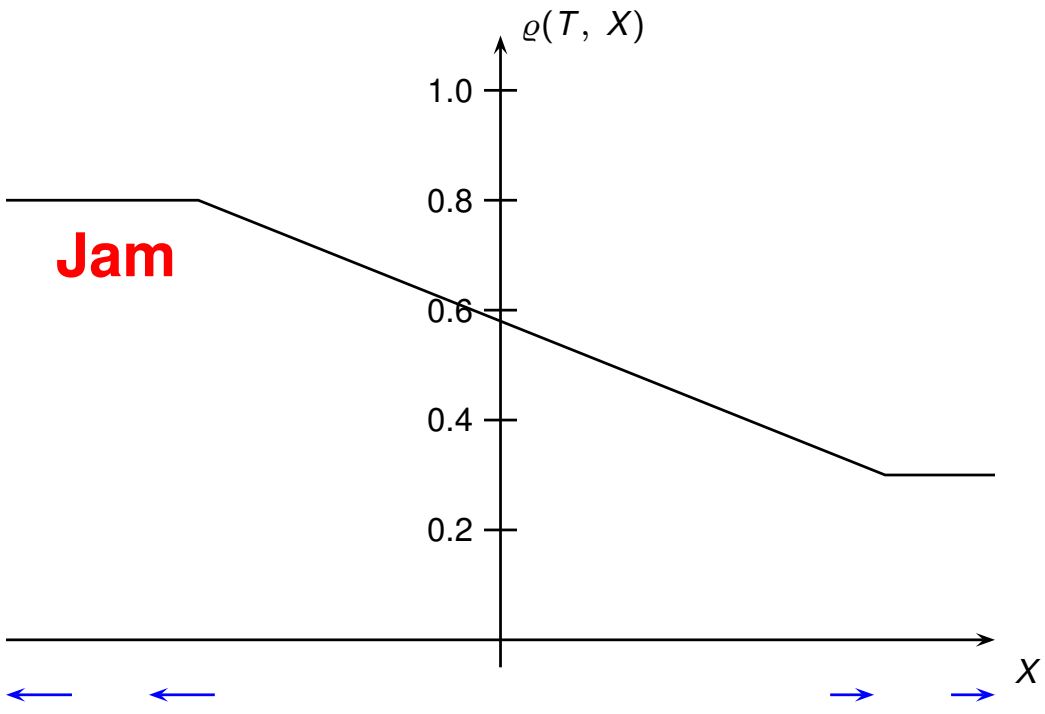
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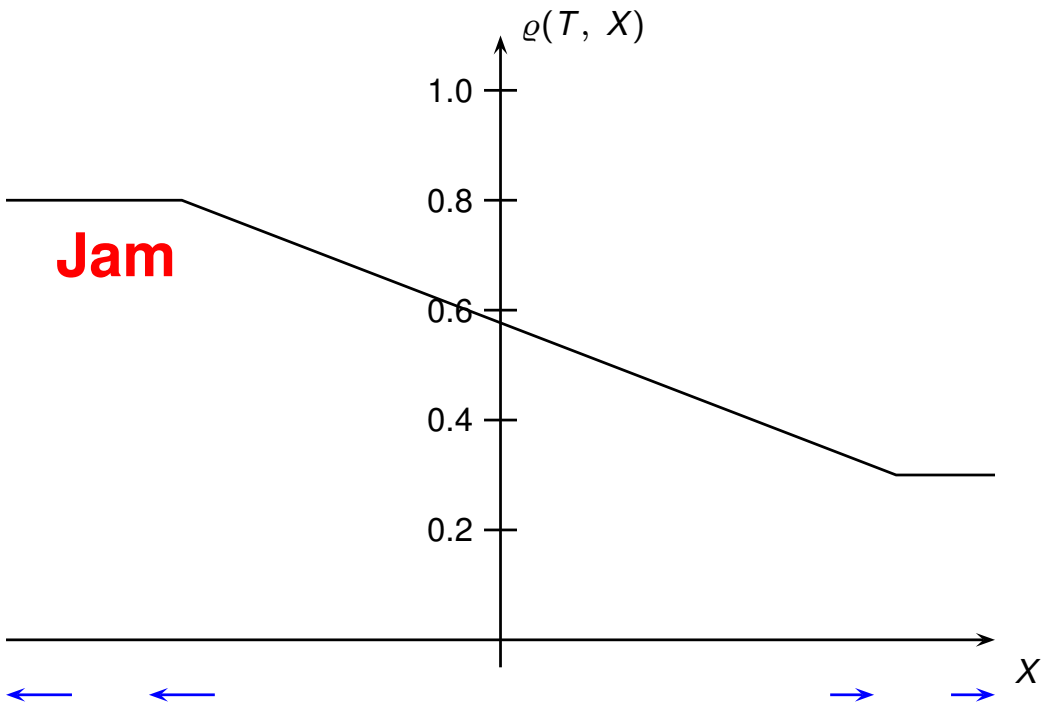
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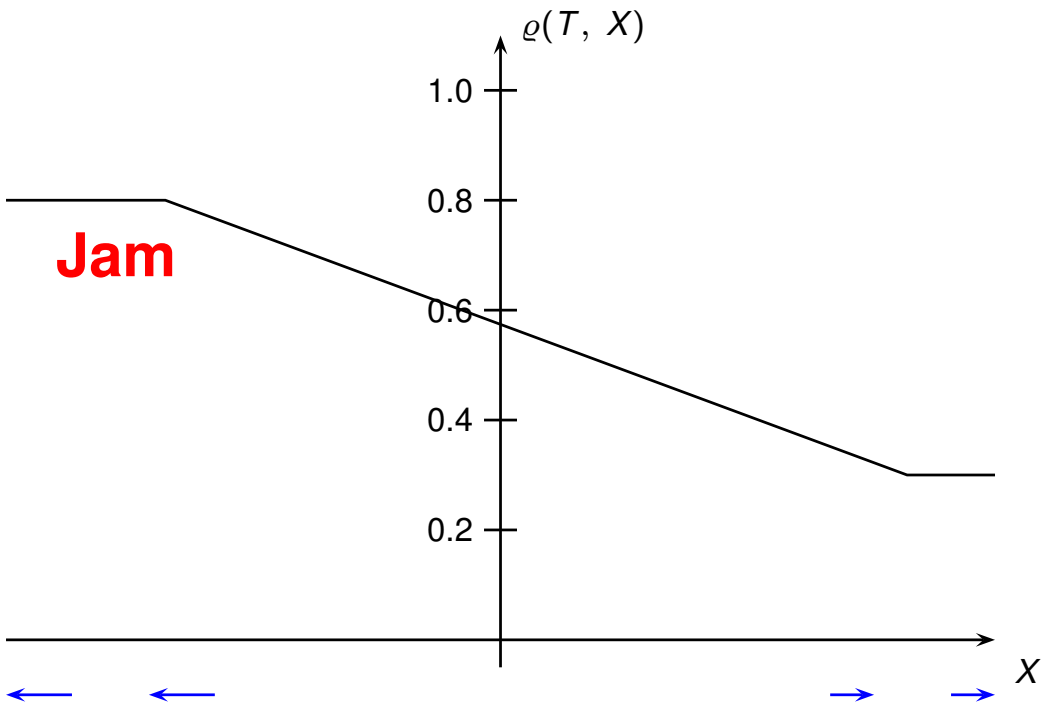
Rescaled version: rarefaction fan



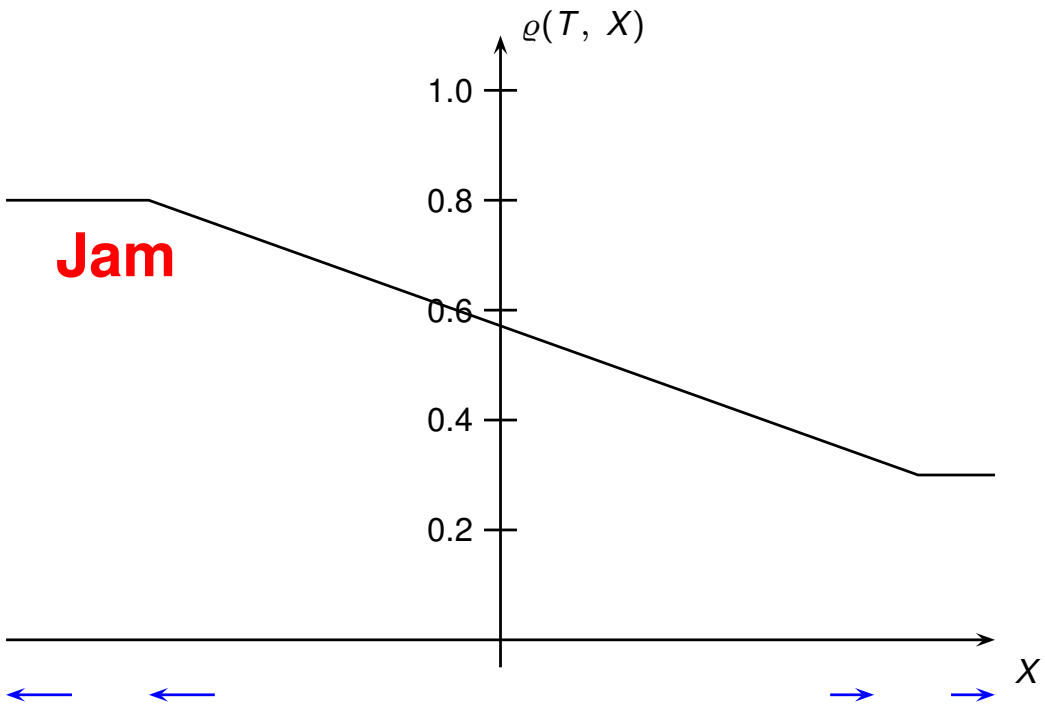
Rescaled version: rarefaction fan



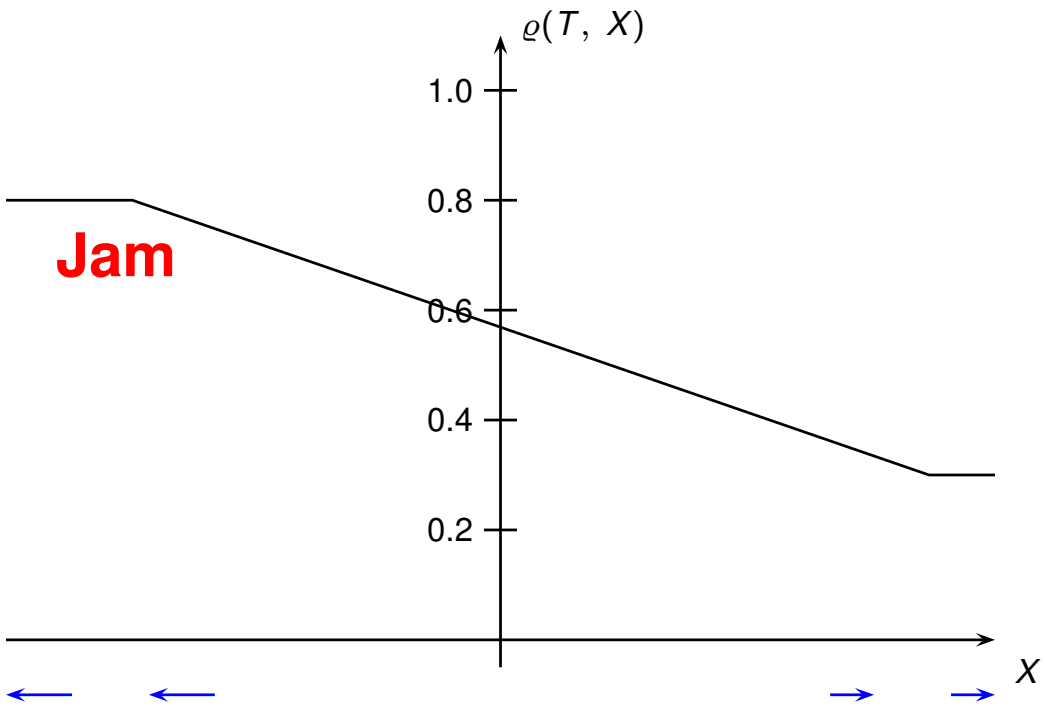
Rescaled version: rarefaction fan



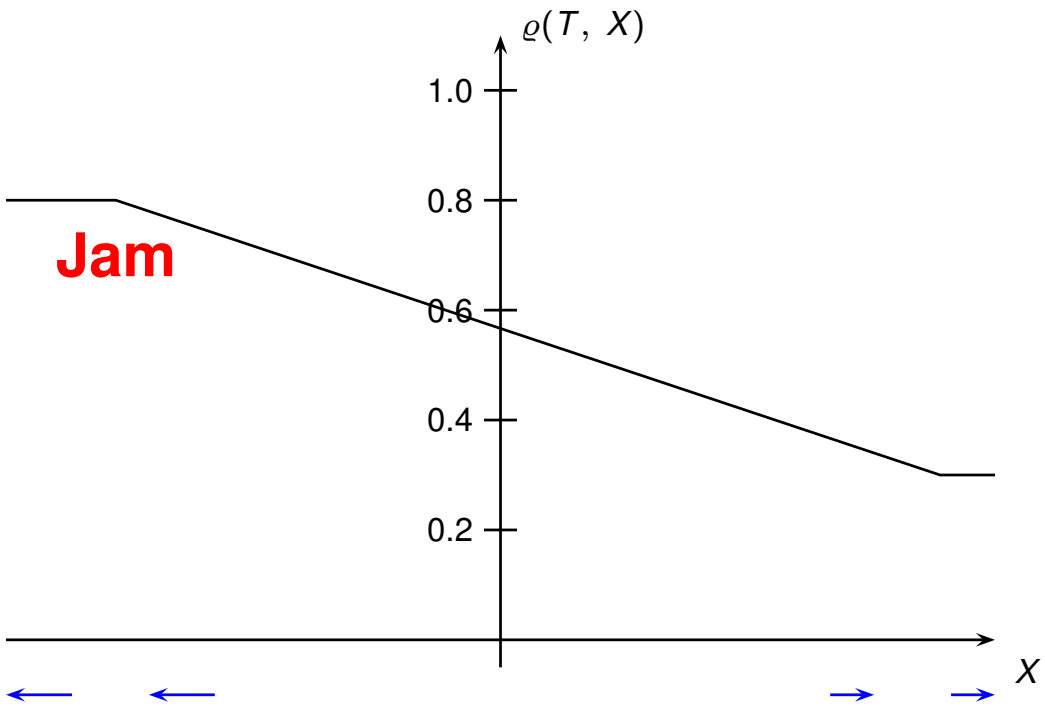
Rescaled version: rarefaction fan



Rescaled version: rarefaction fan

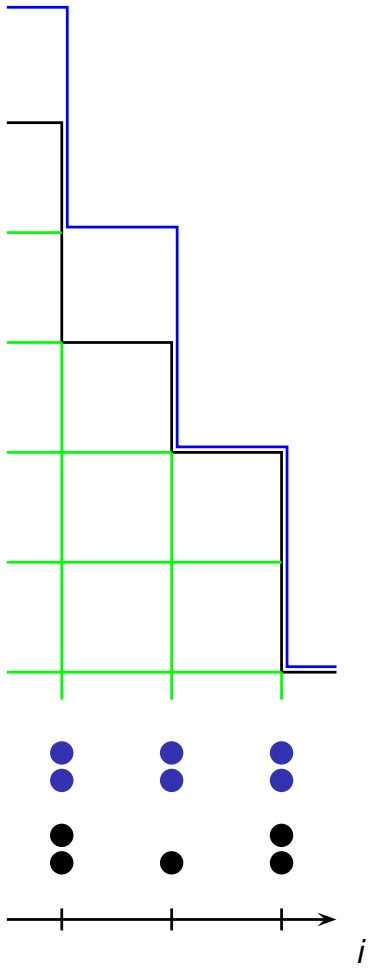


Rescaled version: rarefaction fan



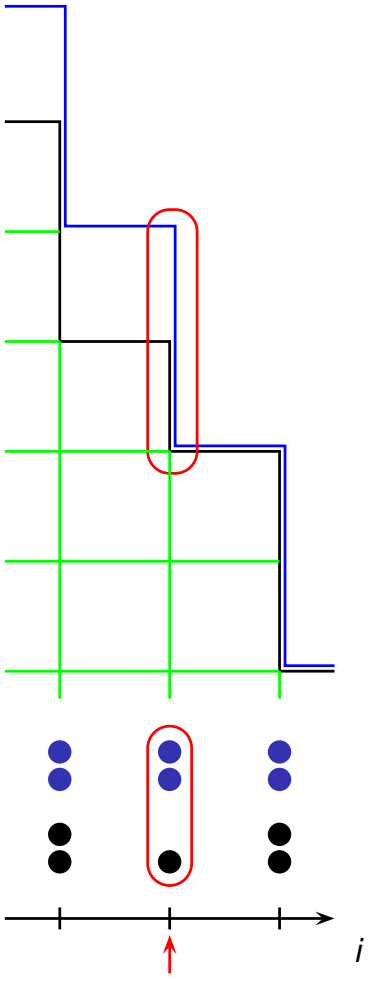
The second class particle

States ω and η only differ at one site.



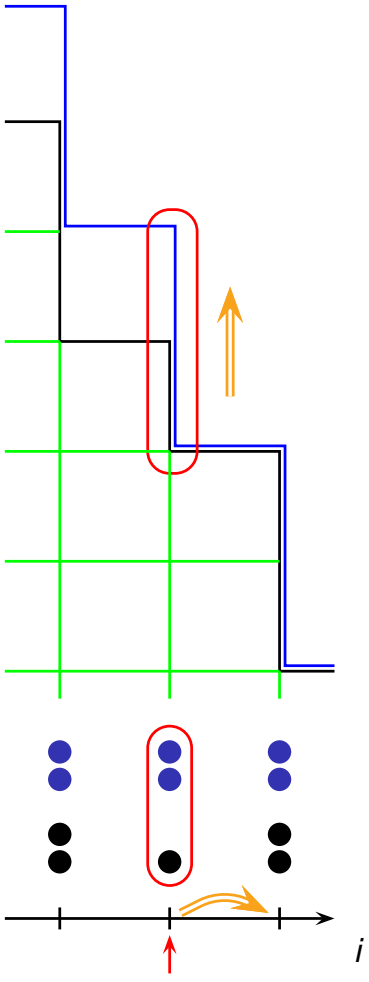
The second class particle

States ω and η only differ at one site.



The second class particle

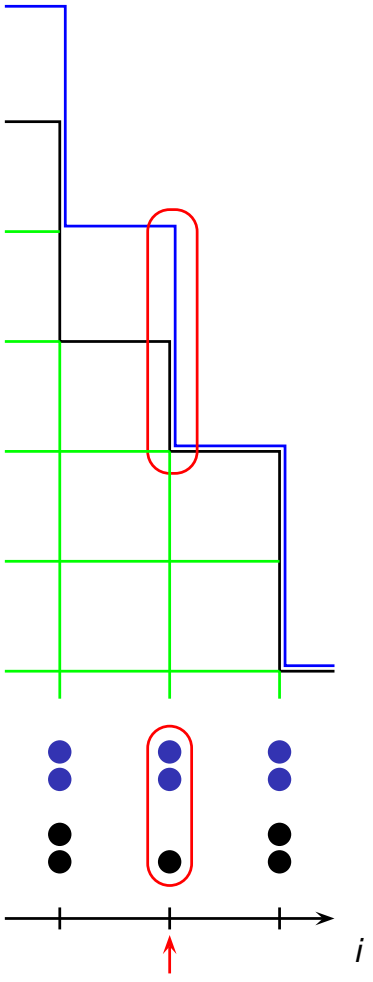
States ω and η only differ at one site.



Growth on the right:
rate$\leq$rate

The second class particle

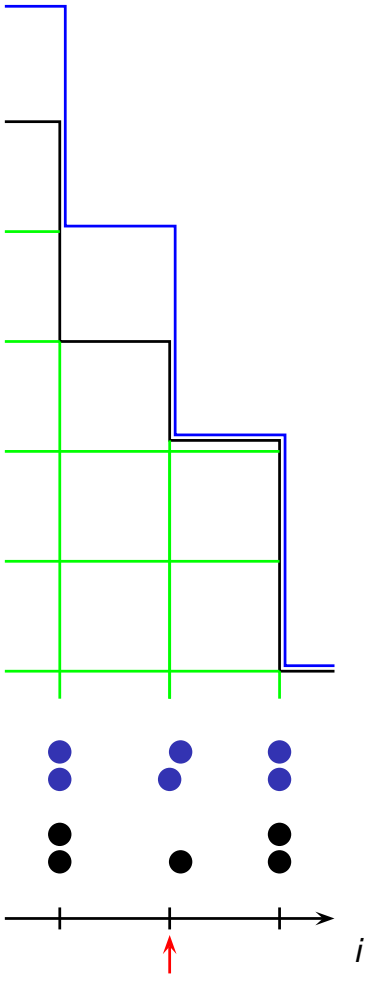
States ω and η only differ at one site.



Growth on the right:
rate$\leq$rate
with rate:

The second class particle

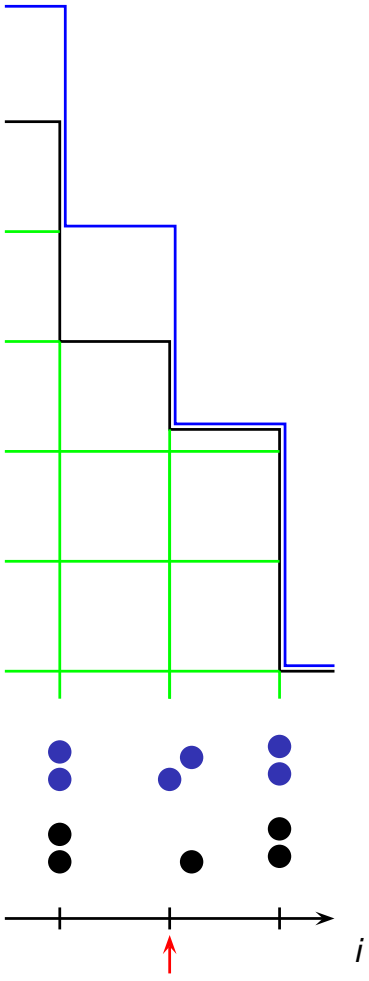
States ω and η only differ at one site.



Growth on the right:
rate$\leq$rate
with rate:

The second class particle

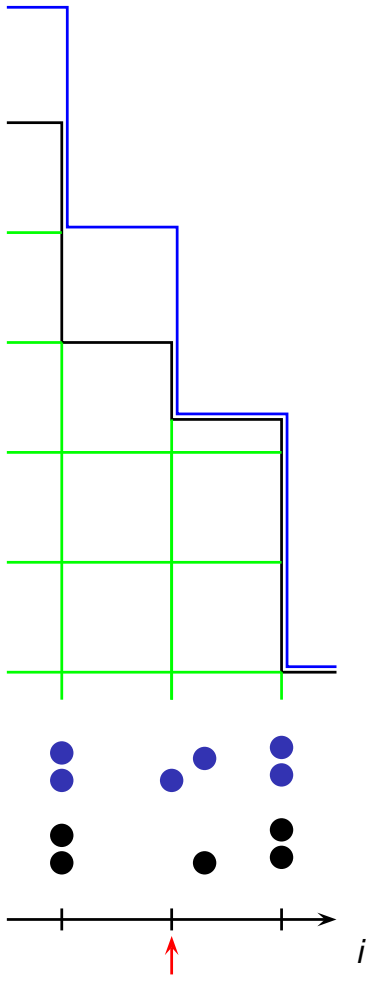
States ω and η only differ at one site.



Growth on the right:
rate$\leq$rate
with rate:

The second class particle

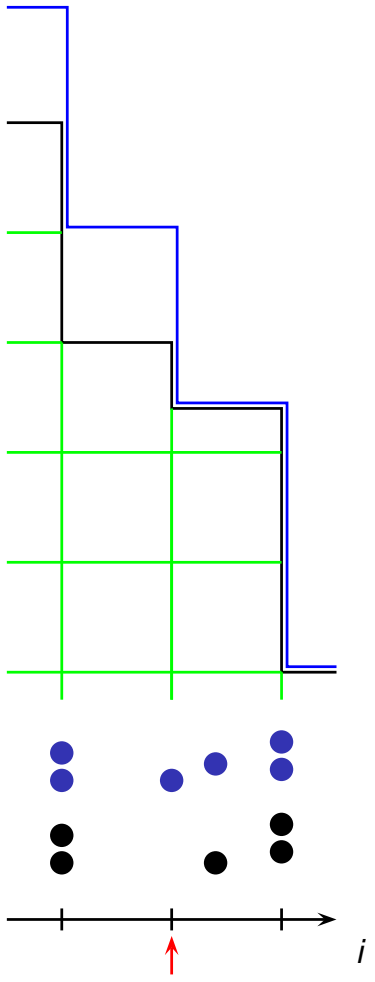
States ω and η only differ at one site.



Growth on the right:
rate$\leq$rate
with rate:

The second class particle

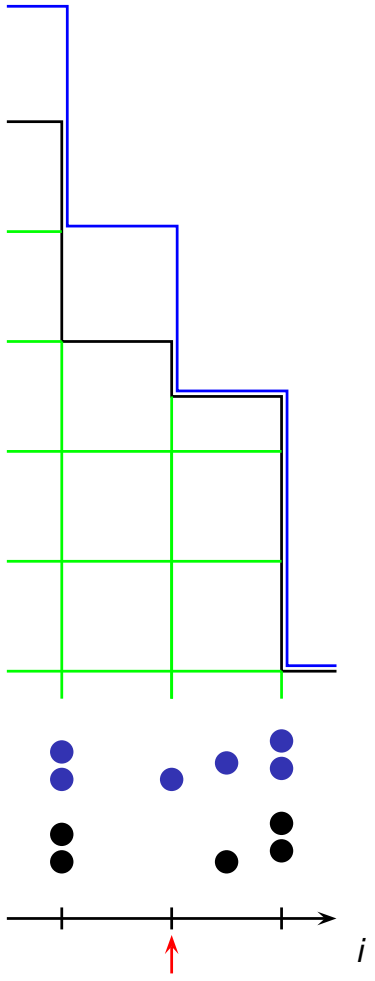
States ω and η only differ at one site.



Growth on the right:
rate$\leq$rate
with rate:

The second class particle

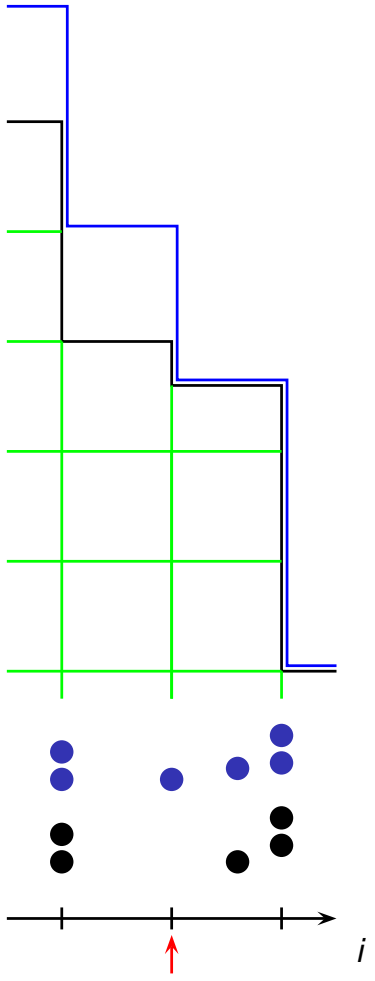
States ω and η only differ at one site.



Growth on the right:
rate$\leq$rate
with rate:

The second class particle

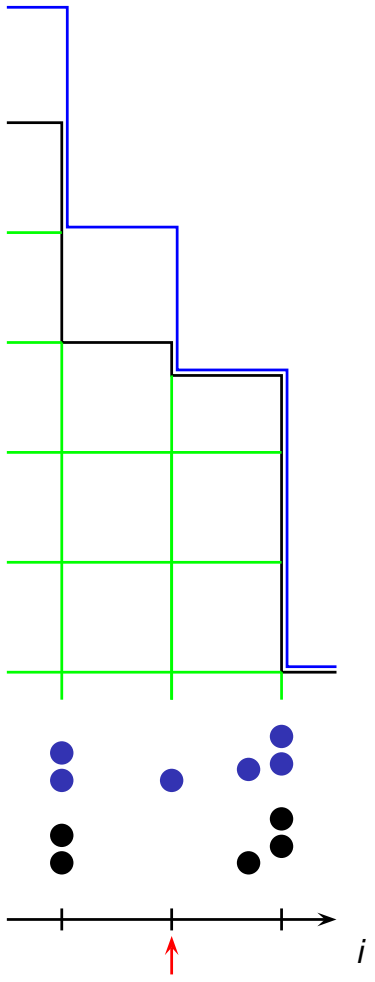
States ω and η only differ at one site.



Growth on the right:
rate$\leq$rate
with rate:

The second class particle

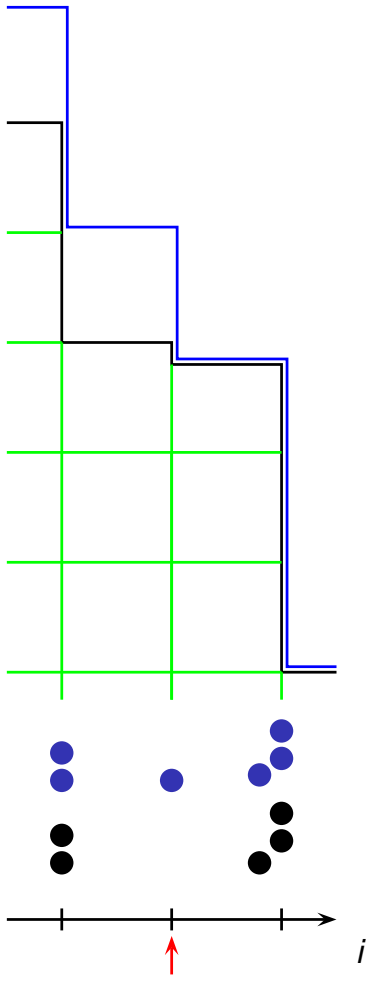
States ω and η only differ at one site.



Growth on the right:
rate$\leq$rate
with rate:

The second class particle

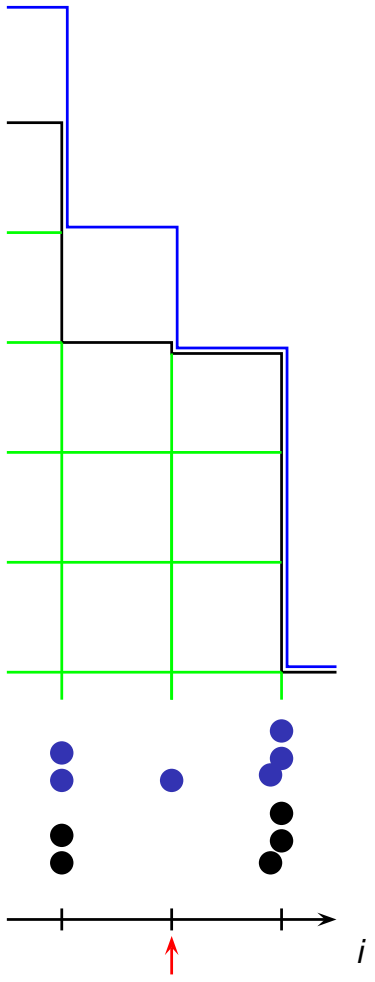
States ω and η only differ at one site.



Growth on the right:
rate$\leq$rate
with rate:

The second class particle

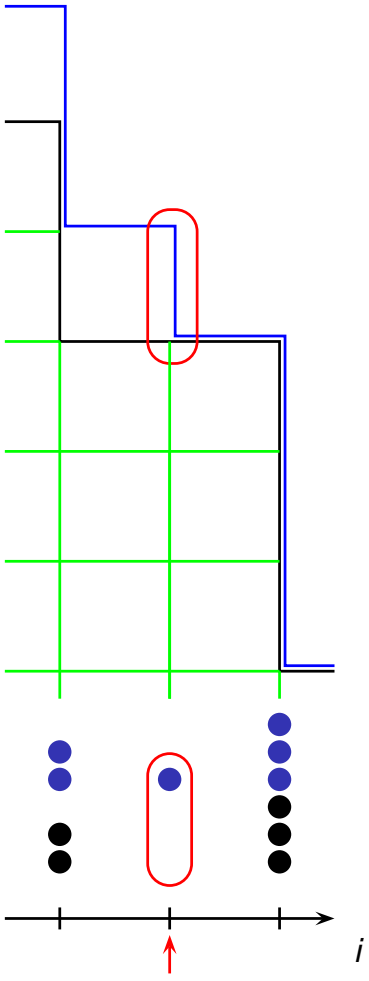
States ω and η only differ at one site.



Growth on the right:
rate$\leq$rate
with rate:

The second class particle

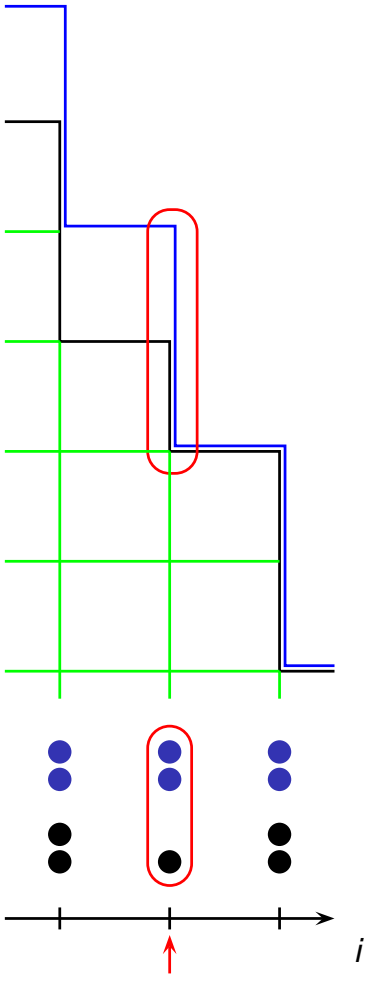
States ω and η only differ at one site.



Growth on the right:
rate$\leq$rate
with rate:

The second class particle

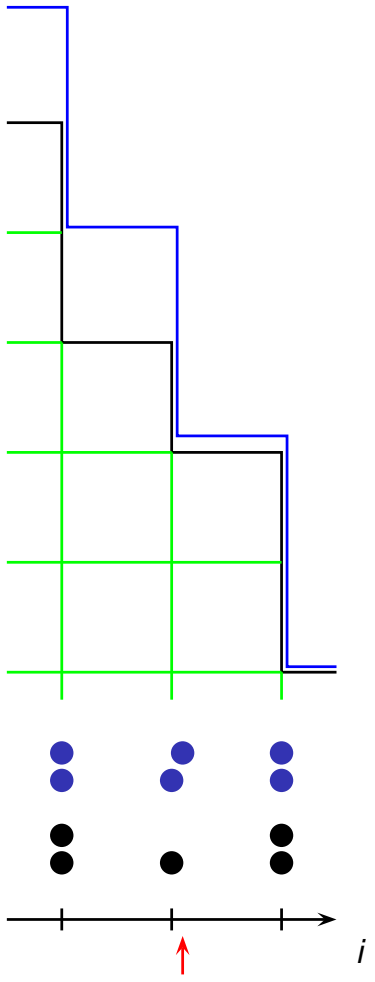
States ω and η only differ at one site.



Growth on the right:
rate$\leq$rate
with rate-rate:

The second class particle

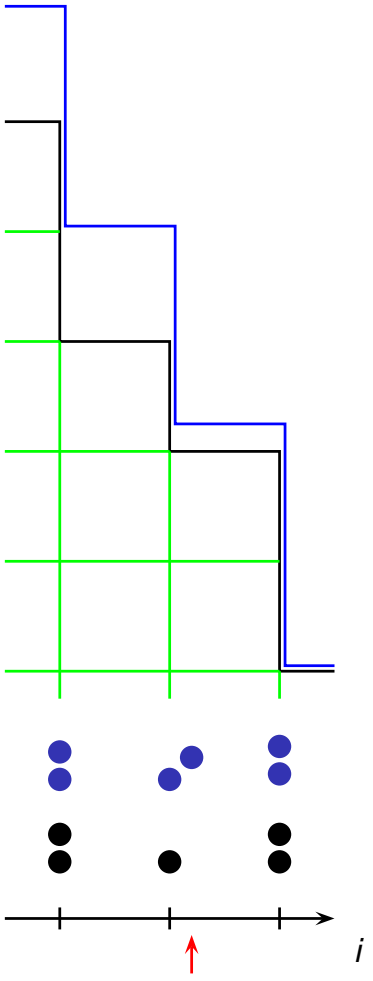
States ω and η only differ at one site.



Growth on the right:
rate$\leq$rate
with rate-rate:

The second class particle

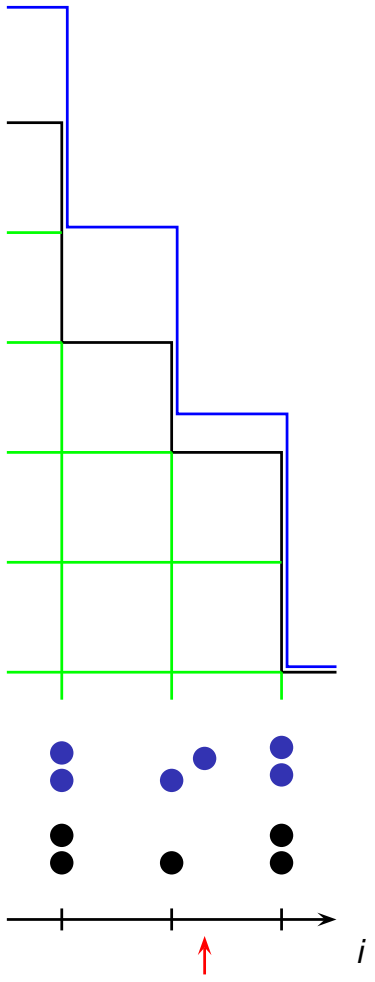
States ω and η only differ at one site.



Growth on the right:
rate$\leq$rate
with rate-rate:

The second class particle

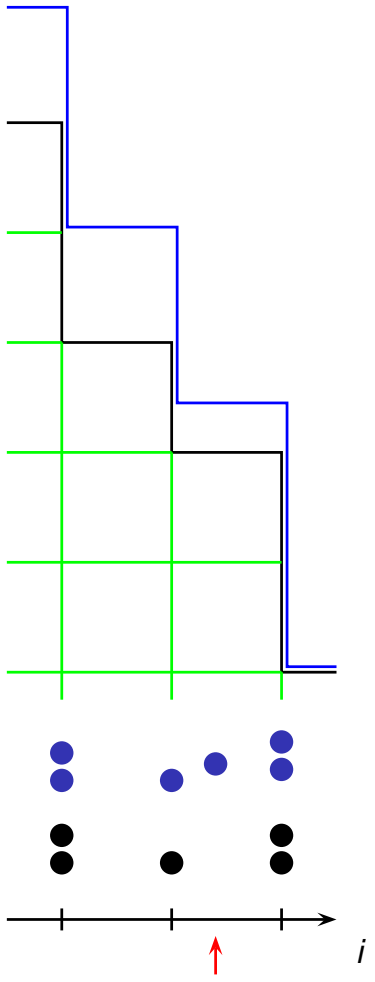
States ω and η only differ at one site.



Growth on the right:
rate$\leq$rate
with rate-rate:

The second class particle

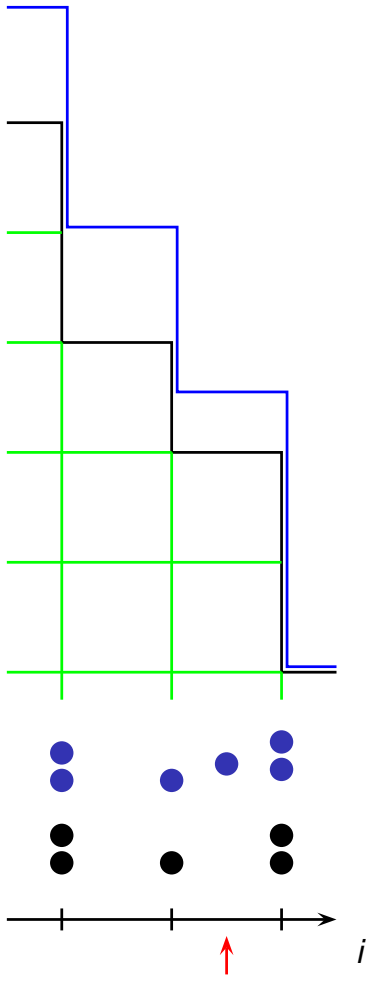
States ω and η only differ at one site.



Growth on the right:
rate$\leq$rate
with rate-rate:

The second class particle

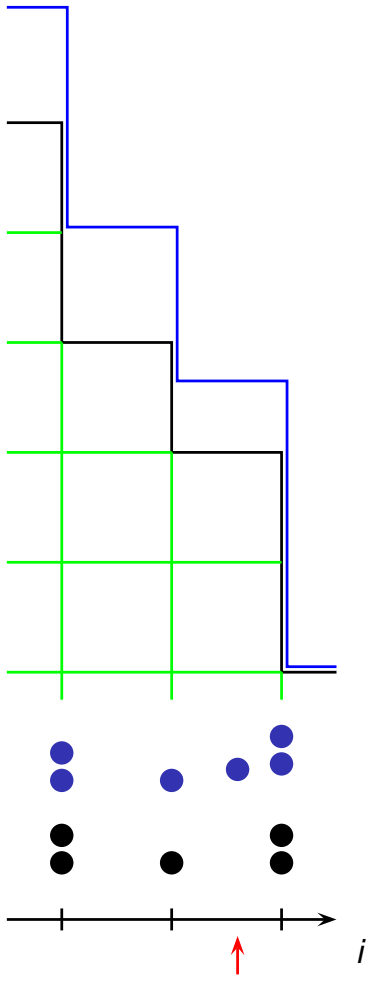
States ω and η only differ at one site.



Growth on the right:
rate$\leq$rate
with rate-rate:

The second class particle

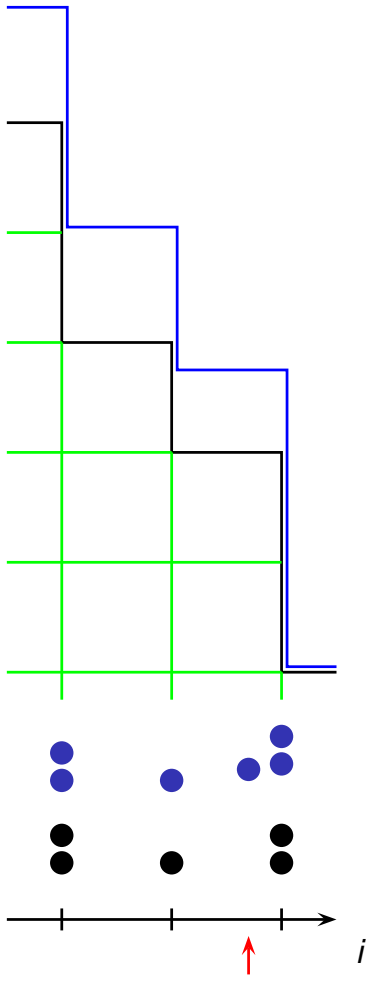
States ω and η only differ at one site.



Growth on the right:
rate$\leq$rate
with rate-rate:

The second class particle

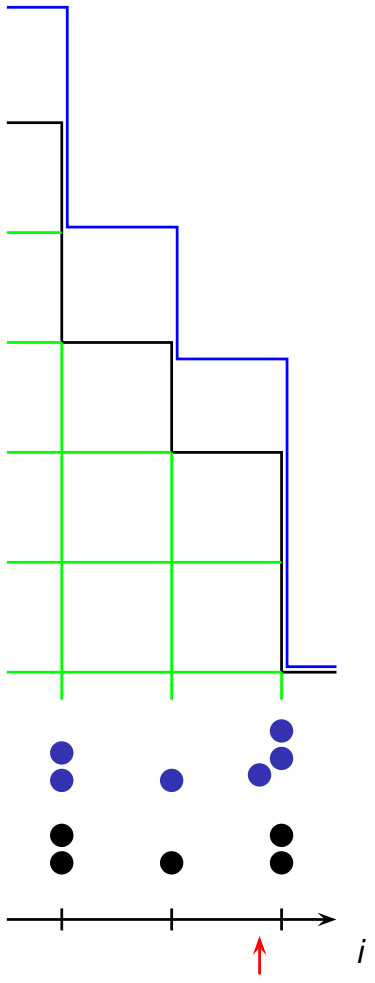
States ω and η only differ at one site.



Growth on the right:
rate$\leq$rate
with rate-rate:

The second class particle

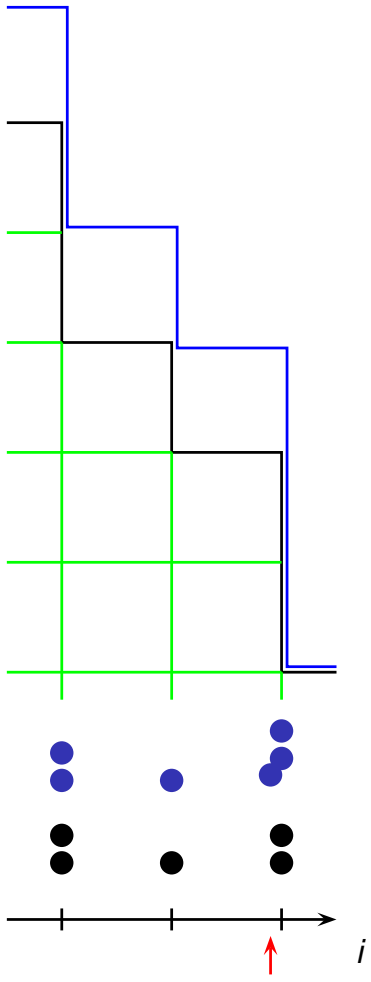
States ω and η only differ at one site.



Growth on the right:
rate$\leq$rate
with rate-rate:

The second class particle

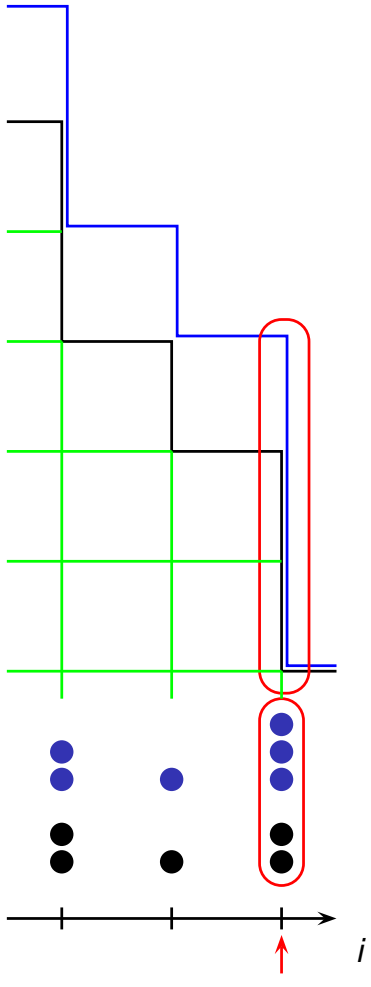
States ω and η only differ at one site.



Growth on the right:
rate$\leq$rate
with rate-rate:

The second class particle

States ω and η only differ at one site.



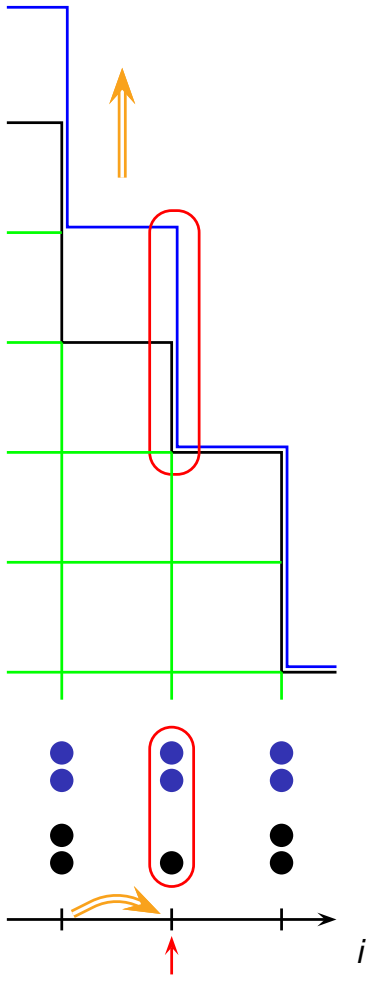
Growth on the right:
rate$\leq$rate
with rate-rate:

The second class particle

States ω and η only differ at one site.

Growth on the left:

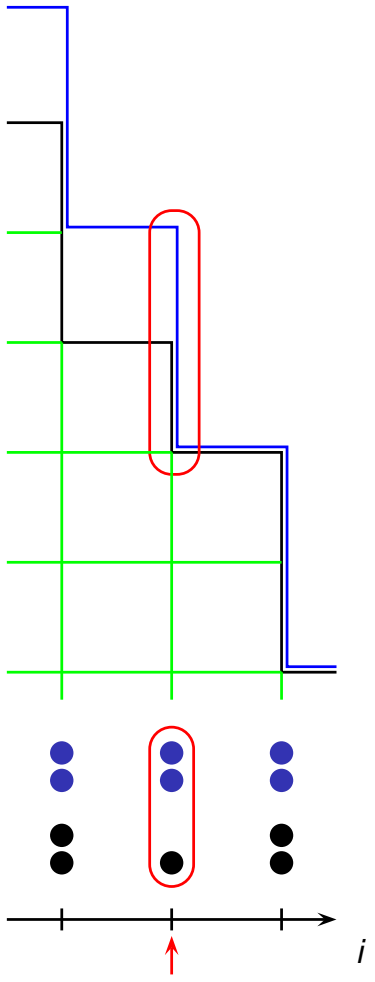
rate$\geq$rate



The second class particle

States ω and η only differ at one site.

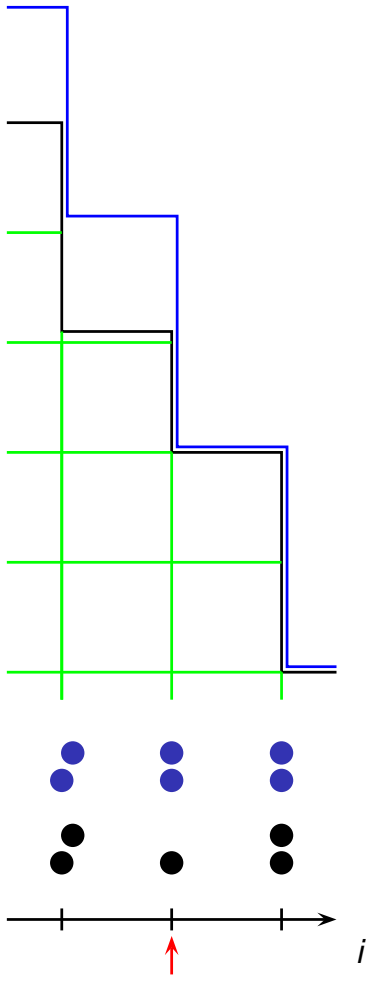
Growth on the left:
rate$\geq$rate
with rate:



The second class particle

States ω and η only differ at one site.

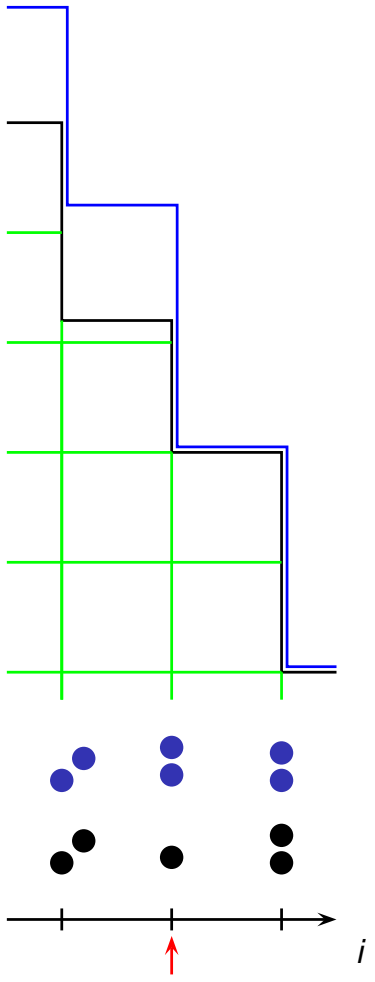
Growth on the left:
rate$\geq$rate
with rate:



The second class particle

States ω and η only differ at one site.

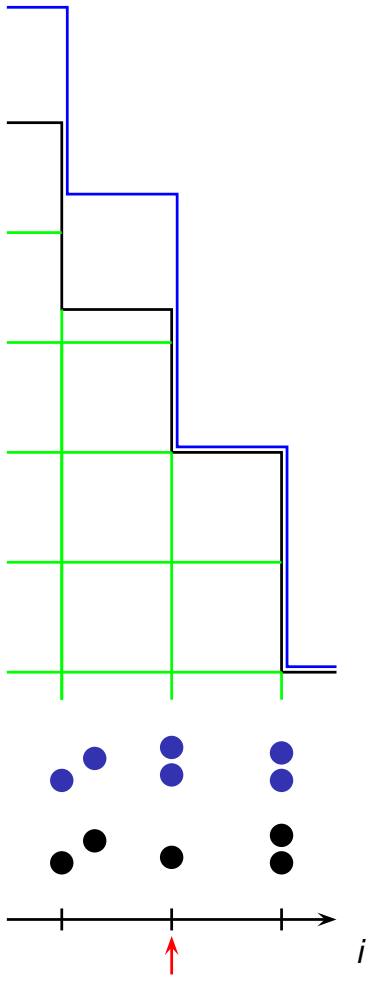
Growth on the left:
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The second class particle

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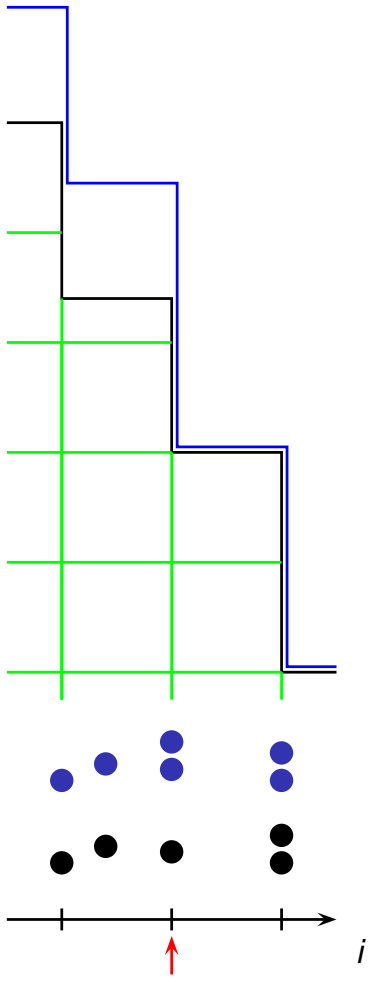
Growth on the left:
rate$\geq$rate
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The second class particle

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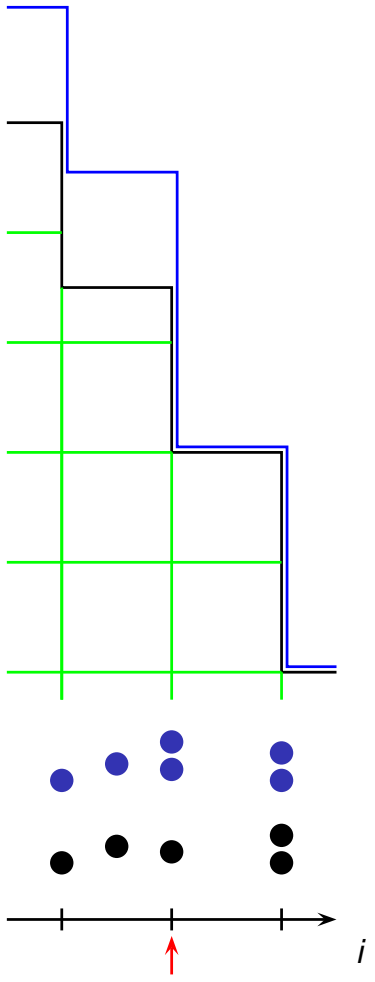
Growth on the left:
rate$\geq$rate
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The second class particle

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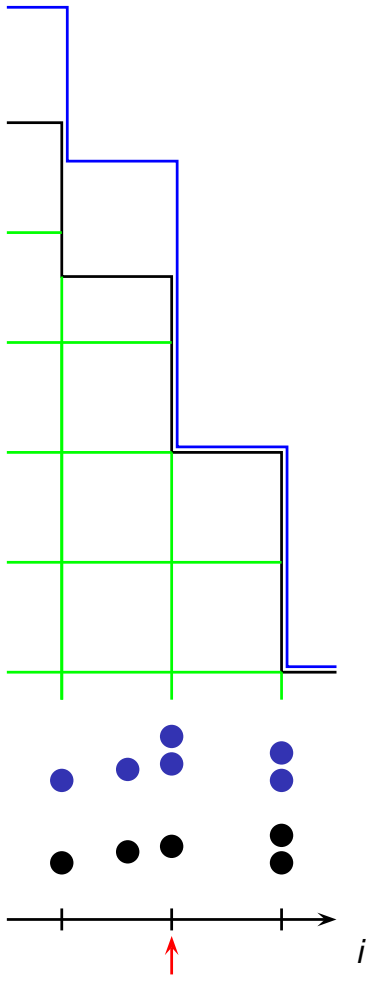
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The second class particle

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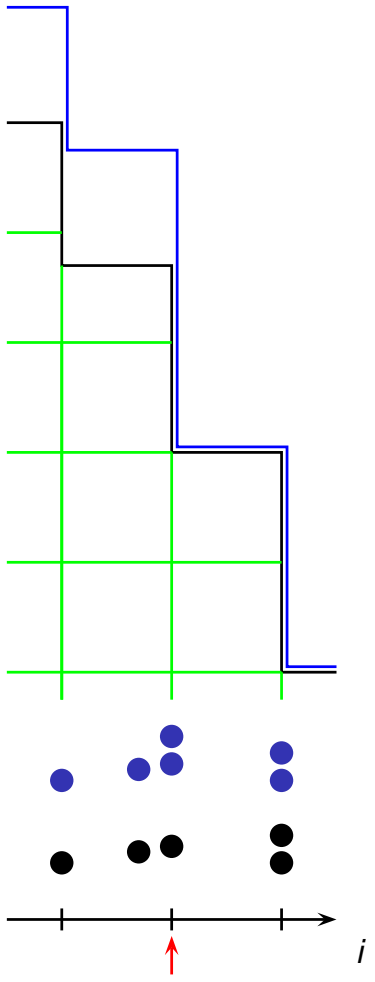
Growth on the left:
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The second class particle

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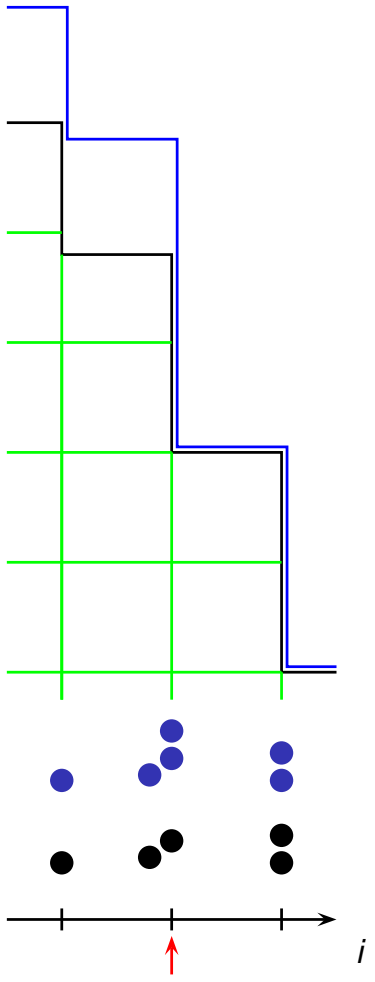
Growth on the left:
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The second class particle

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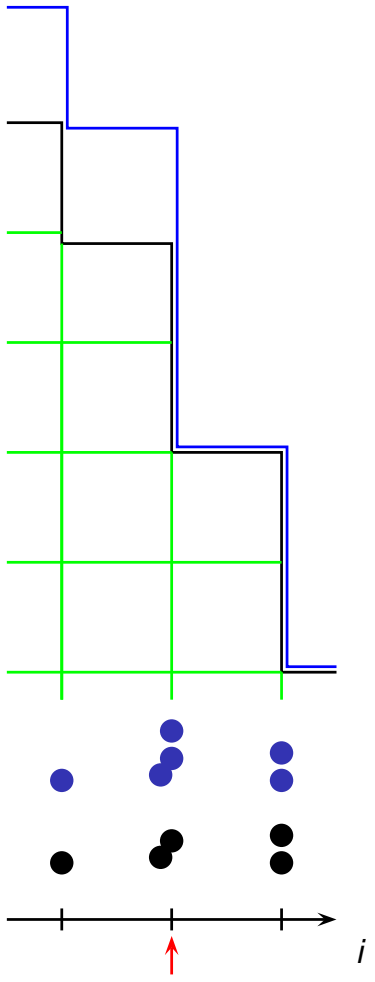
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The second class particle

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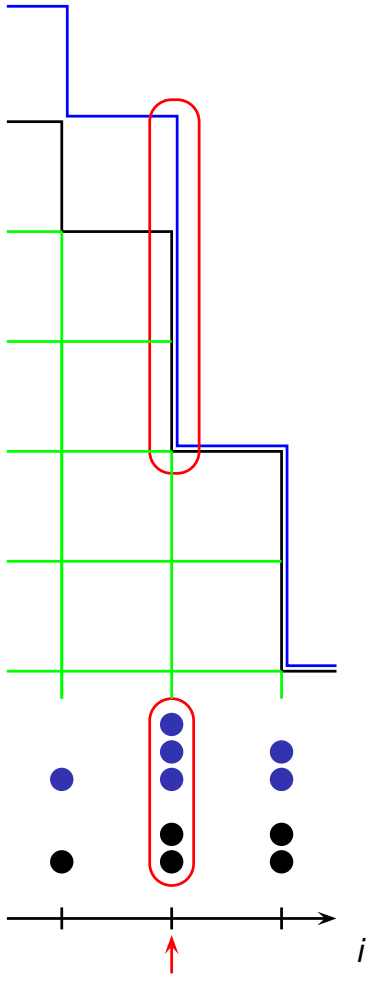
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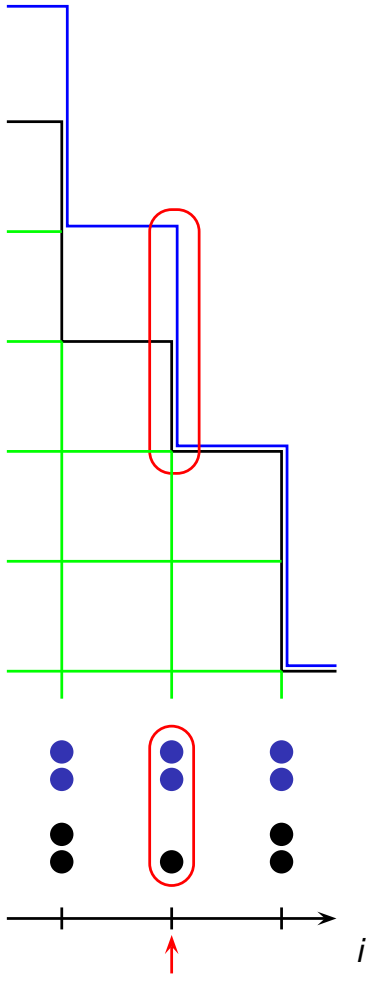
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rate$\geq$rate
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The second class particle

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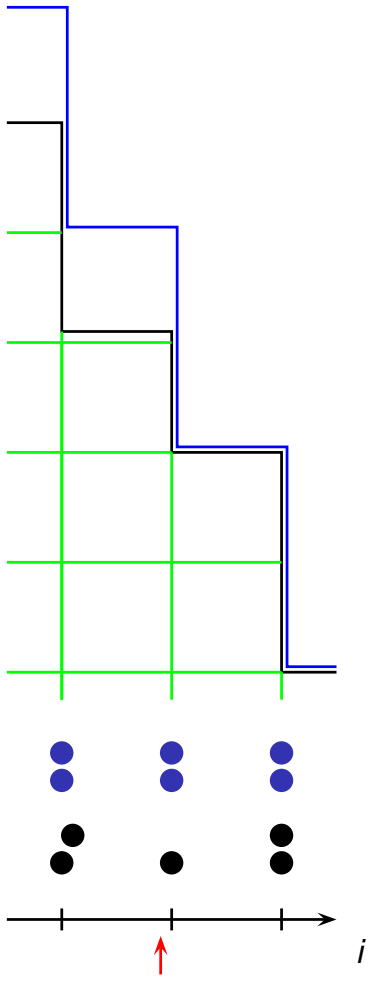
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The second class particle

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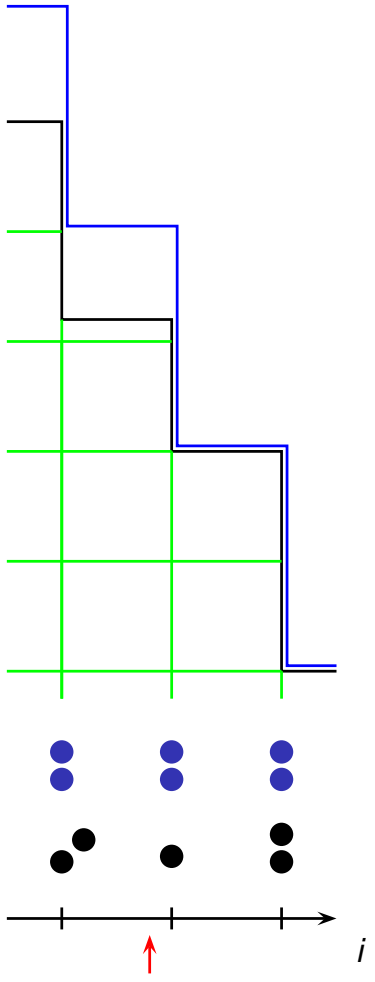
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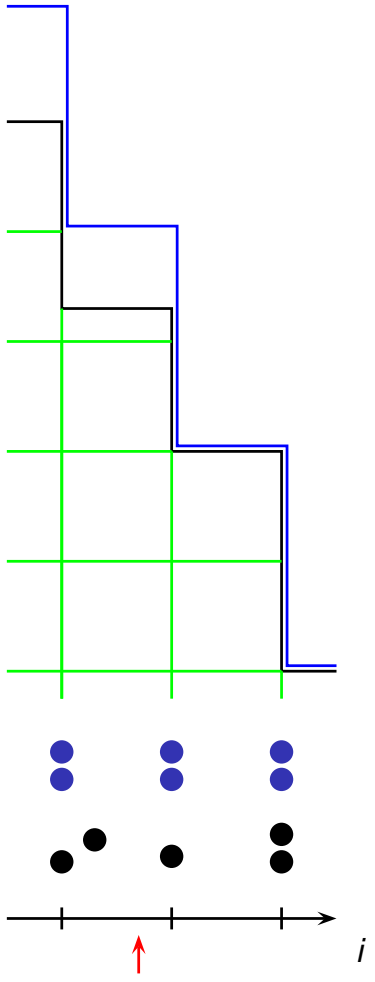
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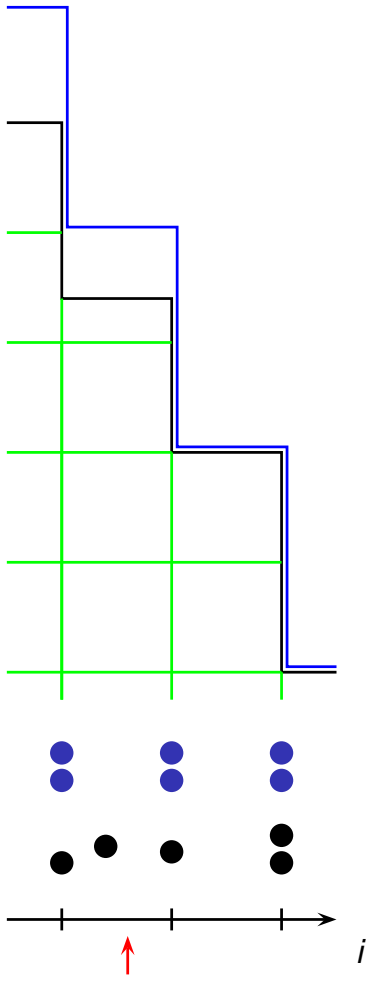
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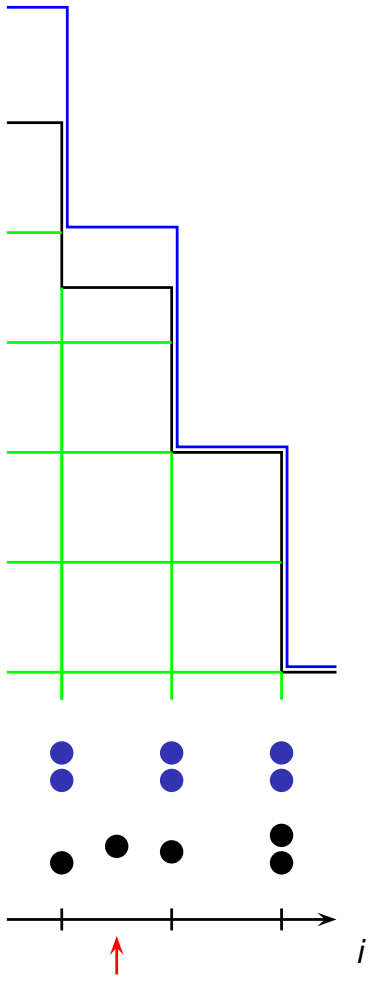
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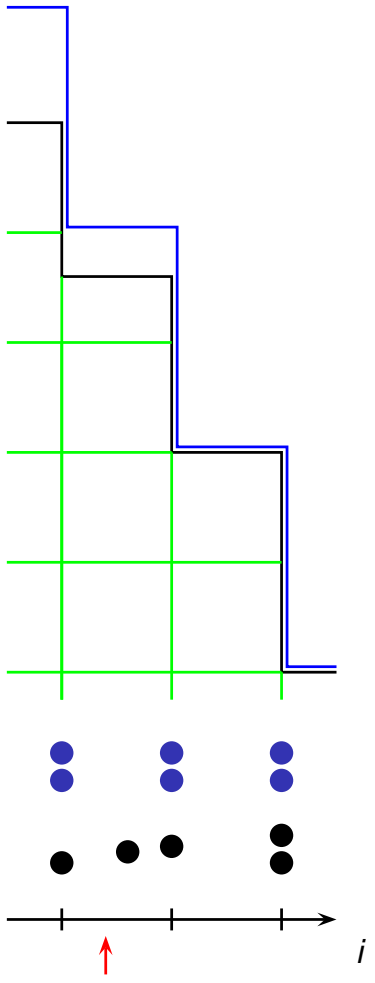
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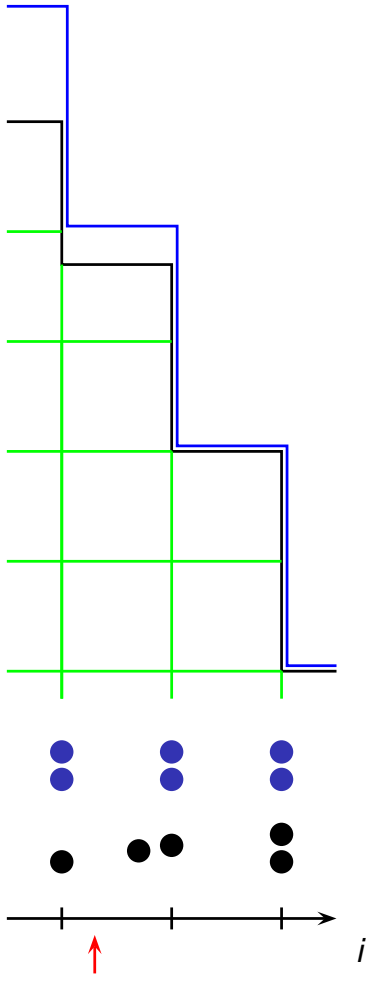
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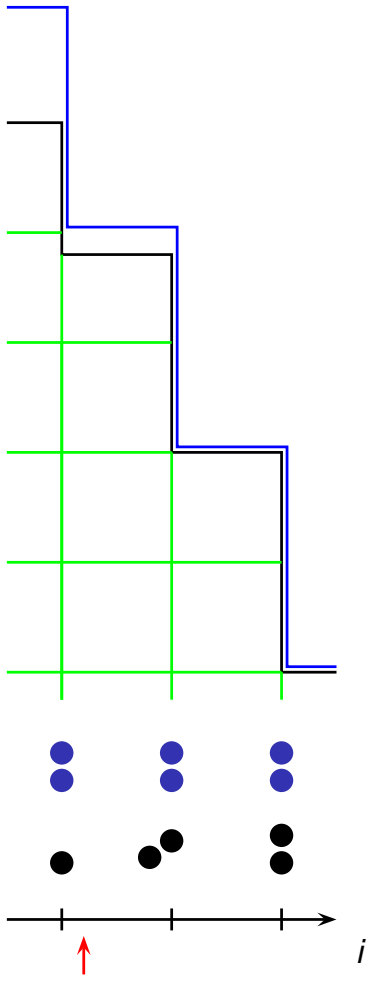
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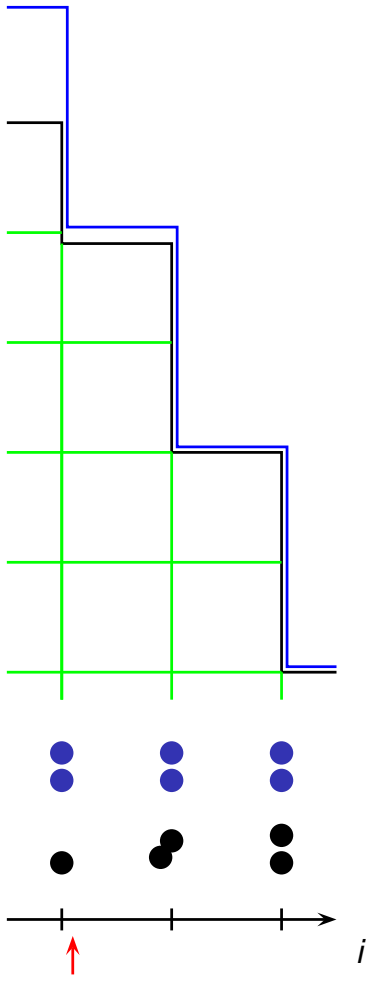
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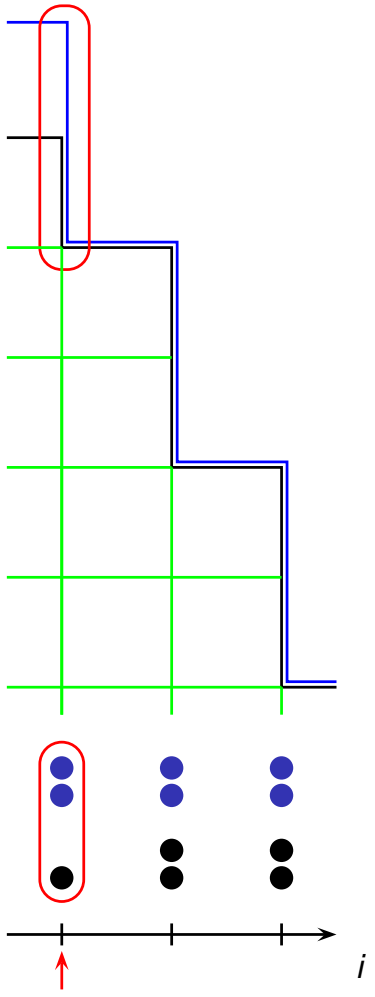
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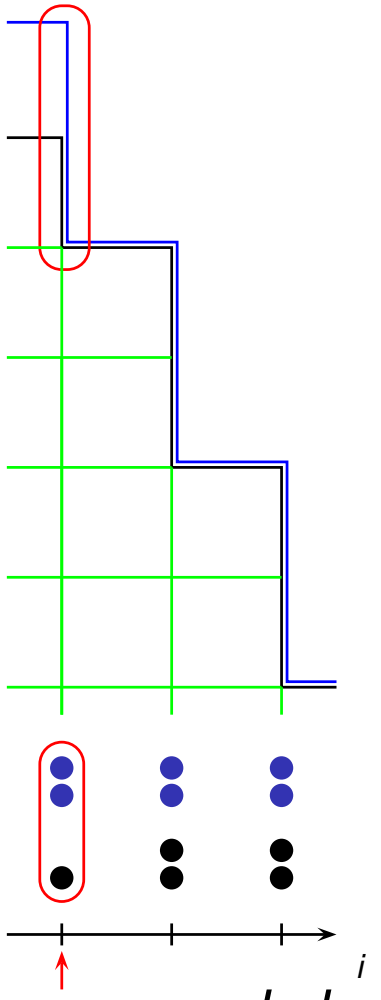
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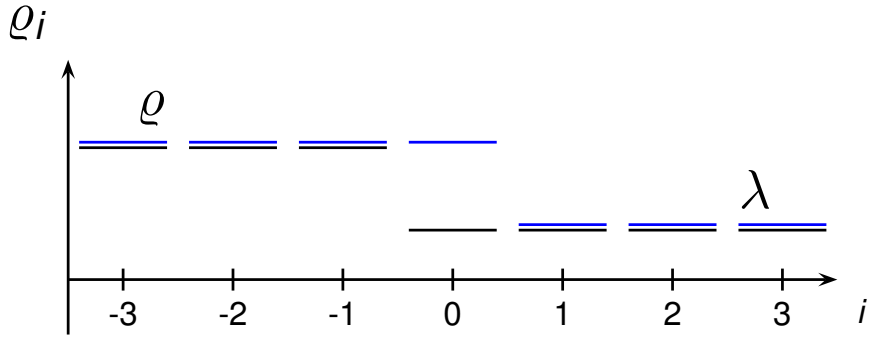


A single discrepancy $\uparrow$, the *second class particle*, is conserved.
Its position at time t is $Q(t)$.

Ferrari-Kipnis '95 for TASEP

Blue TASEP ω:
Bernoulli(ϱ) for sites $\{\ldots, -2, -1, 0\}$,
Bernoulli(λ) for sites $\{1, 2, 3, \ldots\}$.

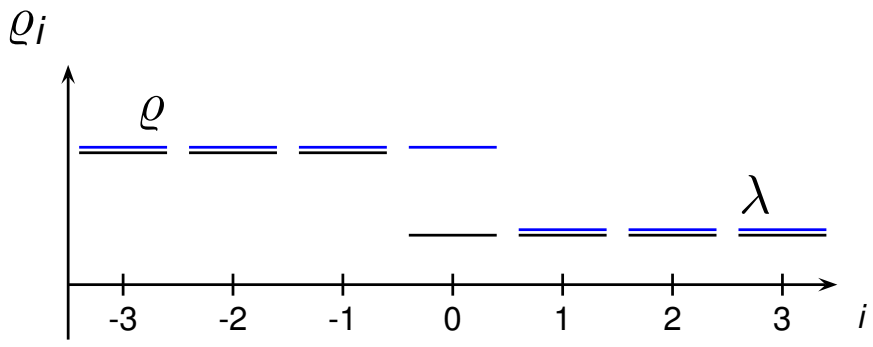
Black TASEP η:
Bernoulli(ϱ) for sites $\{\ldots, -3, -2, -1\}$,
Bernoulli(λ) for sites $\{0, 1, 2, \ldots\}$.



$h_i(t)$, $g_i(t)$ are the respective numbers of particles jumping over the edge $(i, i+1)$ by time t $(i > 0)$.

Ferrari-Kipnis '95 for TASEP, Part 1

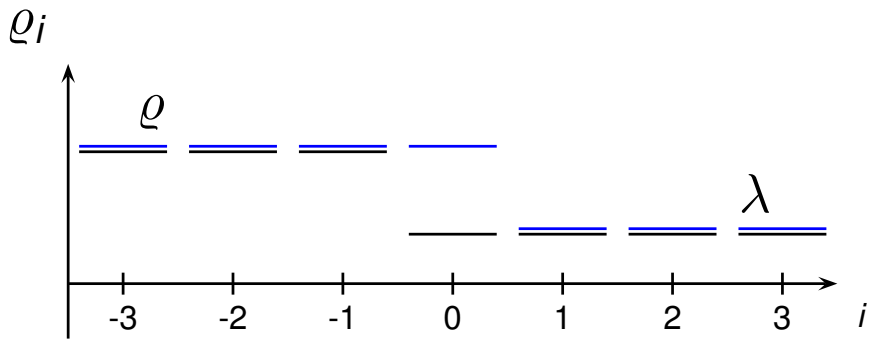
First realization:



Ferrari-Kipnis '95 for TASEP, Part 1

First realization:

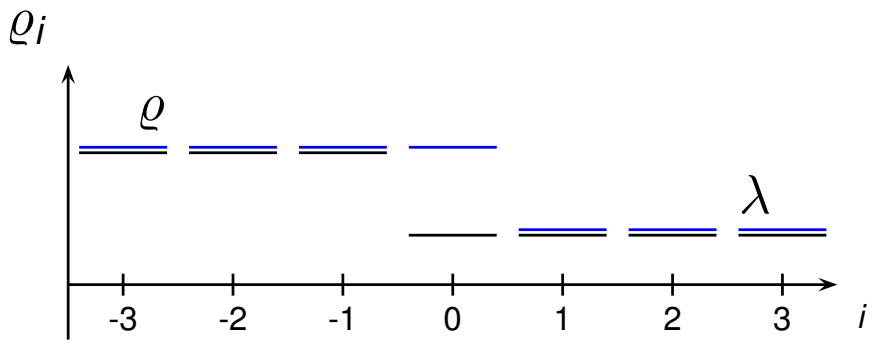
- $\omega_i(0) = \eta_i(0) \sim$ Bernoulli(ϱ) for $i < 0$



Ferrari-Kipnis '95 for TASEP, Part 1

First realization:

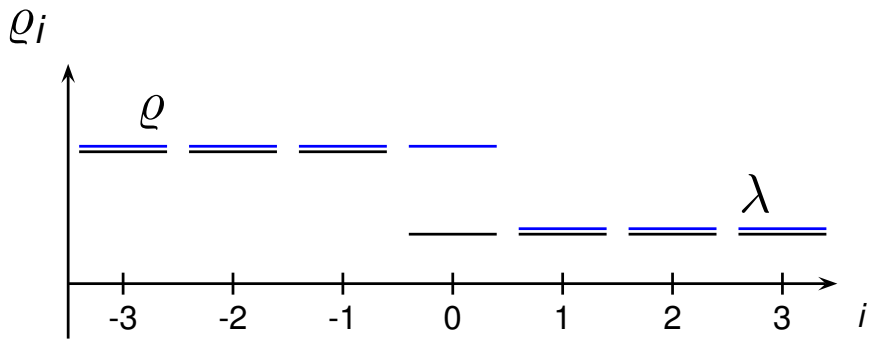
- $\omega_i(0) = \eta_i(0) \sim$ Bernoulli(ϱ) for $i < 0$
- $(\omega_0(0), \eta_0(0)) = (0, 0)$ w. prob. $1 - \varrho$
  $(\omega_0(0), \eta_0(0)) = (1, 0)$ w. prob. $\varrho - \lambda$
  $(\omega_0(0), \eta_0(0)) = (1, 1)$ w. prob. λ



Ferrari-Kipnis '95 for TASEP, Part 1

First realization:

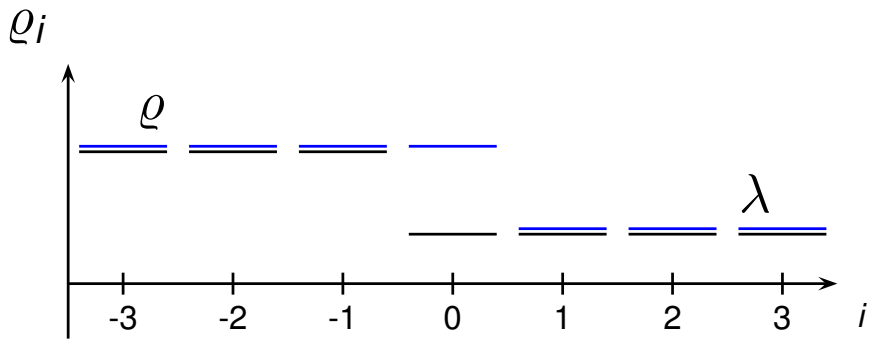
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Ferrari-Kipnis '95 for TASEP, Part 1

First realization:

- $\omega_i(0) = \eta_i(0) \sim$ Bernoulli(ϱ) for $i < 0$
- $(\omega_0(0), \eta_0(0)) = (0, 0)$ w. prob. $1 - \varrho$
  $(\omega_0(0), \eta_0(0)) = (1, 0)$ w. prob. $\varrho - \lambda$ 2nd class particle
  $(\omega_0(0), \eta_0(0)) = (1, 1)$ w. prob. λ
- $\omega_i(0) = \eta_i(0) \sim$ Bernoulli(λ) for $i > 0$

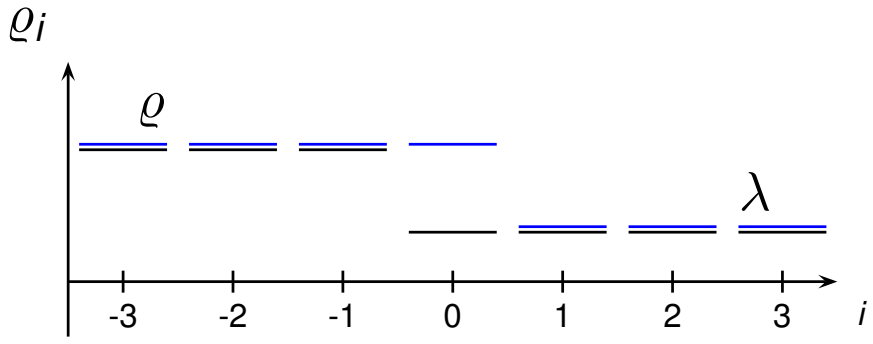


Ferrari-Kipnis '95 for TASEP, Part 1

First realization:

- $\omega_i(0) = \eta_i(0) \sim$ Bernoulli(ϱ) for $i < 0$
- $(\omega_0(0), \eta_0(0)) = (0, 0)$ w. prob. $1 - \varrho$
  $(\omega_0(0), \eta_0(0)) = (1, 0)$ w. prob. $\varrho - \lambda$
  $(\omega_0(0), \eta_0(0)) = (1, 1)$ w. prob. λ
- $\omega_i(0) = \eta_i(0) \sim$ Bernoulli(λ) for $i > 0$

2nd class particle

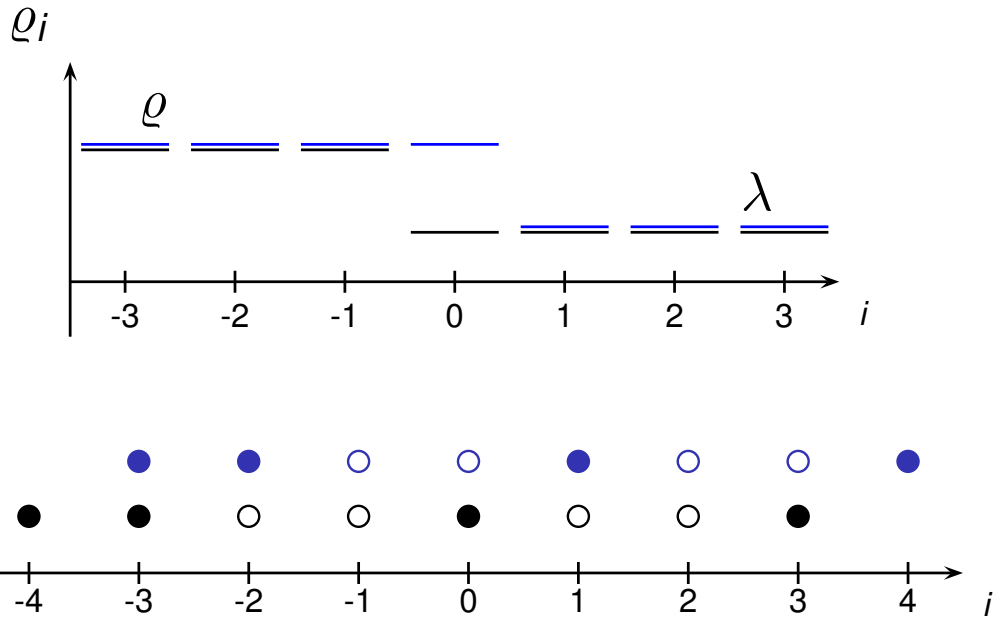


$$\mathbf{E}h_i(t) - \mathbf{E}g_i(t) = \mathbf{E}(h_i(t) - g_i(t)) = (\varrho - \lambda) \cdot \mathbf{P}\{Q(t) > i\}.$$

Ferrari-Kipnis '95 for TASEP, Part 2

Second realization:

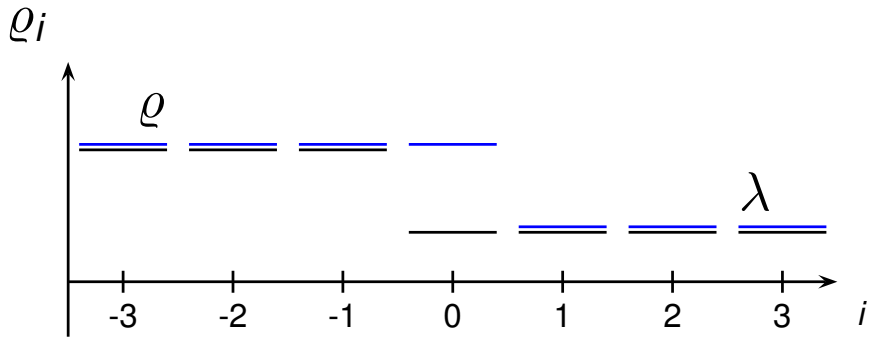
$$\omega_i(t) \equiv \eta_{i-1}(t) \qquad \forall i,\ \forall t.$$

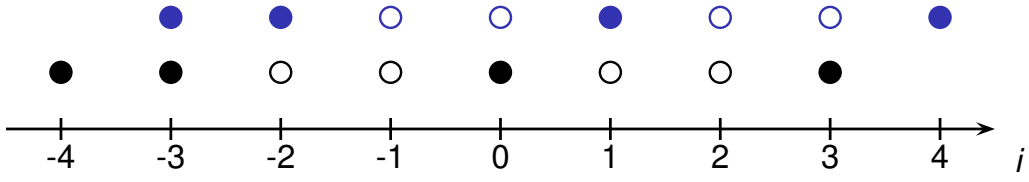


Ferrari-Kipnis '95 for TASEP, Part 2

Second realization:

$$\omega_i(t) \equiv \eta_{i-1}(t) \qquad \forall i,\ \forall t.$$





$$\mathbf{E}h_i(t) - \mathbf{E}g_i(t) = \mathbf{E}(h_i(t) - g_i(t)) = \mathbf{E}(\eta_i(t) - \eta_i(0)) = \mathbf{E}\eta_i(t) - \mathbf{E}\eta_i(0).$$

Ferrari-Kipnis '95 for TASEP

Thus,

$$\mathbf{E}h_i(t) - \mathbf{E}g_i(t) = \mathbf{E}(h_i(t) - g_i(t)) = (\varrho - \lambda) \cdot \mathbf{P}\{Q(t) > i\},$$
$$\mathbf{E}h_i(t) - \mathbf{E}g_i(t) = \mathbf{E}(h_i(t) - g_i(t)) = \mathbf{E}\eta_i(t) - \mathbf{E}\eta_i(0),$$

Ferrari-Kipnis '95 for TASEP

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$$\mathbf{P}\{Q(t) > i\} = \frac{\mathbf{E}\eta_i(t) - \mathbf{E}\eta_i(0)}{\varrho - \lambda}.$$

Ferrari-Kipnis '95 for TASEP

Thus,

$$\mathbf{E}h_i(t) - \mathbf{E}g_i(t) = \mathbf{E}(h_i(t) - g_i(t)) = (\varrho - \lambda) \cdot \mathbf{P}\{Q(t) > i\},$$
$$\mathbf{E}h_i(t) - \mathbf{E}g_i(t) = \mathbf{E}(h_i(t) - g_i(t)) = \mathbf{E}\eta_i(t) - \mathbf{E}\eta_i(0),$$

$$\mathbf{P}\{Q(t) > i\} = \frac{\mathbf{E}\eta_i(t) - \mathbf{E}\eta_i(0)}{\varrho - \lambda}.$$

Combine with hydrodynamics to conclude

$$\frac{Q(t)}{t} \Rightarrow \begin{cases} \text{shock velocity} & \text{in a shock,} \\ \mathrm{U}\big(H'(\varrho),\, H'(\lambda)\big) & \text{in a rarefaction wave.} \end{cases}$$

Let's generalise

Other models have more than 0 or 1 particles per site. How do we start the second class particle?

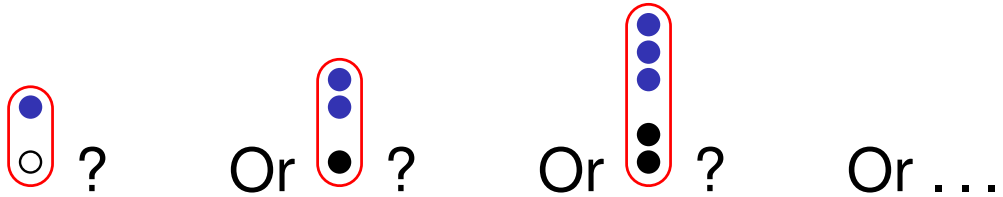
- Shall we do



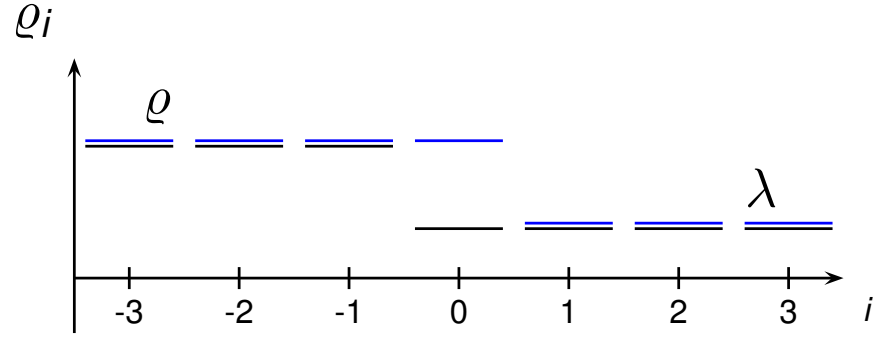
Let's generalise

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- Shall we do



- Recall for TASEP we increased λ to ϱ by adding or not adding a 2nd class particle.
  $(\omega_0(0), \eta_0(0)) = (0, 0)$ w. prob. $1 - \varrho$
  $(\omega_0(0), \eta_0(0)) = (1, 0)$ w. prob. $\varrho - \lambda$
  $(\omega_0(0), \eta_0(0)) = (1, 1)$ w. prob. λ



Let's generalise: problems with coupling

Fix $\lambda < \varrho \leq \lambda + 1$. Is there a joint distribution of (ω_0, η_0) such that

- the first marginal is $\omega_0 \sim$ stati. μ^{ϱ};
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- *Yes for discrete Gaussian (bricklayers with* $r(\omega_i) = \mathrm{e}^{\beta\omega_i}$*).*

Let's generalise

Keep calm and couple anyway.

Find a coupling measure ν with

- first marginal $\omega_0 \sim$ stati. μ^ϱ;
- second marginal $\eta_0 \sim$ stati. μ^λ;
- zero weight whenever $\omega_0 \notin \{\eta_0, \eta_0 + 1\}$.

Not many choices:

$$\begin{aligned} \nu(x, x) &= \mu^\varrho\{-\infty \dots x\} - \mu^\lambda\{-\infty \dots x-1\}, \\ \nu(x+1, x) &= \mu^\lambda\{-\infty \dots x\} - \mu^\varrho\{-\infty \dots x\}, \\ \nu &= \text{zero elsewhere.} \end{aligned}$$

Let's generalise

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Let's generalise

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We can still use the *signed measure* ν formally, as we only care about $\nu(x+1,\, x)$. Scale this up to get the initial distribution at the site of the second class particle:

$$\mu(\omega_0,\, \eta_0) = \mu(\eta_0+1,\, \eta_0) = \frac{\nu(\eta_0+1,\, \eta_0)}{\sum_x \nu(x+1,\, x)} = \frac{\nu(\eta_0+1,\, \eta_0)}{\varrho - \lambda}.$$

Let's generalise

$$\mu(\omega_0, \eta_0) = \frac{\nu(\eta_0 + 1, \eta_0)}{\varrho - \lambda}$$

- is a proper probability distribution;
- actually agrees with the coupling measure ν conditioned on a 2nd class particle when ν behaves nicely (Bernoulli, discr.Gaussian);

Let's generalise

Theorem

Starting in

$$\bigotimes_{i<0} \mu_i^{\varrho} \otimes \mu_0 \otimes \bigotimes_{i>0} \mu_i^{\lambda},$$

$$\lim_{N\to\infty} \mathbf{P}\left\{ \frac{Q(NT)}{N} > X \right\} = \frac{\varrho(X,\, T) - \lambda}{\varrho - \lambda}$$

where $\varrho(X,\, T)$ *is the entropy solution of the hydrodynamic equation with initial data*

ϱ *on the left*

λ *on the right.*

What do we have?

$$\lim_{N\to\infty} \mathbf{P}\left\{ \frac{Q(NT)}{N} > X \right\} = \frac{\varrho(X,\, T) - \lambda}{\varrho - \lambda}$$

$\rightsquigarrow$ The solution $\varrho(X,\, T)$ is the distribution of the velocity for Q.

- Shock: distribution is step function, velocity is deterministic (LLN).
- Rarefaction wave: distribution is continuous, velocity is random (e.g., Uniform for TASEP).

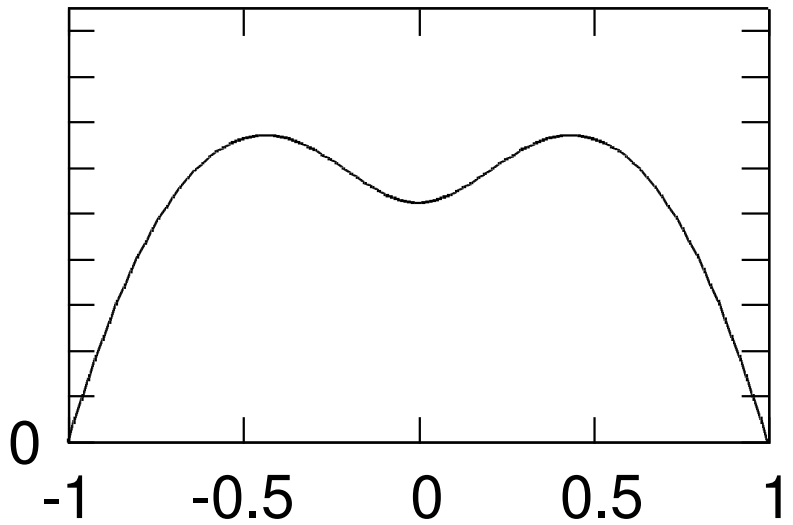
A fun model (B., A.L. Nagy, I. Tóth, B. Tóth)

$\omega_i = -1, 0, 1;$

$$
\begin{aligned}
(0, -1) &\to (-1, 0) && \text{with rate } \frac{1}{2},\\
(1, 0) &\to (0, 1) && \text{with rate } \frac{1}{2},\\
(1, -1) &\to (0, 0) && \text{with rate } 1,\\
(0, 0) &\to (-1, 1) && \text{with rate } c.
\end{aligned}
$$

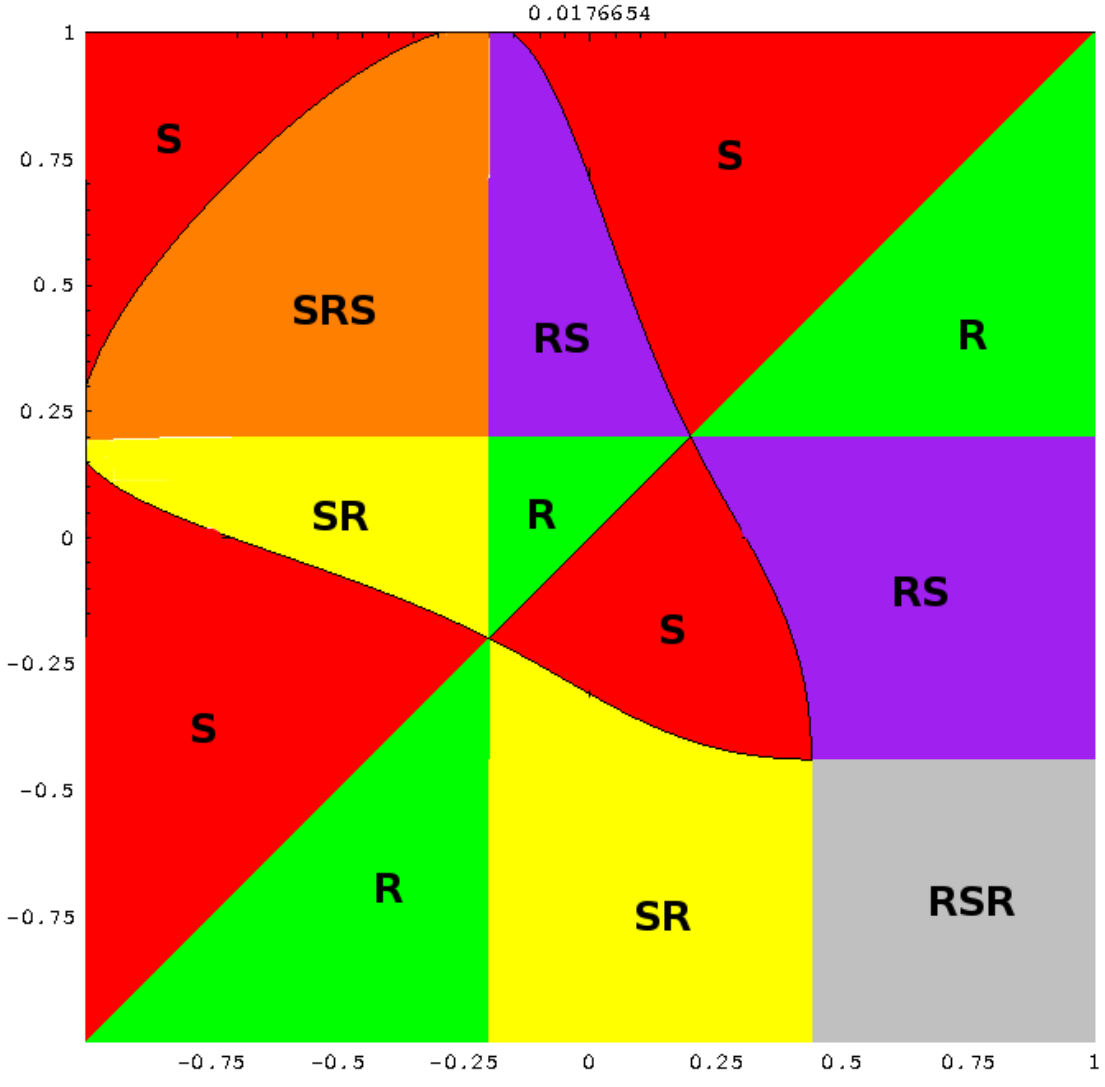
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Hydrodynamic flux $H(\varrho)$, for certain c:



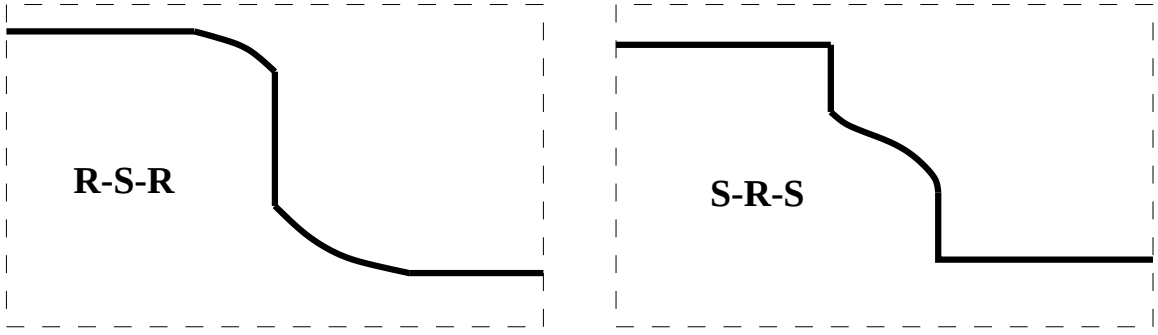
A fun model (B., A.L. Nagy, I. Tóth, B. Tóth)

Here is what can happen (**R**: rarefaction wave, **S**: Shock):



A fun model (B., A.L. Nagy, I. Tóth, B. Tóth)

Examples for $\varrho(T, X)$:



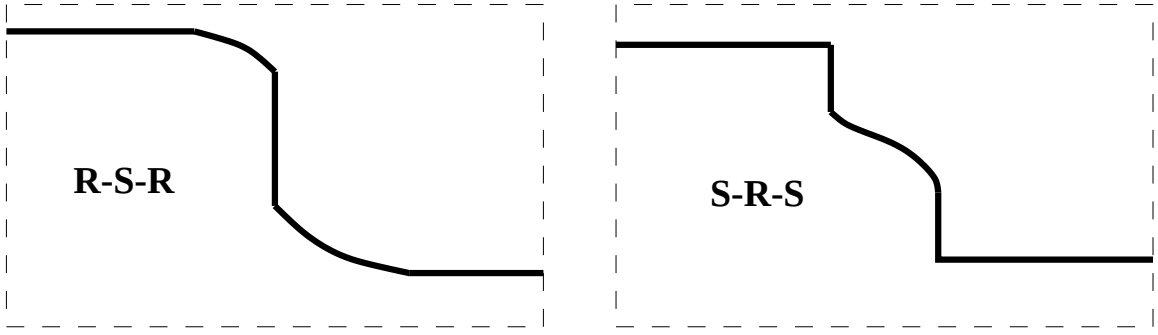
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Thank you.